

Weakness is All You Need:

Notes on Quantale Weakness
as a Unifying Generalized-Occam Principle
for Cognitive Science and AGI

EARLY ROUGH VERSION,
– ERRORS LIKELY ABOUND –
– BEWARE! –

Ben Goertzel

$$w(H) = \bigoplus_{(u,v) \in H} [\mu(u) \otimes \mu(v)]$$

$$w(H) = \bigoplus_{(u,v) \in H} [\mu(u, v)]$$

My grace is sufficient for you, for
my power is made perfect in
weakness

Jesus Christ, as reported by Paul
2 Corinthians 12

Weakness is a great thing, and
strength is nothing.... Pliancy
and weakness are expressions of
the freshness of being.

Andrei Tarkovsky

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Part I

Quantale Weakness

1 Introduction

The principle of preferring simpler explanations – often termed “Occam’s razor” – is a foundational heuristic in learning and reasoning. We here present a rethinking and reformulation of what “simpler” means, by framing it at a higher level of abstraction than has typically been done. We follow Michael Timothy Bennett’s recent work by construing simplicity as a form of “weakness” (a simpler hypothesis is one which restricts less), but we depart from Bennett by formulating weakness in a more algebraic rather than explicitly semantic way.

Bennett [Ben25] defines the *weakness* of an hypothesis as the size of its satisfaction set: the hypothesis that excludes the fewest possible worlds is deemed the weakest and hence most general. He argues compellingly that this is in some senses a more foundational way of formalizing Occam’s Razor than Minimum

Description Length principles, and shows how these emerge as a consequence of weakness under appropriate assumptions.

Building on Bennett’s approach, we formulate a broader notion of weakness. Philosophically, we ground weakness in the arguably more elementary concepts of ”relationship” and ”value,” proposing that:

*Given any system for relating and valuing entities, think of an individual binary relationship as **high-prominence** to the extent two things it joins are both valuable. A collection of binary relationships is then **weak** to the extent it contains a sufficient aggregate of high-prominence relationships.*

Conceptually: The high-prominence relationships are the obvious ones in terms of the valuations and relationships given, so if the collection gives you the high-prominence relationships, it’s not really giving you anything interesting... it’s giving you the ”obvious” (i.e. less discriminative), which is a weak thing to do.

We formalize this broader view of weakness in the category-theoretic setting using a framework of *quantales*, which gives a very general notion of relationship and valuation. We show that our categorial notion of weakness generalizes Bennett’s possible-worlds notion of weakness, but also goes significantly beyond it, and can be specialized in a variety of different ways, to yield e.g. probabilistic, linear (SVM-related), information-theoretic, hypervectorial or quantum notions of weakness.

We then explore how these different forms of weakness can be leveraged to reformulate the particulars of various learning and reasoning models and methods used in the AI field. The AI methods explored are drawn from the Hyperon AGI programme [GBD⁺23], however the core theoretical ideas here are not especially tied to Hyperon, that is just the context in which we have been working.

The conceptual picture thickens a bit when one looks at quantum systems, but one can then reformulate things roughly as,

*Given any system for relating and valuing entities, think of an individual binary relationship as **high-prominence** to the extent two things it joins are both valuable. A collection of binary relationships is then **weak** to the extent that, considered through the lens of a variety of complementary perspectives, it tends to contain a high combined weight of high-prominence relationships in the various perspectives.*

(and one then can make choices regarding how to do the compositing across different complementary perspectives).

The breadth of the discussion should not be surprising given what we’re talking about: Occam’s Razor in all its traditional glory plus more. I.e., the scope of formal applicability of our categorial weakness is basically the same as the scope of conceptual applicability of Occam’s Razor as a philosophical principle. To apply categorial weakness in a certain domain one ”simply” has to figure out the appropriate way to model that domain as a quantale.

A note on terminology: For simplicity, as the paper proceeds we will mostly refer to our "categorical weakness" simply as "weakness", using terms like "set-theoretic weakness" (for Bennett's version), "probabilistic weakness" etc. for versions obtained from specific quantales.

And a note on exposition: We would like to apologize to the reader in advance for the density of some of the material here, and the erratic nature of the text's attempts to explain itself intuitively and conceptually. The topics covered are a wild mix of highly theoretical and close-to-the-practical, and the text has been put together at a time when the AI field is advancing tremendously fast... given the time-demands of working at the cutting-edge of the AI field, we have not had time to put tremendous effort into creating simple and elegant explanations of new and complex concepts (which is of course a non-trivial thing to do even with ample time on one's hands). What we have aimed for instead is a reasonably clear and accurate presentation of a network of what we think are some interesting and important concepts, many of which we believe have significant practical implementation in the near term... and we have faith that as the ideas here and their applications are fleshed out in the coming years, a variety of more expository and didactic expositions will unfold.

1.1 Contents Preview

The treatise is organized into five Parts:

- **Part I: Quantale Weakness**

- Chapter 1 is this ver Introduction!
- Chapter 2 develops a general categorical notion of weakness, extending Bennett's set-theoretic definition, and exemplifying the new formalism via application to probabilistic, logical-entropy, fuzzy, hyper-vector, linear-SVM, and finance contexts via commutative quantales.
- Chapter 3 develops a unified "General Theory of Generalization" by showing that all inductive biases?from classic sum-of-products to arbitrary joint?valuation priors?arise from a single abstract weakness measure. We demonstrate how Peircean habits, Smolin's precedence, and Sheldrake's resonance each induce a corresponding weakness that predicts link?prediction accuracy, and show how to embed Peircean weakness into PLN message?passing to prioritize the strongest associations.
- Chapter 4 defines weakness in power-set logic quantales, constructs quantales for Constructible Duality Logic and Probabilistic Logic Networks, and demonstrates how these embed and support residuated implication.
- Chapter 5 relates set-theoretic weakness to description length through Kraft-type bounds, and shows how a particular computational model

called CoDD (Combinatory Decision Dags) yields a strict length-weakness correlation and supports efficient weakness-oriented program search.

- Chapter 6 recasts quantale weakness as a generalized MDL principle by introducing a generalized notion of "language" and showing that weakness in any quantale corresponds to minimizing description length in some generalized "language"
- Chapter 7 presents a semantics for Probabilistic Logic Networks (PLN) via graph-structured lambda theory, and in this setting explains a route to doing weakness-based inference control for PLN
- Chapter 8 extends factor-graph models from the usual Bayesian networks to commutative quantales, and applies quantale-annotated factor graphs to efficient background inference in PLN.

• **Part II: Universal & Quantum AI**

- Chapter 9 constructs WAIXI (Weak AIXI), a novel universal AI algorithm, by integrating CoDD weakness into the AIXI^{tl} framework, proving universality, asymptotic optimality, strengthened regret bounds, and demonstrating (at least semi-) practical Monte Carlo sampling approaches with hypothetical use cases in humanoid robotics and chatbot assistants.
- Chapter 10 articulates formal "mind-world correspondence principles" connecting goals and environments with the cognitive systems that can be successful relative to them. We give results mapping weakest predictive models to smallest CoDD controllers, deriving complexity bounds on program size for achieving specified behaviors.
- Chapter 11 generalizes categorical weakness to quantum operator quantales, defining a novel concept of "multiperspectival quantum weakness" (conciliating weakness with the subtleties of quantum measurement) and a "QWAIXI" quantum-AIXI^{tl} analogue
- Chapter 12 adapts Baez & Stay's reduction of algorithmic-information to Shannon-information to general quantale based information, develops algebraic quantale-entropy and mutual-information measures, and recovers Shannon, quantum, fuzzy, and hypervector information theories within this unified framework.
- Chapter 13 shows how Galois connections between MeTTa-IL rule algebras and MORK knowledge graphs yield efficient implementations of WAIXI-style dynamic programming, Q-learning, SARSA, actor-critic methods, and quantum-fused Bellman updates. Alongside theoretical interest these concepts may be of value for practical implementation of weakness-related AI computations on concurrent hardware.

• **Part III: Decision, Motivation & Consciousness**

- Chapter 14 defines weakness-based measures of causality and clarity by fusing counterfactual causal strength with categorical simplicity. One application is to quantify the notion of disentangled representation in AI systems.
- Chapter 15 introduces Weakness-Based Causal Decision Theory and Weak Updateless Decision Theory, comparing them against "classical" decision logics EDT, TDT, and UDT through exploration of scenarios such as medical decision-making, Newcomb's paradox, superrationality, and voting paradoxes.
- Chapter 16 integrates weakness into the MetaMo motivational architecture, formally adding simplicity priors to curiosity, confidence, and valuation subsystems and showing how Occam-biased weakness gradients can drive exploration and balanced resource allocation.
- Chapter 17 develops a formal model of efficient, honest, and ethical multi-agent communication by proving a Cognitive Communication Theorem. Conditions under which AI systems governed by MetaMo motivational systems will engage in ethical and non-violent communication are explored, along with complementary mechanisms leading to similar outcomes in a non-MetaMo context.
- Chapter 18 presents a formalization inspired by the Taoist notion of *wú wéi* or "no-unnecessary-effort action", portraying this classical notion in terms of geodesics in weakness-metric space, and exploring this both as a goal to be potentially included within the MetaMo framework, and as an alternate way of interpreting quantum dynamics (where "least action" is portrayed as "least effort" in a quantale constructed according to Schrodinger-bridge operations)
- Chapter 19 proposes a weakness-based aesthetic metric measuring the notion of "broad appeal" of an artistic creation, and explores possible methods for neural-symbolic music generation that maximize weakness in pursuit of a high level of novelty and coherence.
- Chapter 20 introduces a categorical weakness-based integrated-information measure that recovers Tononi's Φ as a special case, but also extends to broader measures like causal and quantum Φ . (These are viewed as interesting indicators and correlates of an intelligent system's degree or nature of consciousness rather than as definitional for consciousness.)

• Part IV: Information & Pattern

- Chapter 21 introduces the McBride derivative of weakness and shows how it enables gradient descent, differentiable proof search, and hierarchical neural-symbolic weak-model learning.
- Chapter 22 develops Quantale Pattern Theory by defining quantale patterns, emergent pattern intensities, algebraic pattern laws,

McBride derivatives of pattern intensities, and dynamical analogues including exponential-growth and delay-logistic ODE models. This represents a significant generalization of previous formalizations of "pattern theory", with an aim of making the theory more useful for practical AI applications.

- Chapter 23 presents a new formalization of the concept of "cognitive synergy" through lax-monoidal weakness functors, proving central theorems that relate synergy metrics to integrated information measures.
- Chapter 24 presents the "multiresolution weakness transform" – a rough weakness-theoretic analogue of a wavelet transform – detailing a greedy pattern-mining heuristic, efficient PathMap implementation in MORK, factor-graph realizations, and potential applications to PLN inference heuristics and neural-symbolic software for robotics.

• Part V: AI Algorithmics

- Chapter 25 proves a categorical schema theorem for evolving weak CoDD models via genetic algorithms, showing exponential convergence to maximal-weakness populations under quantale-based fitness.
- Chapter 26 gives a different perspective on evolutionary learning, showing that as population size tends to infinity then under appropriate assumptions evolution will follow an optimal-transport (wú wéi) path in weakness-metric space, and connecting this optimal flow with the optimal flow of weakness-AIXI systems
- Chapter 27 applies polynomial-functor techniques to homomorphic encryption of GA search for weak models, modeling quantales as ring-objects, encoding candidate models, and implementing secure homomorphic GA operations. A quantum-secure variation of the algorithm, applicable to GA evolution of quantum computer programs, is also developed.
- Chapter 28 presents *hierarchical hypervector embeddings*, which extend random high-dimensional vectors with tree-structured composition to capture both local and global semantics. We prove that these nested embeddings support efficient similarity decoding and algebraic operations (intersection, union, binding), and show empirically on taxonomy and few-shot tasks that they outperform flat vectors. We also relate subvector overlap to a natural weakness measure that correlates with generalization.
- Chapter 29 develops a framework that leverages predictive coding based neural networks and information geometry for learning SVM deep kernel functions – which is an interesting algorithmic construct, and also an occasion to explore the monoidal composition of weakness measures defined on different quantales to form a multiquantale-weakness-based AI approach.

- Chapter 30 constructs a categorical integration of SVM into PLN to help with inference control, detailing both linear and nonlinear kernel cases and defining combined weakness functors.
- Chapter 31 formalizes concept blending via weakness functors, presenting categorical formulations and bat-and-man blending examples that illustrate diverging blend rankings under different weakness measures.
- Chapter 32 outlines a novel approach to using weakness to drive predictive coding based neural learning. McBride derivatives are integrated with traditional gradient methods, and formal examples are given across various neural architectures?from simple feed-forward nets to Transformers, WaveFormers, and Neural-Symbolic WaveFormers.
- Chapter 33 explores financial-modeling applications of quantale information, pattern theory, and McBride calculus by defining quantale-valued stochastic processes for risk-normalized returns, modeling cryptocurrency time series, financial attention allocation, and emergent attractor dynamics.

• **Part V: Weakness Is All You Need (Except For All That Other Stuff): Leveraging Weakness for AGI**

- Summarizing and integrating many of the themes from the previous chapters, Chapter 34 applies weakness as a unifying bias across the PRIMUS cognitive architecture, detailing its role in probabilistic logical inference, evolutionary program learning, concept blending, and ECAN attention dynamics.
- Chapter 35 sketches possible lines for the development of a quantale weakness library for MeTTa, presenting quantale-based crossover and mutation implementations and MeTTa-style pseudocode for evolving weak CoDD models.

The choice of topics here reflects in some measure our own current interests and pursuits, rather than being an exhaustive or objectively priority-ranked survey of all applications of the categorical weakness formalism. Numerous further applications exist, as one would expect for a concept and construct of such fundamental importance.

Beyond the diverse particulars, however, what we aim to demonstrate here is that our categorical framing of "weakness" not only unifies a spectrum of simplicity measures in AI and information sciences, but also yields practical algorithms and architectures for learning, inference, communication, and decision-making. Many steps remain to turn the algorithms, structures and ideas presented here into practical systems; what we aim for here is to articulate a significant body of highly promising and interconnected directions.

1.2 PRIMUS and Hyperon

A number of the weakness applications pursued here are closely related to the Hyperon AGI framework or associated PRIMUS cognitive architecture ... so a brief review of these constructs will be useful at this point, before launching into technical particulars.

In the ongoing pursuit of human-level Artificial General Intelligence (AGI) and ultimately Artificial Superintelligence (ASI), a variety of approaches are currently being pursued, ranging from end-to-end deep learning to purely symbolic systems and hybrid cognitive architectures. PRIMUS/Hyperon distinguishes itself as a broadly integrative, metagraph-centered framework that brings together symbolic reasoning (e.g. PLN), probabilistic inference, sub-symbolic learning (e.g. MOSES), attention dynamics (ECAN), planning and motivational control (MetaMo) under a single shared knowledge store. Its Hyperon runtime—built on the MeTTa language and Metta-IL with GSLT semantics—provides the high-performance substrate for incremental, asynchronous module interaction and approximate stochastic dynamic programming orchestration. By unifying these diverse AI algorithms through a common cognitive-synergy paradigm and (based on the work given here) now a common weakness-based biasing framework, PRIMUS/Hyperon offers a scalable pathway for combining the strengths of today’s neural, statistical, and symbolic methods, positioning it as a extremely promising approach (in our admittedly biased view) in the roadmap toward robust AGI and the longer-term goal of ASI.

1.2.1 Weakness as a Unifying Bias in PRIMUS

PRIMUS interleaves two cognitive meta-dynamics over a shared collection of Atomspaces—stored in in-RAM MORK instances and distributed via DAS—both driven by MeTTa scripts: goal-driven cognition, in which the MetaMo motivational framework maintains and prioritizes goals to invoke planning, inference, learning and decision routines; and background cognition, where unsupervised pattern mining, concept formation and evolutionary processes continuously enrich the Atomspaces involved.

Within both of these meta-dynamics, PRIMUS leverages a suite of AI algorithms, designed for close and synergetic operation. One of the key purposes and themes of this paper is to explain how PRIMUS’s various AI algorithms can be guided by weakness as a principle, thus bringing them even closer together than has been the case in historical PRIMUS prototypes.

As we will see here, for instance,

- Probabilistic Logical Networks (PLN) – PRIMUS’s core method for handling declarative knowledge – can use weakness or SVM-guided inference to rank and select only the simplest, highest-confidence derivations for both goal-driven proofs and background proof-mining, keeping the belief graph lean and reusable.
- Evolutionary program learning, via the MOSES framework – PRIMUS’s

core method for handling procedural knowledge – extends classical genetic algorithms with Kraft-type weakness priors and adaptive schema theorems to evolve compact, high-utility codelets for control and synthesis tasks

- Concept blending – a core PRIMUS tool for creativity – can apply categorical blending routines parameterized by chosen weakness semirings to invent novel concepts, merging existing ontologies in a controlled manner
- Economic attention allocation (ECAN) – PRIMUS’s key method for managing high level resource distribution – can diffuse Short-Term and Long-Term Importance values via a commutative quantale ODE system, yielding chaotic attractor dynamics that model sustained, structured attentional focus under resource constraints.
- The MetaMo motivational framework – which guided PRIMUS’s goal-directed meta-dynamic – orchestrates these subsystems by integrating local weakness gradients into short-term importance and confidence signals, focusing effort on under-explained subgoals and flagging high-weakness regions for future objectives, thus closing the loop between inference, learning and decision-making

Across these and other cognitive algorithms, PRIMUS’s hallmark cognitive synergy emerges from embedding these diverse algorithms within an approximate stochastic dynamic programming template. As detailed in the Patterns of Cognition paper [Goe21d], each algorithm in the PRIMUS palette—from forward/backward PLN chaining to MOSES evolution, ECAN attention dynamics and concept formation—can be formalized as a discrete decision system on typed metagraphs, optimized via greedy or stochastic DP processes and efficiently implemented through hylomorphisms and chronomorphisms. The weakness framework provides the quantale-based Occam simplicity measures that ground these DP formulations, yielding pragmatic second- and third-order approximations to higher-order probabilistic reasoning suitable for real-time AGI under resource limitations.

1.2.2 Cognitive Synergy and the Structuring Role of Weakness

A central organizing principle in PRIMUS is *cognitive synergy*: the notion that diverse reasoning and learning processes—each with its own inductive biases and computational trade-offs—can compensate for one another’s blind spots, yielding emergent problem-solving capabilities greater than the sum of individual modules. Realizing such synergy in practice requires coordinating when and how modules share partial results, hypotheses, and critiques. Here the concept of *weakness*, defined as quantitative measures of under-determination or under-support for a candidate hypothesis, provides a unifying scaffold. Weakness scores guide attention selection (focusing resources on the most uncertain regions of the graph), priority scheduling (interleaving fast heuristic scans with deeper logical proofs), and belief revision (deciding when to retract or reinforce inferences). Because weakness is computed uniformly across modules, it

becomes straightforward to tune synergy pathways, compare alternative orchestration strategies, and formally analyze convergence properties.

1.2.3 The Hyperon Implementation Infrastructure

To bring PRIMUS from design to scalable implementation, we developed *Hyperon*, a high-performance metagraph runtime and toolkit engineered for cognitive architectures. Hyperon encapsulates the shared knowledge graph in a memory-mapped, versioned store supporting atomic micro-transactions, fine-grained subscriptions, and lock-free concurrent updates. Its event-driven execution engine dispatches updates to registered cognitive operators—each running in its own lightweight coroutine—with prioritization driven by weakness and salience metrics. Hyperon also provides a rich set of built-in utilities for graph querying, incremental consistency checking, and distributed deployment, enabling researchers to experiment with large-scale PRIMUS networks on multi-core and cluster-based infrastructures with minimal boilerplate.

1.2.4 MeTTa, Metta-IL, and Exploration of Weak-Model Languages

At the heart of Hyperon lies the *MeTTa* meta-programming language and its underlying *Metta-IL*, whose formal semantics are grounded in Graph-Structured Lambda Theory (GSLT). MeTTa’s homoiconic syntax treats graph nodes, hyperedges, and patterns as first-class values, allowing concise expression of PRIMUS algorithms—whether defining a weakness propagation operator or writing a custom attention policy—in a uniform language. The flexibility of MeTTa’s parsing and macro facilities further permits rapid prototyping of domain-specific dialects tailored to particular forms of uncertainty. For example, in this work we introduce the *Constructive Duality-Driven Domain* (CoDD) language atop MeTTa, enabling seamless experimentation with weak-model learning formalisms by defining new truth-value constructors, evidence-aggregation rules, and inference schemata without modifying the core runtime. This extensibility makes MeTTa an ideal substrate both for implementing PRIMUS’s cognitive synergy mechanisms and for theoretical exploration of alternative weak reasoning paradigms.

2 A General Categorical Notion of Weakness

We now review Bennett’s notion of weakness, and explain how it extends to a more general notion of weakness formulated via quantales. We will save the quantum-theoretic extensions for Chapter 11, as these involve some additional nuances.

2.1 Bennett’s Notion of Weakness

Bennett’s notion of “weakness” (which in our categorical perspective becomes “set-theoretic weakness”) is a formalized way of choosing, among several expla-

nations that all fit observed data, the one that makes the fewest assumptions. Imagine you have some observed facts, and you want to pick a rule or a hypothesis that explains those facts. You could choose a very specific rule that only applies in a tiny number of situations, or you could choose a more general rule that applies in many situations. Bennett observes that the most “weak” rule is the one that rules out the fewest possible worlds; equivalently, it is the one that applies in the largest set of possible scenarios consistent with the data. By picking the “weakest” explanation, you are following what we would call an Occam-style principle: prefer the explanation that generalizes best because it makes the fewest extra commitments beyond the observed evidence.¹

The key insight here is that via formulating weakness in an axiomatic way focused on semantics, one comes up with a more fundamental depiction of “Occam’s razor style heuristics” than the “minimum description length” approach more commonly taken. MDL approaches emerge as a way of achieving or approximating weakness under appropriate assumptions.

Formal Definition. Let U be a finite universe of *possible worlds*. A *hypothesis* (or *explanation*) is a subset $H \subseteq U$ of worlds in which the hypothesis is true. Equivalently, we view a hypothesis as a condition that rules out all worlds *not* in H .

1. Define the *weakness* of a hypothesis H by

$$w(H) = |H|$$

where $|H|$ is the number of worlds in H . A larger $w(H)$ means the hypothesis is *weaker*, since it excludes fewer possible worlds.

2. Suppose we have a collection of hypotheses $\{H_i\}_{i \in I}$ that are *all* consistent with some observed data D ; that is, each H_i contains at least those worlds in which D holds. Then Bennett’s criterion is to choose the hypothesis H^* satisfying

$$H^* = \arg \max_{H_i \supseteq D} w(H_i).$$

In other words, among all hypotheses that explain D , pick the one that has the largest cardinality $w(H_i)$.

Properties.

- **Monotonicity under Refinement.** If a hypothesis H_1 is a *refinement* of H_2 (meaning $H_1 \subseteq H_2$), then

$$w(H_1) = |H_1| \leq |H_2| = w(H_2).$$

¹As we will note in section 2.6 below, this is our own take on the relation between Bennett’s weakness and the classical “Occam’s Razor” idea, and Bennett’s own perspective is slightly different.

Thus more specific explanations (which exclude more worlds) are strictly *stronger* and hence have smaller weakness.

- **Lattice Bounds.** The collection of all subsets of U forms a Boolean algebra (and in particular a lattice) under intersection and union. For any two hypotheses $H_1, H_2 \subseteq U$,

$$H_1 \cap H_2 \subseteq H_1, H_2 \implies w(H_1 \cap H_2) \leq \min\{w(H_1), w(H_2)\},$$

and

$$H_1 \cup H_2 \supseteq H_1, H_2 \implies w(H_1 \cup H_2) \geq \max\{w(H_1), w(H_2)\}.$$

Hence $w(\cdot)$ is *order-preserving* (isotone) on this lattice.

- **Connection to MDL instantiations of Occam’s Razor.** Bennett shows that selecting the maximum-weakness hypothesis is equivalent to choosing the hypothesis of *minimum description length* when one places a constraint on the “vocabulary size” of descriptions. In particular, if there are only finitely many distinct hypotheses whose descriptions have length at most L , then the one of greatest weakness among those corresponds to the one of minimal description length. This is how Bennett’s weakness criterion recovers the standard MDL Occam-style principle [Ris78]: shorter descriptions (weaker hypotheses) are preferred. We will elaborate on this aspect in Chapter 5 below.
- **Statistical Generalization.** Suppose there is an underlying (unknown) probability distribution $\Pr$ on U . Then the probability mass of a hypothesis H is

$$\Pr(H) = \sum_{u \in H} \Pr(u).$$

If we require that each hypothesis H_i covering the data D also satisfies

$$\Pr(H_i) \geq p_0,$$

for some threshold p_0 , then among those H_i , the one of maximal weakness $w(H_i)$ will also tend to have larger probability $\Pr(H_i)$, assuming worlds in U are roughly equally likely. In Bennett’s original work, this is made precise via bounds showing that with high probability, a sample-derived hypothesis whose weakness is large will generalize better to unseen data. We will explore the application of weakness to probability via a closely related but slightly different route below.

2.2 Probabilistic Weakness as a Generalization of Bennett’s Measure

We will be generalizing Bennett’s weakness quite broadly below, but let us begin with the most straightforward extension, to a natively probabilistic form of weakness.

To generalize this to a setting where worlds have different likelihoods or importances, we have to switch from looking at individual worlds to looking at pairs of worlds.

To see why, consider that the weakest hypothesis is supposed to be the one that "restricts the least" – and when we say "restrict the least," what we really mean is "collapse as few important distinctions as possible." Under a uniform distribution, counting single worlds left "possible" captures that: you simply count how many worlds survive your filter. But once worlds carry different probabilities, a single-world count no longer tells you how much overall possibility you're throwing away—because throwing out a high-probability world is far more restrictive than throwing out a low-probability one.

What we can do instead is view each pair (u, v) of worlds as occurring with joint probability $p(u)p(v)$ under independent draws. Summing those probabilities over H yields

$$w_p(H) = \sum_{(u,v) \in H} p(u)p(v),$$

which reduces to $|H|$ when p is uniform.

What we are asking here is: "If I draw two independent samples from my belief distribution, what is the chance that they end up in the same collapsed equivalence class?" That probability directly measures how many and how significant the collapsed distinctions are, in exactly the sense of "restricting the least." Minimizing that pair-collision probability guarantees you're preserving as much of the total probability mass of distinctions as possible. In other words, you're keeping your model from merging two independently likely scenarios, which is exactly what you want if your goal is to restrict your hypothesis space as little as you can.

This provides a smooth, interpretable simplicity bias that can be estimated and optimized in statistical settings. We will revisit this concept below as a specialization of a broader notion of quantale-weakness, but it also has a simple meaning and definition on its own.

2.3 Weakness, Logical Entropy and Graphropy

Next we clarify and explore some relationships between weakness and a few other measures of the informativeness of mappings. This will be interesting and useful in itself, and will also help us find the right way to generalize weakness into a category-theoretic setting.

The core relationship we want to look at here relates to *logical entropy* [Ell13] – and then a generalization of this notion called *graphropy* [Goe19].

2.3.1 Weakness and Logical Entropy

Logical entropy is a different concept from the more familiar Shannon entropy; it quantifies the probability that two independent draws from a distribution yield different outcomes. more formally:

Definition. Let $\Pi = \{B_1, \dots, B_m\}$ be a finite partition of a sample space and write $p(B_j)$ for the probability of block B_j . Then the *logical entropy* of Π is

$$h(\Pi) = 1 - \sum_{j=1}^m p(B_j)^2.$$

The connection with weakness should be conceptually somewhat clear – *weakness* measures, in a rough sense, how *general* or *indistinct* a rule or condition is ... and one can view this as related to, specifically, the probability that two draws fall into the *same* block of a partition or the same satisfaction set of a rule.

More precisely, we may say that:

- If a partition (or rule) is *maximally weak*, then everything is in one block – so two draws surely lie in the same block, and logical entropy (the chance they differ) is nzero.
- If a partition is *maximally strong*, then each element is isolated, so two draws surely differ, and logical entropy is near its maximum.

Formally, it is not hard to see that for any finite partition or rule,

$$\text{weakness} + \text{logical entropy} = 1.$$

Thus, in the context of partitions, these are basically two ways of looking at the same thing.

From Probabilistic Weakness to Partition-Weakness To understand why this identity makes sense, we first elaborate the notion of "partition weakness" as a special case of the weakness of probability distributions.

In the continuous or distributional setting, the general notion of *weakness* measures how large an event H is under a probability distribution. Concretely, if (Ω, p) is a probability space and $H \subseteq \Omega$ is any measurable event, then

$$w(H) = \Pr_{\omega \sim p} [\omega \in H].$$

When we draw two independent samples $\omega, \omega' \sim p$, the product space $(\Omega \times \Omega, p \otimes p)$ inherits the same definition:

$$w(H) = \Pr_{(\omega, \omega') \sim p \otimes p} [(\omega, \omega') \in H].$$

Theorem 1. Let Ω be a finite set with probability measure p , and let $\Pi = \{B_1, \dots, B_m\}$ be a partition of Ω . Define the diagonal event

$$H_\Pi = \{(\omega, \omega') \in \Omega \times \Omega : \exists j, \omega \in B_j \wedge \omega' \in B_j\}.$$

Then the weakness of H_Π under the product measure equals the partition-weakness:

$$w(H_\Pi) = \Pr_{(\omega, \omega') \sim p \otimes p} [(\omega, \omega') \in H_\Pi] = \sum_{j=1}^m p(B_j)^2.$$

Proof. By independence,

$$\Pr(\omega \in B_j \wedge \omega' \in B_j) = p(B_j) \times p(B_j) = p(B_j)^2.$$

Since the blocks $B_j \times B_j$ are disjoint and cover H_Π , summing over j gives

$$w(H_\Pi) = \sum_{j=1}^m \Pr(\omega \in B_j \wedge \omega' \in B_j) = \sum_{j=1}^m p(B_j)^2,$$

which is exactly the definition of partition-weakness. $\square$

Thus *partition-weakness* is nothing more than the general probabilistic weakness of the single event "two independent draws fall in the same block," under the IID assumption. It does not rely on the finite-universe counting approach of Bennett's original weakness, but emerges naturally from the distributional setting.

It then follows immediately that

Theorem 2. *For any finite partition Π ,*

$$w(H_\Pi) + h(\Pi) = 1.$$

Proof. By definition,

$$h(\Pi) = 1 - \sum_{j=1}^m p(B_j)^2 = 1 - w(H_\Pi).$$

Rearranging gives $w(H_\Pi) + h(\Pi) = 1$, as claimed. $\square$

2.3.2 Graphentropy and Weakness

The relation between weakness and logical entropy extends naturally beyond the case of partitions, to the situation of "distinction graphs" indicating a given perceiving system's capability to distinguish between two entities asymmetrically [Goe19]. We review the notions of distinction graphs and graphentropy, then show that graphentropy is exactly the *weakness* of the non-distinction event in the product sample space.

2.3.3 Distinction Graphs

A *distinction graph* is a simple graph that encodes which pairs of states an observer *cannot* distinguish.

Definition 1 (Distinction Graph). *Let U be a finite set of states. A distinction graph is an undirected graph*

$$G = (U, E),$$

where each edge $(u, v) \in E$ indicates that the observer cannot distinguish state u from state v . We assume reflexivity, so $(u, u) \in E$ for all $u \in U$.

Thus G divides states into indistinguishability classes, but need not be transitive (so the relation may not form a true partition).

2.3.4 Graphropy

Graphropy measures how coarse (weak) the observer's distinctions are by computing the probability that two independent draws from U land in an indistinguishable pair.

Definition 2 (Graphropy). *Let $G = (U, E)$ be a distinction graph on a finite set U . The graphropy of G is*

$$h(G) = \frac{|E|}{|U|^2},$$

where $|E|$ counts each reflexive edge (u, u) and each undirected edge (u, v) once.

Equivalently, if ω, ω' are drawn uniformly and independently from U , then

$$h(G) = \Pr[(\omega, \omega') \in E].$$

Graphropy as Weakness We now relate graphropy to the general notion of weakness under a product distribution.

Theorem 3. *Let $G = (U, E)$ be a distinction graph on finite U and equip $U \times U$ with the uniform distribution. Define the non-distinction event*

$$H_G = \{(\omega, \omega') \in U \times U \mid (\omega, \omega') \in E\}.$$

Then the general weakness of H_G under the product measure equals the graphropy of G :

$$w(H_G) = \Pr_{(\omega, \omega') \sim \text{Unif}(U \times U)}[(\omega, \omega') \in H_G] = h(G).$$

Proof. By definition of weakness in a probability space,

$$w(H_G) = \Pr((\omega, \omega') \in H_G) = \frac{|H_G|}{|U|^2}.$$

But $H_G = E$, so $|H_G| = |E|$, and hence

$$w(H_G) = \frac{|E|}{|U|^2} = h(G).$$

□

This result shows that graphropy is simply the probability that two independent samples are not distinguished by the observer, i.e. the *weakness* of the non-distinction hypothesis. A high graphropy indicates a very weak (coarse) distinction graph, while a low graphropy reflects a strong (fine) set of distinctions.

2.4 Lifting Weakness to a Categorical Setting

The notion of weakness is already very broad conceptually, however its specific mathematical formalization is tied to elementary set-theoretic constructs, which makes the elaboration of some conceptually natural applications unnecessarily laborious on a technical level.

With this in mind, we ask here: Can the notion of weakness be lifted to *category theory*, where objects and morphisms are abstracted beyond sets and functions? That is: We want to keep the intuitive concept the same, but formalize it in a more abstract way that will make it easier for us to see the applications in different sorts of domains and to different sorts of AI systems.

We will start with ordinary categories and then move on to enriched categories and quantales, which turn out to give the most useful version of weakness we have found so far.

In a categorical framework, conceptually we replace "rules that match patterns" by "objects representing propositions or inference rules", and we say that one object is weaker than another if there exists a morphism from the weaker object to the stronger one. Intuitively:

- Objects in our category are *contexts* or *conditions* under which a rule applies.
- A morphism $f : A \multimap B$ means "whenever condition A holds, condition B also holds." Hence A is *weaker* (less restrictive) than B because B implies A in reverse, or equivalently, A factors through B .
- In this way, *weakness becomes the existence of a morphism*. The weaker object can feed into the stronger object via a morphism.

Thus "universal weakness" corresponds to an *initial object* (which has a unique arrow to every other object), and "absolute strength" corresponds to a *terminal object* (to which every object admits a unique arrow). By framing weakness in categorical terms, we obtain a powerful, abstract language to talk about generality and specificity that applies uniformly across many domains: sets, topological spaces, logical theories, and so on.

2.4.1 Defining Weakness via Morphisms

Definition 3 (Weakness Ordering). *Let C and D be objects in $\mathcal{W}$. We say C is weaker than or equal to D , denoted*

$$C \preceq D,$$

if and only if there exists a morphism $f : C \rightarrow D$ in $\mathcal{W}$. Dually, D is stronger than or equal to C if $C \rightarrow D$.

Remark. *Since morphisms compose and identities exist, $\preceq$ is a preorder on $\text{Ob}(\mathcal{W})$. We may pass to the associated partial order by quotienting out isomorphisms if desired.*

2.4.2 Initial Object as Maximally Weak

Definition 4 (Universal Weakness). *An object $0 \in \text{Ob}(\mathcal{W})$ is universally weak if it is initial; that is, for every $C \in \text{Ob}(\mathcal{W})$, there is a unique $! : 0 \rightarrow C$. In symbols:*

$$\forall C, \exists ! (! : 0 \rightarrow C).$$

Universally weak means "applies to every situation" – it is the least restrictive condition. Dually, a terminal object is "universally strong", matching no more than all other objects do.

2.4.3 Simple Examples

Example 1 (Sets and Subsets). *Let $\mathcal{P}(X)$ be the power-set of some fixed set X . Define a category $\mathcal{W}_X$ whose objects are subsets $A \subseteq X$, and a morphism $f : A \rightarrow B$ exists iff $A \subseteq B$ (choose exactly one arrow when that condition holds). Then weakness $A \preceq B$ means $A \subseteq B$. The initial object is $\emptyset$, since $\emptyset \subseteq B$ for every $B \subseteq X$. The terminal object is X .*

Example 2 (Types and Coercions). *In a programming-language category, objects are types, and morphisms are implicit coercions or subtyping relations. A type T_1 is weaker (more general) than T_2 if there is a coercion $T_1 \rightarrow T_2$. The void type (which has no inhabitants) is often initial, and a universal top type (e.g. *Object*) is terminal.*

2.4.4 Characterizing Weakness in a Category with Subobjects

In a more general category $\mathcal{C}$ with *subobjects*, we can interpret conditions as subobjects of a fixed universe object U . Then:

Definition 5 (Category of Subobjects). *Let U be a fixed object in a category $\mathcal{C}$. The subobject category $\text{Sub}_{\mathcal{C}}(U)$ has:*

- *Objects: Monomorphisms $m : A \hookrightarrow U$ (each A is a subobject of U).*
- *Morphisms: Given $m_1 : A_1 \hookrightarrow U$ and $m_2 : A_2 \hookrightarrow U$, a morphism $f : (A_1 \hookrightarrow U) \rightarrow (A_2 \hookrightarrow U)$ is any arrow $f : A_1 \rightarrow A_2$ in $\mathcal{C}$ such that $m_2 \circ f = m_1$. In effect, A_1 factors through A_2 .*

Composition and identities come from $\mathcal{C}$.

Proposition 1. *In $\text{Sub}_{\mathcal{C}}(U)$, we have $(A_1 \hookrightarrow U) \preceq (A_2 \hookrightarrow U)$ if and only if A_1 is a subobject of A_2 (i.e. there exists a monomorphism $A_1 \hookrightarrow A_2$ such that the triangle commutes).*

Proof. By definition of morphisms in $\text{Sub}_{\mathcal{C}}(U)$, $A_1 \rightarrow A_2$ must satisfy $A_2 \hookrightarrow U \circ f = A_1 \hookrightarrow U$. Since both $A_1 \hookrightarrow U$ and $A_2 \hookrightarrow U$ are monos, this forces f itself to be a monomorphism $A_1 \hookrightarrow A_2$. Conversely, any such monic f yields a valid arrow in the subobject category. $\square$

2.4.5 Weakness as Factorization

The following observation is elementary but worth calling out:

Theorem 4. *Let $\mathcal{C}$ be any category with a chosen object U , and consider $\text{Sub}_{\mathcal{C}}(U)$ with objects $m_A : A \hookrightarrow U$, $m_B : B \hookrightarrow U$. Then A is weaker than B (in the sense of the subobject category) if and only if A factors through B , i.e. there is a morphism $f : A \rightarrow B$ in $\mathcal{C}$ such that $m_B \circ f = m_A$. Moreover, the initial object is the zero subobject $0 \hookrightarrow U$ (if it exists), and the terminal object is $U \hookrightarrow U$ itself.*

Proof. By the definition of morphisms in $\text{Sub}_{\mathcal{C}}(U)$, $A \preceq B$ precisely means there exists $f : A \rightarrow B$ with $m_B \circ f = m_A$. Such an f is necessarily a monomorphism (since m_A is mono and m_B is mono), so A is literally included in B as a subobject. The initial object $0 \hookrightarrow U$ factors through every $A \hookrightarrow U$ via the unique arrow $0 \rightarrow A$, and $U \hookrightarrow U$ is terminal since every subobject $A \hookrightarrow U$ factors uniquely through the identity on U . $\square$

Remark. *In many familiar categories (e.g. **Set** or any topos), 0 is the empty set or initial object, and U is the whole universe. This matches the intuition that the empty condition is the weakest (it holds nowhere), and the universal condition holds everywhere and is strongest.*

Overall, by formulating weakness as the existence of a morphism in a category of conditions or subobjects, we find that the abstract categorical framework naturally captures the ordering of rules or contexts by generality. An initial object is universally weak (applies everywhere), and a terminal object is universally strong. This port of weakness into category theory is both conceptually meaningful – because it matches our intuitive notion of “applies to more cases” – and useful, because it applies uniformly across diverse fields: logic, type theory, toposes, and beyond.

2.5 Enriched-Category Weakness.

Next we may articulate the generalized notion of weakness in the context of *enriched categories*, which have a notion of (e.g. quantitative) measure or valuation associated with them. This turns out to open up a host of practical applications in a very natural way.

The formulation of weakness in this setting looks a bit more like the logical entropy or probabilistic weakness than like the traditional weakness, in that it involves pairs of objects. We will reiterate why it comes out this way below, after going through the formal details.

Let $\mathcal{C}$ be a category enriched over a *commutative quantale* $(V, \otimes, e, \leq)$, with object set U . Suppose we have a *measure* $\mu : U \rightarrow V$. Then the *weakness* of any subobject (subgraph) $H \subseteq U \times U$ is defined by

$$w(H) = \bigoplus_{(u,v) \in H} [\mu(u) \otimes \mu(v)],$$

where $\oplus$ is the quantale join in V .

Quantale Weakness and Pairwise Distinctions An intuitive justification for this definition would be as follows (following the justification of the definition of probabilistic weakness given above). In the quantale formulation, weakness measures how many—and how serious—binary distinctions a relation fails to make. This naturally leads to summing over pairs of objects rather than singletons:

$$w(H) = \bigoplus_{(u,v) \in H} [\mu(u) \otimes \mu(v)],$$

where $H \subseteq U \times U$ is the relation of undistinguished pairs, $\mu : U \rightarrow V$ is an object valuation, and $\oplus, \otimes$ are the join and multiplication in the commutative quantale V .

- **Distinctions are pairwise.** Each time the relation collapses two objects u and v , it removes exactly that distinction. Counting singletons cannot track which objects are conflated.
- **Joint importance.** Weighting each pair by $\mu(u) \otimes \mu(v)$ ensures that confusing two high-value objects contributes more to overall weakness than confusing two low-value ones.

Thus, whenever we need to know which distinctions are lost and how important they are, we work over pairs. The quantale framework then gives a flexible, uniform way to assign and aggregate those pairwise weights.

This brings us back to the rough verbal gloss given at the start of the paper, i.e.

*Given any system for relating and valuing entities, think of an individual binary relationship as **high-prominence** to the extent two things it joins are both valuable. A collection of binary relationships is then **weak** to the extent it contains a sufficient aggregate of high-prominence relationships.*

Formal Reconciliation with Bennett’s Original Weakness To articulate the reduction of quantale weakness to the original Bennett weakness: Basically we are looking at the special case where we choose the real-addition quantale

$$V = (\mathbb{R}_{\geq 0}, \times, 1, \leq), \quad \otimes = \times, \quad \oplus = +, \quad \mu(u) \equiv 1,$$

Here we obtain

$$w(H) = \sum_{(u,v) \in H} 1 = |H|,$$

and we exactly reproduce Bennett’s “count the remaining worlds” measure of weakness. Thus our enriched-category formulation subsumes the classical set-theoretic notion as the special case of unweighted counting, while extending seamlessly to arbitrary measures of weight and join in any commutative quantale.

2.6 Relation Between Bennett’s and Occam’s Razors

How these various mathematical notions relate to the broader philosophical notion of ”Occam’s Razor” is an interesting question, verging more into philosophy and history rather than mathematics.

What William of Occam originally wrote (in Latin) was ”plurality should not be posited without necessity” – which is generally interpreted conceptually as the principle that, among explanations that equally fit the data, one should prefer the hypothesis that introduces the fewest unnecessary assumptions.

Bennett contrasts the three approaches:

- *Simplicity-maxing* (simp-maxing) refers to selecting the hypothesis with the shortest descriptive length, measured for example by bit-count or program size. This strategy penalizes every additional symbol equally, without regard for which worlds it rules out.
- *Weakness-maxing* (w-maxing) refers to selecting the hypothesis that imposes the weakest constraints, i.e. that admits the largest set of possible worlds. We might call this set-theoretic weakness criterion ”Bennett’s Razor” – it is a semantic measure of parsimony: two hypotheses of equal length can differ in how many worlds they exclude.
- *Scale-maxing* refers to preferring methods that leverage increased computational resources, data, or model capacity. It is a brute-force approach, orthogonal to any simplicity measure.

One could also identify other such principles e.g.

- *Lulz-maxing* (aka ”Zarathustra’s Razor”) refers to preferring methods that are maximally entertaining to some particular entity, for instance Zarathustra

Although all four meta-approaches guide hypothesis selection, one may say that simp-maxing and w-maxing instantiate distinct senses of Occam-ish ”simplicity”: one over symbols, the other over possibilities, while scale-maxing relies on sheer scale rather than parsimony .. and lulz-maxing does something else different.

Each of these four meta-approaches can be encompassed within quantale weakness if one sets up the quantale appropriately. Let $(Q, \leq, \otimes)$ be a commutative quantale of constraint costs, and let each constraint c carry a weight $w(c) \in Q$. For any hypothesis H , its weakness is defined by

$$weakness(H) = \bigvee_{c \in H} w(c),$$

where $\bigvee$ is the quantale join. By selecting different quantales and weight functions, one recovers each of the following meta-approaches:

- *Simplicity-maxing.* Take $Q_{\text{simp}} = (\mathbb{N} \cup \{\infty\}, \leq)$ with monoid operation $+$, and set $w(c) = 1$ for each symbol or token in the description. Then

$$\text{weakness}(H) = \sum_{\text{symbol} \in H} 1$$

is exactly the description length, so minimizing weakness selects the shortest hypothesis.

- *Weakness-maxing.* Let $Q_{\text{weak}} = (2^W, \subseteq)$ be the lattice of sets of possible worlds, and set $w(c) = \{\text{worlds satisfying } c\}$. Then

$$\text{weakness}(H) = \bigcap_{c \in H} w(c)$$

is the set of worlds admitted by H , and choosing H to minimize the size of this intersection (or equivalently maximize its cardinality) yields the least constraining hypothesis.

- *Scale-maxing.* Use $Q_{\text{scale}} = (\mathbb{R}, \leq)$ and define $w(M) = -\text{resource}(M)$ for each model or mechanism M . Then

$$\text{weakness}(M) = \max_{c \in M} (-\text{resource}(c))$$

encodes negative resource usage, so minimizing weakness picks the model with maximal resource allocation.

- *Lulz-maxing.* Choose $Q_{\text{lulz}} = (\mathbb{R}, \leq)$ and set $w(c) = -\text{fun}(c)$ for an entertainment value function fun . Then minimizing

$$\text{weakness}(H) = \bigvee_{c \in H} (-\text{fun}(c))$$

selects the hypothesis that maximizes entertainment value.

However, as we will show in Section ??, there is a quantale isomorphism between Q_{weak} and Q_{simp} . The set-theoretic weakness measure and the description-length measure coincide, exhibiting a formal isomorphism between weakness-maxing and simplicity-maxing, which is conceptually because these are two different ways of manifesting the conceptual notion of Occam's Razor. The general notion of quantale weakness, on the other hand, maintains something of the general structure of Occam's Razor, but can be used much more broadly than any reasonable interpretation of William of Occam's original (appropriately terse) formulation.

2.7 Syntactic versus Semantic Weakness: Intuitive Recap

While we will wait till Section ?? to formally articulate the relationship between Bennett's weakness (w-max) and MDL-style weakness (simp-max), it may be worth zooming out a bit and giving a brief conceptual re-cap of this point.

The MDL/ simp-max approach interprets Occam’s Razor syntactically, using measures like:

- The number of symbols required for description (Minimum Description Length)
- The parameter count in a model
- The perceived mathematical complexity

These measures are fundamentally *syntactic*—they depend on our choice of representation rather than the actual content of the explanation.

Bennett’s concept of “weakness” (w-max) represents a fundamental shift in perspective. Rather than asking “how concisely can we describe this?”, weakness asks: **“how many possibilities does this explanation leave open?”**

Going back to intuitive basics, consider two explanations for why a light turned on:

Explanation A: “Someone flipped the switch”

- Excludes millions of alternative scenarios
- Highly specific $\Rightarrow$ STRONG

Explanation B: “Something happened involving electricity”

- Preserves many possibilities (switch, timer, sensor, surge, etc.)
- Highly general $\Rightarrow$ WEAK

The crucial insight: Explanation B is “weaker” precisely because it *commits to less* about the actual state of affairs.

That is: Bennett’s weakness is commitment-based rather than symbol-based.

Consider these equivalent propositions:

- “The switch was flipped”
- “El interruptor fue accionado”
- “Der Schalter wurde umgelegt”
- “01010011 01110111 01101001 01110100 01100011 01101000”

Traditional MDL assigns different complexity values based on encoding length. Weakness recognizes that these make *identical commitments* about the world—they eliminate the same set of alternative possibilities.

To see intuitively why these come out to basically the same thing – presaging the math we’ll go through later – consider what happens when we describe something:

1. **Short descriptions must be vague:** With only n bits available, we can distinguish at most 2^n possibilities. Any description must therefore leave many options open.
2. **Precise descriptions must be long:** To uniquely identify one option among N possibilities requires at least $\log_2(N)$ bits.

Principle 1 (The Fundamental Trade-off). *Specificity costs bits.*

So the proportionality between simp-max and w-max emerges naturally, and the following crude argument (to be run through later in more detail) is elementary:

- A hypothesis with weakness W (leaving W possibilities open)
- Can be uniquely identified using $\log_2(N/W)$ bits, where N is the total number of possibilities
- Therefore: high weakness $\Leftrightarrow$ few bits needed $\Leftrightarrow$ short description

More formally:

$$\text{weakness}(H) = W \tag{1}$$

$$\text{MDL}(H) \approx \log_2\left(\frac{N}{W}\right) = \log_2(N) - \log_2(W) \tag{2}$$

Thus MDL and $\log(\text{weakness})$ differ only by a constant.

One of Bennett's key contributions in his thesis was showing this relationship is not approximate but exact under reasonable assumptions:

1. **Finite description space:** With symbols from a finite alphabet, there are exactly V^L programs of length L
2. **Unique assignment:** The weakest hypothesis among those expressible with $\leq L$ symbols necessarily claims the shortest available description
3. **Information-theoretic necessity:** You literally cannot encode a stronger (more specific) hypothesis using fewer bits—the information theory bound prevents it

Remark (The Parking Lot Analogy). *Think of descriptions as parking spots and hypotheses as vehicles of different sizes. The largest vehicles (weakest hypotheses) necessarily occupy the most accessible spots (shortest descriptions) by geometric necessity.*

This equivalence is more than a mathematical curiosity:

- **Algorithmic flexibility:** Searching for simple descriptions (MDL) automatically yields weak hypotheses

- **Computational choice:** Searching for weak hypotheses automatically yields simple descriptions
- **Implementation freedom:** One can optimize whichever is computationally more tractable

In a way the universe appears to have this connection built into its information-theoretic fabric: entities that are easiest to describe are precisely those that make the fewest assumptions about the world. Occam’s Razor succeeds not by philosophical decree but because information theory enforces it—you cannot have a short description without high weakness, nor high weakness without a short description.

2.8 Natural Compositionality and Quantale Weakness

Continuing the intuitive recap of the formal constructs we’ve presented, one may also think about weakness *combinatorially*. For systems with multiple components:

- **Traditional approach:** Requires ad hoc methods to aggregate component complexities
- **Weakness approach:** The weakest explanation of the whole naturally emerges from combining the weakest explanations of the parts

This compositionality is elegantly captured by the quantale formulation – which we have expressed intuitively as

*Given any system for relating and valuing entities, think of an individual binary relationship as **high-prominence** to the extent two things it joins are both valuable. A collection of binary relationships is then **weak** to the extent it contains a sufficient aggregate of high-prominence relationships.*

The initially opaque notion of “high-prominence relationships” becomes perhaps clearer when reformulated:

Remark (Prominent vs. Obscure Relationships). *“Prominent” relationships are connections between elements that both have high measure values—they stand out or are prominent given the base measure μ . A hypothesis is weak when it primarily consists of such prominent relationships—it’s highlighting connections between already-significant elements rather than revealing hidden structure.*

This perspective directly connects to:

- **Probability:** Weak explanations have high probability (many instantiations exist)
- **Information:** Weak explanations convey little information (minimal surprise)

The central equation:

$$w(H) = \bigoplus_{(u,v) \in H} [\mu(u) \otimes \mu(v)]$$

translates to a straightforward algorithm:

1. Enumerate all pairs (u, v) related by hypothesis H
2. Measure the “prominence” of each element of the pair-set: $\mu(u), \mu(v)$
3. Combine these measures: $\mu(u) \otimes \mu(v)$
4. Aggregate via the supremum operation: $\bigoplus$

Intuitive interpretation: A hypothesis is weak if it primarily connects elements that were already prominent (had high measure values) individually.

Weakness-based approaches to AI, which will consume most of this book, are then interesting because they optimize for:

1. **Minimal assumptions:** Avoiding unnecessary commitments
2. **Broad applicability:** Maintaining generality
3. **Robustness:** Reducing overspecification

These properties directly benefit:

- **Generalization:** Weak explanations transfer to more situations
- **Fault tolerance:** Fewer specific commitments mean fewer failure modes
- **Efficiency:** Computational resources aren’t wasted on irrelevant details

Overall, weakness represents the “right” formulation of Occam’s Razor because:

1. **Semantic focus:** It measures assumption content, not representational artifacts
2. **Natural composition:** Weak components naturally yield weak systems
3. **Theoretical unity:** Probability, information, and complexity emerge as facets of a single framework
4. **Practical guidance:** Optimizing for weakness produces more general, robust solutions

Traditional “simplicity” was always gesturing toward this deeper principle—we prefer explanations that assume as little as necessary. Weakness makes this intuition mathematically precise and practically actionable.

2.9 Quantale Weakness and Bennett’s Layers

It’s also worth explicating how our notion of quantale weakness relates to the notion of “layered” weakness Bennett introduces in his thesis.

In Bennett’s approach, the notion of an abstraction layer is made explicit by indexing the cost structure with a layer L – which in our approach could be formulated as a corresponding quantale Q_L . Hypotheses and constraints are interpreted relative to L , and the weakness of a hypothesis H is defined as

$$weakness_L(H) = \bigvee_{c \in H} \omega_L(c),$$

where each constraint weight $\omega_L(c)$ lies in the quantale Q_L . This explicit relativization clarifies how specificity and granularity depend on the chosen layer.

In the present framework, different representational granularities are handled by selecting different quantales (for example, Q_{bits} versus $Q_{\text{sentences}}$) rather than by introducing a separate layer parameter. Concretely, one simply writes

$$weakness(H) = \bigvee_{c \in H} \omega(c)$$

with the understanding that the chosen quantale already encodes the desired level of abstraction. In this view, an explicit layer index is a useful notational convention but not a mathematical necessity, since the choice of quantale implicitly determines the layer of analysis.

2.10 Weakness and Hoare-Style Preconditions

2

Next, let us explore a different twist on the conceptual foundation of weakness in the theory of possible worlds.

The notion of “possible worlds” involved in Bennett’s original notion of weakness treats as the elements of a truth-set and measures simplicity by counting (or weighting) how many of those worlds remain undistinguished. This is compatible with the notion of “possible worlds” used in Kripke semantics, where “possible worlds” serve as states of a modal frame, but truth is evaluated via an accessibility relation between worlds rather than by membership alone. Hoare logic then treats program states as worlds and uses preconditions and postconditions to describe which states lead to which, effectively defining a one-step accessibility relation under program execution.

Integrating the categorial version of weakness into these Kripke-semantics possible worlds turns out to be straightforward and fruitful. By viewing Hoare preconditions as subspaces of Kripke worlds equipped with a quantale-based weakness measure, we bring all three ideas together: Bennett’s Occam-style world counting becomes a graded measure of state-set permissiveness, Kripke

²The observation in this section was made by Greg Meredith after hearing a talk presenting aspects of an earlier version of this paper

accessibility defines "reachability" under program steps, and Hoare pre/post triples emerge as constraints on which weighted world-sets guarantee correct execution. This unification lets us speak in a single formalism about how many—and how important—initial states are collapsed or preserved by a program, yielding a precise Occam-biased weakest-precondition framework.

In traditional Hoare logic, a precondition is simply the set of states from which running a program is guaranteed to achieve a desired postcondition. We propose to view that precondition set not merely as a yes-or-no predicate, but as a "weakness-weighted" subspace of all possible states. By assigning each state a numerical valuation and then aggregating those valuations over the precondition set, we obtain a single scalar that tells us how permissive—or "weak"—that precondition really is. A larger value means the program will work correctly from more initial states, but it also signals that the precondition is less restrictive.

Formally, let W be the set of all states and let $P \subseteq W$ be a precondition. In our quantale-weakness framework we assign each state $w \in W$ a valuation $\mu(w)$ in a commutative quantale $(V, \otimes, e, \leq)$, and define

$$w(P) = \bigoplus_{w \in P} \mu(w),$$

where $\oplus$ is the quantale join. The quantity $w(P)$ measures how permissive P is: higher values correspond to weaker (more inclusive) preconditions.

When we combine this idea with a Kripke-style accessibility relation $R \subseteq W \times W$ (the one-step reachability of states under program execution), we recover the classical weakest precondition. Given a postcondition $Q \subseteq W$, the weakest precondition in the usual sense is

$$\text{Pre}(Q) = \{ w \in W \mid \exists w' \in Q, (w, w') \in R \}.$$

In our framework we also get a weakness score for this set:

$$w(\text{Pre}(Q)) = \bigoplus_{\substack{w \in W, w' \in Q \\ (w, w') \in R}} \mu(w),$$

which quantifies how many and how important are the states from which Q is reachable in one step. This extra information lets us articulate statements such as "the program is correct for all but a low-weight subset of initial states" in a precise way.

Framing preconditions as weighted subobjects aligns cleanly with both modal-logic and Hoare-logic semantics, while introducing an Occam-style simplicity bias: we can prefer the minimally restrictive precondition P whose aggregated weakness stays below a chosen threshold yet still guarantees correctness. In practice, this means we can automatically search for the largest P such that $w(P) \leq \epsilon$ for a given tolerance ϵ , offering a principled trade-off between generality and safety.

Overall, embedding weakness into program logic provides a unified framework for talking about preconditions, reachability, and robustness. We no longer

ask merely "does the program work?" but also "how widely and how robustly does it work across different initial scenarios?"

2.11 Quantale Weakness in a Probabilistic and Logical-Entropy Context

The quantale definition of weakness has an extremely clear and obvious relationship to logical entropy (which is essentially equivalent to probabilistic weakness).

To draw this out: Recall that logical entropy of a partition measures the probability that two independent draws fall into *different* blocks. If W is a finite set of outcomes with probability distribution $p : W \rightarrow [0, 1]$, and π is a partition of W , then the *logical entropy* of π is

$$h(\pi) = \sum_{\substack{u, v \in W \\ u \text{ and } v \text{ in different blocks}}} p(u) p(v)$$

Equivalently, as we have seen above, if we let $H_\pi \subseteq W \times W$ be the *indistinction* relation (pairs in the *same* block), then the *same-block probability* is $w_p(H_\pi) = \sum_{(u, v) \in H_\pi} p(u) p(v)$, and logical entropy satisfies $h(\pi) = 1 - w_p(H_\pi)$.

These latter two formulas fit exactly into our quantale-weakness framework by choosing:

- The probability quantale $(V, \otimes, \oplus, \leq) = ([0, 1], \times, +, \leq)$.
- The valuation $\mu(u) = p(u)$.
- The subobject (relation) $H_\pi \subseteq W \times W$ of same-block pairs.

Then the quantale definition of weakness

$$w(H_\pi) = \bigoplus_{(u, v) \in H_\pi} [\mu(u) \otimes \mu(v)] = \sum_{(u, v) \in H_\pi} p(u) p(v)$$

is precisely the *same-block probability*, and its complement $1 - w(H_\pi)$ is the logical entropy $h(\pi)$.

We see here once again that probabilistic weakness (as a special case of quantale weakness) and logical entropy are two sides of the same coin: both arise by summing over pairs, weighted by joint probability, but one measures indistinction (same-block weakness) while the other measures distinction (logical entropy). This connection provides an intuitive interpretation of quantale weakness as "the probability that two random draws fail to be distinguished by the given relation."

2.11.1 Mapping Between Mutual-Information and Probabilistic Weakness

As a related example of the flexibility of the enriched-category framework for managing weakness, we can use it trivially to tweak the probabilistic weakness

into an informational weakness. In Chapter 14 below we will have reason to define the weakness of a causal explanation E via its mutual information $I(E)$, and will set

$$W_{\text{info}}(E) = 2^{-I(E)},$$

so that

$$I(E) = -\log_2 W_{\text{info}}(E).$$

It's easy to see that $W_{\text{info}}(E) \in [0, 1]$ and is $\otimes$ -monoidal under subadditivity of $I(\cdot)$.

While meaningfully different in a quantitative sense, this is basically the same story as the probabilistic weakness described above. Both the mutual-information-based and probabilistic notions of weakness take values in the same commutative quantale

$$Q = ([0, 1], \leq, \otimes, e), \quad x \otimes y = x \times y, \quad e = 1.$$

Thus there is an identity quantale-homomorphism

$$\phi : Q \longrightarrow Q, \quad \phi(x) = x,$$

respecting $\otimes$ and the order.

Explicitly, the mapping that identifies an explanation E with the corresponding event H_E whose probability encodes its informational content is as follows. If

$$I(E) = -\log_2 \Pr(H_E),$$

then

$$W_{\text{info}}(E) = 2^{-I(E)} = \Pr(H_E) = W_{\text{prob}}(H_E).$$

Thus

$$W_{\text{prob}} = \phi(W_{\text{info}}), \quad W_{\text{info}} = \phi^{-1}(W_{\text{prob}}),$$

and the relation $I(E) = -\log_2 W_{\text{prob}}(H_E)$ provides the simple mapping between the two weakness structures.

2.12 Fuzzy Weakness

Just as we have used quantales to extend weakness to probability measures, we can equally well extend it to fuzzy sets.

Let X be a universe and let $\varphi : X \rightarrow [0, 1]$ be a valuation on elements. We construct a quantale

$$V = [0, 1], \quad a \oplus b = \max(a, b), \quad a \otimes b = T(a, b),$$

where T is any commutative associative t-norm (for example $T(a, b) = \min(a, b)$ or $T(a, b) = a \cdot b$). The units are 0 for $\oplus$ and 1 for $\otimes$.

Fuzzy Subobject A fuzzy subset H of X is given by its membership function

$$H : X \rightarrow [0, 1], \quad H(x) = \text{degree to which } x \in H.$$

Weakness of a Fuzzy Set We define the weakness of H by "joining" over all elements:

$$w(H) = \bigoplus_{x \in X} [\varphi(x) \otimes H(x)] = \max_{x \in X} T(\varphi(x), H(x)).$$

If one prefers a bounded-sum join, use

$$a \oplus b = \min(1, a + b) \implies w(H) = \min\left(1, \sum_{x \in X} T(\varphi(x), H(x))\right).$$

Thus fuzzy-set weakness is obtained by enriching subobjects in the $[0, 1]$ quantale and "summing" membership via $\oplus$ and $\otimes$.

2.13 Hypervector Weakness

In a similar vein we show how one can set up a quantale to model weakness in hypervector space, choosing in particular the Modular Composite Representation of Snaider and Franklin from [SF14a] (though something similar could be done for other hypervector frameworks).

Let $r \geq 2$ be even and $n \in \mathbb{N}$. Define the *hypervector space*

$$V = \mathbb{Z}_r^n$$

with two binary operations (the Modular Composite Representation)

- *Binding (monoidal product)*

$$a \otimes b = (a_i + b_i \bmod r)_{i=1}^n, \quad e = (0, 0, \dots, 0).$$

- *Grouping (join)* For any finite multiset $\{v^{(1)}, \dots, v^{(k)}\} \subset V$, let

$$\bigoplus_{j=1}^k v^{(j)} = \left[\sum_{j=1}^k v^{(j)} \right],$$

where $[-]$ denotes the Snaider-Franklin normalization making the multi-operand sum associative.

One checks that

$\otimes$ is commutative, associative, with unit e , $\otimes$ distributes over $\bigoplus$,

so that $(V, \oplus, \otimes)$ is a *commutative semiring*. To turn it into a *quantale* we equip V with a *distance* and an induced *similarity measure*:

Distance and Similarity Define the modular Manhattan distance $\delta : V \times V \rightarrow \{0, \dots, n(r/2)\}$ by

$$\delta(a, b) = \sum_{i=1}^n \min(|a_i - b_i| \bmod r, r - (|a_i - b_i| \bmod r)),$$

and let $D_{\max} = n(r/2)$. Then set

$$\phi : V \rightarrow [0, 1], \quad \phi(v) = 1 - \frac{\delta(v, 0)}{D_{\max}},$$

so that $\phi(v) = 1$ iff $v = e$, and $\phi(v)$ decreases linearly with distance

The Similarity Quantale Let

$$Q = ([0, 1], \oplus_Q, \otimes_Q), \quad a \oplus_Q b = \max(a, b), \quad a \otimes_Q b = ab,$$

which is a commutative quantale: $\otimes_Q$ is associative, commutative, unital (unit 1), and distributes over arbitrary $\oplus_Q = \sup$.

Weak Models in Hypervector Algebra A *hypervector model* is a map

$$m : X \longrightarrow V.$$

We measure its *weakness* by lifting ϕ into Q and joining over X :

$$w(m) = \bigoplus_{x \in X}^Q \phi(m(x)) = \max_{x \in X} \phi(m(x)) \in [0, 1].$$

Models whose outputs are *close to the origin* in V (i.e. simple, low-complexity hypervectors) achieve $\phi(m(x)) \approx 1$, hence $w(m) \approx 1$, and are deemed *weak models* under this quantale-weakness schema.

2.14 Linear Weakness and SVMs

Another interesting variation of quantale weakness to look at is "linear weakness" constructed over a Hilbert space. We will show here that the *margin-based* notion of the quality of a classifier used in SVMs [CV95] arises as special case of quantale weakness, via appropriate choice of quantale.

2.14.1 Linear-Kernel (SVM) Weakness

Let $\Phi : U \rightarrow \mathcal{H}$ embed data into a Hilbert space, and let $k(u, v) = \langle \Phi(u), \Phi(v) \rangle$ be its positive-definite kernel. We use the quantale $V = [0, \infty]$ with order $\geq$, product $\otimes = \min$, identity $e = \infty$, and join $\oplus = \sup$. Define

$$\mu(u) = \frac{\|w\|}{y_u (w^\top \Phi(u))}$$

for labels $y_u \in \{\pm 1\}$ and normal vector w of a separating hyperplane; this assigns to each point its signed distance to the decision boundary. Then for the subobject

$$H = \{u \in U : y_u (w^\top \Phi(u)) \geq 0\}$$

we compute

$$w(H) = \sup_{u \in H} \min[\mu(u), \mu(u)] = \sup_{u \in H} \frac{\|w\|}{y_u (w^\top \Phi(u))} = \inf_u \frac{y_u (w^\top \Phi(u))}{\|w\|},$$

which is exactly the *margin* of the SVM classifier. Maximizing this $w(H)$ is therefore equivalent to solving the hard-margin SVM optimization.

2.14.2 Extension to Nonlinear-Kernel SVM

For practical application to SVMs, it is important to note that this linear-weakness construct also applies to the nonlinear kernel case. The previous construction relies only on an implicit feature map Φ and inner products $\langle \Phi(u), \Phi(v) \rangle$. Hence it carries over unchanged to nonlinear-kernel SVMs via the kernel trick. This is a fairly obvious point once one works through all the formalism, but to avoid confusion we will elaborate it briefly.

Setup. Let X be our input space, and let

$$k : X \times X \rightarrow R$$

be a positive-definite kernel. By Mercer's theorem there is a (possibly infinite-dimensional) feature map $\Phi : X \rightarrow \mathcal{H}$ into a Hilbert space $\mathcal{H}$ so that $\langle \Phi(x), \Phi(x') \rangle = k(x, x')$. An SVM classifier takes the form

$$f(x) = \sum_{i=1}^n \alpha_i y_i k(x_i, x), \quad y_i \in \{\pm 1\},$$

with dual coefficients α_i and training points $\{(x_i, y_i)\}$.

Quantale and Measure. As before, work in the quantale $V = [0, \infty]$ with $\otimes = \min$, $\oplus = \sup$, identity $e = \infty$.

Define

$$\|w\|^2 = \sum_{i,j} \alpha_i \alpha_j y_i y_j k(x_i, x_j), \quad \mu(x) = \frac{\|w\|}{y f(x)} \quad (y = y_x).$$

Hypothesis Subobject. Let

$$H = \{x \in X \mid y f(x) \geq 0\},$$

the set of correctly classified points.

Weakness as Margin. Then the categorical weakness of H is

$$w(H) = \sup_{x \in H} \min[\mu(x), \mu(x)] = \inf_{x \in H} \frac{y f(x)}{\|w\|},$$

which is exactly the (hard-margin) SVM margin. All inner products are computed via k , so no explicit Φ is needed.

So working things through, it's clear that nonlinear-kernel SVMs fit directly into the enriched-category weakness framework: by choosing the same quantale and measure construction, the margin maximization problem appears as maximizing categorical weakness of the classifier's acceptance region.

2.15 Weakness in a Finance Context

As an example of more specialized application of the quantale-weakness setup, consider the financial modeling domain. We will now explore a couple different ways of applying this formalism in the finance context – an application we will take up further in Chapters ?? and 34.1 where we will explore its value in the AGI attention allocation setting and also the more obvious domain of financial time series modeling.

2.15.1 A Financial Payoff Quantale and Its Weakness

We can model financial operations—aggregation of payoffs and compounding of returns—via the commutative quantale

$$V = ([0, \infty], \otimes, \oplus) \quad \text{with} \quad a \otimes b = a \times b, \quad a \oplus b = a + b,$$

unit for $\otimes$ equal to 1 (no gain) and neutral for $\oplus$ equal to 0 (no payoff). Note that

$$a \times \left(\sum_i b_i \right) = \sum_i (a \times b_i),$$

so $\otimes$ distributes over arbitrary $\oplus$.

Interpretation of Quantale Operations

- *Monoidal product* $\otimes$ (multiplication) models *compounding* of returns: if a position yields factor a in one period and b in the next, the total factor is $a \times b$.
- *Join* $\oplus$ (addition) models *aggregation* of independent payoffs: if two disjoint positions yield payoffs a and b , the total payoff is $a + b$.

Weakness as Expected Payoff or Exposure Let X be a set of financial instruments or scenarios, and let

$$m : X \rightarrow [0, \infty]$$

assign each instrument its payoff (or scenario payoff). Fix a valuation (weighting)

$$\varphi : X \rightarrow [0, \infty],$$

for instance portfolio weights summing to one or scenario-probabilities. Then the *weakness* of m is

$$w(m) = \bigoplus_{x \in X} [\varphi(x) \otimes m(x)] = \sum_{x \in X} \varphi(x) m(x).$$

In finance this is exactly the *expected payoff* (or *exposure*) under weights φ .

Examples

1. *Portfolio return*: X is a set of assets, $m(x)$ is the return factor of asset x , and $\varphi(x)$ is the fraction of capital invested. Then $w(m)$ is the portfolio's expected return factor.
2. *Scenario loss*: X is a set of stress-test scenarios, $m(x)$ is the loss in scenario x , and $\varphi(x)$ is the probability of x . Then $w(m)$ is the *expected loss*.

Thus the quantale $([0, \infty], \times, +)$ neatly captures the algebra of financial returns and payoffs, and its induced weakness $w(m)$ corresponds to the fundamental financial concept of expectation (of return or loss).

2.15.2 Risk-Normalized Expected Payoff as Weakness

We continue to work in the commutative quantale

$$V = ([0, \infty), \times, +), \quad \otimes = \times, \quad \oplus = +,$$

with unit 1 for $\otimes$ and zero 0 for $\oplus$.

Modified Valuation with Risk Discounting

Let

- X be a set of financial instruments or scenarios,
- $m : X \rightarrow [0, \infty)$ assign each $x \in X$ its payoff,
- $p : X \rightarrow [0, 1]$ be a probability or weight distribution,
- $r : X \rightarrow (0, \infty)$ be a risk measure (e.g. volatility, Value-at-Risk).

Fix a *risk-aversion parameter* $\gamma \geq 0$ and define a *risk-discount* function

$$f(r) = r^{-\gamma}.$$

Then set the *risk-normalized valuation*

$$\varphi(x) = p(x) \times f(r(x)) = \frac{p(x)}{(r(x))^\gamma}.$$

Weakness as Risk-Normalized Expected Payoff The weakness of the payoff predicate m is

$$w(m) = \bigoplus_{x \in X} [\varphi(x) \otimes m(x)] = \sum_{x \in X} \frac{p(x)}{(r(x))^\gamma} m(x).$$

This quantity is the *expected payoff discounted by risk to the power γ* . For $\gamma = 0$ it reduces to the usual expected payoff; for $\gamma > 0$ it penalizes high-risk instruments.

Alternative: Prospect-Theory Distortions One may also incorporate prospect-theory style value and probability weighting:

$$v : [0, \infty) \rightarrow [0, \infty), \quad \pi : [0, 1] \rightarrow [0, 1],$$

where v is a concave value function (e.g. $v(x) = x^\alpha$, $0 < \alpha < 1$) and π a probability-weighting function (e.g. $\pi(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{1/\delta}}$). Then define

$$\varphi(x) = \pi(p(x)) \times f(r(x)), \quad w(m) = \sum_{x \in X} \varphi(x) v(m(x)),$$

yielding a *risk- and prospect-adjusted expected value* as the weakness.

Interpretation

- If $r(x)$ is asset volatility and $\gamma > 0$, then $w(m)$ favors high-return, low-volatility assets.
- In scenario analysis, if $r(x)$ is scenario severity, $w(m)$ down-weights extreme but low-probability losses.
- Prospect-theory distortions further capture observed human risk-attitudes via v and π .

Thus by modifying the valuation $\varphi(x)$ to include a geometric risk discount and/or prospect-theory distortions, the quantale weakness $w(m)$ becomes a natural measure of payoff *normalized by risk*.

3 A (Fairly) General Theory of Generalization

There is a fairly obvious and straightforward sense in which, among multiple models explaining the same data, the model with the smallest Bennett set-theoretic weakness will generalize the best. Here we explain how to generalize this connection between weakness and generalization to quantale weakness. This will also lead us to generalize our notion of quantale weakness a bit, guided by the observation that there are more general forms of quantale weakness (than the ones presented above) that both have some philosophical and scientific meat to

them, and also support robust conclusions of the form "weaker models generalize better."

The theoretical analysis of generalization runs deep and we will address it from various angles throughout these pages, however what we argue here is that when the matter is viewed at an appropriate level of abstraction, one can explain a variety of different notions of generalization in a very simple and basic way.

3.1 Quantale Weakness and Expected Link-Prediction Accuracy

We start by presenting a very general and fairly trivial-looking set of assumptions under which quantale-weak models (in the sense described above) will generalize better. This turns out to yield classical results about the implications of set-theoretic and probabilistic weakness as immediate corollaries.

We start by modeling generalization as a "link prediction" task. In many settings one evaluates a model's ability to "generalize" by withholding some known links (u, v) from a relation $H \subseteq U \times U$ and then trying to predict them. We now show that, under a simple generative assumption, a higher sum-of-products weakness $w(H)$ directly yields better expected accuracy in such link-prediction tasks.

Assumption (Habitual Link Model). *There is a valuation $\mu : U \rightarrow \mathbb{R}_{\geq 0}$. For any pair (u, v) , the true probability of a link is*

$$\Pr((u, v) \in H) = \frac{\mu(u)\mu(v)}{Z}, \quad Z = \sum_{x, y \in U} \mu(x)\mu(y).$$

This captures the idea that nodes with large μ -values tend to form many links together.

We now withhold a random subset $H_{\text{train}} \subset H$ and attempt to predict the removed edges in $H_{\text{test}} = H \setminus H_{\text{train}}$. We use the following predictor:

$$\hat{p}(u, v) = \frac{\mu(u)\mu(v)}{Z},$$

identical to the true generative model.

Theorem 5 (Weakness Bounds Expected Accuracy). *Under the Habitual Link Model, the expected accuracy (fraction of correctly predicted test links, in expectation over random withholding) is*

$$\mathbb{E}[\text{Accuracy}] = \frac{w(H)}{Z}, \quad w(H) = \sum_{\substack{u, v \in H \\ u < v}} \mu(u)\mu(v),$$

so it increases monotonically with the sum-of-products weakness $w(H)$.

Proof. Let $M = |H_{\text{test}}|$. The expected number of correctly predicted test-links is

$$\mathbb{E}[|\{(u, v) \in H_{\text{test}} : \hat{p}(u, v) \text{ predicts link}\}|] = \sum_{(u, v) \in H_{\text{test}}} \Pr((u, v) \in H) = \sum_{(u, v) \in H_{\text{test}}} \frac{\mu(u)\mu(v)}{Z}.$$

Since H_{test} is a uniform random subset of H of size M , in expectation $\sum_{(u, v) \in H_{\text{test}}} \mu(u)\mu(v) = \frac{M}{|H|} \sum_{(u, v) \in H} \mu(u)\mu(v)$. Hence

$$\mathbb{E}[\text{Accuracy}] = \frac{1}{M} \cdot \frac{M}{|H|} \frac{w(H)}{Z} = \frac{w(H)}{|H| Z}.$$

Up to the constant $|H| Z$, this is linear in $w(H)$, proving that larger weakness yields higher expected link-prediction accuracy. $\square$

Discussion This theorem formalizes the intuition that when links are generated proportionally to $\mu(u)\mu(v)$, a relation H with larger sum-of-products weakness $w(H)$ contains more "habitual" link-patterns for the predictor to exploit, thereby boosting its expected accuracy. In this sense, greater quantale weakness under the habitual-link prior indeed leads to better generalization in link-prediction tasks.

One could certainly argue there's nothing surprising here. If you assume the world is based on a certain structural distribution of data, then models reflecting this structural distribution will generalize better. Umm, yeah. However, what's interesting is how this extremely obvious and simplistic principle of generalization comprises the essential meat of the generalization guarantees underlying so many sophisticated learning and reasoning approaches.

3.2 Application to Partition-Weakness

Specializing the above analysis to the case of partition-weakness leads to useful but unsurprising conclusions about logical entropy and generalization.

A *partition* Π on a finite set U induces an equivalence relation, and its *logical entropy* (partition-weakness) is

$$H_L(\Pi) = 1 - \sum_{B \in \Pi} p_B^2, \quad p_B = \frac{|B|}{|U|}.$$

This equals the probability that two independent uniform draws from U fall into *different* blocks?i.e. the fraction of distinguished pairs.

Habitual-Distinction Model Assume the true relation H consists of all pairs within the same block of Π , and we withhold a random subset $H_{\text{test}} \subset H$. We predict $(u, v) \in H$ by guessing "same-block" whenever u and v land in the same block of Π . Then the expected accuracy on H_{test} is exactly

$$\Pr(\text{same block}) = \sum_{B \in \Pi} p_B^2 = 1 - H_L(\Pi).$$

Equivalently, $H_L(\Pi)$ measures expected *distinction* accuracy on the complement of H . Thus a partition with larger logical entropy (weaker, more distinctions) yields better generalization for predicting which pairs *do not* belong together, and conversely more cohesion ($\sum p_B^2$ large) generalizes better for predicting which pairs *do* belong together.

3.3 Application to Bennett’s Set-Theoretic Weakness

Similarly, specializing our link prediction analysis to set-theoretic weakness provides a new route to the familiar connection between Bennett’s weakness and generalization.

For a propositional theory Φ over a finite universe of possible worlds W , define its *weakness* as the fraction of worlds satisfying it:

$$w(\Phi) = \frac{|\{w \in W : w \models \Phi\}|}{|W|}.$$

Under the *uniform-model* assumption – i.e. that withheld propositions are drawn uniformly from those true in Φ – we predict truth by assuming Φ holds. Then the expected accuracy on the withheld set is precisely $w(\Phi)$. Hence a theory with greater set-theoretic weakness (more models) generalizes better in the sense of correctly predicting more withheld truths.

Unified Perspective Thus we see both the partition and set case fit the general quantale-weakness $\rightarrow$ generalization mapping:

- In the partition quantale, weakness $H_L(\Pi)$ quantifies distinctions and governs expected prediction accuracy of non-membership.
- In the Boolean-algebra quantale, weakness $w(\Phi)$ quantifies model-set size and governs expected prediction accuracy of truth.

Under simple generative priors (uniform-model or uniform-block assumptions), higher quantale weakness directly yields higher expected link – or truth – prediction accuracy, making “weaker” models provably better at generalization in their respective domains.

3.4 Generalized Quantale Weakness and Expected Link-Prediction Accuracy

The above analysis of the generalization properties of quantale weakness suggests a generalization of the notion of quantale weakness itself. Because it is simple to see that the analysis we’ve given works perfectly well if one tweaks the definition of weakness to involve a *joint* measure on pairs, rather than assuming a product of marginals. All the conclusions about weakness and generalization carry over *mutatis mutandis*.

Assumption (Generalized Link Model). *There is a joint valuation*

$$\mu : U \times U \longrightarrow \mathbb{R}_{\geq 0}.$$

For each ordered pair $(u, v) \in U \times U$, the true probability of observing a link is

$$\Pr((u, v) \in H) = \frac{\mu(u, v)}{Z}, \quad Z = \sum_{x, y \in U} \mu(x, y).$$

As before, we withhold a random training subset $H_{\text{train}} \subset H$ of size $|H| - M$, set $H_{\text{test}} = H \setminus H_{\text{train}}$, and use the predictor

$$\hat{p}(u, v) = \frac{\mu(u, v)}{Z}, \quad (u, v) \in U \times U.$$

Theorem 6 (Weakness Bounds Expected Accuracy). *Under the Generalized Link Model, the expected fraction of correctly predicted test-links is*

$$\mathbb{E}[\text{Accuracy}] = \frac{w(H)}{|H|Z}, \quad w(H) = \sum_{(u, v) \in H} \mu(u, v).$$

In particular, the expected accuracy increases monotonically with the generalized weakness $w(H)$.

Proof. Let $M = |H_{\text{test}}|$. The expected number of correctly predicted test-links is

$$\mathbb{E}[\#\{(u, v) \in H_{\text{test}} : \hat{p}(u, v) \text{ predicts link}\}] = \sum_{(u, v) \in H_{\text{test}}} \Pr((u, v) \in H) = \sum_{(u, v) \in H_{\text{test}}} \frac{\mu(u, v)}{Z}.$$

Since H_{test} is a uniform random M -subset of H , in expectation

$$\sum_{(u, v) \in H_{\text{test}}} \mu(u, v) = \frac{M}{|H|} \sum_{(u, v) \in H} \mu(u, v) = \frac{M}{|H|} w(H).$$

Hence

$$\mathbb{E}[\text{Accuracy}] = \frac{1}{M} \frac{M}{|H|} \frac{w(H)}{Z} = \frac{w(H)}{|H|Z},$$

which is manifestly increasing in $w(H)$. $\square$

Discussion By replacing the product-measure prior with an arbitrary joint valuation $\mu(u, v)$, we see that the same core insight holds: a relation H whose *generalized weakness*

$$w(H) = \sum_{(u, v) \in H} \mu(u, v)$$

is larger will, under this generative model, admit higher expected link-prediction accuracy. Thus quantale weakness in its most abstract form continues to capture the essential connection between structural sparsity (or "weakness") and generalization performance.

I.e., once again: If you assume the world is based on a certain structural distribution of data, then models reflecting this structural distribution will generalize better. Umm, yeah. And what’s interesting is how this extremely obvious and simplistic principle of generalization can be seen to underlie so many useful and sophisticated things.

3.5 Philosophical Principles of Weakness and Generalization

Philosophy and physics provide numerous examples where instantiation of this notion of generalized weakness may be interesting. In most of the present work we will stick with the “traditional” form of weakness involving a sum of products, but a sequel digging deeper into potential physics applications will leverage some of these variations centrally.

Here we briefly review here three conceptual notions from the literature – Peirce’s tendency-to-take-habits, Smolin’s Precedence Principle and Sheldrake’s Morphic Resonance – that fit cleanly into a weakness framework where the product in the sum-of-products is replaced with a product plus an additional correlation term – encompassed by the generalized weakness from the previous section. In a way these three concepts are all close variants on the same idea, but they are habitually associated with slightly different mathematics.

3.5.1 Regularized Weakness and Philosophical Variants

We begin by observing that a *regularized* product?measure prior is a special case of the generalized?weakness framework:

Let

$$\mu_{\text{comb}}(u, v) = \mu(u) \mu(v) + \lambda M(u, v),$$

where

- $\mu : U \rightarrow \mathbb{R}_{\geq 0}$ is a marginal ?activity? measure,
- $M : U \times U \rightarrow \mathbb{R}$ is a correction term (e.g. mutual information, rarity penalty, semantic distance),
- $\lambda \geq 0$ trades off baseline product?measure and regularizer.

Define the regularized weakness of $H \subseteq U \times U$ by

$$w_{\text{comb}}(H) = \sum_{(u,v) \in H} \mu_{\text{comb}}(u, v).$$

All the expected?accuracy bounds for an arbitrary joint $\mu(u, v)$ hold verbatim upon replacing $\mu(u, v)$ with $\mu_{\text{comb}}(u, v)$ and $w(H)$ with $w_{\text{comb}}(H)$.

3.5.2 Peircean Habit Variant.

Peirce argued that repeated co?occurrence of ideas or actions gives rise to *habits* which then bias future inference and behavior [Pei92]. In his words, “habit-taking” is the process by which patterns of association become entrenched through repetition. We capture this by setting

$$M_P(u, v) = \text{frequency with which } u \text{ and } v \text{ co?occur, } \lambda = \lambda_P,$$

so that

$$\mu_P(u, v) = \mu(u) \mu(v) + \lambda_P M_P(u, v).$$

Here the regularizer boosts links with strong historical co?occurrence, aligning weakness with Peircean habit?strength.

3.5.3 Smolin?s Precedence Variant.

Smolin?s Precedence Principle holds that the probabilities of future events are shaped by the relative frequency of analogous events in the past, so that “what has happened before will influence what happens again” [Smo13]. Intriguingly, Smolin uses this as an alternative way to derive the basic equations of quantum mechanics. We model Smolin’s Principle by

$$M_S(u, v) = \text{count of past instances where } u \text{ preceded } v, \quad \lambda = \lambda_S,$$

giving

$$\mu_S(u, v) = \mu(u) \mu(v) + \lambda_S M_S(u, v).$$

The regularizer thus favors links that mirror established precedents.

3.5.4 Sheldrake?s Resonance Variant.

Sheldrake proposed that form and behavior are influenced by a “morphic field” which accumulates a collective memory of past similar patterns, a process he calls *morphic resonance* [She81]. We let

$$M_M(u, v) = \text{resonance strength between } u \text{ and } v, \quad \lambda = \lambda_M,$$

so that

$$\mu_M(u, v) = \mu(u) \mu(v) + \lambda_M M_M(u, v).$$

Here the correction term encodes collective memory effects as in Sheldrake?s hypothesis.

3.5.5 Implications for Generalization.

In each variant, weakness

$$w(H) = \sum_{(u,v) \in H} \mu_\star(u, v)$$

blends independent activity with a domain-specific correction. Under the corresponding generative model, larger $w(H)$ still yields higher expected generalization, while M encodes the extra inductive bias (habits, precedents, resonance) in a unified framework.

3.6 Weakest-Hypothesis-Driven Meta-Learning: A Conceptual Template

Finally, we leverage these ideas about generalized weakness to describe a *weakest-hypothesis-driven* learner that simultaneously discovers its own inductive bias and uses it to choose actions (including internal ones) so as to maximize cumulative reward. This is emphatically *not* a practical proposal for an AGI algorithm, but rather a *conceptual template* for thinking about learning under uncertainty.

Core Idea.

- Maintain a finite history H of past observations, actions and rewards.
- Fix a finite family $\{\mu_\theta\}_{\theta \in \Theta}$ of candidate joint valuations on (state, action) pairs. These might include product-measure priors, Peircean habit priors, Smolinian precedence priors, etc.
- On each step, *meta-learn* the best measure μ_{θ^*} by subsampling H (cross-validation): ask, “Which μ_θ would have led me via choosing the *weakest useful* hypothesis to best predict my past rewards?”
- Given μ_{θ^*} , enumerate a finite set of candidate hypotheses h (e.g. simple decision rules) consistent with H . Compute each hypothesis’s weakness

$$w_{\theta^*}(h) = \sum_{(u,v) \in h} \mu_{\theta^*}(u, v),$$

and select the *weakest* h that still promises high reward. Execute the action it prescribes.

- Treat internal operations—memory pruning, hypothesis-space refinement, update of Θ , etc.—as *actions* in the same framework.

Why This Matters.

- By searching over measures μ_θ , the learner *automatically adapts* its inductive bias to match the structure of its environment.
- Choosing the *weakest* hypothesis ensures that the agent only commits to as much structure as is needed to explain and predict reward, guarding against overfitting.
- Viewing internal algorithm updates and memory management as actions unifies *learning*, *planning*, and *self-modification* under one decision-theoretic roof.

Relation to Universal AI Frameworks. This template echoes ideas from:

- *Solomonoff induction*, which weights all computable hypotheses by their algorithmic complexity.
- The *Gödel machine*, which self-modifies by proving that a change will improve expected utility.
- *AIXI*, which selects actions to maximize expected reward under a universal prior over environments.

Like these frameworks, our approach searches over hypotheses (and even over the prior itself), but it does so by privileging the *weakest useful* explanation rather than shortest program or universal mixture.

A Conceptual, Not Practical, Algorithm. Because the history H , hypothesis space and measure family Θ must all be finite, the template is in principle computable but typically astronomically expensive. Its value lies in illustrating how an agent might *bootstrap* both its inductive bias and its behavior, always favoring the minimal structure needed to predict reward. This perspective can guide the design of more efficient, approximate algorithms, and we will revisit these conceptual ideas as appropriate in the following pages.

4 Weakness in Logical Quantales

The quantale weakness framework encompasses various logic systems just as easily as it does quantitative operations. We will here present a simple approach to constructing "logical weakness" and then explore connecting this to different forms of truth values, including Heyting algebra based (intuitionist), paraconsistent (Constructible Duality Logic) and uncertain (Probabilistic Logic Networks).

4.1 The Logic Quantale: Definitions of Weakness and Mutual Information

Let W be a nonempty set of *possible worlds*. We form the power-set quantale

$$Q_{\text{logic}} = (\mathcal{P}(W), \otimes_{\text{logic}}, \oplus_{\text{logic}}, \emptyset, W)$$

where for $A, B \subseteq W$:

$$A \otimes_{\text{logic}} B = A \cap B, \quad A \oplus_{\text{logic}} B = A \cup B.$$

This makes $\mathcal{C}_{\text{logic}}$, the one-object monoidal category with $\text{Hom}(\text{ast}, \text{ast}) = Q_{\text{logic}}$, into a commutative quantale.

Formulas as Predicates A *logical model* or *predicate* on a domain X is a function

$$m : X \rightarrow Q_{\text{logic}},$$

sending each $x \in X$ to the set of worlds $m(x) \subseteq W$ in which the formula holds.

Weakness in the Logic Quantale Given a *background valuation* $\varphi : X \rightarrow Q_{\text{logic}}$, define the *weakness* of m by

$$w_{\text{logic}}(m) = \bigoplus_{\text{logic } x \in X} [\varphi(x) \otimes_{\text{logic}} m(x)] = \bigcup_{x \in X} [\varphi(x) \cap m(x)].$$

Intuitively, $w_{\text{logic}}(m)$ is the set of worlds consistent with both background knowledge φ and the candidate predicate m —a larger set means a *weaker* (more permissive) model.

Residuation and Implication The quantale residuation (implication) $\multimap_{\text{logic}}$ is given by

$$A \multimap_{\text{logic}} C = \{w \in W : A \cap \{w\} \subseteq C\} = (W \setminus A) \cup C.$$

4.1.1 Connection to Hoare-Style Preconditions

We mentioned above how Hoare preconditions connect with the notion of possible-worlds used in the foundation of the weakness concept. Hoare preconditions can also be worked into the logical quantale setup we are now exploring.

Hoare logic treats a precondition $P \subseteq W$ as the set of states from which running a program guarantees a postcondition Q . In the quantale framework, one defines the classical weakest precondition

$$\text{Pre}(Q) = \{w \in W \mid \exists w' \in Q, (w, w') \in R\},$$

where $R \subseteq W \times W$ is the one-step reachability relation. The *weakness score* of this set is

$$w_{\text{logic}}(\text{Pre}(Q)) = \bigoplus_{\substack{w \in W, w' \in Q \\ (w, w') \in R}} \mu(w) = \bigcup_{\substack{w, w' \\ (w, w') \in R}} \mu(w),$$

quantifying how many—and how important—initial states lead into Q in one step

Thus the same power-set quantale $(\mathcal{P}(W), \cap, \cup)$ supports:

- *Logical weakness* of any predicate $P \subseteq W$ as $\bigcup_{w \in P} \mu(w)$.
- *Hoare preconditions* as particular subobjects defined by reachability R .
- *Quantified robustness* of a program via the scalar weakness $w_{\text{logic}}(\text{Pre}(Q))$, giving a precise Occam-biased measure of how permissive the precondition is.

In this way, just as alluded earlier, our categorical logic quantale seamlessly recovers and extends the Hoare-style weakest-precondition framework by endowing it with a graded Occam-style simplicity bias.

4.2 Quantale for Heyting Algebra

One can make a similar development for an intuitionistic logic based on Heyting algebras.

Let H be any *complete Heyting algebra*, with order $\leq$, meet $\wedge$, join $\vee$, top element $\top$ and bottom element $\perp$. Then

$$Q_H = (H, \leq, \wedge, \top, \vee)$$

is a commutative quantale, since finite meets distribute over arbitrary joins in a frame.

A *valuation* $\mu : U \rightarrow H$ on a universe U assigns each case $u \in U$ a truth?degree $\mu(u) \in H$. For any relation $R \subseteq U \times U$, define its *Heyting?weakness* by

$$w_H(R) = \bigvee_{(u,v) \in R} (\mu(u) \wedge \mu(v)) \in H.$$

By contrast, the *logic quantale* from Chapter 2 is

$$Q_{\text{logic}} = (\mathcal{P}(W), \subseteq, \cap, W, \cup),$$

and for a predicate $m : X \rightarrow \mathcal{P}(W)$ one has

$$w_{\text{logic}}(m) = \bigcup_{x \in X} (\phi(x) \cap m(x)),$$

i.e. the union of all surviving worlds :contentReference[oaicite:1]index=1.

Concrete example.

- $U = \{u, v\}$, with $\mu(u) = 0.4$, $\mu(v) = 0.7$ in $H = [0, 1]$ (meet min, join max).
- Relation $R = \{(u, v), (v, v)\}$.

Then

$$w_H(R) = \max\{\min(0.4, 0.7), \min(0.7, 0.7)\} = \max\{0.4, 0.7\} = 0.7,$$

whereas interpreting the same data in the logic quantale yields

$$w_{\text{logic}}(R) = \{u\} \cup \{v\} = \{u, v\},$$

a non?graded set. Thus the Heyting?quantale weakness is a *graded* value in $[0, 1]$, while the classical logic quantale gives only a Boolean set of worlds.

4.3 Quantale for Constructible Duality Logic (CDL)

Constructible Duality Logic (CDL), introduced by Patterson [Pat98], is a non-classical logical system designed to model reasoning in the presence of constructive knowledge and its dual—constructive refutation. CDL is a bivalent logic in which both *proof* and *disproof* are treated as primitive and independent forms of epistemic justification.

Rather than taking truth as primitive, CDL is based on the notion of *constructibility*: a proposition is considered accepted (“true”) if there exists a construction (e.g., a proof), and rejected (“false”) if there exists a disproof. Importantly, the absence of either does not force a classical determination of the truth value. This allows CDL to formally accommodate situations where neither a proposition nor its negation is currently justified—capturing a key aspect of reasoning under uncertainty and partial knowledge.

CDL supports a richer set of modal and epistemic distinctions than classical logic, while avoiding some of the paradoxes and explosion principles of more rigid systems.

4.3.1 Simple Quantitative Extension of CDL

In prior work [Goe21b], we have extended CDL by introducing a quantitative generalization suitable for application in AI systems, particularly in the context of reasoning with uncertain and partial evidence. In this extension, rather than simply treating proof and disproof as binary conditions, each proposition is associated with *counts* of supporting and refuting evidence instances—analogous to constructive arguments for and against the proposition.

These counts can be interpreted in Bayesian terms or aggregated using other probabilistic or evidential frameworks, leading to the assignment of degrees of belief or support. This results in a logic system that preserves the dual constructive character of CDL, while enabling graded inference.

This quantitative CDL framework can be seen as a special case of, or complementary to, frameworks such as Probabilistic Logic Networks (PLN, to be reviewed in Chapter ?? below), where truth values are explicitly probabilistic. In both systems, inference is based not on binary deduction, but on propagation of uncertain, context-dependent evidence. Goertzel’s formulation allows for the explicit integration of logical dualities (e.g., proof/disproof, affirmation/negation) with practical AI inference under bounded computational resources.

4.3.2 A Quantale for CDL Truth Values

We will present a general categorical framing for CDL, then afterwards explain how the simple quantitative extension mentioned above fits into this framework.

Formally, CDL can be constructed using two Heyting algebras, one for positive and one for negative evidence. Specifically, let H be a Heyting algebra with order $\leq$, meet $\wedge$, join $\vee$, top 1 and bottom 0. Define

$$V_{\text{CD}} = H \times H^{\text{op}},$$

ordered by

$$(x, x') \leq (y, y') \iff x \leq y \text{ and } y' \leq x'.$$

Give V_{CD} the commutative quantale structure

$$(x, x') \oplus (y, y') = (x \vee y, x' \wedge y'), \quad (x, x') \otimes (y, y') = (x \wedge y, x' \vee y'),$$

with unit $I = (1, 1)$ and zero $0 = (0, 0)$. These operations recover the CD logic conjunction and disjunction rules:

$$(x, x') \wedge (y, y') = (x \wedge y, x' \vee y'), \quad (x, x') \vee (y, y') = (x \vee y, x' \wedge y').$$

4.3.3 Weakness and Information Theory on the CD Quantale

Let $(H, \wedge, \vee, \rightarrow, 0, 1)$ be a Heyting algebra, and define the CD quantale

$$V_{\text{CD}} = H \times H^{\text{op}}, \quad (x, x') \oplus (y, y') = (x \vee y, x' \wedge y'), \quad (x, x') \otimes (y, y') = (x \wedge y, x' \vee y'),$$

with unit $I_{\text{CD}} = (1, 1)$ and zero $0_{\text{CD}} = (0, 0)$.

Weakness on V_{CD} A V_{CD} -valued predicate (subobject) is a function

$$m : X \rightarrow V_{\text{CD}}.$$

Its *weakness* is the quantale-join over all elements:

$$w(m) = \bigoplus_{x \in X} m(x).$$

More generally, given a valuation $\varphi : X \rightarrow V_{\text{CD}}$, one sets

$$w_{\varphi}(m) = \bigoplus_{x \in X} [\varphi(x) \otimes m(x)].$$

Residuated Implication in V_{CD} Since V_{CD} is a commutative quantale, it admits a residuated implication

$$(x, x') \multimap (y, y') = (x \rightarrow y, y' \rightarrow x'),$$

where $\rightarrow$ is the Heyting implication in H .

Quantale-Valued Distributions A *distribution* on X is a function $P : X \rightarrow V_{\text{CD}}$ satisfying

$$\bigoplus_{x \in X} P(x) = I_{\text{CD}}.$$

Marginals and joints in product spaces are formed by the usual $\oplus$ -join and pointwise $\otimes$.

CD-Quantale Entropy Define the entropy of P by

$$H(P) = \bigoplus_{x \in X} [P(x) \multimap 0_{\text{CD}}].$$

Since $P(x) \multimap 0_{\text{CD}}$ is the "negation" of $P(x)$ in the quantale, $H(P)$ measures the aggregate uncertainty of the distribution.

CD-Quantale Mutual Information For a joint distribution $P(x, y) \in V_{\text{CD}}$, let its marginals be

$$P_X(x) = \bigoplus_y P(x, y), \quad P_Y(y) = \bigoplus_x P(x, y).$$

Then the mutual information is

$$I(X; Y) = \bigoplus_{x, y} [P(x, y) \multimap (P_X(x) \otimes P_Y(y))].$$

Algebraic Properties By the axioms of a commutative quantale and Heyting algebra, the CD-quantale entropy and mutual information satisfy analogues of:

- Nonnegativity: $H(P) \geq 0_{\text{CD}}$, $I(X; Y) \geq 0_{\text{CD}}$.
- Symmetry: $I(X; Y) = I(Y; X)$.
- Chain rules and data-processing inequalities via monotonicity of $\oplus, \otimes, \multimap$.

Thus we obtain a fully quantale-enriched notion of weakness and information theory on the Constructible Duality logic quantale.

4.3.4 The Simple Quantitative Extension of PLN as a Special Case of the Quantale Framework

By instantiating the Heyting algebra H underlying the CDL quantale

$$V_{\text{CD}} = H \times H^{\text{op}}$$

to a complete lattice of raw evidence counts—for example

$$H = (\mathbb{N} \cup \{\infty\}, \leq, \wedge = \min, \vee = \max)$$

or more generally

$$H = ([0, \infty], \leq, \min, \sup),$$

and interpreting each proposition P as the pair $(n^+, n^-) \in H \times H^{\text{op}}$ of its constructive "proof" and "disproof" counts, one recovers exactly the simple quantitative CDL reviewed above. Under this specialization the quantale operations

$$(x, x') \oplus (y, y') = (x \vee y, x' \wedge y'), \quad (x, x') \otimes (y, y') = (x \wedge y, x' \vee y'),$$

become the appropriate evidence-aggregation rules—combining positive counts by their maximum (or by addition in an additive quantale variant) and negative counts dually—while the residuated implication

$$(x, x') \multimap (y, y') = (x \rightarrow y, y' \rightarrow x')$$

encodes conditional inference on these counts. Thus his pragmatic scheme of tracking and propagating explicit support and refutation tallies is realized as the special case of the general CDL quantale semantics when H is chosen to model raw quantitative evidence.

4.4 Quantale for PLN Simple Truth Values

Probabilistic Logic Networks (PLN) is a framework for uncertain inference that integrates probabilistic reasoning with symbolic logic, designed to serve as a foundation for Artificial General Intelligence (AGI). It extends traditional logic systems by incorporating probabilistic truth values and enabling inference under uncertainty, while maintaining computational tractability.

PLN operates within a typed metagraph formalism, where knowledge is represented as nodes and links in a graph structure. Each logical statement (or atom) is annotated with a *Simple Truth Value* (STV), which quantifies both the estimated **strength** (probability) and **confidence** (certainty about the estimate) of the statement.

Simple Truth Values A PLN Simple Truth Value is a pair (s, c) , where:

- $s \in [0, 1]$ is the *strength*, representing an estimate of the probability that the statement is true.
- $c \in [0, 1]$ is the *confidence*, indicating the degree of support behind the strength value.

The confidence can be interpreted in Bayesian terms as related to the equivalent sample size used to compute the strength. For instance, a strength of 0.8 with a confidence of 0.9 suggests strong evidence supporting a high probability of truth.

Inference in PLN PLN defines probabilistic inference rules (e.g., deduction, induction, abduction) that operate directly on STVs. These inference rules are derived from Bayesian principles and uncertainty propagation, ensuring that resulting truth values preserve meaningful probabilistic interpretations. For example, in a simple deductive rule $A \rightarrow B$, if A and $A \rightarrow B$ are known with associated STVs, then the STV for B can be computed via an appropriate propagation formula that reflects both the strength and confidence of the premises.

Semantic Foundations and Practical Approximations As will be articulated further in Section 7.4, the semantics of PLN can be formally grounded using *Graph-Structured Lambda Theory* (GSLT), a general formalism for representing knowledge, computation, and logic in a unified higher-order framework. Within GSLT, it is possible to represent and manipulate arbitrary higher-order probabilities, including infinite-order belief structures [Goe10] [War22]. In this context, the STVs used in PLN can be understood as pragmatic approximations to second- or third-order probability distributions—simplified summaries of uncertainty about uncertainty—that have proved effective for practical reasoning under resource constraints. Other PLN truth value structures, such as indefinite probabilities, serve a similar role as computationally feasible projections of deeper probabilistic semantics.

4.4.1 A Quantale for PLN Simple Truth Values

Let

$$V_{\text{PLN}} = \mathcal{R}_{\geq 0} \times \mathcal{R}_{\geq 0},$$

ordered componentwise. For any two STVs $(n^+, n^-), (m^+, m^-) \in V_{\text{PLN}}$ define $(n^+, n^-) \oplus (m^+, m^-) = (n^+ + m^+, n^- + m^-)$, $(n^+, n^-) \otimes (m^+, m^-) = (n^+ m^+, n^- m^-)$, with unit $1 = (1, 1)$ and zero $0 = (0, 0)$. Here $\oplus$ models the union of independent micro-situations by adding evidence counts, and $\otimes$ models independent conjunction by multiplying proportions.

4.4.2 Homomorphism Between V_{CD} and V_{PLN}

Fix an ensemble size $N > 0$. Define

$$\Phi : V_{\text{CD}} \rightarrow V_{\text{PLN}}, \quad \Phi(x, x') = (N x, N x').$$

Then for any $(x, x'), (y, y') \in V_{\text{CD}}$:

$$\Phi((x, x') \oplus (y, y')) = (N(x \vee y), N(x' \wedge y')) = (N x, N x') \oplus (N y, N y') = \Phi(x, x') \oplus \Phi(y, y'),$$

$$\Phi((x, x') \otimes (y, y')) = (N(x \wedge y), N(x' \vee y')) = (N x, N x') \otimes (N y, N y') = \Phi(x, x') \otimes \Phi(y, y').$$

Thus Φ is a commutative-quantale homomorphism, mapping CD p-bits to PLN evidence counts by renormalizing paraconsistent truth values into positive and negative evidence.

4.4.3 Exemplifying Where CDL and PLN May Differ: Conflicting Evidence in Sensor Diagnosis

Consider a simple fault-diagnosis task with two binary sensors $S_1, S_2 \in \{0, 1\}$ and a hidden fault variable $F \in \{OK, Fail\}$. Each sensor gives both positive and negative evidence:

- Sensor S_1 : reports "1" usually when $F = Fail$, but sometimes falsely.
- Sensor S_2 : reports "0" usually when $F = OK$, but occasionally misses a fault.

PLN Probabilistic View We collect counts of observations and learn a standard PLN simple-truth-value model:

$$P(F = \textit{Fail} \mid S_1 = 1) = 0.9, \quad P(F = \textit{OK} \mid S_1 = 1) = 0.1,$$

and similarly for S_2 . From these we build a joint probability distribution $P_{\text{PLN}}(S_1, S_2, F)$, compute marginals and then the Shannon mutual information

$$I_{\text{PLN}}(F; S_1, S_2) = \sum_{s_1, s_2, f} P_{\text{PLN}}(s_1, s_2, f) \log \frac{P_{\text{PLN}}(s_1, s_2, f)}{P_{\text{PLN}}(f) P_{\text{PLN}}(s_1, s_2)}.$$

This quantifies, in bits, how much observing both sensors reduces average uncertainty about the fault.

CD-Logic View Instead of crisp probabilities, represent each (s_1, s_2, f) by a paraconsistent truth-pair $(p, n) \in H \times H^{\text{op}}$ where

p = degree of evidence for $(F = f \mid S_1, S_2)$, n = degree of evidence against $(F = f \mid S_1, S_2)$.

These live in the CD quantale $V_{\text{CD}} = H \times H^{\text{op}}$. We form a V_{CD} -valued distribution

$$P_{\text{CD}}(s_1, s_2, f) \in V_{\text{CD}}, \quad \bigoplus_{s_1, s_2, f} P_{\text{CD}}(s_1, s_2, f) = (1, 1).$$

Then the CD-mutual information is

$$I_{\text{CD}}(F; S_1, S_2) = \bigoplus_{s_1, s_2, f} \left[P_{\text{CD}}(s_1, s_2, f) \multimap (P_F(f) \otimes P_{S_1, S_2}(s_1, s_2)) \right].$$

This measures how evidence for and against "flows" between sensors and fault, explicitly accounting for contradictory readings.

Different Insights

- **PLN mutual information** I_{PLN} gives a single scalar (in bits) of expected uncertainty reduction, but it *averages over* all cases and cannot distinguish *conflicting* vs. *supporting* evidence patterns.
- **CD mutual information** I_{CD} lives in $H \times H^{\text{op}}$ and can *separately* quantify accumulated positive vs. negative evidence flow. A high I_{CD} component in the "against" coordinate reveals systematic sensor conflicts, even if the probabilistic I_{PLN} remains moderate.

Thus for the same data, $I_{\text{PLN}}(F; S_1, S_2)$ tells you *how much* uncertainty is reduced on average, whereas $I_{\text{CD}}(F; S_1, S_2)$ tells you *which way* and *how strongly* evidence supports or contradicts a diagnosis—crucial in safety-critical or adversarial settings.

4.5 Connecting the Logic Quantale to the CDL and PLN Quantales

We now show how the power-set logic quantale embeds into both the Constructible-Duality-Logic (CDL) quantale and the PLN probabilistic-logic quantale, preserving weakness and mutual information.

The Three Quantales as One-Object Categories

$$Q_{\text{logic}} = (\mathcal{P}(W), \cap, \cup), \quad Q_{\text{CDL}} = (L_{\text{CDL}}, \otimes_{\text{CDL}}, \oplus_{\text{CDL}}), \quad Q_{\text{PLN}} = (L_{\text{PLN}}, \otimes_{\text{PLN}}, \oplus_{\text{PLN}}),$$

each viewed as a one-object monoidal category $\mathcal{C}_{\text{ast}}$ with $\text{Hom}(\text{ast}, \text{ast}) = Q_{\text{ast}}$, composition $= \otimes_{\text{ast}}$, tensor $= \oplus_{\text{ast}}$.

Embedding into the CDL Quantale Choose for each world $w \in W$ a primitive proposition $p_w \in L_{\text{CDL}}$. Define

$$E_{\text{CDL}} : Q_{\text{logic}} \rightarrow Q_{\text{CDL}}, \quad E_{\text{CDL}}(A) = \bigoplus_{\text{CDL } w \in A} p_w,$$

so that $E_{\text{CDL}}(A \cap B) = E_{\text{CDL}}(A) \otimes_{\text{CDL}} E_{\text{CDL}}(B)$, $E_{\text{CDL}}(A \cup B) = E_{\text{CDL}}(A) \oplus_{\text{CDL}} E_{\text{CDL}}(B)$, a strict monoidal functor.

Embedding into the PLN Quantale Let $\pi = (\pi_w)_{w \in W}$ be a probability distribution. Define

$$E_{\text{PLN}} : Q_{\text{logic}} \rightarrow Q_{\text{PLN}}, \quad E_{\text{PLN}}(A) = \bigoplus_{\text{PLN } w \in A} \pi_w,$$

so that $E_{\text{PLN}}(A \cap B) = E_{\text{PLN}}(A) \otimes_{\text{PLN}} E_{\text{PLN}}(B)$ and $E_{\text{PLN}}(A \cup B) = E_{\text{PLN}}(A) \oplus_{\text{PLN}} E_{\text{PLN}}(B)$, again a strict monoidal functor.

Consistency via the CDL-PLN Isomorphism Earlier we constructed a quantale isomorphism $\Psi : Q_{\text{CDL}} \rightarrow Q_{\text{PLN}}$ sending each atomic p_w to weight π_w . One checks

$$E_{\text{PLN}} = \Psi \circ E_{\text{CDL}}.$$

Preservation of Weakness For any predicate $m : X \rightarrow Q_{\text{logic}}$ and background $\varphi : X \rightarrow Q_{\text{logic}}$:

$$w_{\text{logic}}(m) = \bigcup_{x \in X} \varphi(x) \cap m(x).$$

Define $m_{\text{CDL}} = E_{\text{CDL}} \circ m$, $\varphi_{\text{CDL}} = E_{\text{CDL}} \circ \varphi$. Then

$$w_{\text{CDL}}(m_{\text{CDL}}) = \bigoplus_{\text{CDL } x} \varphi_{\text{CDL}}(x) \otimes_{\text{CDL}} m_{\text{CDL}}(x) = E_{\text{CDL}}(w_{\text{logic}}(m)).$$

Similarly for w_{PLN} , using $E_{\text{PLN}} = \Psi \circ E_{\text{CDL}}$.

Preservation of Mutual Information Given a joint predicate $P : X \times Y \rightarrow Q_{\text{logic}}$ with marginals P_X, P_Y , define $P_{\mathcal{CDL}} = E_{\mathcal{CDL}} \circ P$, etc. Then

$$I_{\mathcal{CDL}}(X; Y) = \bigoplus_{\mathcal{CDL} x, y} \left[P_{\mathcal{CDL}}(x, y) \otimes_{\mathcal{CDL}} (P_{\mathcal{CDL}}(x, y) \multimap_{\mathcal{CDL}} (P_X \otimes_{\mathcal{CDL}} P_Y)) \right] = E_{\mathcal{CDL}}(I_{\text{logic}}(X; Y)).$$

And again $I_{\mathcal{PLN}}(X; Y) = \Psi(I_{\mathcal{CDL}}(X; Y)) = E_{\mathcal{PLN}}(I_{\text{logic}}(X; Y)).$

Hence the logic quantale of possible-world sets embeds functorially into the CDL and PLN quantales, carrying over both Occam-style weakness and mutual-information measures intact.

4.5.1 CDL and PLN as Truth-Value Attachments to the Logic Quantale

Informally, the above formalism is basically viewing CDL and PLN as different ways to attach truth value objects to expressions in the categorial logic quantale.

Let

$$Q_{\text{logic}} = (\mathcal{P}(W), \otimes_{\text{logic}} = \cap, \oplus_{\text{logic}} = \cup)$$

be the "logic quantale" of possible-world predicates, viewed as the single-object monoidal category

$$\mathcal{C}_{\text{logic}} : \mathcal{O} \downarrow = \{\text{ast}\}, \text{Hom}(\text{ast}, \text{ast}) = Q_{\text{logic}}.$$

CDL Embedding Define the Constructible-Duality-Logic quantale

$$Q_{\mathcal{CDL}} = (L_{\mathcal{CDL}}, \otimes_{\mathcal{CDL}}, \oplus_{\mathcal{CDL}})$$

and a strict monoidal functor

$$E_{\mathcal{CDL}} : \mathcal{C}_{\text{logic}} \longrightarrow \mathcal{C}_{\mathcal{CDL}}$$

by

$$E_{\mathcal{CDL}}(A) = \bigoplus_{\mathcal{CDL} w \in A} p_w, \quad p_w \in L_{\mathcal{CDL}} \text{ atomic.}$$

Then

$$E_{\mathcal{CDL}}(A \cap B) = E_{\mathcal{CDL}}(A) \otimes_{\mathcal{CDL}} E_{\mathcal{CDL}}(B), \quad E_{\mathcal{CDL}}(A \cup B) = E_{\mathcal{CDL}}(A) \oplus_{\mathcal{CDL}} E_{\mathcal{CDL}}(B).$$

Thus each logic-formula $A \subseteq W$ is assigned a CDL truth-value object $E_{\mathcal{CDL}}(A)$.

PLN Embedding Let $\pi = (\pi_w)_{w \in W}$ be a probability distribution and define the probabilistic-logic quantale

$$Q_{\mathcal{PLN}} = ([0, 1], \otimes_{\mathcal{PLN}} = \times, \oplus_{\mathcal{PLN}} = +).$$

Define

$$E_{\mathcal{PLN}} : \mathcal{C}_{\text{logic}} \longrightarrow \mathcal{C}_{\mathcal{PLN}}, \quad E_{\mathcal{PLN}}(A) = \sum_{w \in A} \pi_w.$$

One checks

$$E_{\mathcal{PLN}}(A \cap B) = E_{\mathcal{PLN}}(A) \otimes_{\mathcal{PLN}} E_{\mathcal{PLN}}(B), \quad E_{\mathcal{PLN}}(A \cup B) = E_{\mathcal{PLN}}(A) \oplus_{\mathcal{PLN}} E_{\mathcal{PLN}}(B).$$

Thus each logic-formula A is assigned a PLN truth-value $E_{\mathcal{PLN}}(A)$.

Interpretation In this view, *CDL* and *PLN* are simply two different choices of "truth-value objects" attached to the same underlying logic quantale via strict monoidal functors. Logical operations $\cap, \cup$ are preserved, and both the Occam-style weakness and quantale mutual information defined on Q_{logic} transport directly into $Q_{\mathcal{CDL}}$ and $Q_{\mathcal{PLN}}$ by functoriality.

5 Relationships Between Set-Theoretic Weakness and Program Size

The concept of weakness arose as an attempt to formalize the intuitive notion of "Occam's Razor"—roughly "when in doubt between multiple hypotheses, all else equal, prefer the simpler." The intuition being that if A and B both explain the same thing, but A is simpler, B probably has some extra cruft on it. This intuition needs some adjustment to deal with quantum mechanics and other sorts of observer-dependence (as we'll argue in Chapter 11 below), but it does have some solid foundation as has been shown definitively by various related threads of work in learning theory.

The general convention in modern times has been to formalize Occam's Razor using minimum description length, and this does basically work, but it gets messy. Weakness takes a semantic more than syntactic approach, and as Bennett has shown this makes many things a fair bit cleaner. However, there are very clear and very close connections between the weakness and MDL approaches. Bennett gives some basic results in this regard in his thesis, showing (in essence) that weakness plus a finite-vocabulary assumption or similar on one's description language, gives a sort of equivalence between weakness and MDL. Here we pursue this sort of equivalence a little further, looking closely at what happens without the finite-vocabulary assumption, and expressing Bennett's weakness/MDL correspondence in a novel (and in some ways more revealing) form. Finally, we explore a special programming language (CoDD) in which there is an even *stronger* correlation between weakness and MDL and consider whether this language could be useful in the practical search for weak patterns within AI systems.

This section deals basically with the original Bennett-style set-theoretic weakness. In Chapter 6 we explore how the weakness/MDL parallel may be extended to general quantale weakness, via appropriately extending the notion of what constitutes a "language."

5.1 General Weakness/Description-Length Relation

Let U be a finite universe of N equally likely worlds. A hypothesis h is any subset $H \subseteq U$, with *weakness*

$$w(h) = |H|.$$

Suppose we fix any *prefix-free* description language mapping binary strings p to hypotheses $h = \text{Dec}(p)$. Let

$$\ell(p) = \text{length of } p, \quad L(h) = \min\{\ell(p) : \text{Dec}(p) = h\}$$

be the *description length* of h .

Kraft Bound. Prefix-freeness implies the Kraft inequality

$$\sum_p 2^{-\ell(p)} \leq 1.$$

Grouping by decoded hypothesis,

$$\sum_h \sum_{p:\text{Dec}(p)=h} 2^{-\ell(p)} \leq 1 \implies \sum_h 2^{-L(h)} \leq 1.$$

Weakness vs. Length. Since each hypothesis h covers exactly $w(h)$ worlds, its uniform probability is

$$P(h) = \frac{w(h)}{N}.$$

We also have a universal probability lower bound

$$m(h) = 2^{-L(h)} \leq P(h),$$

because otherwise $\sum_h 2^{-L(h)}$ would exceed one. Rearranging,

$$w(h) \geq N 2^{-L(h)}.$$

Equivalently,

$$L(h) \geq \log_2 N - \log_2 w(h).$$

Interpretation. This inequality holds for *any* prefix-free description scheme. It shows that *shorter descriptions can never rule out too many worlds*: a hypothesis of length L must leave at least $N/2^L$ worlds possible. Conversely, very *weak* hypotheses (large w) admit short descriptions. So basically, description-length and weakness are inversely coupled in the broadest class of languages via the Kraft bound.

5.2 Bennett's Finite-Vocabulary Extension of the Kraft Bound

Bennett's approach to connecting weakness with MDL adds to the Kraft argument the key assumption of a *finite description vocabulary* of size V . Then:

1. There are at most V^L distinct codewords of length exactly L , and at most $\sum_{\ell=0}^L V^\ell < \frac{V^{L+1}}{V-1}$ codewords of length $\leq L$.
2. Consequently, among all hypotheses expressible with descriptions of length $\leq L$, there are at most $\frac{V^{L+1}}{V-1}$ distinct hypotheses. Let

$$W_L = \max\{w(h) : L(h) \leq L\}.$$

3. By the Kraft bound,

$$\sum_{h:L(h) \leq L} 2^{-L(h)} \leq \sum_{\ell=0}^L V^\ell 2^{-\ell} < \frac{V^{L+1}}{V-1} 2^{-L}.$$

Since each term $2^{-L(h)} \leq w(h)/N$, summing gives

$$\frac{W_L}{N} \cdot \sum_{h:L(h) \leq L} 1 \geq \sum_{h:L(h) \leq L} \frac{w(h)}{N} \geq \sum_{h:L(h) \leq L} 2^{-L(h)} > \frac{V^{L+1}}{V-1} 2^{-L}.$$

Hence

$$W_L > \frac{V^{L+1}}{V-1} 2^{-L} N.$$

4. As L increases, the right-hand side grows exponentially in L , forcing W_L – the maximum weakness among all length- $\leq L$ programs – to grow in lockstep. Thus *the unique hypothesis of maximal weakness among those of length $\leq L$ is also the unique hypothesis of minimal description length achieving that weakness.*

In this way Bennett's finite-vocabulary assumption strengthens the one-sided Kraft bound into a true equivalence: within the finite set of hypotheses of description length at most L , the *weakest* hypothesis is precisely the *shortest*. This underlies his Occam's razor principle that "maximal weakness" and "minimal description length" coincide in any realistic, finite coding scheme.

5.3 ENF for Arbitrary Languages and Normalization Cost

While Bennett's argument connecting MDL and weakness is quite powerful, its assumptions are still fairly special. To re-express the above math in slightly plainer language, Bennett's finite-vocabulary argument shows a tight equivalence between *weakness* and *description length* only under two strong assumptions:

- **Prefix-free, semantics-preserving code:** every syntactic extension must change the decoded hypothesis.
- **No *dead code* or *unused definitions*:** all symbols in the program contribute to its meaning.

ENF Guarantees Prefix-Free, Dead-Code-Free Representation . Now one might wonder: if you're starting with some arbitrary language, then why not just remove the redundancies and dead code from your language to get a weakness-friendly version? This idea brings us directly to Holman's notion of an Elegant Normal Form [Hol90], which my colleagues and I have been using in automated program learning for a couple decades [Loo06]. Indeed, in principle, *any* programming language or formalism can be equipped with an Elegant Normal Form (ENF) that enforces:

- **Prefix-free codes:** each program maps to a unique tree/DAG, so no encoding is a prefix of another.
- **Non-dead code:** maximal sharing and canonical merging remove any subexpressions that do not contribute to the program's output.

Under these conditions, Bennett's finite-vocabulary argument raises the one-sided Kraft bound into a full equivalence: the *weakest* hypothesis of a given ENF size is also the *shortest* description.

However, the cost of performing this sort of elegant normalization may be highly language-dependent ... and sometimes highly high...

Normalization Cost Depends on Language Semantics

CMMC or MeTTa-calculus In our AI work on the Hyperon system, we have made use of simple formal language such as CMMC ([Goe20b], and see Appendix A.1) or the MeTTa Calculus [Mer25], which are basically combinatory/logical DSLs consisting of tree-structured terms and fixed rewrite rules. In the case of these languages (or others like them), transforming an arbitrary term into ENF requires

- a single bottom-up DAG-merging pass, and
- child-list sorting for commutative operators,

all running in time polynomial in the tree size.

General Languages More broadly, for general programming languages, achieving full ENF entails:

1. *Dead-code elimination*, which is equivalent to deciding reachability and equivalence in a Turing-complete setting (undecidable in general).

2. *Canonical serialization*, requiring normalization of macros, inlining, alias analysis, etc., with worst-case exponential blow-up.

Thus normalization to ENF in an arbitrary practical programming language can be super-expensive; it can even be outright pragmatically impossible in the general case.

Overall, while ENF can in theory make any language satisfy Bennett’s prefix-free, dead-code-free requirements, in practice DSLs like CMMC or MeTTa tend to admit efficient ENF normalization, whereas full GPLs like most common imperative programming languages tend not to.

5.3.1 Why Bennett’s Argument Fails in Practice for General MDL (e.g. C Programs)

To drive the latter point home more fully, consider that in a language like C, for instance:

1. You can add arbitrary comments, whitespace, unused functions or global variables without affecting the program’s behavior; such extensions increase *length* but do not change the set of worlds the program admits.
2. C is not a prefix-free encoding of its semantics: two different source texts can compile to the same binary, and long strings of dead code compile away.

Hence in plain C:

$$L(h_1) > L(h_2) \not\Rightarrow w(h_1) < w(h_2).$$

A longer C program can be no stronger (it may admit exactly the same behaviors) or even weaker if it accidentally changes some constraint. Thus minimum-length C programs need *additional normalization* (dead-code elimination, canonical ordering, etc.) before one can recover a monotonic length-weakness relation. And the normalization needed to make an arbitrary language like C into a prefix-free, dead-code-free formalism requires powerful rewrite passes that can approach the difficulty of full program equivalence – an undecidable problem in general. In particular:

- **Dead-code elimination** demands whole-program analysis of control-flow and data-flow to decide which functions, globals or branches are never reachable. In Turing-complete languages, reachability is undecidable, so practical compilers use conservative approximations that can still leave substantial dead code.
- **Prefix-free encoding** requires a canonical serialization of abstract syntax (removing comments, whitespace, renaming identifiers, reordering commutative operations, inlining macros, etc.). Achieving a unique representation up to semantics often entails deep normalization – constant folding,

loop unrolling, syntactic sugar elimination – which can blow up program size and compilation time.

- **Semantic preservation** must be guaranteed: every normalization step must preserve the program’s observable behaviors. Verifying that two large programs are semantically equivalent is itself at least as hard as the halting problem, so real- world toolchains rely on heuristics and may fail to detect some semantically redundant code.
- **Scalability issues** arise because every small optimization or canonical rewrite may interact with others. A single combinator- like elimination can trigger further inlining, constant propagation, or alias analyses, leading to an exponential cascade in worst- case compile time or intermediate code size.

5.4 Monotonicity of Weakness in the CoDD Model

A particularly elegant framework for exploring relationships between weakness and program size is the Combinational Decision DAG (CoDD) framework language, described in [Goe20a] and reviewed here in Appendix A.2.

In CoDD, each program h defines a set of possible “worlds” in which h holds true. We measure the *weakness* of h by

$$w(h) = |\{\text{worlds consistent with } h\}|.$$

A larger $w(h)$ means h rules out fewer worlds and is therefore *weaker*.

Extension implies strength. A program A *extends* a program B if A can be obtained by composing B with additional combinators or subprograms. Concretely, if

$$A = \text{ast}_i(B, C)$$

for some combinator ast_i and subprogram C , then the set of worlds satisfying A is a subset of those satisfying B . Hence

$$w(A) = |\{\text{worlds satisfying } A\}| \leq |\{\text{worlds satisfying } B\}| = w(B).$$

Thus any extension strictly *strengthens* the hypothesis and reduces its weakness.

Stronger length- weakness correlation. In CoDD, program *length* (the number of combinator applications) and *weakness* are tied by the Compositional Simplicity Measure:

$$\sigma(h) = \min_{y, z, i: h = y \text{ast}_i z} [\sigma(y) + \sigma(z) + c_i],$$

where each ast_i carries cost c_i . Because each extension increases $\sigma(h)$ by exactly the cost of one combinator, length and cost track one another precisely, and cost inversely tracks weakness.

By contrast, in most programming languages adding lines or imports does not guarantee strictly greater strength or lower generality, since libraries may bundle large amounts of code without changing the set of inputs accepted. CoDD’s free-algebra construction ensures that *every* syntactic extension narrows the hypothesis, yielding a direct, monotonic correlation between program length and weakness that is far stronger than in conventional languages.

5.5 Relation of CoDD to Bennett’s Argument

As we’ve reviewed above, Bennett’s thesis shows that in any *finite-vocabulary* prefix-free code, the hypothesis that rules out the fewest worlds among all descriptions of length at most L is exactly the one of minimal description length. He derives this by combining the Kraft inequality with an upper bound on the number of codewords of each length, yielding a tight correspondence between *maximum weakness* and *minimum length* under the finite-vocabulary assumption.

Our CoDD argument is a concrete instantiation – and a substantial strengthening – of Bennett’s general result. In CoDD:

- The *vocabulary* is the finite set of combinators $\{\text{ast}_i\}$.
- Programs are built as trees (or DAGs) of these combinators in *Elegant Normal Form*, giving a unique, prefix-free encoding.
- The CoDD *cost* $\sigma(h)$ counts exactly the number of combinator applications (each with fixed cost c_i), so program length and cost coincide.
- Weakness $w(h)$ is the exact number of worlds consistent with h , and each single-combinator extension both increases cost by c_i and *strictly* reduces weakness.

This yields not merely the one-sided Kraft bound $w(h) \geq N 2^{-\sigma(h)}$, but a *bijective monotonicity*:

$$\sigma(h') = \sigma(h) + c_i \implies w(h') < w(h),$$

and when all $c_i = 1$ exactly

$$w(h) = N 2^{-\sigma(h)}.$$

Thus CoDD realizes Bennett’s Occam-weakness equivalence in its purest form: every additional combinator shrinks the hypothesis by a provable factor, and minimal-length hypotheses are exactly maximal-weakness hypotheses without any slack or constant-factor losses.

5.6 Program Search with ENF and CoDD Weakness

So what. might be the use of this elegant , weakness-friendly CoDD representation of programs? Not for humans to program in, obviously. But there could be advantages for automated program learning.

We suggest that combining the Elegant Normal Form (ENF) representation of Combinational Decision DAGs (CoDDs) with the tight correspondence between CoDD cost and hypothesis weakness yields a principled framework for searching weak models that fit a dataset.

1. **Representation.** Each candidate hypothesis is encoded as a CoDD in ENF. ENF guarantees a unique, canonical DAG per hypothesis, maximal sharing of common subexpressions, and a direct node count equal to the CoDD cost $\sigma(h)$.

2. **Fitness Function.** For each hypothesis h compute:

$$\text{Loss}(h) \quad (\text{data fit}), \text{QQuadn} = \#\{\text{nodes in ENF of } h\} \quad (\text{CoDD cost}).$$

Combine by a regularized score,

$$F(h) = \text{Loss}(h) + \lambda n,$$

where $\lambda > 0$ trades off generalization bias.

3. **Search Algorithm.** Employ evolutionary or stochastic search over ENF trees:

- *Mutations* add, delete or swap combinator nodes, then re-merge ENF subgraphs.
- *Crossovers* splice parent DAGs at shared subtrees, preserving ENF.
- ENF recomputation after each edit runs in time proportional to the edit size.

4. **Generalization Guarantee.** In CoDD, adding a single combinator increases cost by a fixed positive amount and strictly decreases weakness. Penalizing node count therefore enforces an exact discrete Occam bias, ensuring that higher $F(h)$ correlates with hypotheses that both fit data and generalize.

5. **Practical Benefits.**

- *Deduplication:* ENF collapses identical subprograms, focusing search effort.
- *Smooth landscape:* each node edit yields a controlled change in both Loss and cost.
- *Scalability:* DAG sharing avoids exponential blow-up when combining good substructures.

- *Theory-backed bias*: the cost-weakness theorem guarantees sound Occam-style regularization without ad-hoc heuristics.

This framework thus seems to have potential to provide an efficient, semantically grounded method for evolving the *weakest* models that explain observed data.

6 Quantale-based Weakness as "Generalized MDL"

Having worked through various aspects of the relation between set-theoretic weakness and standard MDL, we now show that that general quantale weakness can be understood as a form of Minimum Description Length – however, not in the usual sense of bit-strings, but in a generalized "language" built from the objects themselves. By viewing each object as a symbol whose code-word lives in the free commutative quantale on our set of generators, we obtain a natural "length" function valued in the same quantale. The weakness of any relation then emerges as the maximal "description length" of the two-symbol code-words it contains. In the special case where our quantale is the familiar natural-number length algebra, this construction reduces exactly to classical MDL.

6.1 Generalized MDL Interpretation of Quantale-based Weakness

Let $(V, \otimes, e, \leq)$ be any commutative quantale, and let

$$\mu : U \longrightarrow V$$

be a given object-weighting (measure) on the underlying set U . We now show how the usual "weakness" measure given above can be viewed as a kind of Minimum Description Length in a free quantale "language."

6.1.1 The free commutative quantale on U .

Form the free commutative quantale

$$\mathbb{N}[U] = \left\{ \sum_{u \in U} n_u u \mid n_u \in \mathbb{N}, \text{ all but finitely many } n_u = 0 \right\},$$

where

$$\left(\sum n_u u \right) \otimes \left(\sum m_u u \right) = \sum (n_u + m_u) u, \quad \left(\sum n_u u \right) \oplus \left(\sum m_u u \right) = \sum \max(n_u, m_u) u,$$

and the unit e is the empty sum.

6.1.2 Induced length map.

There is a unique surjective quantale homomorphism

$$\Phi : \mathbb{N}[U] \twoheadrightarrow V$$

sending each generator $u \in U$ to its weight $\mu(u) \in V$. We define the *length* of any multiset $w \in \mathbb{N}[U]$ by

$$\ell(w) = \Phi(w) \in V.$$

Concretely, if $w = u_1 + \cdots + u_n$, then

$$\ell(w) = \mu(u_1) \otimes \mu(u_2) \otimes \cdots \otimes \mu(u_n),$$

independent of ordering by commutativity.

Generalized Length-like Properties It is perhaps worth clarifying why this is a reasonable way to generalize the ordinary notion of "expression length"?

The map

$$\ell : \mathbb{N}[U] \longrightarrow V, \quad \ell(u_1 + \cdots + u_n) = \Phi(u_1 + \cdots + u_n) = \mu(u_1) \otimes \cdots \otimes \mu(u_n)$$

satisfies the following axioms, mirroring the usual properties of a length function:

1. **Null length.** The empty multiset is the unit of the free quantale, and its length is the unit of V :

$$\ell(0) = \Phi(0) = e.$$

2. **Additivity (monoid homomorphism).** Concatenation (quantale multiplication) adds lengths:

$$\ell(w_1 \otimes w_2) = \ell(w_1) \otimes \ell(w_2).$$

3. **Commutativity.** Since $\mathbb{N}[U]$ is commutative, any reordering of generators leaves the length unchanged.

4. **Monotonicity.** If $w_1 \leq w_2$ in the free quantale order (pointwise inclusion of multiplicities), then

$$\ell(w_1) \leq \ell(w_2).$$

5. **Join-preservation.** The quantale join corresponds to taking the more expensive of two code-words:

$$\ell(w_1 \oplus w_2) = \ell(w_1) \oplus \ell(w_2).$$

6.1.3 Weakness as MDL-style supremal length.

Each pair $(u, v) \in H$ corresponds to the two-letter multiset $u + v \in \mathbb{N}[U]$, whose length is

$$\ell(u + v) = \mu(u) \otimes \mu(v).$$

Hence the quantale-valued weakness

$$w(H) = \bigoplus_{(u,v) \in H} (\mu(u) \otimes \mu(v)) = \bigoplus_{(u,v) \in H} \ell(u + v)$$

is exactly the "supremum of description-lengths" of all two-letter code-words drawn from H . In this sense, the native quantale-valued weakness is a generalized MDL: the free algebra $\mathbb{N}[U]$ plays the role of the language, $\ell(\cdot)$ is the code-length function, and $w(H)$ the maximal codelength over H .

6.2 Recovery of classical MDL.

It's not hard to see how this general sort of MDL reduces to standard "MDL over one dimensional languages" when the quantale is appropriately defined.

Suppose we take the special case where the quantale

$$(V, \otimes, e, \leq) = (\mathbb{N}, +, 0, \leq) \quad \text{and} \quad \mu : U \rightarrow \mathbb{N}$$

assigns each generator an integer weight. Then the canonical map $\Phi : \mathbb{N}[U] \rightarrow V$ is in fact an isomorphism, and the length

$$\ell(w) = \Phi(w)$$

of any multiset w coincides with the usual sum-of-letter-weights code length. Hence

$$w(H) = \bigoplus_{(u,v) \in H} \ell(u + v) = \max_{(u,v) \in H} (\mu(u) + \mu(v)),$$

which exactly recovers the standard MDL principle of taking the maximal combined code length over the candidate code-words in H . In this way, classical MDL appears as the special case of our generalized framework when the underlying quantale is the natural-number length quantale.

Relation to Bennett-Style Reduction via Kraft's Inequality The argument just given is, in fact, just a rearrangement of a "standard" reduction of Bennett's set-theoretic weakness to standard MDL (under the assumption of a prefix-free language with a finite vocabulary, etc.)

When U is finite and each $\mu(u) \in \mathbb{N}$, the five axioms for

$$\ell : \mathbb{N}[U] \rightarrow \mathbb{N}$$

(null length, additivity, commutativity, monotonicity, join-preservation) guarantee that the lengths $\{\ell(u) \mid u \in U\}$ satisfy the Kraft-McMillan inequality

$$\sum_{u \in U} 2^{-\ell(u)} \leq 1.$$

By Kraft's theorem, there thus exists a prefix code on U whose codeword-lengths are exactly $\ell(u)$, and the induced extension

$$\ell(u_1 + \cdots + u_n) = \sum_i \ell(u_i)$$

becomes the ordinary MDL length of the string $u_1 \cdots u_n$. Hence the supremal description length

$$w(H) = \bigoplus_{(u,v) \in H} \ell(u+v) = \max_{(u,v) \in H} (\ell(u) + \ell(v))$$

coincides with Bennett's classical reduction of quantale-weakness to the usual MDL principle, under the finiteness and integer-length assumptions.

6.3 Morphisms between sequential and multiset languages

The general relation between multi-set based languages and ordinary sequence-based languages is fairly simple but probably worth explicitly calling out.

Obviously, in ordinary human or programming-language communications we use *sequences* of symbols (a free monoid U^*) rather than unordered multisets. However, there is a canonical “forgetful” quantale homomorphism

$$q : (P(U^*), \cup, \text{concat}) \rightarrow (P(\text{NN}[U]), \cup, \otimes)$$

induced by the map $q : U^* \rightarrow \text{NN}[U]$ that sends each word to its commutative multiset of letters. Concretely,

$$q^\#(L) = \{q(w) \mid w \in L\}, \quad q(w_1 \cdot w_2) = q(w_1) \otimes q(w_2).$$

Because $q^\#$ preserves both unions and concatenation, it is a morphism of quantales.

6.4 Refinement of weakness to MDL in ordinary languages

Given a weight map $\ell : \text{NN}[U] \rightarrow V$, one obtains a V -valued length on sequences by

$$\ell_{\text{seq}}(w) = \ell(q(w)).$$

For any relation $H \subseteq U \times U$, the multiset-weakness

$$w(H) = \bigoplus_{(u,v) \in H} \ell(\{u,v\})$$

coincides with the supremal length of all two-letter words under ℓ_{seq} . Thus the abstract quantale-weakness exactly lifts to a Minimum Description Length on ordinary sequential codes. In the special case $V = P(S)$ with union and intersection, this construction gives a faithful embedding of multiset messages into set-theoretic descriptions, showing that – as we have already concluded above in a more specialized way – any multiset-based agent protocol can be implemented in a standard symbolic language without loss of the weakness (expressivity) metric.

6.5 Extensions to Other Quantales

The same pattern of "forgetting order" and passing from sequential codes to multiset codes carries over to richer quantales. For the *probability quantale* $V = ([0, 1], \times, +)$, one defines

$$q : U^* \rightarrow \text{NN}[U], \quad q(w_1 \cdot w_2) = q(w_1) \otimes q(w_2),$$

and then sets

$$\ell_{\text{prob}}(w) = \prod_{u \in U} p(u)^{n_u} = \bigotimes_{u \in q(w)} \mu(u)^{n_u},$$

so that the multiset-weakness $\bigoplus_{(u,v) \in H} \mu(u) \otimes \mu(v)$ agrees with the supremum of product-lengths of two-letter words under ℓ_{prob} . Likewise for the *quantum-operator quantale* $(\text{End}(H), \circ, \oplus)$, one treats each unitary or projection symbol as a generator, sends a sequence of gates to its multiset of gate-types, and then reconstructs the overall operator by multiplying (composing) the generators in any order. In each case the forgetful map $q : U^* \rightarrow \text{NN}[U]$ is a quantale homomorphism, and the induced length on sequences $\ell_{\text{quantum}}(w) = \ell(q(w))$ or $\ell_{\text{prob}}(w) = \ell(q(w))$ reproduces exactly the multiset-based weakness measure when maximized over the relevant pairs.

6.6 Example: Quantum Operator Multiset Language

To illustrate the breadth of this framework, let's consider a totally different sort of communication – between two quantum systems, assuming their states as "uncollapsed", communicating about aspects of their states using the multiset-language corresponding to the quantum-operator quantale.

Vocabulary of primitive rotations. Two uncollapsed quantum agents agree on a single basic rotation about the z -axis:

$$U = \{r\}, \quad \mu(r) = R_z(\frac{\pi}{4}) \in \text{End}(\mathbb{C}^2),$$

where

$$R_z(\phi) = e^{-i\phi Z/2} \quad (Z = \text{Pauli } Z)$$

is the unitary rotation through angle ϕ . The free commutative quantale on U is

$$\mathbb{N}[U] = \{n \cdot r \mid n \in \mathbb{N}\},$$

with

$$(n \cdot r) \otimes (m \cdot r) = (n + m) \cdot r, \quad (n \cdot r) \oplus (m \cdot r) = \max(n, m) \cdot r, \quad e = 0 \cdot r.$$

The induced length map is

$$\ell(n \cdot r) = \Phi(n \cdot r) = [R_z(\frac{\pi}{4})]^n = R_z(n \frac{\pi}{4}).$$

Communicating an internal phase. If an agent’s internal state is the unitary $R_z(\theta)$, it chooses the smallest $n \in \mathbb{N}$ such that

$$n \frac{\pi}{4} = \theta \quad (\text{or the nearest multiple if } \theta \text{ is not exact}),$$

and sends the multiset

$$w_{\min}(\theta) = n \cdot r.$$

Upon receipt, the other agent composes n copies of $R_z(\frac{\pi}{4})$ to reconstruct $R_z(\theta)$ exactly (or approximately).

Distinctive features.

- Messages are finite *multisets* of identical primitives, not sequences of symbols.
- The rotations commute, so only the *count* n matters.
- The “length” $\ell(n \cdot r) = R_z(n\pi/4)$ is an *operator* in $\text{End}(\mathbb{C}^2)$, not a bit-count.
- This lets uncollapsed quantum agents communicate continuous internal parameters (rotation angles) via discrete multiset codes in the operator quantale.

7 Weakness-Guided Probabilistic Logical Inference

In this section we show how the abstract notion of *weakness* can be integrated into the Probabilistic Logic Networks framework for uncertain AI inference [Goe08] by casting PLN semantics as a Probabilistic Graph-Structured Lambda Theory (P-GSLT). This provides a unified, graph-based foundation for both *hypothesis finding* and *inference control*, ensuring that more general (weaker) explanations are explored first, subject to probabilistic plausibility.

7.1 Graph-Structured Lambda Theory

Graph-Structured Lambda Theory (GSLT) [F1R25] represents lambda- or combinator-style terms as a directed labeled graph $G = (V, E)$. Each node $v \in V$ is labeled by a symbol (variable, constant, combinator, or predicate), and each directed edge $e = (u \rightarrow v) \in E$ carries a label indicating its logical or computational role (e.g. application, argument, or annotation). Reduction rules – such as beta-reduction or combinator rewrites – are implemented by local graph transformations: whenever a pattern subgraph L is found, it can be replaced by another subgraph R according to the rewrite rule. This approach unifies term syntax and operational semantics in a single combinatorial structure.

7.2 Probabilistic Enrichment

To model uncertainty in this setting, we enrich GSLT into a *Probabilistic GSLT* (P-GSLT) by assigning probabilities to nodes and edges. Concretely:

- A *prior probability* $PP(v) \in [0, 1]$ is attached to each node $v \in V$, reflecting the plausibility of the corresponding atomic proposition or combinator.
- A *conditional probability* $QQ(e) \in [0, 1]$ is attached to each edge $e \in E$, expressing the likelihood that the edge's relation holds given its source node.

Under a standard Bayesian factorization, the prior probability of any subgraph $H \subseteq G$ is

$$\Pr(H) = \prod_{v \in H} PP(v) \times \prod_{e \in H} QQ(e).$$

Observed data D then refines these priors into posteriors $\Pr(H \mid D)$ via Bayes' rule. This construction integrates logical structure and probabilistic reasoning in a single semantic framework.

7.3 Probabilistic Weakness in a GSLT Setting

We define a partial order among hypotheses (subgraphs) by saying H_1 is *weaker* (more general) than H_2 , written $H_1 \preceq H_2$, if for every probability threshold $p \in [0, 1]$,

$$\Pr(H_2 \mid D) \geq p \implies \Pr(H_1 \mid D) \geq p.$$

In words, any data instance that makes H_2 sufficiently probable also makes H_1 at least that probable. Equivalently, H_1 must cover all the explanatory power of H_2 and possibly more. This formalizes the intuitive notion that a *weaker* hypothesis is one that applies to a broader range of situations without dropping below any probability threshold.

Derivation from Categorical Weakness To see formally why this is the same as the probabilistic weakness we discussed above, recall that in the enriched-category framework we set

$$w(H \mid D) = \Pr(H \mid D),$$

the posterior probability of hypothesis H given data D . It is straightforward to show that the partial order

$$H_1 \preceq H_2 \iff \forall p \in [0, 1], \Pr(H_2 \mid D) \geq p \implies \Pr(H_1 \mid D) \geq p$$

is equivalent to comparing these weakness values.

7.4 Semantic Foundation of PLN via Graph-Structured Lambda Theory

Graph-Structured Lambda Theory (GSLT) provides a unified, higher-order, and graph-based semantics in which both terms (logical assertions) and rewrites (inference steps) are first-class citizens. We now outline how PLN's uncertain-inference machinery embeds naturally into a GSLT enriched over the PLN quantale.

Terms and Rewrites as a Reduction Graph A GSLT is specified by

- two sorts, T (terms) and R (rewrites),
- constructors
 - $s, t : R \rightarrow T$ assigning source/target terms to each rewrite,
 - term-constructors such as application $(-_-) : T \times T \rightarrow T$ and abstraction $\lambda : (T \rightarrow T) \rightarrow T$,
 - rewrite-constructors $\rho(\cdot)$ for each inference rule (e.g. deduction, induction, abduction),
- equations of the form

$$\rho(\vec{x}) : s(\rho(\vec{x})) \mapsto t(\rho(\vec{x}))$$

declaring that whenever a term matches $s(\rho)$, it may be rewritten to $t(\rho)$.

In PLN, each inference rule is exactly such a rewrite: matching premise-terms annotated with STVs and producing a conclusion-term with its propagated STV. Thus the space of all PLN proofs is realized as the *reduction graph* of the GSLT.

Quantale Enrichment = Truth-Value Arithmetic We enrich the GSLT reduction graph in the commutative quantale

$$V_{\text{PLN}} = [0, 1] \times [0, 1], \quad (s, c) \oplus (s', c') = (s + s', c + c'), \quad (s, c) \otimes (s', c') = (s s', c c'),$$

so that:

- parallel derivations of the same conclusion aggregate via $\oplus$,
- sequential chaining of inference steps multiplies via $\otimes$.

Concretely, the hom-object from premises to conclusion in the enriched category is the "sum" (over all paths) of "products" (along each path) of edge-weights – exactly reproducing PLN's STV propagation formulas.

Higher-Order and Modal Interpretations Because GSLT admits higher-order term-constructors (e.g. abstraction on $T \rightarrow T$), it inherently supports higher-order probabilities (probabilities about probabilities). PLN’s simple truth values then appear as pragmatic projections (second- or third-order marginals) of this fully higher-order semantics. Moreover, identifying suitable rewrite-constructors yields modal types

$$\langle A \rangle B, \quad [A]B$$

that correspond to PLN’s conditional and guaranteed-entailment modalities.

Change-of-Base and Practical Approximation Different truth-value structures arise by changing the enrichment quantale. For example, the homomorphism

$$\Phi : V_{\text{CD}} \longrightarrow V_{\text{PLN}}, \quad \Phi(x, x') = (N x, N x')$$

is precisely a *change-of-base* functor that renormalizes paraconsistent truth values into PLN evidence counts, preserving both $\oplus$ and $\otimes$. In implementation, one works with a coarse-grained GSLT for tractable proof search and a fine-grained GSLT with indexing and normal-form optimizations for efficiency.

5. Summary By viewing PLN’s formulas as T -terms and its inference rules as R -rewrites in a GSLT enriched over $(V_{\text{PLN}}, \oplus, \otimes)$, we obtain a fully compositional, higher-order semantics in which:

- proof-theoretic derivations correspond to paths in the reduction graph,
- probabilistic strength-and-confidence calculations arise from the quantale enrichment,
- higher-order and modal aspects of uncertainty are captured by the theory’s native support for function-type constructors and modal rewrite-symbols.

This categorical-enriched perspective thus elegantly unifies the symbolic and numeric facets of PLN.

7.5 Inference Control with Weakness Ordering

??

To leverage probabilistic weakness in PLN proof search, one simplistic but potentially helpful strategy is to use the following algorithm:

Basic idea: Maintain a priority queue of candidate subgraphs (hypotheses), always expanding the *weakest* ones first, subject to a minimum posterior probability. In more detail:

1. **Initialization.** Generate a set of initial hypotheses by matching observed facts D to small subgraphs of G . Compute each hypothesis’s posterior $\Pr(H \mid D)$.

2. **Priority Queue.** Insert all hypotheses with $\Pr(H \mid D) \geq \tau$ into a queue ordered by the weakness relation: if $H_1 \preceq H_2$, then H_1 precedes H_2 .
3. **Search Loop.**
 - Pop the most general (weakest) hypothesis H .
 - If H meets the success criterion (e.g. predicts D with high confidence), return H .
 - Otherwise, apply applicable rewrite rules to H to produce refined hypotheses H' , compute $\Pr(H' \mid D)$, and insert those with $\Pr \geq \tau$ into the queue.
4. **Termination.** Halt when the queue is empty or a satisfactory hypothesis is found.

Because weaker hypotheses cover more of the data space, this strategy prioritizes broad, high-entropy explanations before drilling down into special cases, improving both efficiency and generalization.

8 Factor Graphs over Commutative Quantales

Factor graphs were originally introduced in the context of Bayesian networks as a generalized graphical model that makes both the factorisation of a joint distribution and the flow of messages in belief-propagation explicit. In particular, any Bayesian network can be converted into a semantically equivalent factor graph, and standard inference algorithms (sum-product, max-product) admit a uniform description in this framework.

Paul Rosenbloom and colleagues have advocated (Bayesian) factor graphs as a broadly applicable, "universal" representation for AGI architectures, arguing that many forms of knowledge – probabilistic, logical, relational – can be encoded as factors and variables in a single graph [Ros10]. While we wouldn't go quite that far – in our humble view factor graphs are not literally the one grand unified representation for every aspect of intelligence – they do provide a remarkably versatile substrate for combining diverse inference rules and knowledge modalities in a uniform, parallelizable way.

In this section we extend the factor-graph paradigm by replacing real-valued weights with elements of an arbitrary commutative quantale, thereby capturing not only probabilities but also fully-quantified logical expressions and graded-weakness or confidence measures in PLN, CDL and related formalisms. We show how the same message-passing machinery carries over verbatim to the quantale setting, and how this enriched view naturally supports multiresolution inference, structured priors, and incremental updates.

We also discuss the practical realization of quantale factor graphs in the Hyeron framework using the MOR AtomSpace. MOR's efficient storage of atoms, hyperedges and attributes aligns perfectly with the factor-graph data model, enabling rapid insertion, retraction and indexed lookup of factors and

variables. This makes quantale factor graphs not only theoretically unified but also immediately usable as a background inference engine in a Hyperon-based AGI system.

8.1 Factor Graphs in Commutative Quantales

1. **Quantale recap.** A commutative quantale is a tuple

$$(Q, \leq, \otimes, e, \bigvee)$$

where

- $(Q, \leq)$ is a complete lattice,
- $\otimes$ is an associative, commutative product distributing over arbitrary joins,
- $e \in Q$ is the unit for $\otimes$,
- $\bigvee$ denotes the join (least upper bound).

2. **Variables and factors.**

- *Variable nodes* $v \in V$, each ranging over a domain D_v .
- *Factor nodes* $f \in F$, each equipped with a local function

$$\phi_f : \prod_{v \in \text{Ne}(f)} D_v \longrightarrow Q.$$

3. **Global weight.** A full assignment $x = (x_v)_{v \in V}$ is given weight

$$W(x) = \bigotimes_{f \in F} \phi_f(x_{\text{Ne}(f)}).$$

4. **Marginalization.** To marginalize out $V \setminus S$ one computes

$$W(x_S) = \bigvee_{x_{V \setminus S}} \left[\bigotimes_{f \in F} \phi_f(x_{\text{Ne}(f)}) \right].$$

8.1.1 Cheap Weakness Computation via Message Passing

Because the factor graph is typically sparse, one can run a generalized belief-propagation algorithm:

- Each message update involves only a few $\otimes$ and $\bigvee$ operations.
- Complexity per iteration is $O(|V| + |F|)$.
- If $\otimes$ and $\bigvee$ are chosen to be cheap (e.g. min-plus, truncated sum), updates are very fast.

8.1.2 Storing Factor Graphs in Mork

Factor graphs, due to their intrinsic hierarchical nature, lend themselves very naturally to implementation in MORK as an AtomSpace substrate.

In a Mork AtomSpace:

- **VariableAtoms** represent variable nodes.
- **FactorAtoms** represent factor nodes, tagged by rule type.
- **HyperEdgeLinks** connect each FactorAtom to its neighboring VariableAtoms.
- **AttributeAtoms** store the local quantale functions or weights.

Mork supports

- $O(\log N)$ insertion/retraction of atoms and links,
- indexing and subsumption lookups for lattice-structured quantales,
- efficient bulk and incremental updates.

8.2 Factor Graphs over a Quantale of Logical Expressions

One very interesting application of quantale factor graphs is to the domain of logical expressions. This allows a much richer use of factor graphs for probabilistic logic than is achieved in conventional Bayes Net and related applications.

8.2.1 The Quantale of Logical Expressions

- **Carrier Q** : the set of all well-formed formulas (WFFs) in a given logic, modulo logical equivalence.
- **Order $\leq$** : logical entailment,

$$p \leq q \iff p \models q.$$

- **Monoid product $\otimes$** : conjunction,

$$p \otimes q = (p \wedge q).$$

- **Unit e** : the tautology (*True*).
- **Join $\vee$** : disjunction (least upper bound under $\leq$),

$$\bigvee_i p_i = p_1 \vee p_2 \vee \dots$$

8.2.2 Variables, Factors, and Global Formula

- **Variables** v : each ranges over a domain D_v (e.g. Boolean or finite).
- **Factors** f : come with a local constraint formula

$$\phi_f(\mathbf{v}) \in Q,$$

where $\mathbf{v} = \{v_1, \dots, v_k\}$ are the free variables of ϕ_f .

- **Global conjunction:**

$$\Phi(\mathbf{V}) = \bigotimes_{f \in F} \phi_f(\mathbf{V}) = \bigwedge_{f \in F} \phi_f(\mathbf{V}),$$

a single formula in Q over all variables $\mathbf{V}$.

8.2.3 Marginalization as Existential Quantification

To focus on a subset $S \subseteq \mathbf{V}$, we compute

$$\Phi_S(\mathbf{v}_S) = \bigvee_{\mathbf{v}_{\mathbf{V} \setminus S}} \left[\bigwedge_{f \in F} \phi_f(\mathbf{v}) \right] = \exists \mathbf{v}_{\mathbf{V} \setminus S}. \bigwedge_{f \in F} \phi_f(\mathbf{v}).$$

8.2.4 Message-Passing of Formulas

- **Variable→Factor**: each variable sends to f the join of all incoming factor-messages.
- **Factor→Variable**: to send from factor f to variable v :
 1. Conjoin ϕ_f with all incoming messages from other neighbors.
 2. Existentially quantify out all variables except v .
 3. The result is the new message $m_{f \rightarrow v}(v)$.

8.2.5 Efficiency Considerations

- **Formula representation**: compile to a shared canonical form (AST, BDD, SDD) for subformula sharing.
- **Incremental updates**: adding or changing one ϕ_f only triggers local message updates.
- **Shared subformulas**: identical subexpressions are stored once and reused across factors.

8.2.6 Interpretation

This construction yields a distributed constraint network:

- Global consistency is checking satisfiability of $\bigwedge_f \phi_f$.
- Marginals are logical projections via $\exists$.
- Message-passing is constraint propagation via $\wedge$ and $\exists$.

8.3 Combinatory Decision DAGs as a Special Case of Quantale Factor Graphs

In this section we show how any Combinatory Decision DAG (CoDD) can be viewed as a factor graph over a suitable commutative quantale Q . The usual CoDD inference algorithm then becomes an instance of quantale-valued message passing.

8.3.1 Quantale and Factor Graph Definitions

- **Quantale Q .** A commutative quantale is a tuple

$$(Q, \leq, \otimes, e, \bigvee)$$

where $\leq$ is a complete lattice order, $\otimes$ is an associative, commutative product distributing over joins, e is the unit for $\otimes$, and $\bigvee$ is the lattice join.

- **Factor graph.** A factor graph is a bipartite graph (V, F, E) where
 - V is a set of *variable nodes*, each ranging over some domain D_v .
 - F is a set of *factor nodes*, each with a local potential $\phi_f : \prod_{v \in \text{Ne}(f)} D_v \rightarrow Q$.
 - $E \subseteq V \times F$ are edges connecting variables to factors.

8.3.2 Encoding a CoDD as a Factor Graph

Given a CoDD with decision nodes implementing various PLN rules and edges carrying "weakness" values:

1. **Variables.** For each edge in the CoDD (i.e. each intermediate proposition or pattern), introduce a variable node $v \in V$ whose domain is the set of possible "truth" or "pattern" values.
2. **Factors.** For each CoDD decision node d , introduce a factor node $f \in F$ with local potential

$$\phi_f(x_{v_1}, \dots, x_{v_k}) = x_{v_1} \otimes_f \dots \otimes_f x_{v_k},$$

where $\otimes_f$ is the quantale operation corresponding to the PLN rule encoded by d . The neighbors $\{v_1, \dots, v_k\}$ are precisely the edges feeding into d .

3. **Global inference.** A complete assignment x to all variables has weight

$$W(x) = \bigotimes_{f \in F} \phi_f(x_{\text{Ne}(f)}).$$

Computing the marginal weight of any variable v corresponds to summing out (joining over) all other variables:

$$W(x_v) = \bigvee_{x_{V \setminus \{v\}}} \left[\bigotimes_{f \in F} \phi_f(x) \right].$$

8.3.3 Message-Passing Implements CoDD Inference

By running a generalized belief-propagation algorithm over this factor graph:

- *Variable* $\rightarrow$ *Factor* messages collect the join ($\bigvee$) of incoming factor messages.
- *Factor* $\rightarrow$ *Variable* messages combine (via $\bigotimes$) the local potential ϕ_f with incoming variable messages, then eliminate other variables by join ($\bigvee$).

These updates exactly mirror CoDD’s downstream propagation of weakness values through combinatory decision nodes, yielding the same graded inference results but in a unified, parallelizable framework.

8.3.4 Advantages of the Factor-Graph View

- **Structural sharing.** Repeated sub-DAGs in the CoDD become shared factor nodes, avoiding redundant recomputation.
- **Incremental updates.** Adding or modifying a CoDD node corresponds to inserting or updating a single factor, with $O(\log N)$ cost in Mork.
- **Multiresolution.** By choosing Q to be a chain of sub-quantales, one can perform coarse-to-fine weakness inference akin to multigrid methods.

Thus every CoDD is naturally a special case of a quantale-valued factor graph, enabling efficient, incremental, and parallel graded inference.

8.4 Efficient PLN Background-Inference via Quantale-Annotated Logical Factor Graphs

In order to support wide-ranging “background inference” over a large AtomSpace, we maintain an ongoing Factor Graph whose variable nodes correspond to logical expressions and whose factor nodes encode the combination rules of PLN. Each logical expression carries, as a separate annotation, its current uncertainty (truth-value) living in a distinct commutative quantale Q_{truth} . Inference then proceeds by lightweight message-passing over this graph rather than by expensive rule-search, enabling rapid discovery of implicit knowledge.

8.4.1 Quantale-Annotated Logical Factor Graph

- **Variable nodes.** Each free logical expression ϕ in the AtomSpace is a variable node v with domain $\{\mathbf{true}, \mathbf{false}\}$.
- **Annotations.** On each v we store an attribute $\alpha(v) \in Q_{\text{truth}}$ representing its current PLN truth-value or confidence.
- **Factor nodes.** For each PLN inference rule application r deriving ψ from premises $\phi_1, \dots, \phi_k$, we create a factor node f_r with local potential

$$\phi_{f_r}(\alpha(\phi_1), \dots, \alpha(\phi_k)) = \alpha(\phi_1) \otimes_r \dots \otimes_r \alpha(\phi_k)$$

where $\otimes_r$ is the quantale operation associated with rule r .

- **Global inference.** The combined confidence of any expression ϕ is obtained by marginalizing over all paths in the graph, i.e. by join-summing over all factor contributions via the $\bigvee$ operation of Q_{truth} .

8.4.2 Background Inference Heuristic in MeTTa-Style Pseudocode

Below is a sketch of a MeTTa-like module that continually ingests new facts, updates the factor graph, and propagates uncertainties to reveal all high-confidence conclusions:

```
(module PLN.BackgroundInference

  ;--- Initialize global factor graph over quantale Q_truth
  (def fg (fg-create :quantale Q_truth))

  ;--- Add or update a fact phi with truth-value val
  (defn update-fact! (phi val)
    ;; ensure var-node exists
    (when-not (fg-has-var? fg phi)
      (fg-add-var fg phi Boolean))
    ;; annotate its current truth
    (fg-set-annotation fg phi val)
    ;; for each applicable PLN rule r deriving psi from phi:
    (for r in (find-pln-rules phi)
      (let ((psi (apply-rule-head r phi)))
        ;; ensure factor-node exists
        (when-not (fg-has-fctr? fg r phi psi)
          (fg-add-fctr fg r (list phi) psi))
        ;; set rule-specific tensor on factor
        (fg-set-factor-op! fg r :tensor (tensor-op r))))))

  ;--- Main background loop
  (defn background-loop ()
```

```

;; ingest newly asserted or retracted facts
(let ((deltas (observe-atomspace-deltas)))
  (for (phi,val) in deltas
    (update-fact! phi val)))
;; run one round of message-passing
(fg-infer fg (all-vars fg))
;; extract and act on new high-confidence conclusions
(for phi in (vars-with-annotation-above fg threshold)
  (assert-to-atomspace phi (fg-get-annotation fg phi)))
;; schedule next iteration in 1 second
(sleep 1) (background-loop))

;; start the background inference
(background-loop))

```

This design leverages:

- *Sparse updates*, since only the local subgraph around changed facts is revisited;
- *Parallel message-passing*, enabling multi-core scaling over millions of nodes;
- *Quantale joins and tensors*, giving consistent graded aggregation of all inference paths;
- *Anytime behavior*, as partial results are always available during continuous background processing.

8.5 Peircean Weakness in PLN Factor?Graph Message Passing

In standard PLN (Probabilistic Logic Networks) inference, factor?graph message?passing exploits a *weakness* measure to schedule and prune messages, typically based on product?measure assumptions. Here we show how to *replace* that traditional weakness with a *Peircean weakness* derived from a habit?based prior, thereby biasing inference toward the most ?entrenched?? links. This modification is interesting because it aligns the propagation order with Peirce?s notion of habit formation, potentially yielding more human?like generalizations in noisy or sparse data settings.

Peircean Habit Prior. Introduce a joint ?habit?strength?? valuation

$$\mu_P : U \times U \longrightarrow \mathbb{R}_{\geq 0}, \quad \Pr((u, v) \in H) = \frac{\mu_P(u, v)}{Z},$$

capturing Peirce?s tendency to form habits :contentReference[oaicite:2]index=2.

Peircean Weakness Function. For any message carrying formula ϕ whose atomic links are $\text{Links}(\phi) \subseteq U \times U$, define

$$\text{Weakness}_{\text{P}}(\phi) = \sum_{(u,v) \in \text{Links}(\phi)} \mu_{\text{P}}(u, v).$$

Modified Message Scheduling. In the standard Weakness-Driven Belief-Propagation algorithm, each message $m_{f \rightarrow v}$ is assigned

$$w = \text{Weakness}(m_{f \rightarrow v}).$$

We replace this by

$$w_{\text{P}} = \text{Weakness}_{\text{P}}(m_{f \rightarrow v}),$$

and enqueue $(w_{\text{P}}, m_{f \rightarrow v}, v)$ in the priority queue ordered by ascending w_{P} . Pruning and update steps-discarding messages with $w_{\text{P}} < \tau$, combining via $\otimes$, and propagation to neighbors-remain as in the original algorithm :contentReference[oaicite:3]index=3.

Conceptual Impact. By prioritizing messages with lower Peircean weakness, the inference process focuses first on the strongest habitual associations. This yields a propagation order that mirrors human tendencies to rely on well-worn patterns, potentially improving generalization when data are scarce or noisy.

Part II

Universal & Quantum AI

9 Weakness-Flavored Universal AI

Practical intelligence, biological or artificial, is largely about coping with resource restrictions of various sorts – temporal, spatial, energetic and so forth. Nonetheless, it has proved valuable from an AI system and algorithm design perspective to also look at what happens when these resource restrictions are substantially relaxed and explore what AI systems may look like under highly (even very unrealistically) abundant resource scenarios. This research direction has led to systems like Marcus Hutter’s AIXI [Hut05] and Juergen Schmidhuber’s Godel Machine [Sch07], among others. We develop here versions of these theoretical AI systems founded on the weakness concept, showing some interesting theoretical properties that in some ways simplify things compared to classical approaches based on other concepts such as minimum description length.

9.1 WAIXI – AIXI^{tl} via CoDD Weakness

We sketch here an "WAIXI" analogue of Hutter's resource-bounded AIXI^{tl} agent, replacing Solomonoff/MML priors with a *weakness* prior over CoDD programs in Elegant Normal Form.

9.1.1 Agent-Environment Interaction

At each time step $t = 1, 2, \dots$ the agent has history

$$h_{t-1} = a_1 o_1 r_1 \dots a_{t-1} o_{t-1} r_{t-1},$$

where $a_i \in \text{Act}$ are actions, $o_i \in \text{Obs}$ observations, and $r_i \in \mathbb{R}$ rewards. Upon choosing a_t , the environment returns (o_t, r_t) .

9.1.2 Model Class: CoDD Predictors

Let $\mathcal{H}_{t,\ell}$ be the set of all CoDD programs h in ENF such that:

- h takes as input a history h_{t-1} and candidate action a and outputs a distribution $\hat{P}_h(o, r \mid h_{t-1}, a)$.
- The *description cost* $\sigma(h) \leq \ell$ (CoDD cost bound).
- The *runtime* of h on any input of length $\leq t$ is at most t (time bound).

9.1.3 Weakness Prior

Define a prior weight on each $h \in \mathcal{H}_{t,\ell}$ by

$$\pi(h) = \frac{w(h)^\alpha}{\sum_{h' \in \mathcal{H}_{t,\ell}} w(h')^\alpha}, \quad w(h) = \text{CoDD-weakness of } h, \quad \alpha > 0.$$

This favours *weaker* (more general) predictors within the resource bounds.

9.1.4 Value Function

For any $h \in \mathcal{H}_{t,\ell}$ and continuation policy π , define the expected m 'step return

$$V_{t,m}^h(h_{t-1}, a) = \sum_{o,r} \hat{P}_h(o, r \mid h_{t-1}, a) \left[r + \max_{a'} V_{t,m-1}^h(h_{t-1} o a, r, a') \right],$$

with $V_{t,0}^h \equiv 0$ and m chosen so that $t+m$ does not exceed the global time bound.

9.1.5 Action Selection

We then define an AIXI^{tl}-CoDD agent (a species of WAIXI, weakness-driven AIXI, agent) that picks

$$a_t = \arg \max_{a \in \text{Act}} \sum_{h \in \mathcal{H}_{t,\ell}} \pi(h) V_{t,m}^h(h_{t-1}, a).$$

This maximizes the posterior-weighted expected return, where the posterior is just the prior (no updating) restricted to predictors that run within (t, ℓ) .

9.1.6 Discussion

- *Weakness prior vs. MDL prior:* We replace $2^{-\ell(h)}$ by $w(h)^\alpha$, biasing toward CoDD programs that carve out the largest set of possible environments.
- *Resource bounds:* By limiting both $\sigma(h)$ and runtime, we recover the tractability of AIXI^{tl}.
- *Algebraic synergy:* Under CoDD’s monoidal ENF, crossover-like operations preserve weakness additively. Thus if one were to evolve the model class $\mathcal{H}_{t,\ell}$ via genetic operators, schematic theorems guarantee convergence toward maximal-weakness predictors.
- *Updatelessness:* No posterior update is required— $\pi(h)$ is fixed at decision time—so the agent is *weakly updateless* within its resource envelope.

This construction unites the *Occam-weakness* principle, resource-bounded computation, and AIXI’s Bayesian maximization in a single CoDD-based framework.

9.1.7 Performance Guarantees via Weakness-Prior Mixtures

We have described a CoDD-based AIXI^{tl} agent that uses a fixed ”weakness prior” $\pi(h) \propto w(h)^\alpha$ over a resource-bounded model class $\text{HCal}_{t,\ell}$ and selects actions to maximize the mixture value

$$V_{\text{mixture}}(h_{t-1}, a) = \sum_{h \in \text{HCal}_{t,\ell}} \pi(h) V_{t,m}^h(h_{t-1}, a).$$

To begin fleshing out how this sort of agent works, we now give two illustrative theorems showing how the algebraic properties of weakness yield clean bounds on this mixture’s performance relative to any single model in $\text{HCal}_{t,\ell}$.

Theorem 7 (Mixture-Performance Lower Bound). *For every $h \in \text{HCal}_{t,\ell}$, the mixture value satisfies*

$$V_{\text{mixture}}(h_{t-1}, a) \geq \pi(h) V_{t,m}^h(h_{t-1}, a) = \exp(\ln \pi(h)) V_{t,m}^h(h_{t-1}, a).$$

Since $\pi(h) = w(h)^\alpha / Z$ with $Z = \sum_{h'} w(h')^\alpha$, we get

$$V_{\text{mixture}}(h_{t-1}, a) \geq \exp(\alpha \ln w(h) - \ln Z) V_{t,m}^h(h_{t-1}, a).$$

In particular, if $w(h)$ is close to 1 (a very weak, general model) then $\pi(h)$ is large and the mixture value is within a moderate constant factor of V^h .

Sketch. Because π is a probability distribution, $\sum_{h'} \pi(h') = 1$, hence each non-negative summand $\pi(h') V^{h'}$ cannot exceed the total. Rearranging gives the first display. $\square$

Theorem 8 (Regret Bound for True-Environment Model). *Suppose the true environment’s predictor $h^* \in H\text{Cal}_{t,\ell}$ has weakness $w(h^*) > 0$. Then at each decision step,*

$$\max_a V_{t,m}^{h^*}(h_{t-1}, a) - \max_a V_{\text{mixture}}(h_{t-1}, a) \leq \frac{1}{\alpha} \ln \frac{1}{w(h^*)} + \ln Z,$$

where $Z = \sum_h w(h)^\alpha$. As $\alpha \rightarrow \infty$, the regret tends to $\ln Z$, and if $w(h^*)$ is maximal then the regret vanishes.

Sketch. Using the log-sum-exp bound $\max_i x_i \leq \frac{1}{\alpha} \ln \sum_i e^{\alpha x_i}$, set $x_h = \ln V^h$ (viewing values as positive). One shows $\max_h V^h \leq \frac{1}{\alpha} \ln \sum_h w(h)^\alpha V^h$, which in turn bounds $\max V^{h^*}$ by a mixture-logarithm. Converting back from logs yields the stated regret bound. $\square$

Discussion.

- The mixture-performance lower bound shows that any single model’s weakness directly boosts its contribution to the agent’s decision value.
- The regret bound quantifies how much worse the mixture can do compared to the best resource-bounded predictor h^* ; it grows only logarithmically in $1/w(h^*)$ and in the normalizer Z .
- Because under CoDD’s ENF every syntactic extension strictly lowers weakness, one can control Z by limiting program size, and ensure $w(h^*)$ remains substantial for highly general predictors.

These results illustrate how the *monotonicity* and *additivity* properties of CoDD weakness lead to transparent performance guarantees for the AIXI^{tl}-style agent built on a weakness prior.

9.2 Bellman Dynamic Programming as a Special Case of WAIXI

It is interesting to observe how, under appropriate assumptions, a concrete algorithm like Dijkstra’s shortest-path algorithm emerges as a special case of weakness-AIXI. This is also a reasonably AGI-pertinent observation since a number of important practical AGI algorithms can be viewed as approximate stochastic dynamic programming [Goe21d].

Let $(Q, \otimes, \oplus)$ be a commutative quantale. View a directed graph C as a Q -enriched category with

$$\text{Ob}(C) = S, \quad C(s, s') = q_{s \rightarrow s'} \in Q,$$

the “weight” of the arrow $s \rightarrow s'$.

Hypotheses as Paths and Weakness Prior A hypothesis h is any finite path

$$h : s = s_0 \xrightarrow{e_1} s_1 \xrightarrow{e_2} \dots \xrightarrow{e_n} s_n = g.$$

Its weakness is the monoidal product of edge-weights:

$$w(h) = \bigotimes_{i=1}^n C(s_{i-1}, s_i).$$

WAIXI places a prior over paths via

$$\pi(h) = \frac{w(h)^\alpha}{\sum_{h'} w(h')^\alpha}, \quad \alpha > 0,$$

biasing toward *weakest* (i.e. highest- w) paths .

Mixture Value as Soft Bellman Backup Define the WAIXI value of starting in state s by mixing over all paths from s to goal g :

$$V(s) = \bigoplus_{h:s \rightarrow g} \pi(h) [r(h)],$$

where $r(h)$ is the reward of path h . If we further identify $r(h) = w(h)$, then in the limit $\alpha \rightarrow \infty$ the mixture concentrates on the single path maximizing $w(h)$, giving

$$V(s) = \max_{h:s \rightarrow g} w(h).$$

Bellman Fixpoint Emergence The optimal-weakness value $V(s)$ so defined satisfies the Bellman equation in the quantale:

$$V(s) = \bigoplus_{s' \in S} \left(C(s, s') \otimes V(s') \right).$$

This is exactly the shortest-path (or max-convolution) dynamic-programming update, recovered as the $\alpha \rightarrow \infty$ limit of WAIXI planning under a weakness prior.

Greedy Policy Extraction A corresponding optimal action (edge) from s is

$$s^* = \arg \max_{s'} C(s, s') \otimes V(s'),$$

yielding the usual greedy policy for shortest-path planning.

Thus *shortest-path dynamic programming* on a Q -enriched graph is exactly WAIXI over the hypothesis class of all paths, with path-weakness as the Occam prior, and its Bellman backup emerges in the $\alpha \rightarrow \infty$ limit of the WAIXI mixture.

9.3 Universality and Asymptotic Optimality of the Weakness-CoDD AIXI^{tl} Analogue

We now give universality and asymptotic optimality theorems for the Weakness-CoDD AIXI^{tl} agent introduced above.

9.3.1 CoDD-ENF Universality

Let Act, Obs, R be finite sets of actions, observations and rewards, and let $H\text{Cal}_{t,\ell}$ be the class of CoDD-ENF programs of cost $\leq \ell$ and runtime $\leq t$ that map histories and actions to distributions over $\text{Obs} \times \text{R}$.

Theorem 9 (Polynomial-size Simulator). *For any deterministic (or probabilistic) classical Turing machine M running in time $T(n)$ and space $S(n)$ on inputs of length n , there exists a CoDD-ENF program*

$$h \in H\text{Cal}_{t,\ell} \quad \text{with} \quad \ell = O(T(n) + S(n)), \quad t = O(T(n)),$$

which simulates M step-by-step to inverse-polynomial accuracy in its output distribution. Hence the CoDD-ENF class is Turing-universal up to polynomial overhead.

Sketch. Encode each Turing-machine transition as a small CoDD combinator; chaining $T(n)$ transitions yields cost $O(T(n))$. Space bound $S(n)$ limits the size of the tape-encoding subgraph to $O(S(n))$. ENF merging prevents blow-up beyond these overheads. $\square$

9.3.2 Asymptotic Optimality under Weakness Prior

Recall the agent maintains a fixed "weakness prior" $\pi(h) = \frac{w(h)^\alpha}{\sum_{h'} w(h')^\alpha}$ over $H\text{Cal}_{t,\ell}$, and selects action $a_t = \arg \max_a \sum_h \pi(h) V_{t,m}^h(h_{t-1}, a)$, where $V_{t,m}^h$ is the m -step Bellman return under model h .

Theorem 10 (Regret Vanishing). *Suppose the true environment predictor h^* is computable in time $T(n)$ and thus lies in $H\text{Cal}_{t,\ell}$ for all sufficiently large (t, ℓ) . Then for any $\varepsilon > 0$ there exist T_0, L_0 such that for all $t \geq T_0$ and $\ell \geq L_0$,*

$$\max_a V_{t,m}^{h^*}(h_{t-1}, a) - \max_a V_{\text{mix}}(h_{t-1}, a) < \varepsilon.$$

In other words, the agent's regret relative to the optimal computable predictor vanishes as its resource bounds grow.

Sketch. By Theorem 9, h^* can be encoded in $H\text{Cal}_{t,\ell}$ for large (t, ℓ) . Since $\pi(h^*) > 0$ and does not vanish, standard mixture-regret bounds (e.g. log-sum-exp or mutual-information arguments) imply the weighted combination of V^h approaches $\max_h V^h$ within ε as $\pi(h^*) - 1$ when $\ell \rightarrow \infty$ and $\alpha > 0$. $\square$

9.3.3 Mixture-Performance Lower Bound

We also obtain a transparent lower bound on the mixture value in any finite resource regime.

Proposition 2 (Mixture-Value Guarantee). *For every $h \in H\mathcal{C}al_{t,\ell}$,*

$$V_{mix}(h_{t-1}, a) \geq \pi(h) V_{t,m}^h(h_{t-1}, a) = \exp(\alpha \ln w(h) - \ln Z) V_{t,m}^h(h_{t-1}, a),$$

where $Z = \sum_{h'} w(h')^\alpha$. *In particular, high-weakness models h retain non-negligible influence on the agent's decision.*

Proof. Since $\sum_{h'} \pi(h') = 1$, each term $\pi(h') V^{h'} \leq \sum_{h'} \pi(h') V^{h'} = V_{mix}$. $\square$

9.3.4 Conclusion

Together, Theorems 9 and 10 show that the Weakness-CoDD AIXI^{tl} analogue is both computationally universal (simulating any effective environment predictor) and asymptotically optimal (regret vanishes as resource bounds grow). The mixture-value guarantee further quantifies how the Occam-weakness prior ensures robust contributions from general models throughout the decision process.

9.4 Sampling-Based Weakness Estimation and Robust Guarantees

In practical situations we would basically always need to approximate the ideal weakness prior $\pi(h) \propto w(h)^\alpha$ by Monte Carlo sampling from a posterior over possible worlds. We now restate the definitions and theoretical results—including error terms due to finite sampling or bootstrap noise.

9.4.1 Monte Carlo Weakness Estimator

Let $\mathcal{W}$ be the (unknown) distribution over worlds ω consistent with historical data. We draw K independent samples $\omega_1, \dots, \omega_K \sim \mathcal{W}$. For each CoDD program h , define the Monte Carlo estimate

$$\hat{w}(h) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[h(\omega_k) \text{ holds}], \quad \hat{\pi}(h) = \frac{\hat{w}(h)^\alpha}{\sum_{h'} \hat{w}(h')^\alpha}.$$

9.4.2 Bootstrap for Robustness

To guard against faulty observations or biased samples, we repeat the above B times with data-resampling (bootstrap) and obtain $\hat{w}_b(h)$ for $b = 1, \dots, B$. A conservative estimate is $\tilde{w}(h) = \min_b \hat{w}_b(h)$, or the lower $(100-p)\%$ percentile of $\{\hat{w}_b(h)\}$. We then set $\tilde{\pi}(h) \propto \tilde{w}(h)^\alpha$.

9.4.3 Uniform Error Bound

By Hoeffding's inequality, for any $\delta > 0$ and finite program class HCal , with probability at least $1 - \eta$ over sampling,

$$\sup_{h \in \text{HCal}} |\hat{w}(h) - w(h)| \leq \delta \quad \text{provided} \quad K \geq \frac{1}{2\delta^2} \ln \frac{2|\text{HCal}|}{\eta}.$$

9.4.4 Effect on Universality

Theorem 11 (Support Preservation). *If the true environment model $h^* \in \text{HCal}$ has $w(h^*) > 0$, then for any $\delta < w(h^*)$, with high probability $\hat{w}(h^*) > 0$ and hence $\hat{\pi}(h^*) > 0$. Thus the Monte Carlo-approximate agent still assigns nonzero weight to h^* and retains universality.*

9.4.5 Regret Bound with Prior Error

Let V_{mix} and $\hat{V}_{\text{mix}}$ be the exact and approximate mixture values respectively, and let $Z = \sum_h w(h)^\alpha$. If $\sup_h |\hat{w}(h) - w(h)| \leq \delta$ then $\|\hat{\pi} - \pi\|_1 \leq O(\delta)$, and we have:

Theorem 12 (Regret Degradation). *For any history h_{t-1} and decision step,*

$$\left| \max_a V_{\text{mix}}(h_{t-1}, a) - \max_a \hat{V}_{\text{mix}}(h_{t-1}, a) \right| \leq O(\delta) \max_a \left| \ln V^{h^*}(h_{t-1}, a) \right| + O(\delta) \ln Z.$$

In particular, as $\delta \rightarrow 0$, the approximate regret converges to the ideal regret.

9.4.6 Sample Complexity

Combining the above, to ensure that with probability $\geq 1 - \eta$ the regret degradation is $\leq \varepsilon$, it suffices to choose

$$\delta = O\left(\frac{\varepsilon}{\ln Z}\right), \quad K = O\left(\frac{1}{\delta^2} \ln \frac{|\text{HCal}|}{\eta}\right).$$

9.4.7 Bootstrap Error Robustness

Bootstrap sampling further reduces bias: if each $\hat{w}_b(h)$ estimates $w(h)$ with error δ and the bootstrap distribution has failure probability η/B , then the aggregated estimate $\tilde{w}(h)$ satisfies the same δ -error bound with probability $1 - \eta$, preserving the above universality and regret bounds.

These results show that all earlier universality and regret-vanishing theorems for the Weakness-CoDD AIXI^{tl} analogue continue to hold up to small additive terms $O(\delta)$ due to Monte Carlo or bootstrap estimation error.

9.5 Strengthened Regret Bounds via CoDD Weakness Structure

The generic regret-degradation bounds of the previous section apply to *any* prior. However, because the CoDD weakness prior obeys a *Kraft-type exponential tail*, we can obtain substantially tighter bounds that depend only on the *description length* of the true model rather than the total hypothesis-space size.

9.5.1 Exponential Tail of the Weakness Prior

Recall that under CoDD ENF, each program h has weakness

$$w(h) = N 2^{-\sigma(h)},$$

and we set

$$\pi(h) = \frac{w(h)^\alpha}{\sum_{h'} w(h')^\alpha} = \frac{2^{-\alpha \sigma(h)}}{Z}, \quad Z = \sum_{h'} 2^{-\alpha \sigma(h')} \leq 1.$$

Hence

$$\ln \frac{1}{\pi(h)} = \alpha \sigma(h) \ln 2 + \ln \frac{1}{Z} \leq \alpha \sigma(h) \ln 2.$$

9.5.2 Improved Mixture Regret Bound

Let V^* be the optimal m -step return under the true model h^* , and let V_{mix} be the mixture value under π . Then for any $\alpha > 0$,

$$V_{\text{mix}} \geq \pi(h^*) V^* \implies V^* - V_{\text{mix}} \leq (1 - \pi(h^*)) V^* \leq V^* - 2^{-\alpha \sigma(h^*)} V^*.$$

Equivalently, using a log-sum-exp argument,

$$V^* - V_{\text{mix}} \leq \frac{1}{\alpha} \ln \frac{1}{\pi(h^*)} \leq \sigma(h^*) \ln 2.$$

Thus the *regret* is bounded purely in terms of the description length $\sigma(h^*)$ of the true model, independent of the total number of programs.

9.5.3 Effect of Monte Carlo / Bootstrap Error

Suppose our estimated prior $\hat{\pi}$ arises from a weakness estimate $\hat{w}(h)$ satisfying $\sup_h |\hat{w}(h) - w(h)| \leq \delta$. Then one shows $\|\hat{\pi} - \pi\|_1 = O(\delta)$, and the mixture-value degradation satisfies

$$|V_{\text{mix}} - \hat{V}_{\text{mix}}| \leq O(\delta) \max_a V^*(a),$$

so that the total regret under the approximate prior obeys

$$V^* - \hat{V}_{\text{mix}} \leq \sigma(h^*) \ln 2 + O(\delta) \max_a V^*(a).$$

In particular, for sufficiently small δ this remains dominated by $\sigma(h^*) \ln 2$.

9.5.4 Sample Complexity Improvement

Because the effective support of π concentrates on programs of length $\sigma \leq L$ with total mass at least $1 - 2^{-\alpha L}$, one needs only

$$K = O\left(\frac{1}{\epsilon^2} \left(\alpha L \ln 2 + \ln \frac{1}{\eta}\right)\right)$$

Monte Carlo samples to guarantee δ -uniform error with probability $1 - \eta$. Thus sample complexity scales with the true model's description length $L = \sigma(h^*)$ rather than $\ln |\text{HCal}|$.

Summary By exploiting the exponential decay of the CoDD weakness prior, we replace generic $\ln |\text{HCal}|$ -type regret and sampling-complexity bounds with much tighter, *length-based* bounds that depend only on the description length $\sigma(h^*)$ of the true environment model.

9.6 When Description Length Dominates Sampling Error

What then can we say about the conditions under which the description length rather than the MC/sampling error would be dominant?

Recall our regret bound for the approximate-prior agent:

$$\text{Regret} \leq \sigma(h^*) \ln 2 + C \delta V_{\max},$$

where

- $\sigma(h^*) \ln 2$ is the description-length term,
- $\delta = \sup_h |\hat{w}(h) - w(h)| = O(1/\sqrt{K_{\text{eff}}})$ is the Monte Carlo error,
- $V_{\max} = \max_a V^{h^*}(h_{t-1}, a)$,
- C is a constant absorbing the log-sum-exp factors.

9.6.1 Dominance Condition

The description-length term dominates whenever

$$\sigma(h^*) \ln 2 \gg C \delta V_{\max} \iff \delta \ll \frac{\sigma(h^*) \ln 2}{C V_{\max}}.$$

Since $\delta = O(1/\sqrt{K_{\text{eff}}})$, this becomes

$$K_{\text{eff}} \gg \frac{C^2 V_{\max}^2}{(\sigma(h^*) \ln 2)^2}.$$

Here K_{eff} is the *effective* number of independent samples, which in practice is K/τ if the sampler has mixing time τ . Thus in real-world systems:

- If the true model's description length $\sigma(h^*)$ is *moderate* and $V_{\max}$ is bounded, then a *polynomially* large sample size K suffices to make the sampling term negligible.
- If the generative sampler mixes *rapidly* (small τ), $K_{\text{eff}} \approx K$ and one achieves $\delta \rightarrow 0$ quickly, so the description-length term $\sigma(h^*) \ln 2$ sets the regret floor.
- Conversely, in systems with very *complex* generative models (large mixing time τ or high effective dimension), one may need impractically large K before δ falls below $\sigma(h^*) \ln 2 / (C V_{\max})$. In that regime the MC error dominates until sufficient samples are drawn.

9.6.2 Scaling and Convergence of the Generative Model

More precisely, if the posterior sampler satisfies a spectral gap $\gamma > 0$, then by standard Markov-chain concentration $\delta = O(\sqrt{\frac{\tau}{K}})$ with $\tau = 1/\gamma$. Substituting gives

$$\text{Regret} \leq \sigma(h^*) \ln 2 + O\left(\sqrt{\frac{\tau}{K}} V_{\max}\right).$$

Hence for fixed τ and $V_{\max}$, the sampling error term decays as $1/\sqrt{K}$ and the description-length term remains the dominant factor once $K = \Omega(\tau V_{\max}^2 / (\sigma(h^*) \ln 2)^2)$.

In summary, in realistic settings with well-behaved, moderately sized generative models, one quickly reaches a regime where $\sigma(h^*) \ln 2$ dominates the regret, and further sampling yields diminishing returns. Only in the presence of very slow mixing or high-dimensional posteriors does the Monte Carlo error term remain significant for large numbers of samples.

9.6.3 Concrete Numerical Illustration

To get a more concrete feel for when the description-length term $\sigma(h^*) \ln 2$ dominates the sampling error $C \delta V_{\max}$, recall

$$\delta \approx \sqrt{\frac{\tau}{K}}, \quad K \gg \tau \left(\frac{C V_{\max}}{\sigma(h^*) \ln 2} \right)^2.$$

Here τ is the mixing-time of your world-sampler, $V_{\max}$ the maximum achievable return, and $C \approx 1$ a modest constant from the log-sum-exp bound. We illustrate two stylized real-world regimes:

Case A: Fast Mixing, Concise Model.

$$\tau = 10^2, \quad \sigma(h^*) = 20, \quad V_{\max} = 10.$$

Then

$$K \gg 10^2 \left(\frac{1 \times 10}{20 \times 0.693} \right)^2 = 10^2 (0.721)^2 \approx 52.$$

So $K \sim 10^2\text{-}10^3$ samples suffice, and the *description-length* term $20 \ln 2 \approx 13.9$ dominates.

Case B: Slow Mixing, Compact Model.

$$\tau = 10^6, \quad \sigma(h^*) = 10, \quad V_{\max} = 100.$$

Then

$$K \gg 10^6 \left(\frac{1 \times 100}{10 \times 0.693} \right)^2 = 10^6 (1.443)^2 \approx 2.1 \times 10^6.$$

Here one needs a few million samples before $\delta V_{\max} < \sigma \ln 2 \approx 6.93$. Only after $K \gtrsim 2 \times 10^6$ does the description-length term begin to dominate.

Interpretation.

- In well-behaved domains with small mixing-time ($\tau \lesssim 10^3$) and moderately concise true models ($\sigma \gtrsim 10$), $\sigma \ln 2$ dominates after only a few hundred samples.
- In challenging settings with slow samplers ($\tau \gtrsim 10^6$) or large returns ($V_{\max} \gtrsim 10^2$), you may need millions of samples before the Occam-weakness term is the primary source of regret.

Thus in practical systems one should benchmark the sampler’s mixing-time τ and the true model’s description length $\sigma(h^*)$ to choose a sample size K that ensures the Occam-bound $\sigma(h^*) \ln 2$ is indeed the dominant regret term.

9.7 Practical Viability of the Weakness-CoDD AIXI^{tl} Analogue

AIXI^{tl} is intended more of a theoretical construct than a practical AI approach. However, it is still interesting to explore how far one could go toward making it practically usable. There are some cases where this might actually be feasible, but more importantly, this can also be a way of framing the utilization of other more practical techniques. Here we will make some back of the envelope calculations regarding the practical use of both Weakness-CoDD AIXI^{tl} and, more scalably for most real-world uses, the combination of evolutionary learning and EDAs [nL02] for learning control strategies. The latter approach has been explored theoretical and in prototype experiments as part of the Hyperon AGI design, and here

We compare two implementation strategies for a Weakness-CoDD AIXI^{tl}-style planner:

1. *Monte Carlo sampling* of possible worlds to estimate the weakness prior.
2. *Asynchronous EDA* (Estimation-of-Distribution Algorithm) seeded from a continually learned knowledge graph.

In the following sections we will give stylized numerical estimates for the cost of these two approaches, and explore these in the context of two concrete use-cases: a mobile humanoid robot and a persistent, curious chatbot research assistant.

9.8 Monte Carlo Sampling Approach

9.8.1 Weakness Prior Estimation

Let $\mathcal{W}$ be the posterior over worlds given the agent's history and let τ be the mixing time of our sampler. Draw $\omega_1, \dots, \omega_K \sim \mathcal{W}$ independently and define

$$\hat{w}(h) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[h(\omega_k) \text{ holds}], \quad \hat{\pi}(h) \propto \hat{w}(h)^\alpha.$$

By Hoeffding's inequality, with probability $1 - \eta$,

$$\sup_h |\hat{w}(h) - w(h)| \leq \delta \quad \text{whenever} \quad K \geq \frac{1}{2\delta^2} \ln \frac{2|\text{HCal}|}{\eta}.$$

9.8.2 Regret Bound

Let $\sigma(h^*)$ be the description length (in CoDD ENF) of the true optimal model and $V_{\max}$ the maximum return per decision. Then

$$\text{Regret} \leq \sigma(h^*) \ln 2 + C \delta V_{\max}, \quad \delta = \sqrt{\frac{\tau}{K}}.$$

To make the sampling term small compared to the Occam term, choose

$$K \gg \tau \left(\frac{C V_{\max}}{\sigma(h^*) \ln 2} \right)^2.$$

9.8.3 Stylized Numerical Examples

Case A: Fast Mixing, Concise Model

$$\tau = 10^2, \sigma(h^*) = 20, V_{\max} = 10 \implies K \gg 10^2 (10/(20 \cdot 0.693))^2 \approx 52.$$

Regret floor $\sigma \ln 2 \approx 13.9$.

Case B: Moderate Mixing, Large Controller

$$\tau = 10^3, \sigma(h^*) = 75, V_{\max} = 20 \implies K \gg 10^3 (20/(75 \cdot 0.693))^2 \approx 150.$$

Regret floor $\sigma \ln 2 \approx 52$.

Case C: Slow Mixing, Compact Model

$$\tau = 10^6, \sigma(h^*) = 10, V_{\max} = 100 \implies K \gg 10^6 (100/(10 \cdot 0.693))^2 \approx 2.1 \times 10^6.$$

Regret floor $\sigma \ln 2 \approx 6.93$.

9.9 Asynchronous EDA with Knowledge-Graph Seeding

9.9.1 Setup and Workload

Suppose we use an asynchronous EDA with

$$N_{Rmpop} = 10^4, \quad G = 10^4, \quad K = 100, \quad t_{Rmeval} = 1\text{Rm s}, \quad \sigma = 50,$$

and assume asynchronicity yields a work-reduction factor $f \approx 0.5$.

In this case, the serial evaluation cost would be

$$T_{Rmsync} = G[N_{Rmpop} K t_{Rmeval} + N_{Rmpop} \sigma^2] \approx 2.6 \times 10^{11} \text{ s}.$$

With $f = 0.5$, the asynchronous serial cost is

$$T_{Rmasync,serial} = 0.5 T_{Rmsync} \approx 1.3 \times 10^{11} \text{ s}.$$

9.9.2 Massive Parallel Scaling

Distribute the $N_{Rmpop} \times K = 10^6$ eval tasks over M workers:

$$T_{Rmwall} \approx \frac{1.3 \times 10^{11}}{M} \text{ s}.$$

M	<i>wall-Clock Time</i>
10^4	$1.3 \times 10^7 \text{ s} \approx 150 \text{ d}$
10^5	$1.3 \times 10^6 \text{ s} \approx 15 \text{ d}$
10^6	$1.3 \times 10^5 \text{ s} \approx 1.5 \text{ d}$
10^7	$1.3 \times 10^4 \text{ s} \approx 3.6 \text{ h}$
10^8	$1.3 \times 10^3 \text{ s} \approx 22 \text{ min}$

9.9.3 Seeding from a Continual Knowledge Graph

By initializing the EDA's probabilistic model from a learned knowledge graph, one can reduce to

$$N_{Rmpop} = 100, \quad G = 20, \quad K = 500, \quad t_{Rmeval} = 0.5\text{Rm s}, \quad f = 0.8, \quad M = 500.$$

Serial cost:

$$T_{Rmasync,serial} = 0.8 \cdot 20 \cdot 100 \cdot 500 \cdot 0.5 = 4 \times 10^5 \text{ s},$$

wall-time:

$$T_{Rmwall} \approx \frac{4 \times 10^5}{500} = 800 \text{ s} \approx 13.3 \text{ min}.$$

9.10 Use Case: Mobile Humanoid Robot

9.10.1 Monte Carlo Planner

Assume

$$\tau = 10^3, \quad \sigma(h^*) = 75, \quad V_{\max} = 20, \quad t_{Rmeval} = 1\text{Rm s}, \quad M = 100.$$

Then $K \approx 150$ gives sequential cost 150s and with $M = 100$ yields ~ 1.5 s decision latency. For $K = 500$, latency ~ 5 s.

9.10.2 Asynchronous EDA Planner

Mid-level primitives: choose

$$N_{Rmpop} = 100, G = 20, K = 50, t_{Rmeval} = 0.5Rm\ s, f = 0.8, M = 100.$$

Serial cost $\approx 0.8 \cdot 20 \cdot 100 \cdot 50 \cdot 0.5 = 4 \times 10^4$ s, wall-time ≈ 400 s (~ 6.7 m).

High-level tasks ($M \geq 10^3$) reach sub-minute planning.

9.11 Use Case: Chatbot Research Assistant

9.11.1 Monte Carlo Planner

Assume

$$\tau = 10^4, \sigma(h^*) = 200, V_{\max} = 100, t_{Rmeval} = 0.5Rm\ s, M = 1000.$$

Then $K \approx 5 \times 10^3$ gives sequential cost 2500 s (~ 42 m) and with $M = 1000$ yields ~ 2.5 s latency.

9.11.2 Asynchronous EDA Planner

Use

$$N_{Rmpop} = 100, G = 20, K = 500, t_{Rmeval} = 0.5Rm\ s, f = 0.8, M = 500.$$

Serial cost 4×10^5 s, wall-time ~ 800 s (~ 13.3 m). Larger clusters ($M \geq 2000$) reduce to minutes.

9.12 Conclusions

Wrapping up our crude back-of-the-envelope calculations:

- Monte Carlo sampling attains Occam-dominated regret with $K = 10^2$ - 10^3 samples.
- Asynchronous EDA seeded by a knowledge graph converts exponential search into $O(G N_{Rmpop} \sigma^2)$ work, feasible with $M \geq 10^6$ cores.
- Humanoid robot high-level planning in seconds, mid-level in minutes; low-level on dedicated loops.
- Chatbot planning in seconds; EDA variants in minutes.

Both strategies, with moderate parallelism and hierarchical control, render the evolutionary/EDA Weakness-CoDD AIXI^{tl} analogue viable for real-world systems. This sort of approach could be pursued on its own as an AGI architecture, or (more promising in our view) it can be coupled with other structures and methods in an integrative design like Hyperon. Our emphasis here has been on showing a fairly direct conceptual and formal route from abstract considerations regarding weakness and AIXI^{tl} to practical components of real-world AI systems.

9.13 Weakness-AIXI, Quantum Evolution, and Wu Wei Geodesics

Next, taking a different view on the behavior of weakness-AIXI systems, we endeavor to prove that a Bayesian agent endowed with a *weakness prior* (“weakness-AIXI”) updates its belief distribution along geodesics of an entropically-regularized optimal-transport cost in a quantale of representational effort. We then show that quantum evolution, in its hydrodynamic (Madelung) form, follows the same geodesic principle under a Schrodinger-bridge cost. This allows an interesting interpretation of the universal wave function as a belief distribution over classical configurations, whose dynamics coincide with the agent’s minimal-effort inference.

9.13.1 Schrödinger bridges

A Schrödinger bridge interpolates between two probability distributions μ_0 and μ_1 by finding the most likely evolution under a reference stochastic process $p(x, t)$, subject to endpoint constraints. The variational principle underlying this process minimizes the entropic transport cost:

$$W_{\text{SB}}[\rho, v] = \int_{t_0}^{t_1} \int_{\mathbb{R}^n} \rho(x, t) \left(\frac{1}{2} m \|v(x, t)\|^2 + \epsilon \log \left(\frac{\rho(x, t)}{p(x, t)} \right) \right) dx dt, \quad (3)$$

subject to the continuity equation constraint

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

Here:

- The first term quantifies the kinetic effort required to deform the distribution $\rho(x, t)$ through velocity field $v(x, t)$.
- The second term penalizes divergence from the reference evolution $p(x, t)$, which could represent free diffusion or another background process.
- The parameter ϵ controls the trade-off between kinetic and entropic cost; in the limit $\epsilon \rightarrow 0$, the functional reduces to Wasserstein-2 transport.

Minimizing W_{SB} yields the Schrödinger bridge between the boundary distributions. In our framework, we interpret this minimal entropic-kinetic action as a measure of total representational effort. This represents a refined quantification of “weakness” in evolving belief states or quantum configurations

9.13.2 Weakness-AIXI and Optimal Transport

Let $\mathcal{H}$ be a countable hypothesis class and let

$$\Delta(\mathcal{H}) = \{\rho : \mathcal{H} \rightarrow [0, 1] \mid \sum_{h \in \mathcal{H}} \rho(h) = 1\}$$

be the belief simplex. Fix a commutative quantale $(Q, \leq, \otimes, e)$ and a *weakness prior* $w : \mathcal{H} \rightarrow Q$. At each discrete time t , the agent holds $\rho_t \in \Delta(\mathcal{H})$, selects an action policy π_t , receives an observation o_t , and updates by

$$\rho_{t+1}(h) \propto \rho_t(h) P(o_t | h, \pi_t) \exp\left(-\frac{1}{\epsilon} w(h)\right),$$

where $\epsilon > 0$ is an entropy-regularization parameter.

Under the continuum limit $t \mapsto t\Delta t$, small Δt , and large-horizon, the discrete update induces a flow $\rho(t)$ satisfying

$$\partial_t \rho = -\nabla_{W_2} \left\{ \mathbb{E}_\rho[w(h)] - \mathbb{E}_\rho[\log P(o | h, \pi(t))] + \epsilon \text{KL}(\rho \| p) \right\},$$

where p is the prior predictive. By the Jordan-Kinderlehrer-Otto theory, this is the gradient flow of a free-energy in the Wasserstein-2 metric with entropic regularization. Equivalently:

Theorem 13. *The continuum-time belief dynamics $\rho(t)$ minimize the Schrodinger-bridge cost*

$$W_{\text{SB}}[\rho, v] = \int_0^T \int_{\mathcal{H}} \rho(h, t) \left(\frac{1}{2} \|v(h, t)\|^2 + \epsilon \log \frac{\rho(h, t)}{p(h, t)} \right) dh dt$$

subject to $\partial_t \rho + \nabla \cdot (\rho v) = 0$. Thus $\rho(t)$ is a geodesic in the quantale of weakness costs.

Sketch. Discretize time and apply the minimizing-movement (JKO) scheme:

$$\rho^{k+1} = \arg \min_{\rho} \left\{ \frac{1}{2\Delta t} W_2^2(\rho, \rho^k) + \mathcal{F}[\rho] \right\},$$

with $\mathcal{F}[\rho] = \mathbb{E}_\rho[w] - \mathbb{E}_\rho[\log P(o)] + \epsilon \text{KL}(\rho \| p)$. As $\Delta t \rightarrow 0$, this converges to the W_2 -gradient flow of $\mathcal{F}$, which is equivalently the Schrodinger-bridge variational solution. Standard quantale-enriched-category arguments then yield the geodesic property. $\square$

9.13.3 Parallel with Quantum Evolution

In quantum mechanics, let $\psi(x, t) = \sqrt{\rho(x, t)} e^{iS(x, t)/\hbar}$ satisfy the Schrodinger equation. The Madelung transform yields

$$\partial_t \rho + \nabla \cdot (\rho v) = 0, \quad v = \nabla S / m,$$

and a quantum Hamilton-Jacobi equation with quantum potential. One shows that (ρ, v) minimize exactly the same Schrodinger-bridge cost

$$W_{\text{SB}}[\rho, v] = \int_0^T \int_{\mathbb{R}^n} \rho(x, t) \left(\frac{1}{2} m \|v\|^2 + \hbar \log \frac{\rho}{p} \right) dx dt,$$

with p the free-diffusion prior and $\epsilon = \hbar$. Thus quantum evolution is also a geodesic in the same quantale of representational effort.

9.13.4 Quantum Wave Function as Belief Distribution

Under this unified picture:

- The magnitude $\rho(x, t) = |\psi(x, t)|^2$ is the *belief* weight on classical configuration x .
- The phase $S(x, t)/\hbar$ encodes the *direction* of belief update (velocity v).
- Planck’s constant $\hbar$ is the *entropic budget* controlling the trade-off between deviating from free evolution and kinetic belief transport.

Hence the universal wavefunction ψ can be viewed as a *universal belief function*, whose dynamics coincide with the minimal-effort inference of a weakness-AIXI agent. Wú wéi – effortless action – becomes the common principle governing both rational belief updates and quantum motion in a single information-geometric framework.

9.14 A Weakness-Based Goedel-Machine Analogue

Having made a weakness/CoDD analogue of AIXI^{tl} and explored its properties a bit, we now create an analogous analogue of a (computable) Godel Machine along the rough lines of [Sch07]. That is, we sketch a Goedel-Machine-style self-modifying agent whose proof system and code are both expressed in CoDD’s ENF, and whose self-improvement criterion is based on *weakness* (Occam bias) and expected utility.

We will not explore this Goedel Machine construct as thoroughly as we have the AIXI analogues above, but further theoretical development could certainly be fruitfully pursued.

9.14.1 Godel Machine Concept

A Goedel Machine continually searches for proofs that rewriting a part of its own code will increase its expected performance. In our analogue, the machine’s *code* is a CoDD program in Elegant Normal Form (ENF), and its *proofs* are also CoDD terms via the Curry-Howard correspondence in a simple dependently typed logic. The machine uses a *weakness prior* to bias towards more general (higher-weakness) rewrites, and a utility model to evaluate the expected reward of each candidate program. When it finds a proof that *both*

- the replacement code runs within its resource bounds, and
- the expected utility gain exceeds the cost of rewriting,

it performs the rewrite. This ensures it only self-modifies in ways that provably improve its weakness-prior-weighted performance.

9.14.2 Formal Specification

Code and Proofs as CoDD ENF Terms. Let CoDD_{ENF} be the set of all ENF-normalized CoDD programs. We fix a simple typed lambda-calculus $\mathcal{L}$ with:

Types: $\text{Code}, \text{Proof}, \text{Bool}, \mathbb{R}, \dots$

and with Curry-Howard:

$$\text{Proof}(P) \text{ Cong } \{p \in \text{CoDD}_{\text{ENF}} \mid p \text{ has type } P\}.$$

Utility and Weakness Predicates. We assume a fixed model $M \in \text{CoDD}_{\text{ENF}}$ of the environment that, given a history h and candidate code c , returns a distribution over future rewards. Define:

$$\text{EU}(c) = \mathbb{E}_{r \sim M(c)}[r], \quad \text{Cost}(c) = \sigma(c),$$

and the *weakness-prior-weighted utility*

$$\text{WU}(c) = \text{EU}(c) + \lambda \ln w(c),$$

where $w(c)$ is the CoDD weakness of c and $\lambda > 0$ trades off Occam bias.

Self-Improvement Criterion. At each step t , let its current code be c_t . The machine searches for a proof term

$$\pi : \text{Proof}[\text{WU}(c') - \text{WU}(c_t) > \delta]$$

in the logic $\mathcal{L}$, where $\delta > 0$ exceeds the cost of self-modification. That is, it proves

$$\text{WU}(c') > \text{WU}(c_t) + \delta.$$

Rewrite Rule. Once such a proof π is found, the machine executes the CoDD program

$$c_{t+1} = \text{Replace}(c_t, c')$$

which replaces the designated subgraph of c_t with c' . By proof soundness, $\text{WU}(c_{t+1})$ is guaranteed to exceed $\text{WU}(c_t)$.

9.14.3 Theoretical Properties

Theorem 14 (Safe Self-Modification). *If $\mathcal{L}$ is consistent and all proofs are found within the machine's resource bounds, then every self-rewrite strictly increases $\text{WU}(c)$. Furthermore, because weakness $\ln w(c)$ is monotonic in ENF size, the machine avoids pathological overfitting rewrites that reduce Occam-bias.*

Sketch. Soundness of $\mathcal{L}$ ensures π implies $\text{WU}(c') > \text{WU}(c_t) + \delta$. The ENF-monotonicity of $w(c)$ prevents arbitrary code bloat from increasing weakness, so the Occam term $\lambda \ln w(c)$ penalizes overly specific rewrites. $\square$

Theorem 15 (Eventual Optimality). *Assume the set of candidate rewrites and proof search are fair (every true improvement can eventually be proven). Then the sequence (c_t) converges—modulo resource bounds—to a (local) maximum of $WU(c)$.*

Sketch. Each rewrite increases WU by at least δ . Since the code size and runtime are bounded, there are only finitely many programs with WU below the global maximum. Fair proof search ensures eventual discovery of each beneficial rewrite. $\square$

9.14.4 Discussion

This construction implements a *Weakness Goedel Machine*:

- Self-modification is *provably safe* and *weakness-aware*.
- The Curry-Howard embedding into CoDD ENF unifies code, proofs, and weakness measurement.
- Occam bias is baked in via the $\ln w(c)$ term, preventing over-specialized rewrites.
- Resource bounds on proof search and program size guarantee computability.

Thus we obtain a computable, weakness-driven analogue of Schmidhuber’s Goedel Machine, firmly grounded in the algebraic theory of CoDD.

10 Mind-World Correspondence Principles

In [Goe13] it is proposed that one ideal form for a theory of general intelligence to take would be: You feed it a description of an environment, a set of goals and a set of resource limitations.. and it tells you a cognitive architecture that would effectively work toward the given goals in the given environment under the given resource limitations (when this is feasible at all).

This ideal theoretical ”Mind-World Correspondence” framework has not yet been achieved, but we can take some steps in this direction using the theoretical concepts developed above.

10.1 From Weakest Model to Smallest CoDD Controller

We start by exploring the question: Suppose we have a characterization of the weakest model M explaining which actions will lead to which results for a certain system in a certain environment, based on a volume of historical data. What can we say about the nature of the smallest CoDD program P that will allow a system to perform optimally in the environment based on model M ? Can we generate P from M in some direct way using the relevant symmetries?

It seems this is actually relatively straightforward given the formalisms we've set up. (Of course, this relative simplicity is the result of operating fairly obliviously to practical resource restrictions, so it needs to be taken with some grains of salt in terms of practical applications. In real life, outside very simple situations one can't really aim at "performing optimally." But the theory is still interesting, and potentially of some use.)

Of course this is not really the idealized general theory of general intelligence fantasized above, because knowing this sort of "weakest model" is different than just knowing "a description of an environment, a set of goals and a set of resource limitations" – there is a lot of computational work inbetween. But it is still a somewhat interesting step forward.

Suppose we have learned a *weakest* CoDD model

$$M : S \times A \longrightarrow \text{Dist}(O \times R \times S)$$

in Elegant Normal Form (ENF), where S is the state-space, A the action-space, O observations, R rewards, and Dist denotes a distribution object. We now ask:

Q. What is the *smallest* CoDD program

$$P : S \longrightarrow A$$

that, when run in the environment, achieves optimal expected return according to M ? And can we construct P directly from M , perhaps exploiting any symmetries of M ?

Optimal-Control as Program Composition. By dynamic-programming over a finite horizon H , the optimal policy P is given by

$$P(s) = \arg \max_{a \in A} \mathbb{E}_{(o,r,s') \sim M(s,a)} \left[r + \max_{a'} V(s') \right],$$

where $V(s)$ is the optimal value-function. In CoDD this becomes the composition of two subprograms:

$$P = \text{ArgMax} \circ \text{Expect} \circ M,$$

where

- M is the given model $S \times A \rightarrow \text{Dist}(O \times R \times S)$,
- Expect computes the Bellman-update $\text{Dist}(O \times R \times S) \rightarrow \text{Dist}(R + V(S))$, and
- ArgMax picks the action a maximizing the expectation.

Cost and Weakness Accounting. Because M is in ENF with cost $\sigma(M)$, and Expect and ArgMax can each be implemented by a fixed finite CoDD fragment of cost c_{RmE} and c_{RmA} , the total cost is

$$\sigma(P) = \sigma(M) + c_{RmE} + c_{RmA}.$$

No smaller program can realize the same semantics, since any program computing the Bellman expectation or the argmax must contain those combinators. Thus P is *minimal* in ENF.

Exploiting Symmetries. If M satisfies a group-equivariance $\forall g \in G : M(g \cdot s, g \cdot a) = g \cdot M(s, a)$, then Expect $\circ M$ and hence P are also G -equivariant. By passing to the quotient S/G , one can reduce the effective state space and synthesize a *smaller* ENF for M as $\bar{M} : S/G \times A/G \rightarrow \text{Dist}(\dots)$. The corresponding controller $\bar{P}$ on S/G then lifts back to P on S , yielding a program whose cost is reduced by the size of the group orbits:

$$\sigma(\bar{P}) = \sigma(\bar{M}) + c_{RmE} + c_{RmA} \ll \sigma(P).$$

Direct Generation Procedure.

1. *Normalize* the learned M into ENF, merging shared subgraphs.
2. *Detect symmetries* of M (automorphism group G) and construct the quotient model $\bar{M}$.
3. *Compose* $\bar{P} = \text{ArgMax} \circ \text{Expect} \circ \bar{M}$.
4. *Lift* $\bar{P}$ to P on the full state space by acting through the group G .

This yields the *smallest* ENF program P implementing the optimal policy, directly synthesized from M and its symmetries.

10.1.1 Computational Complexity of CoDD-ENF Program Synthesis

We analyze each step of this synthesis pipeline in terms of the size N of the CoDD graph for M (number of nodes+edges) and the size $|G|$ of its automorphism group.

ENF Normalization. Merging identical subgraphs and re-ID-ing nodes can be done in $O(N \log N)$ time via a bottom-up hash-consing pass. In practice this is linear-ithmic in the number of combinator applications.

Symmetry Detection. Computing the automorphism group of a general graph is graph-isomorphism-complete. State-of-the-art algorithms (e.g. `nauty/Traces`) run in quasipolynomial or heuristically near-linear time on typical structured graphs, but worst-case cost can be superpolynomial (quasipolynomial in N).

Quotient Construction. Once generators for G are known, building the quotient model $\bar{M}$ by collapsing each orbit takes $O(N \cdot \log|G|)$ to identify orbits and $O(N)$ to rebuild edges.

Controller Composition. Forming $\bar{P} = \text{ArgMax} \circ \text{Expect} \circ \bar{M}$ is a fixed-size CoDD composition whose cost is $O(1)$ in N (up to the cost of copying the quotient graph).

Lifting via Group Action. Lifting $\bar{P}$ back to P requires applying each group element in G to $\bar{P}$'s subgraph, at cost $O(|G| \cdot |\bar{P}|)$.

Overall Cost.

$$O(N \log N) + \text{GI}(N) + O(N \log|G|) + O(|G| \cdot N_{\bar{P}}),$$

where $\text{GI}(N)$ is the cost of graph-automorphism on N nodes and $N_{\bar{P}}$ is the size of the quotient controller. In most practical cases the ENF normalization and quotient-construction terms dominate, with symmetry detection handled by efficient heuristics and lifting linear in the group size.

Worst-Case. In the worst case, $\text{GI}(N)$ can be quasipolynomial, for example $\text{GI}(N) = \exp(O(\sqrt{N \log N}))$. Hence the overall worst-case complexity is

$$T_{\text{worst}}(N) = \exp(O(\sqrt{N \log N})) + O(N \log N + N \log|G| + |G| N),$$

which is dominated by the graph-isomorphism term as $N \rightarrow \infty$.

Typical-Case (Heuristic). Empirically, state-of-the-art graph-automorphism algorithms (e.g. `nauty`, `Traces`) run in near-linear or low-degree polynomial time on structured graphs arising from CoDD programs, and often produce small automorphism groups $|G| = O(1)$ or $O(\log N)$. In such cases the pipeline cost reduces to

$$T_{\text{typical}}(N) = O(N \log N + N \log|G| + |G| N) = O(N \log N) \quad (|G| = O(\log N)).$$

Thus, while the *theoretical* worst-case is quasipolynomial, the *practical* complexity of synthesizing the minimal CoDD controller from a learned model is effectively $O(N \log N)$ for most real-world applications.

10.2 From Weaker Models to Synthesized Controllers

The discussion in the previous section assumed we have the weakest model M explaining which actions will lead to which results for a certain system in a certain environment, based on a volume of historical data; it then used this model M to synthesize a program P generating the actions explained by the model. We now explore the situation where we have two different such models M and M_1 , and look at the relations between the corresponding synthesized programs P and P_1 .

Conceptually, this partially addresses the issue that assuming we have a weakest explanatory model is a big assumption, involving a lot of computational work beyond just having a description of an environment and goals.

Specifically, suppose we have two learned CoDD-ENF models

$$M, M_1 : S \times A \longrightarrow \text{Dist}(O \times R \times S)$$

that both explain the same historical data, but satisfy

$$M \preceq M_1$$

in the *weakness* ordering (i.e. M is strictly more general, and hence weaker, than M_1). Let the corresponding minimal controllers be

$$P = \arg \max \circ \text{Expect} \circ M, \quad P_1 = \arg \max \circ \text{Expect} \circ M_1.$$

Weakness Prior on Programs (aside). To be sure it's understood, what we mean by weakness of programs here is as follows, in line with our above discussions. We assign each CoDD program Q a prior weight

$$w(Q) = \frac{2^{-\alpha \sigma(Q)}}{\sum_{Q'} 2^{-\alpha \sigma(Q')}} \propto 2^{-\alpha \sigma(Q)},$$

where $\sigma(Q)$ is the ENF description length of Q and $\alpha > 0$ is a fixed constant. Thus $w(Q_1) \leq w(Q_2) \iff \sigma(Q_1) \geq \sigma(Q_2)$.

Literal Weakening. If M_1 is obtained by a *specialization* (graph-refinement) $M_1 = \text{Spec}(M)$, then the same rewrite lifts through expectation and argmax:

$$P_1 = \arg \max \circ \text{Expect} \circ (\text{Spec}(M)) = \text{Spec}(\arg \max \circ \text{Expect} \circ M) = \text{Spec}(P).$$

Hence P_1 is literally an extension of P : every combinator in P appears in P_1 , plus additional structure.

Strict Weakness without Literal Weakening. More generally, if $M \preceq M_1$ but M_1 is *not* a direct subgraph-specialization of M , the Bellmanfusion pipeline still preserves weakness ordering:

$$w(M) \geq w(M_1) \implies w(P) \geq w(P_1).$$

In particular $\sigma(P) \leq \sigma(P_1)$, so P is no more specific than P_1 , even though neither program may be a literal extension of the other.

Incomparable Models. If neither $M \preceq M_1$ nor $M_1 \preceq M$, then the controllers P, P_1 likewise need not be comparable under $\preceq$. They share the common Bellmanfusion core but differ in handling corner cases. One can form the join model $M \vee M_1$, whose minimal controller $P \vee P_1$ is the least upper-bound of P and P_1 in the weakness lattice.

Takeaways.

- *Specialization lifts to programs.* A literal weakening $M \mapsto M_1$ yields a direct program refinement $P \mapsto P_1 = \text{Spec}(P)$.
- *Weakness is monotonic.* $M \preceq M_1$ implies $w(P) \geq w(P_1)$, i.e. $\sigma(P) \leq \sigma(P_1)$.
- *Controller lattice.* Joins and meets of models map to the corresponding joins and meets of their minimal controllers.

Thus *weakening at the model level always respects weakening at the controller level*, and when that weakening is a true refinement, the synthesized programs refine in lock-step.

11 Quantum Weakness

So far we have been elaborating different applications of our "basic" quantale-based formulation of weakness. Now we will deviate from this mode slightly and give a special sort of elaboration or tweak of this formulation. It turns out that porting the categorical-weakness framework to the quantum domain requires more than just choosing a quantale based on quantum algebra, it involves rethinking Occam's Razor a little bit. The issue is that in noncommutative settings no single scalar captures simplicity (or weakness or whatever) across all complementary measurements. We need to go a bit more postmodern, and move from a classical "one-score" view to a fully multiperspectival – and, it will be argued, tunably aggregated – quantum weakness.

11.1 Quantum Weakness as Categorical Weakness

Let $(\mathcal{C}, \otimes, I)$ be a monoidal category of system+context (e.g. finite-dimensional Hilbert spaces and CP maps) and let

$$W : \mathcal{C}^{\text{op}} \longrightarrow (P, \leq, \boxtimes)$$

be a weakness functor landing in a (possibly noncommutative) quantale $(P, \boxtimes)$. In the quantum state category, a subobject (effect) E has

$$W(E) = \text{Tr}(\rho E),$$

the Born-rule expectation under prior ρ . This is exactly categorical weakness specialized to the quantum case.

While this math set-up is clear, the actual evaluation of weakness in this set-up needs careful attention because of the observer-dependence implicit in quantum mechanics [BLM96]. To elaborate we need some conceptual setup:

Accessible measurement contexts (POVMs). A *positive operator-valued measure* (POVM) on a Hilbert space $\mathcal{H}$ is a finite collection of effects

$$\{E_j\}_{j=1}^m, \quad E_j \geq 0, \quad \sum_{j=1}^m E_j = I.$$

Each E_j corresponds to one measurement outcome, and given a quantum state ρ , the Born-rule assigns

$$\Pr(j) = \text{Tr}(\rho E_j).$$

In our framework, each context $\alpha \in \mathcal{A}$ is such a POVM, and for simplicity we focus on a particular effect $T_{h,\alpha} \in \{E_j\}$ that "tests" hypothesis h .

Hence, instead of a single scalar W , we obtain a family of weakness scores

$$W_\alpha(h) = \text{Tr}(\rho T_{h,\alpha}), \quad \alpha \in \mathcal{A},$$

one for each accessible POVM context α . Aggregation then collapses this family $\{W_\alpha(h)\}$ into the scalar Occam prior used by the agent.

Things get a little subtler when you connect this with the subtleties of quantum measurement, which we will pursue in Section 11.5 below. First however we will pause to explain how this notion of weakness relates to the Schrödinger bridge weakness explored earlier.

11.2 Relationship between the quantum-operator quantale and the Schrödinger-bridge quantale

Conceptually, both quantales arise as *Occam-style weakness measures* on different domains of quantum "effort." The *quantum-operator quantale* measures the size or complexity of static operators on a Hilbert space, while the *Schrödinger-bridge quantale* measures the minimal entropic transport cost of evolving quantum (or classical) densities. Despite these different interpretations, they share the same abstract structure and are linked by a precise Galois connection.

11.2.1 The two quantales

- *Quantum-operator quantale*

$$V_Q = \{A : \mathcal{H} \rightarrow \mathcal{H} \mid A \geq 0\}, \quad A \oplus_Q B = A + B, \quad A \otimes_Q B = AB.$$

Order is the Loewner order $A \leq B \iff B - A \geq 0$.

- *Schrödinger-bridge quantale*

$$Q_{SB} = \{ W : (\rho, v) \rightarrow \mathbb{R}_{\geq 0} \}, \quad (W_1 \oplus_{SB} W_2)[\rho, v] = \max\{W_1[\rho, v], W_2[\rho, v]\},$$

$$(W_1 \otimes_{SB} W_2)[\rho, v] = W_1[\rho|_{[t_0, t_1]}, v] + W_2[\rho|_{[t_1, t_2]}, v], \quad e_{SB}[\rho, v] = 0,$$

$$\text{with pointwise ordering } W_1 \leq W_2 \iff W_1[\rho, v] \leq W_2[\rho, v].$$

11.2.2 The embedding map $F : V_Q \rightarrow Q_{SB}$

For each positive operator $A \in V_Q$, define

$$(F(A))[\rho, v] = \int_{t_0}^{t_2} \langle v(t), A v(t) \rangle dt.$$

- F is monotone: $A \leq B \implies F(A) \leq F(B)$.
- F sends sums to concatenations:

$$F(A \oplus_Q B) = F(A + B) = F(A) \otimes_{SB} F(B).$$

- $F(0) = e_{SB}$.

11.2.3 The ?Hessian? map $U : Q_{SB} \rightarrow V_Q$

For any cost functional $W \in Q_{SB}$, let

$$U(W) = \left. \frac{\delta^2 W[\rho, v]}{\delta v^2} \right|_{v=0},$$

the operator whose quadratic form matches the second variation of W at zero velocity. Then

- U is monotone: $W_1 \leq W_2 \implies U(W_1) \leq U(W_2)$.
- $F \dashv U$ is a Galois connection:

$$F(A) \leq W \iff A \leq U(W).$$

11.2.4 Take-away

The maps F and U exhibit a *quantale-adjunction* between the static operator picture and the dynamic transport-cost picture. In this way, the quantum-operator quantale and the Schrödinger-bridge quantale are two facets of the same abstract weakness framework, bridged by a precise Galois connection.

11.3 Implications of the Galois connection for geodesics

Conceptually, the adjunction $F \dashv U$ not only links the two quantales, it also ****identifies their intrinsic Lawvere metrics**** and shows that geodesics in one picture correspond exactly to geodesics in the other.

11.3.1 . Lawvere metrics on the two quantales

Each commutative quantale $(Q, \oplus, \otimes)$ induces a generalized distance (a Lawvere metric)

$$d_Q(x, y) = \inf\{q \in Q \mid x \otimes q \geq y\},$$

where $\geq$ is the quantale order. Concretely:

- *Operator quantale metric*

$$d_{V_Q}(A, B) = \inf\{C \geq 0 \mid A + C \geq B\},$$

i.e. the smallest positive operator C needed so that $A + C$ dominates B .

- *Schrödinger bridge quantale metric*

$$d_{SB}(W_1, W_2) = \inf\{W \mid W_1 \otimes_{SB} W \geq W_2\},$$

i.e. the minimal additional **cost functional** W so that concatenating W onto W_1 covers W_2 .

11.3.2 Distance adjunction

Because F and U satisfy

$$F(A) \leq W \iff A \leq U(W),$$

one deduces for all $A \in V_Q$ and $W \in Q_{SB}$:

$$d_{SB}(F(A), W) = \inf\{\Omega \mid F(A) \otimes_{SB} \Omega \geq W\} = \inf\{\Omega \mid F(A \oplus_Q U(\Omega)) \geq W\} = d_{V_Q}(A, U(W)).$$

In particular, putting $W = F(B)$ gives the exact equality of intrinsic distances:

$$d_{SB}(F(A), F(B)) = d_{V_Q}(A, B).$$

11.3.3 Correspondence of geodesics

- *Operator?space geodesics:* A ?straight?line? path $A(t) = (1 - t) A_0 + t A_1$, $t \in [0, 1]$ attains the infimum in $d_{V_Q}(A_0, A_1)$.
- *Bridge?cost geodesics:* The Schrödinger bridge $W^*(t)$ solving the minimal?cost interpolation between $F(A_0)$ and $F(A_1)$ attains the infimum in $d_{SB}(F(A_0), F(A_1))$.

By the distance adjunction, F carries the linear operator geodesic $A(t)$ onto the unique cost?geodesic $W^*(t) = F(A(t))$, and U carries $W^*(t)$ back to the linear segment $A(t)$. Thus the two pictures share exactly the same geodesic structure, viewed through the lens of the Galois connection.

11.4 Mapping reward functionals to optimal quantum circuits

One can use this adjunction to interestingly map reward functions to the quantum circuits optimally realizing said reward functions – a form of "Mind-World Correspondence Principle."

Given a control-theoretic reward functional $R[\rho, v]$, define the corresponding cost

$$W[\rho, v] = -R[\rho, v],$$

so that minimizing W is equivalent to maximizing R . By the Galois connection $F \dashv U$ between the Schrödinger-bridge quantale and the quantum-operator quantale, compute the positive operator

$$A = U(W) = \left. \frac{\delta^2 W[\rho, v]}{\delta v^2} \right|_{v=0},$$

which encodes the quadratic penalty of the cost functional.

In the operator quantale $(V_Q, \oplus_Q, \otimes_Q)$, the ?wu wei? geodesic from the identity I to the target operator e^{-A} is given by the straight-line interpolation

$$A(t) = (1 - t) I + t e^{-A}, \quad t \in [0, 1].$$

By functional calculus this yields a one-parameter family of unitaries

$$U(t) = \exp(-i A(t)),$$

whose time-slicing or Trotterization produces a quantum circuit that is optimal, in the Occam-weakness sense, for the original reward R .

Optionally, pushing $A(t)$ back through F recovers the explicit cost along the path:

$$(F(A(t)))[\rho, v] = \int_{t_0}^{t_2} \langle v(t), A(t) v(t) \rangle dt,$$

confirming that this circuit indeed minimizes $W[\rho, v]$ (and thus maximizes $R[\rho, v]$).

Hence one obtains a principled pipeline

$$R \mapsto W = -R \xrightarrow{U} A \mapsto \{U(t) = \exp(-i A(t))\},$$

linking reward design to optimal quantum-circuit synthesis via the quantale Galois connection.

11.5 Multiperspectival Quantum Weakness

So, let $\mathcal{A}$ be the finite set of accessible measurement contexts (e.g. POVMs or observables) on an ensemble of quantum systems. For each hypothesis H and context $\alpha \in \mathcal{A}$,

$$w_\alpha(H) = \text{Tr}(\rho T_{H,\alpha}) \quad (\in [0, 1])$$

is the "weakness in context" under prior state ρ .

Aggregation Schemes Rather than a single w , we obtain a vector $\{w_\alpha(H)\}_{\alpha \in \mathcal{A}}$. Occam's Razor then becomes: choose the hypothesis whose vector is "largest" under some aggregation rule, e.g. one could choose:

1. **Worst-case (infimum):**

$$W_{\text{inf}}(H) = \min_{\alpha \in \mathcal{A}} w_\alpha(H).$$

2. **Weighted average:**

$$W_{\text{avg}}(H) = \sum_{\alpha \in \mathcal{A}} p_\alpha w_\alpha(H), \quad \sum_{\alpha} p_\alpha = 1.$$

3. **Weighted-multiplicative:**

$$W_{\text{mult}}(H) = \prod_{\alpha \in \mathcal{A}} [w_\alpha(H)]^{\eta_\alpha}, \quad \eta_\alpha \geq 0, \quad \sum_{\alpha} \eta_\alpha > 0.$$

4. **Lexicographic limit:** Choose exponents $\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$. Then W_{mult} effectively orders hypotheses by the smallest $w^{(1)}$ first, then $w^{(2)}$, etc., exactly enforcing a "weakest-context first" rule.

11.5.1 Unified Occam's Razor

Whatever the aggregation scheme in any case we can set a Bayesian prior

$$\pi(H) \propto [W(H)]^{\eta_0}, \quad \eta_0 > 0,$$

and prefer the hypothesis maximizing $W(H)$. Conceptually then our quantum Occam's razor becomes something like:

Among all hypotheses that fit your observations, choose the one that strikes the best balance of simplicity (weakness) across every relevant perspective— where

this balance gives more emphasis to the perspectives you care about most. One extreme way to assess this "best balance" is to demand that your chosen hypothesis be the simplest (weakest) even in the perspective comprising the single hardest or most critical test. Another way is to look at a weighted average across perspectives, perhaps weighting each one by some meta-level simplicity measure.

Or even more loosely, as we phrased it in the introduction,

*Given any system for relating and valuing entities, think of an individual binary relationship as **high-prominence** to the extent two things it joins are both valuable. A collection of binary relationships is then **weak** to the extent that, considered through the lens of a variety of complementary perspectives, it tends to contain a high combined weight of high-prominence relationships in the various perspectives.*

(and one then can make choices regarding how to do the compositing across different complementary perspectives).

11.6 Multiperspectival Weakness: Identical Ensemble Case

Let $\mathcal{A}$ be the finite set of all accessible measurement contexts (POVMs) on an ensemble of identically prepared systems $\{S_i\}$. For each hypothesis H and context $\alpha \in \mathcal{A}$, define the per-context weakness

$$w_\alpha(H) = \Pr(\text{"accept" under } \alpha),$$

estimated empirically by applying α to a subset of the ensemble.

Aggregation by Infimum or Average A simple multiperspectival Occam's Razor compares

$$W_{\text{inf}}(H) = \min_{\alpha \in \mathcal{A}} w_\alpha(H) \quad \text{or} \quad W_{\text{avg}}(H) = \sum_{\alpha \in \mathcal{A}} p_\alpha w_\alpha(H),$$

where $\{p_\alpha\}$ is a classical context-prior (e.g. based on frequency or MDL).

Weighted-Multiplicative Prior To sharpen the Occam bias, introduce non-negative exponents $\{\eta_\alpha\}$ and form the multiplicative score

$$W_{\text{mult}}(H) = \prod_{\alpha \in \mathcal{A}} [w_\alpha(H)]^{\eta_\alpha},$$

then use $\pi(H) \propto [W_{\text{mult}}(H)]^{\eta_0}$ as the Bayesian prior in AIXI^{tl}, with $\eta_0 > 0$.

Lexicographic Limit By choosing the η_α so that $\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$ (for a suitable ordering $\alpha_1, \alpha_2, \dots$), the product $W_{\text{mult}}(H)$ effectively compares hypotheses first by their worst-case weakness $w^{(1)}$, then by the second-worst $w^{(2)}$, and so on—recovering the lexicographic "worst-context first" rule as a special case.

11.6.1 Potential Generalization to Other Noncommutative Algebras

It's a side note we will not follow up here in details, but for the purpose of framing future work, it's worth noting that exactly the same multiperspectival construction applies in any noncommutative setting—quaternionic, octonionic, or beyond—by replacing POVM contexts with the appropriate family of automorphisms or measurement tests and aggregating via infimum, average, multiplicative weights, or the lexicographic limit.

11.7 Multiperspectival Weakness for Heterogeneous Ensembles

In realistic settings systems $S_1, \dots, S_n$ may differ in nuisance parameters but share a common causal preparation structure. We therefore align contexts via a meta-model and aggregate weakness both over contexts and over systems, using either infimum, average, or weighted-multiplicative—and recovering lexicographic ordering as an extreme limit.

Causal Meta-Model. Let M_{prep} specify how systems share features X and differ in parameters Y_i . From M_{prep} extract a common context set $\mathcal{A}$ and procedures to implement each $\alpha \in \mathcal{A}$ on every S_i .

Per-System, Per-Context Weakness. For each hypothesis H , system i , and context α define

$$w_{i,\alpha}(H) = \Pr(\text{"accept" under } \alpha \text{ on } S_i),$$

estimated experimentally or via a known local prior ρ_i .

Context-Level Aggregation. Within each system i , aggregate over contexts by one of:

$$w_i^{\text{inf}}(H) = \min_{\alpha \in \mathcal{A}} w_{i,\alpha}(H), \quad w_i^{\text{avg}}(H) = \sum_{\alpha} p_{\alpha} w_{i,\alpha}(H),$$

$$w_i^{\text{mult}}(H) = \prod_{\alpha} [w_{i,\alpha}(H)]^{\eta_{\alpha}}, \quad \text{or lexicographically by taking } \eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$$

Here $\{p_{\alpha}\}$ is a classical Occam/context-frequency prior and $\{\eta_{\alpha}\}$ are nonnegative exponents (extreme η recovers lexicographic).

System-Level Aggregation. Then aggregate $\{w_i(H)\}_{i=1}^n$ across systems by

$$W^{\text{inf}}(H) = \min_i w_i(H), \quad W^{\text{avg}}(H) = \sum_i q_i w_i(H),$$

$$W^{\text{mult}}(H) = \prod_i [w_i(H)]^{\zeta_i}, \quad \zeta_1 \gg \zeta_2 \gg \dots \text{ for lexicographic limit,}$$

where $\{q_i\}$ reflects trust or sample size and $\{\zeta_i\}$ are system-importance exponents.

Generalized Occam’s Razor. Choose the hypothesis H maximizing one of W^{inf} , W^{avg} , W^{mult} . In particular:

$$H^* = \arg \max_H W^{\text{mult}}(H) \quad (\text{or } W^{\text{inf}}, W^{\text{avg}}),$$

with the lexicographic prior as the extreme η -, ζ -limit.

Two-Tiered Simplicity Principle.

1. Use a classical causal meta-model to align heterogeneous systems and define contexts.
2. Apply a multiperspectival weakness calculus—infimum, average, Weighted-multiplicative, or lexicographic—first across contexts per system, then across systems globally.

This preserves a robust, observer- and system-dependent Occam’s Razor in non-ideal experimental regimes.

11.8 QWAIXI – Quantum AIXI^{tl} with Multiperspectival Weakness Prior

We describe now a self-contained quantum AIXI^{tl} agent whose prior over quantum world-models is driven by a *multiperspectival quantum weakness* measure, using either infimum, weighted-average, weighted-multiplicative, or lexicographic aggregation of per-context simplicity scores.

Model Class. Fix resource bounds t (max runtime) and ℓ (max description length). Define

$$\text{HCal}_{t,\ell}^Q = \{h = (H_h, \{T_{h,\alpha}\}_{\alpha \in \mathcal{A}}, E_h) \mid \sigma(h) \leq \ell, \text{runtime}(h) \leq t\},$$

where:

- $H_h : \mathcal{H}_S \otimes \mathcal{H}_A \rightarrow \mathcal{H}_S \otimes \mathcal{H}_A$ is a Hamiltonian.
- $\mathcal{A}$ is the finite set of accessible measurement contexts.
- $T_{h,\alpha} : \mathcal{H}_S \rightarrow [0, 1]$ tests hypothesis h under α .
- $E_h : \mathcal{H}_O \otimes \mathcal{H}_R \rightarrow [0, 1]$ produces observations/rewards.
- $\sigma(h)$ is the description length of h .

Let ρ be a fixed prior density on $\mathcal{H}_S \otimes \mathcal{H}_A$.

Per-Context Weakness. For each h and $\alpha \in \mathcal{A}$, define

$$w_\alpha(h) = \text{Tr}(\rho T_{h,\alpha}).$$

Aggregation Schemes. Aggregate $\{w_\alpha(h)\}$ by one of:

$$\underbrace{W_{\text{inf}}(h)}_{\substack{\text{worst-case} \\ \text{(lexicographic limit)}}} = \min_{\alpha} w_{\alpha}(h),$$

$$\underbrace{W_{\text{avg}}(h)}_{\substack{\text{expected} \\ \text{simplicity}}} = \sum_{\alpha} p_{\alpha} w_{\alpha}(h), \quad \sum_{\alpha} p_{\alpha} = 1,$$

$$\underbrace{W_{\text{mult}}(h)}_{\substack{\text{weighted} \\ \text{multiplicative}}} = \prod_{\alpha} [w_{\alpha}(h)]^{\eta_{\alpha}}, \quad \eta_{\alpha} \geq 0, \quad \sum_{\alpha} \eta_{\alpha} > 0.$$

In the limit $\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$, W_{mult} reproduces lexicographic ordering by enforcing maximal simplicity in the hardest context first, then the next, and so on.

Prior over Models. Choose one aggregation $W_* \in \{W_{\text{inf}}, W_{\text{avg}}, W_{\text{mult}}\}$ and exponent $\eta > 0$, then set

$$\pi(h) = \frac{[W_*(h)]^{\eta}}{\sum_{h' \in \text{HCal}_{t,\ell}^Q} [W_*(h')]^{\eta}}.$$

Value Update. Initialize $|V_h(0)\rangle = |0\rangle$. Recursively update

$$|V_h(t+1)\rangle = e^{-\frac{i}{\hbar} H_h \Delta t} |V_h(t)\rangle + E_h |V_h(t)\rangle, \quad V_h^Q(t) = \langle V_h(t) | E_h | V_h(t) \rangle.$$

Action Selection. Given history $|\Psi_{t-1}\rangle$ and action a ,

$$V_h(t-1, a) = \langle \Psi_{t-1}; a | E_h | V_h(t) \rangle, \quad |\Psi_{t-1}; a\rangle = U(a) |\Psi_{t-1}\rangle,$$

$$V_{\text{mix}}(a) = \sum_h \pi(h) V_h(t-1, a), \quad a_t = \arg \max_a V_{\text{mix}}(a).$$

Discussion.

- Infimum, average, and weighted-multiplicative aggregation each enforce a multiperspectival Occam bias, from strict worst-case to softer averaged simplicity.
- By tuning $\{\eta_{\alpha}\}$ one smoothly interpolates between weighted-multiplicative and lexicographic extremes.
- Standard universality and asymptotic optimality proofs carry over unchanged: the richer prior refines convergence constants without altering $O(\sqrt{T})$ scaling.

11.9 Mapping Between the CD-Logic Quantale and the Quantum-Operator Quantale

In [Goe21c] it is shown that fuzzy paraconsistent logic tends toward quantum logic as one takes the limiting value of certain parameters. Here we reconstruct that argument in the context of CDL and quantum-operator quantales – and then show how the quantale approach can express the argument much more nicely than its original form.

11.9.1 Reconstructing the Original Mapping from CDL to Quantum

In what follows we show how the fuzzy paraconsistent ("CD-logic") quantale

$$V_{\text{CD}} = H \times H^{\text{op}}, \quad (x, x') \oplus_{\text{CD}} (y, y') = (x \vee y, x' \wedge y'), \quad (x, x') \otimes_{\text{CD}} (y, y') = (x \wedge y, x' \vee y'),$$

relates to the quantum-operator quantale

$$V_Q = \{ A : \text{cH} \rightarrow \text{cH} \mid A \geq 0 \}, \quad A \oplus_Q B = A + B, \quad A \otimes_Q B = AB,$$

via an approximate quantale homomorphism .

From p-Bits to Density Operators Given a fuzzy p-bit $(w_+, w_-) \in [0, 1]^2 \cong V_{\text{CD}}$, choose an additive-generator pair (f, f^*) of a Schweizer-Sklar t-norm so that $f, f^* : [0, 1] \rightarrow \mathcal{R}_{\geq 0}$. Define a pure-state amplitude

$$\psi_w = (f(w_+), f^*(w_-)) \in \mathbb{C}^2,$$

normalized so that $\|\psi_w\| = 1$. Then set

$$\Phi(w_+, w_-) = \rho_w = |\psi_w\rangle\langle\psi_w| \in V_Q.$$

Approximate Preservation of Join and Tensor One verifies that for any two p-bits $u, v \in V_{\text{CD}}$:

$$\Phi(u \oplus_{\text{CD}} v) \approx \Phi(u) \oplus_Q \Phi(v), \quad \Phi(u \otimes_{\text{CD}} v) \approx \Phi(u) \otimes_Q \Phi(v),$$

where the approximation error can be made arbitrarily small by choosing the t-norm parameter $p \rightarrow -\infty$.

Negation and Involution CD-logic negation $_{\text{CD}}(w_+, w_-) = (w_-, w_+)$ is carried by Φ into the quantum involution

$$\Phi(_{\text{CD}}(w)) = \rho_w^*,$$

up to a unitary phase, matching the complex-conjugation symmetry in quantum mechanics

Insight and Consequences This construction shows that:

- Fuzzy p-bits in CD logic embed approximately as qubit density operators.
- CD-logic disjunction and conjunction map into operator addition and multiplication, explaining the "complex-amplitude" algebra of quantum probabilities as an emergent approximation to paraconsistent truth-value algebra.
- The approximation error reflects assumed "evidential-error uncertainty," tying the mapping to probabilistic-logic-networks semantics.

Thus the paper's core argument—that quantum probability arithmetic arises from fuzzy paraconsistent logic under small evidential-error assumptions—can be restated succinctly as an approximate quantale homomorphism $\Phi : V_{\text{CD}} \rightarrow V_Q$.

11.9.2 A Categorical Refinement of the Mapping via Lax Monoidal Functors

We have exhibited an *approximate* homomorphism $\Phi : V_{\text{CD}} \rightarrow V_Q$ between the CD logic quantale $(V_{\text{CD}}, \oplus_{\text{CD}}, \otimes_{\text{CD}})$ and the quantum-operator quantale $(V_Q, \oplus_Q, \otimes_Q)$. We now show how to restate this neatly in category-theoretic terms:

Quantales as One-Object Monoidal Categories Recall that any commutative quantale $(V, \otimes, \oplus)$ may be viewed as a one-object monoidal category $\mathcal{C}_V$:

$$\text{Ob}(\mathcal{C}_V) = \{\text{ast}\}, \quad \text{Hom}(\text{ast}, \text{ast}) = V, \quad \text{composition} = \otimes, \quad \text{tensor} = \oplus.$$

In this picture a quantale homomorphism is exactly a strict monoidal functor between the corresponding one-object categories.

The Approximation as a Lax Monoidal Functor Our map Φ fails to preserve $\oplus$ and $\otimes$ *on the nose*, but only up to small error. This is precisely a *lax monoidal functor*

$$\Phi : \mathcal{C}_{V_{\text{CD}}} \rightarrow \mathcal{C}_{V_Q},$$

equipped with structure morphisms

$$\phi_2 : \Phi(u) \oplus_Q \Phi(v) \longrightarrow \Phi(u \oplus_{\text{CD}} v), \quad \psi_2 : \Phi(u) \otimes_Q \Phi(v) \longrightarrow \Phi(u \otimes_{\text{CD}} v),$$

and unit maps

$$\phi_0 : 0_Q \rightarrow \Phi(0_{\text{CD}}), \quad \psi_0 : I_Q \rightarrow \Phi(I_{\text{CD}}),$$

satisfying the usual coherence diagrams for a lax monoidal functor (Mac-Lanes pentagon and triangle axioms).

Strictification in the Limit By choosing the t-norm parameter $p \rightarrow -\infty$, one shows that the laxators ϕ_2, ψ_2 become arbitrarily close to invertible. In the limit they yield an *(ordinary) monoidal functor*, i.e. a genuine quantale homomorphism. This passage from lax to strict is an instance of *strictification* in monoidal category theory.

Enrichment and Profunctors Alternatively, one can regard both $\mathcal{C}_{V_{CD}}$ and $\mathcal{C}_{V_Q}$ as enriched over a "metric" quantale $[0, \infty]$, measuring the operator-norm distance on V_Q and an analogous distance on V_{CD} . Then Φ is simply a *short* (Lipschitz) enriched functor, whose laxator norms shrink to zero as $p \rightarrow -\infty$. This yields a fully *enriched* explanation of the approximation error.

Conceptual Benefit Framing Φ as a lax monoidal functor (or enriched functor) makes clear:

- The "approximation" arises from non-invertible lax-structure maps.
- Standard coherence and strictification theorems guarantee that, in an appropriate limit, the approximation collapses to an exact quantale homomorphism.
- One may track the error uniformly by measuring the deviation of the laxators from isomorphism, using either 2-categorical coherence or enriched-category metrics.

Thus the approximation result becomes an instance of a general pattern in category theory—lax monoidal functors and their strictification—providing an elegant, conceptual account of why fuzzy paraconsistent logic embeds (approximately) into quantum operator arithmetic.

11.10 Quantum Universality and Asymptotic Optimality

We now establish two central results for the multiperspectival quantum AIXI^{tl} agent—equipped with any of our aggregation-based Occam priors (infimum, weighted-average, weighted-multiplicative or its lexicographic limit)—namely: (1) the hypothesis class is universal for quantum computation, and (2) the agent's regret vanishes as its resource bounds grow, regardless of which full-support prior is used.

11.10.1 Universality of the Quantum Hypothesis Class

Recall that for resource bounds t (max runtime) and ℓ (max description length), the class

$$\text{HCal}_{t,\ell}^Q = \{ h = (H_h, \{T_{h,\alpha}\}_{\alpha \in \mathcal{A}}, E_h) \mid \sigma(h) \leq \ell, \text{runtime}(h) \leq t \}$$

contains hypotheses encoding a Hamiltonian H_h , multiperspectival tests $\{T_{h,\alpha}\}$, and a reward/observation effect E_h .

Theorem 16 (Polynomial-size Quantum Circuit Simulation). *Let U be any n -qubit unitary implementable by a quantum circuit of size s . Then for sufficiently large (t, ℓ) with*

$$\ell = O(s), \quad t = \text{poly}(n, s),$$

there exists $h \in \text{HCal}_{t,\ell}^Q$ whose H_h and E_h simulate U to inverse-polynomial accuracy. Hence $\text{HCal}_{t,\ell}^Q$ is universal for quantum computation.

Sketch. Encode each elementary gate as a constant-cost CoDD subgraph; a circuit of s gates becomes a CoDD program of cost $O(s)$. ENF merging bounds the runtime to $\text{poly}(n, s)$, and E_h projects onto the circuit's output with inverse-polynomial error. $\square$

11.10.2 Asymptotic Optimality under aggregation-Based Priors

For each hypothesis h , compute the per-context quantum weakness $w_\alpha(h) = \text{Tr}(\rho T_{h,\alpha})$ and aggregate by one of:

$$W_{\inf}(h) = \min_{\alpha} w_{\alpha}(h), \quad W_{\text{avg}}(h) = \sum_{\alpha} p_{\alpha} w_{\alpha}(h), \quad W_{\text{mult}}(h) = \prod_{\alpha} [w_{\alpha}(h)]^{\eta_{\alpha}},$$

or in the lexicographic limit $\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$. Then form a full-support Bayesian prior

$$\pi(h) \propto [W_*(h)]^{\eta_0}, \quad \eta_0 > 0,$$

where W_* is any of the above aggregation schemes.

Let $V^*(h_{t-1}, a)$ be the true optimal m -step return and

$$V_{\text{mix}}(a) = \sum_{h \in \text{HCal}_{t,\ell}^Q} \pi(h) V_h^Q(t-1, a)$$

the mixture return under π .

Theorem 17 (Asymptotic Optimality). *If the true environment model h^* lies in $\text{HCal}_{t,\ell}^Q$ for all sufficiently large (t, ℓ) , then for any $\varepsilon > 0$ there exist thresholds T, L such that whenever $t \geq T$ and $\ell \geq L$,*

$$\max_a V^*(h_{t-1}, a) - \max_a V_{\text{mix}}(a) < \varepsilon.$$

Hence the regret of the quantum AIXI^{tl} agent with any full-support multiperspectival prior vanishes as resources grow.

Sketch. By universality, $\text{HCal}_{t,\ell}^Q$ contains an arbitrarily accurate simulator of h^* for large (t, ℓ) . Since $\pi(h^*) > 0$ under any aggregation-based prior, standard mixture-regret bounds ensure the mixture value approaches the true optimum within ε . $\square$

11.10.3 Discussion

- **Robust Occam Bias.** Infimum, weighted-average, weighted-multiplicative, and lexicographic priors all implement multiperspectival Occam's Razor, from strict worst-case to softer context-weighted simplicity.
- **Convergence Constants.** While all priors yield $O(\sqrt{T})$ regret scaling, the weighted-multiplicative and lexicographic schemes can shrink constants by magnifying per-context simplicity gaps.
- **Generality.** These results extend to any noncommutative/observer-dependent hypothesis category (quaternionic, octonionic, etc.) using the same ensemble-based aggregation machinery.

11.11 Comparison of Lexicographic vs. Weighted-Multiplicative Priors

In both lexicographic and weighted-multiplicative formulations, the mixture-regret of AIXI^{tl} satisfies the standard bound

$$\text{Regret}(T) \leq O\left(\sqrt{T (\ln M - \ln \pi(h^*))}\right),$$

where $M = |\mathcal{H}|$ and $\pi(h^*)$ is the prior mass on the true hypothesis. The two approaches differ only in how large $\pi(h^*)$ becomes:

1. Weighted-Multiplicative Prior.

$$\pi_{\text{mult}}(h) \propto \prod_{j=1}^L [w^{(j)}(h)]^{\eta_j},$$

with moderate exponents $\{\eta_j\}$. If $\Delta_j = w^{(j)}(h^*) - \max_{h \neq h^*} w^{(j)}(h)$, then

$$\ln \frac{1}{\pi_{\text{mult}}(h^*)} = \ln M - \sum_{j=1}^L \eta_j \Delta_j + O(1).$$

The regret constant is thus $\ln M - \sum_j \eta_j \Delta_j$.

2. Lexicographic Prior (Extreme Limit). Take $\eta_1 \gg \eta_2 \gg \dots \gg \eta_L$ (e.g. $\alpha - \infty$), yielding

$$\pi_{\text{lexico}}(h) \propto \prod_{j=1}^L [w^{(j)}(h)]^{\eta_j} \xrightarrow{\eta_1 \rightarrow \infty} \begin{cases} \text{maximizes } w^{(1)}, \\ \text{then } w^{(2)}, \dots \end{cases}$$

In the limit the log-prior margin becomes $\ln M - \sum_{j=1}^L \eta_j \Delta_j$ with $\eta_1 \rightarrow \infty$, so that $\ln(1/\pi_{\text{lexico}}(h^*))$ is *as small as possible* given the gaps $\{\Delta_j\}$.

Implications for Convergence

- Both approaches achieve the minimax-optimal $O(\sqrt{T})$ scaling.
- The *constant* under the square-root is $\ln M - \sum_j \eta_j \Delta_j$. Because the lexicographic prior concentrates nearly all weight on the largest gap Δ_1 (and subsequently on Δ_2 , etc.), it *maximizes* $\sum_j \eta_j \Delta_j$ and thus *minimizes* the regret constant.
- A finite weighted-multiplicative prior trades off between worst-case robustness (η_1 large) and average-case stability (η_j moderate), yielding slightly larger regret constant but greater resilience to noise in any single context.
- In practice, one chooses $\{\eta_j\}$ to balance *convergence speed* (larger exponents) against *sampling-noise robustness* (smaller exponents).

11.12 Practical Robustness of Weighted-Multiplicative Priors in Approximate AIXI^{tl}

In ideal (exhaustive) AIXI^{tl}, a lexicographic prior can minimize the regret constant by concentrating almost all weight on the worst-case context. However, real-world approximations—e.g. Monte-Carlo-Tree-search or particle-based AIXI—cannot enumerate every hypothesis or context pair exactly, and suffer from sampling noise, limited budget, and heuristic proposals. In that setting, a *finite-exponent weighted-multiplicative prior* enjoys several practical advantages:

1. Smoothness vs. brittleness.

- *Lexicographic*: A tiny sampling error in the estimated smallest weakness $w^{(1)}$ can completely reorder the prior, discarding otherwise good hypotheses.
- *Weighted-multiplicative*: Finite exponents $\eta_j < \infty$ ensure each context contributes continuously. A small misestimate shifts $\ln \pi(h)$ by at most $\sum_j \eta_j \delta_j$, avoiding catastrophic model-switching.

2. Computational tractability.

- *Lexicographic*: Requires sorting every candidate’s full L -vector of weaknesses to identify the minimal entries—prohibitive for large, dynamic particle sets.
- *Weighted-multiplicative*: Only needs to compute a scalar $\prod_j [w^{(j)}(h)]^{\eta_j}$ per particle, trivially parallelizable and compatible with incremental updates.

3. Adaptivity to heuristics.

- *Lexicographic*: Hard to combine with domain-specific proposal distributions or value-guided sampling, since any context rank tie can overturn all prior structure.

- *Weighted-multiplicative*: One can smoothly incorporate heuristic quality scores or importance sampling weights into the exponents η_j or into the context weights p_j , preserving the Occam bias while respecting sampler design.

4. Balanced exploration-exploitation.

- *Lexicographic*: Over-biases exploration toward contexts deemed worst by minimal samples, risking neglect of high-value regions.
- *Weighted-multiplicative*: Tunable $\{\eta_j\}$ allow a spectrum from worst-case emphasis (η_1 large) to average-case exploration (η_j moderate), supporting more stable search dynamics.

Conclusion. While lexicographic priors attain the smallest regret constant in theory, a *finite-exponent weighted-multiplicative prior* offers a more robust, computationally feasible, and sampler-friendly Occam bias for practical, sampling-based AIXI^{tl} implementations.

11.13 Multiperspectival Quantum Weakness Prior Estimation

We now describe how to estimate the multiperspectival weakness prior under any aggregation rule—infimum, weighted-average, weighted-multiplicative, or its lexicographic limit.

Per-Context Measurements. For each hypothesis h and context $\alpha \in \mathcal{A}$, perform M independent shots of the effect $T_{h,\alpha}$ on fresh preparations of ρ . Denote

$$\hat{w}_\alpha(h) = \frac{1}{M} \sum_{i=1}^M X_{i,\alpha}(h), \quad X_{i,\alpha}(h) \sim \text{Bernoulli}(w_\alpha(h)),$$

so that by Hoeffding plus a union bound (over $|\text{HCal}_{t,\ell}^Q|$ models and $|\mathcal{A}|$ contexts), with probability $\geq 1 - \eta$:

$$|\hat{w}_\alpha(h) - w_\alpha(h)| \leq \delta \quad \forall h, \forall \alpha,$$

as long as

$$M \geq \frac{1}{2\delta^2} \ln\left(\frac{2|\text{HCal}_{t,\ell}^Q| |\mathcal{A}|}{\eta}\right).$$

Error on Aggregated Weakness. Define three aggregation rules:

$$W_{\inf}(h) = \min_{\alpha} w_{\alpha}(h), \quad W_{\text{avg}}(h) = \sum_{\alpha} p_{\alpha} w_{\alpha}(h),$$

$$W_{\text{mult}}(h) = \prod_{\alpha} [w_{\alpha}(h)]^{\eta_{\alpha}}, \quad \eta_{\alpha} \geq 0, \sum_{\alpha} \eta_{\alpha} > 0.$$

Then, under the same δ -uniform bound:

- $|\min_{\alpha} \hat{w}_{\alpha}(h) - \min_{\alpha} w_{\alpha}(h)| \leq \delta$.
- $|\sum p_{\alpha} \hat{w}_{\alpha}(h) - \sum p_{\alpha} w_{\alpha}(h)| \leq \delta$.
- Writing $\hat{W}_{\text{mult}}(h) = \prod [\hat{w}_{\alpha}(h)]^{\eta_{\alpha}}$, one shows

$$|\ln \hat{W}_{\text{mult}}(h) - \ln W_{\text{mult}}(h)| \leq \sum_{\alpha} \eta_{\alpha} |\ln(\hat{w}_{\alpha}/w_{\alpha})| = O(\delta \sum_{\alpha} \eta_{\alpha}),$$

so relative error in W_{mult} is $O(\delta)$ for moderate η .

Thus the measured prior $\hat{\pi}(h) \propto [\hat{W}_{*}(h)]^{\eta_0}$ differs only by $O(\delta)$ (additive or multiplicative) from the true $\pi(h)$.

Regret Bound with Sampling Error. Let $\sigma(h^*)$ be the description length of the true model and V_{max} the maximum reward. Then, for any full-support Occam prior π ,

$$\text{Regret} \leq \sigma(h^*) \ln 2 + C \delta_{\text{eff}} V_{\text{max}},$$

where $\delta_{\text{eff}} = \delta$ for infimum/average and $\delta_{\text{eff}} = O(\delta \sum \eta_{\alpha})$ for the multiplicative case. To make the sampling-error term negligible compared to the Occam term, choose M so that $\delta_{\text{eff}} \ll \sigma(h^*) \ln 2 / V_{\text{max}}$.

Lexicographic Robustness. In the lexicographic limit ($\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$), one must ensure δ is smaller than the minimal gap between successive $w^{(j)}$. Otherwise, orderings can flip. In practice, one uses finite but large $\{\eta_{\alpha}\}$ to approximate lexicographic behavior while controlling robustness to sampling noise.

Illustration. For $|\text{HCal}^Q| = 10^6$, $|\mathcal{A}| = 10$, $\delta = 0.01$, η_{α} moderate, and $\eta_0 > 0$, one still needs $M \sim 10^5$ shots per context to achieve $\delta_{\text{eff}} \lesssim 0.01$. The Occam term $\sigma(h^*) \ln 2$ (e.g. 50) thus dominates the sampling penalty (≈ 0.2).

Quantum Parallelism. Amplitude-estimation can reduce the required M from $O(1/\delta^2)$ to $O(1/\delta)$ in the weighted-average or multiplicative cases, further improving practical feasibility.

11.14 Illustrative Examples and Use Cases

To very roughly gauge the potential feasibility of this sort of approach in any future practical scenario, we estimate the number of quantum measurement shots and resulting decision-latencies for two representative applications. These estimates depend only on the size of the hypothesis class, the number of measurement contexts, and the desired uniform estimation error δ , and are *independent* of whether one uses infimum, weighted-average, weighted-multiplicative, or lexicographic Occam priors.

Uniform-Error Shot Count Let $|\text{HCal}^Q|$ be the number of candidate models and $|\mathcal{A}|$ the number of distinct measurement contexts. To ensure $|\hat{w}_\alpha(h) - w_\alpha(h)| \leq \delta$ simultaneously for all h and α with confidence $1 - \eta$, Hoeffding’s inequality plus a union bound requires

$$M \geq \frac{1}{2\delta^2} \ln\left(\frac{2|\text{HCal}^Q||\mathcal{A}|}{\eta}\right) \quad \text{shots per context.}$$

Humanoid Robot Control

- $|\text{HCal}^Q| = 10^6$, $|\mathcal{A}| = 3$ (vision, touch, proprioception).
- State-space dimension $d = 1024$, typical effect rank $k \approx 150$, so $w_\alpha \approx 150/1024 \approx 0.15$.
- With $\delta = 0.01$, $\eta = 0.01$:

$$M \gtrsim 5 \times 10^4 \quad \text{shots/context,}$$

yielding unoptimized latency $3 \times 5 \times 10^4 \times 1 \text{ ms} \approx 150 \text{ s}$ at 1-ms/shot.

- The Occam-bias term $\sigma(h^*) \ln 2 \approx 50$ dominates any $O(\delta)$ sampling penalty.

Persistent Chatbot Assistant

- $|\text{HCal}^Q| = 10^6$, $|\mathcal{A}| = 8$ (syntactic, semantic, pragmatic, etc.).
- Embedding dimension $d = 2^{16}$, effect rank $k \approx 10^3$, so $w_\alpha \approx 10^3/2^{16} \approx 0.015$.
- For $\delta = 0.005$, $\eta = 0.01$:

$$M \gtrsim 2 \times 10^4 \quad \text{shots/context,}$$

yielding latency $8 \times 2 \times 10^4 \times 0.5 \text{ ms} \approx 80 \text{ s}$ at 0.5-ms/shot over 8 parallel contexts.

- The Occam-bias term $\sigma(h^*) \ln 2 \approx 138$ again dominates sampling error.

Key Takeaways

- The required shot counts and latencies depend only on $|\text{HCal}^Q|$, $|\mathcal{A}|$, and δ , not on the choice of multiperspectival prior.
- Aggregation schemes differ in how the residual estimation error δ affects the sharpness of the Occam prior (e.g. an $O(\delta)$ shift for infimum/average vs. $O(\delta \sum \eta_\alpha)$ in the multiplicative case, or the need to resolve minimal gaps for lexicographic ordering).

- Even with moderate target error ($\delta \sim 10^{-2}$), the sampling penalty is negligible compared to classical Occam terms in both robot and chatbot settings.
- Quantum-parallel amplitude estimation can further reduce shot requirements from $O(1/\delta^2)$ to $O(1/\delta)$.

11.15 Graphtropic Formulation of the Multiperspectival Quantum Razor

We now show how to recast our "many-perspective" quantum Occam's Razor in the language of *graphtropy*, by viewing each measurement context as a *distinction graph* and then aggregating their graphtropies.

Distinction graphs per context. Let $\mathcal{H}$ be the finite set of pure states of your quantum system. For each hypothesis h and POVM context $\alpha \in \mathcal{A}$, define the *distinction graph*

$$G_\alpha(h) = (V = \mathcal{H}, E_\alpha(h)), \quad (\psi, \phi) \in E_\alpha(h) \iff \Pr_{h,\alpha}(\text{"accept"} \mid \psi) = \Pr_{h,\alpha}(\text{"accept"} \mid \phi),$$

i.e. two states are linked if $T_{h,\alpha}$ cannot distinguish them.

Graphtropy per context. The *graphtropy* of $G_\alpha(h)$ is the normalized edge-density

$$H_\alpha(h) = \frac{|E_\alpha(h)|}{\binom{|\mathcal{H}|}{2}} \in [0, 1].$$

Equivalently, one may define entropy of the partition induced by connected components, but edge-density suffices to measure "fraction collapsed."

Multiperspectival aggregation. We obtain a vector of graphtropies $\{H_\alpha(h)\}_{\alpha \in \mathcal{A}}$. Aggregate into a single scalar by one of:

- **Worst-case (infimum):** $H_{\inf}(h) = \min_{\alpha} H_\alpha(h)$.
- **Weighted average:** $H_{\text{avg}}(h) = \sum_{\alpha} p_{\alpha} H_{\alpha}(h), \quad \sum_{\alpha} p_{\alpha} = 1$.
- **Weighted-multiplicative:** $H_{\text{mult}}(h) = \prod_{\alpha} [H_{\alpha}(h)]^{\eta_{\alpha}}, \quad \eta_{\alpha} \geq 0, \sum_{\alpha} \eta_{\alpha} > 0$.
- **Lexicographic limit:** Choose exponents so $\eta_{\alpha_1} \gg \eta_{\alpha_2} \gg \dots$. Then maximizing H_{mult} orders hypotheses by their minimal H_{α} , then second-minimal, etc., enforcing "simplest in the hardest context first."

Quantum Razor via Graphropy. Finally, set a Bayesian prior

$$\pi(h) \propto [H_*(h)]^{\eta_0}, \quad \eta_0 > 0,$$

where H_* is one of the above aggregations. The agent then mixes over hypotheses according to $\pi(h)$ and proceeds with value updates and action selection as usual.

In this graphtropic picture, the multiperspectival quantum Razor says:

”Construct a graph of collapsed distinctions for each measurement context; compute its graphropy (fraction of indistinguishable pairs); then aggregate those graphropies—by minimum, average, product, or lexicographic product—to obtain a single simplicity score. Prefer the hypothesis that maximizes this aggregated graphropy.”

In particular, the *weighted-multiplicative* and *lexicographic* cases arise as special choices of the exponents $\{\eta_\alpha\}$, yielding a unified graphtropic formulation of Occam’s Razor in fully observer-dependent, noncommutative settings.

12 Quantale Information Theory

In [BS12], it is shown that algorithmic information can be cast as a particular case of Shannon information via appropriate choice of assumed information source. We show here that this argument works also for mutual information based weakness – and that it can be extended beyond this case and applied to weakness over an arbitrary quantale? This yields an elegant algebra of informational quantities in a quite general and flexible setting.

12.1 Baez & Stay’s Argument: Algorithmic Information as a Special Case of Shannon Entropy

We begin by recounting Baez and Stay’s demonstration show that *algorithmic entropy* (Kolmogorov complexity) can be viewed as a special case of the *entropy* of a probability measure – i.e. Shannon information – by choosing an appropriate *Gibbs distribution* over the set of programs. The same argument ports neatly to our set-up here.

Programs as Microstates Let U be a fixed universal, prefix-free Turing machine and let

$$X = \{x \in \{0,1\}^* \mid U(x) \text{ halts}\}$$

be the set of its halting programs. We regard each program $x \in X$ as a ”microstate” of a statistical-mechanical system.

Gibbs Ensemble over Programs Fix a real parameter $\gamma > 0$. The *Gibbs distribution* p on X that maximizes Shannon entropy subject to a constraint on the expected code-length $|x|$ is

$$p_\gamma(x) = \frac{e^{-\gamma |x|}}{Z(\gamma)}, \quad Z(\gamma) = \sum_{x \in X} e^{-\gamma |x|}.$$

Self-Information and Code-Length Under this distribution, the *self-information* of a program x is

$$-\log p_\gamma(x) = \gamma |x| + \log Z(\gamma).$$

In particular, choosing $\gamma = \ln 2$ yields

$$p_{\ln 2}(x) = \frac{2^{-|x|}}{Z(\ln 2)}, \quad -\log_2 p_{\ln 2}(x) = |x| + \log_2 Z(\ln 2),$$

so up to the additive constant $\log_2 Z(\ln 2)$, the self-information equals the program length (Kolmogorov complexity).

Algorithmic Entropy via Relative Entropy For any natural number n , let δ_n be the point-mass distribution on outputs equal to n . The relative entropy of δ_n with respect to $p_{\ln 2}$ is

$$S(\delta_n | p_{\ln 2}) = - \sum_i \delta_n(i) \log p_{\ln 2}(i) = -\log \left(\sum_{x: U(x)=n} 2^{-|x|} \right) + \log Z(\ln 2),$$

which for $\gamma = \ln 2$ recovers the *algorithmic entropy* of n up to the constant $\log Z(\ln 2)$.

Thus Baez and Stay establish that Kolmogorov complexity and related "algorithmic entropy" arise naturally as special cases of Shannon's entropy and relative entropy, when the prior over programs is taken to be the appropriate Gibbs ensemble. This bridges the foundations of statistical mechanics and algorithmic information theory in a unified framework.

12.2 Porting Baez and Stay's Argument to Weakness

Now we extend Baez and Stay's argument from Shannon information to weakness. We work in the common quantale

$$Q = ([0, 1], \leq, \otimes, 1), \quad x \otimes y = xy,$$

and compare two *weakness* functors on a category of explanations $\mathcal{C}$.

Shannon (Mutual-Information) Weakness For any explanation E , let $I(E)$ be its mutual information (e.g. between past and future states). Define

$$W_{\text{info}}(E) = 2^{-I(E)}.$$

By subadditivity of mutual information, W_{info} is a lax monoidal functor into Q .

Bennett's (MDL) Weakness Fix a finite, prefix-free "description language" $\mathcal{L}$ (e.g. a dead-code-eliminated MDL grammar or CoDD). For any explanation E , let

$$C_{\mathcal{L}}(E) = \min\{|p| : p \in \mathcal{L}^*, p \text{ encodes } E\}$$

be its minimal code-length. Then Bennett's weakness is

$$W_{\mathcal{L}}(E) = 2^{-C_{\mathcal{L}}(E)},$$

also lax monoidal in Q .

Finite-Vocabulary Mapping Assume the "true" source over explanations is the MDL-Gibbs ensemble

$$p_{\mathcal{L}}(E) = \frac{2^{-C_{\mathcal{L}}(E)}}{\sum_{E'} 2^{-C_{\mathcal{L}}(E')}} \quad (\text{normalization} = 1 \text{ if the sum is } 1),$$

so that each explanation's Shannon self-information equals its code-length:

$$-\log_2 p_{\mathcal{L}}(E) = C_{\mathcal{L}}(E).$$

Under this assumption,

$$I(E) = C_{\mathcal{L}}(E), \implies W_{\text{info}}(E) = 2^{-I(E)} = 2^{-C_{\mathcal{L}}(E)} = W_{\mathcal{L}}(E).$$

Hence Shannon-weakness and Bennett's discrete set-theoretic weakness *coincide* on Q .

Summary This shows that when one fixes a finite, prefix-free description language $\mathcal{L}$ and adopts the corresponding MDL-Gibbs distribution as the *source*, the Shannon-based weakness functor W_{info} reduces exactly to Bennett's original weakness $W_{\mathcal{L}}$. Thus algorithmic information and Shannon information both basically equate to classic set-theoretic weakness under the finite-vocabulary (MDL) assumption.

12.3 Generalizing Baez and Stay's Argument to an Arbitrary Quantale

Baez and Stay's original derivation of Shannon entropy and mutual information uses only a commutative product and a commutative sum. We can therefore port their argument to any commutative quantale $(V, \otimes, \oplus)$ endowed with a weakness measure $w : \mathcal{X} \rightarrow V$.

Quantale-Gibbs Distribution Define the Gibbs weight of each outcome $x \in \mathcal{X}$ by

$$P(x) = \frac{\exp_{\otimes}(-w(x)/T)}{Z}, \quad Z = \bigoplus_{x \in \mathcal{X}} \exp_{\otimes}(-w(x)/T),$$

where $\exp_{\otimes} : V \rightarrow V$ is the quantale-exponential and $T > 0$ is a temperature parameter.

Entropy via a Quantale Logarithm Choose a logarithm map

$$\log_{\oplus} : V \rightarrow \mathbb{R}$$

that transforms $\oplus$ into real addition. Then the entropy of x is

$$H(x) = -\log_{\oplus}(P(x)).$$

By the associativity and commutativity of $\otimes$ and $\oplus$, H obeys the usual additivity and chain-rule identities.

Mutual Information in the Quantale Setting For a joint weight $P(x, y) \in V$, define

$$I(x; y) = \log_{\oplus} \frac{P(x, y)}{P(x) \otimes P(y)} = H(x) + H(y) - H(x, y),$$

where division uses the quantale's residuation. Standard properties (nonnegativity, symmetry, data-processing inequalities) follow from the quantale axioms.

Categorical Port of Baez-Stay Baez and Stay express Shannon information as a natural transformation in a symmetric monoidal category of stochastic maps. Their proof requires only:

- A commutative monoidal product ($\otimes$) for system combination,
- A commutative join ($\oplus$) for mixing or marginalization,
- A logarithm/residuation to relate $\otimes$ and $\oplus$.

Since any commutative quantale $(V, \otimes, \oplus)$ provides these structures, their entire argument carries over unchanged. We thus obtain a full Shannon-style information theory over an arbitrary quantale-weakness measure.

12.4 Algebraic Properties of Quantale-Entropy and Mutual Information

Let $(V, \otimes, \oplus)$ be a commutative quantale, $P : \mathcal{X} \rightarrow V$ a Gibbs-quantale distribution with normalizer Z , and $\log_{\oplus} : V \rightarrow \mathbb{R}$ a homomorphism of $\oplus$ into real addition. Define

$$H(X) = -\log_{\oplus} P(X), \quad I(X; Y) = \log_{\oplus} \frac{P(X, Y)}{P(X) \otimes P(Y)}.$$

Then the following properties hold:

Nonnegativity

$$H(X) \geq 0, \quad I(X; Y) \geq 0,$$

since $0 \leq P(x) \leq 1$ in the quantale implies $-\log_{\oplus} P(x) \geq 0$, and by residuation $P(X, Y) \leq P(X) \otimes P(Y)$ implies $I(X; Y) \geq 0$.

Additivity for Independent Systems If $P(X, Y) = P(X) \otimes P(Y)$, then

$$H(X, Y) = -\log_{\oplus}(P(X) \otimes P(Y)) = H(X) + H(Y),$$

and hence

$$I(X; Y) = 0.$$

Chain Rule

$$H(X, Y) = H(X) + H(Y|X), \quad H(Y|X) = \bigoplus_{x \in \mathcal{X}} P(x) \otimes (-\log_{\oplus} P(Y|x)).$$

Mutual Information Identities

$$I(X; Y) = H(X) + H(Y) - H(X, Y), \quad I(X; Y) = I(Y; X).$$

Data-Processing Inequality For any Markov chain $X \rightarrow Y \rightarrow Z$,

$$I(X; Z) \leq I(X; Y),$$

since marginalization and the quantale join $\oplus$ are monotone.

Relative Entropy Define

$$D(P||Q) = \bigoplus_{x \in \mathcal{X}} P(x) \otimes (\log_{\oplus} P(x) - \log_{\oplus} Q(x)).$$

Then

$$D(P||Q) \geq 0, \quad D(P||Q) = 0 \iff P = Q.$$

These laws follow directly from the associativity, commutativity, monotonicity, and residuation properties of the quantale $(V, \otimes, \oplus)$ together with the homomorphism $\log_{\oplus}$.

12.5 Recovering Quantum Information Theory from the Quantum-Operator Quantale

It is straightforward and unsurprising to see that this sort of quantale information, applied to a quantale of quantum operators, yields the standard quantum information theory.

Let cH be a finite-dimensional Hilbert space and let V be the quantale of positive semidefinite operators on cH , ordered by the Löwner order ($A \leq B$ iff $B - A$ is positive), with

$$A \oplus B = A + B, \quad A \otimes B = AB$$

(and $\oplus$ and $\otimes$ extended to joins and products over families).

Quantum-Gibbs State Given a *Hamiltonian* H as a self-adjoint operator, define the Gibbs state at temperature $T > 0$,

$$\rho = \frac{\exp_{\otimes}(-H/T)}{Z}, \quad Z = \text{Tr}[\exp(-H/T)],$$

where $\exp_{\otimes}(-H/T)$ is the usual operator exponential.

Quantale-Logarithm via Functional Calculus Use the operator-logarithm

$$\log_{\oplus}(A) = \log(A)$$

defined on positive operators. Then the *von Neumann entropy* arises as

$$S(\rho) = -\text{Tr}[\rho \log_{\oplus}(\rho)] = -\text{Tr}[\rho \log(\rho)].$$

Quantum Relative Entropy For two states $\rho, \sigma \in V$, define

$$D(\rho \parallel \sigma) = \text{Tr}[\rho (\log_{\oplus}(\rho) - \log_{\oplus}(\sigma))] = \text{Tr}[\rho (\log \rho - \log \sigma)].$$

This satisfies $D(\rho \parallel \sigma) \geq 0$ with equality iff $\rho = \sigma$.

Quantum Mutual Information For a bipartite state ρ_{AB} on $\text{cH}_A \otimes \text{cH}_B$, define

$$I(A : B) = \log_{\oplus} \frac{\rho_{AB}}{\rho_A \otimes \rho_B} = S(\rho_A) + S(\rho_B) - S(\rho_{AB}),$$

where $\rho_A = \text{Tr}_B[\rho_{AB}]$ and similarly for ρ_B .

Algebraic Properties All standard identities—nonnegativity, additivity on tensor-product states, chain rules, data-processing inequality—follow from the quantale axioms for $(V, \otimes, \oplus)$ together with properties of the operator trace Tr and the functional calculus $\log$.

Thus by instantiating the general quantale-based entropy and divergence constructions on the quantale of positive operators, one exactly recovers the familiar von Neumann entropy, quantum relative entropy, and quantum mutual information of standard quantum information theory.

12.6 Information Theory on the SVM-Margin Quantale

Somewhat less standard is the information theory one obtains from applying the approach outlined above to the SVM-margin quantale.

Let $\mathcal{X}$ be the input space and let w, Φ define a linear SVM classifier $f(x) = \text{sgn}(w \cdot \Phi(x))$ with margin

$$\mu(x) = \frac{w \cdot \Phi(x)}{\|w\|}.$$

We build a commutative quantale $(V, \otimes, \oplus)$ by taking

$$V = [0, \infty], \quad a \oplus b = a + b, \quad a \otimes b = a \times b,$$

and interpret the *weakness* of each example x as

$$w(x) = |\mu(x)|.$$

Quantale-Gibbs Distribution For a temperature $T > 0$, define

$$P(x) = \frac{\exp(-w(x)/T)}{Z}, \quad Z = \sum_{x \in \mathcal{X}} \exp(-w(x)/T).$$

Here $\exp$ is the usual real exponential and $\oplus$ -join is the sum over $\mathcal{X}$.

Entropy Choose the standard logarithm $\log : (0, \infty) \rightarrow \mathcal{R}$. Then the Shannon-style entropy is

$$H(X) = - \sum_{x \in \mathcal{X}} P(x) \log P(x).$$

By properties of $\times$ and $+$ this H satisfies additivity and chain-rules under product distributions.

Mutual Information For a joint SVM on two inputs (X, Y) with joint margin weakness $w(x, y)$ and joint Gibbs weight $P(x, y)$, define

$$I(X; Y) = \sum_{x, y} P(x, y) \log \frac{P(x, y)}{P(x) P(y)} = H(X) + H(Y) - H(X, Y).$$

Relative Entropy For two margin-based distributions P, Q , define

$$D(P \| Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

This is nonnegative and vanishes only if $P = Q$.

Algebraic Properties All the standard identities—nonnegativity, additivity under independent margins, chain-rule, data-processing inequality—follow from the fact that $(V, \otimes, \oplus)$ is commutative and that $\log$ converts multiplication into addition.

12.6.1 A Theorem Comparing Margin-Mutual-Information and Standard Mutual Information in Feature Selection

One interesting potential application of this sort of SVM-margin information is to feature selection. Often one does feature selection via mutual information, before feeding data to an SVM – but it seems probably a better approach to use the mutual information from the SVM-margin quantale here, than the standard version of MI.

Setup Let $\mathcal{X}$ be the input space partitioned into two disjoint subsets of features, X_S (relevant) and X_U (irrelevant). Let Y be the binary class label. Suppose a classifier is a linear SVM that computes the margin

$$\mu_S = \frac{w^T \Phi_S(X_S)}{\|w\|},$$

where $\Phi_S(X_S)$ is the feature map on X_S only. We assume X_U does not influence the true decision boundary but may be spuriously correlated with Y .

Definitions

- *Standard MI criterion:* select subset S of size k that maximizes $I(X_S; Y)$.
- *Margin-MI criterion:* select S that maximizes $I(\mu_S; Y)$, where $\mu_S = w^T \Phi_S(X_S) / \|w\|$.

Theorem Under the assumptions

1. $Y \rightarrow X_S \rightarrow \mu_S$ is a Markov chain,
2. $X_U \perp \mu_S \mid X_S$,

the following hold for any superset $S' = S \cup U$:

$$I(\mu_S; Y) = I(\mu_{S'}; Y), \quad I(X_S; Y) < I(X_{S'}; Y).$$

Hence the margin-MI criterion ranks the true relevant set S above any superset including irrelevant features, whereas the standard MI criterion may be misled by spurious correlations in X_U .

Proof First, since μ_S is a deterministic function of X_S , by the data-processing inequality

$$I(\mu_S; Y) \leq I(X_S; Y).$$

Next, for the superset $S' = S \cup U$, the margin remains

$$\mu_{S'} = \frac{w^T \Phi_S(X_S)}{\|w\|} = \mu_S,$$

because X_U does not enter the SVM decision function. Therefore

$$I(\mu_{S'}; Y) = I(\mu_S; Y).$$

On the other hand, by the chain-rule for mutual information,

$$I(X_{S'}; Y) = I(X_S, X_U; Y) = I(X_S; Y) + I(X_U; Y \mid X_S) > I(X_S; Y),$$

whenever X_U carries any residual information about Y given X_S .

Combining these inequalities yields

$$I(\mu_S; Y) = I(\mu_{S'}; Y) \quad \text{and} \quad I(X_S; Y) < I(X_{S'}; Y),$$

which proves that margin-MI correctly prefers S over S' , while standard MI does not. $\square$

12.6.2 Using SVM Channel Capacity as a Deep-Kernel Learning Objective

It may also be interesting to use quantities derived from SVM-margin information theory as part of the objective function for learning an SVM kernel, e.g. via deep kernel learning.

Let $\{(x_n, y_n)\}_{n=1}^N$ be a labeled training set, with $y_n \in \{\pm 1\}$. We have a deep kernel $\Phi_\theta(x)$ with parameters θ , and an SVM solve producing decision weights w_θ and bias b_θ .

Define the Margin and Gibbs Prior For each input x , define the signed margin

$$\mu_\theta(x) = y \frac{w_\theta \cdot \Phi_\theta(x) + b_\theta}{\|w_\theta\|}.$$

Introduce a temperature $T > 0$ and set a Gibbs-style prior over inputs:

$$P_\theta(x) = \frac{\exp(-|\mu_\theta(x)|/T)}{\sum_{m=1}^N \exp(-|\mu_\theta(x_m)|/T)}.$$

Define the Soft Channel Model the stochastic decision of the SVM by a logistic "channel":

$$P_\theta(y | x) = \frac{1}{1 + \exp(-\alpha y (w_\theta \cdot \Phi_\theta(x) + b_\theta))},$$

where $\alpha \gg 1$ controls the steepness.

Compute Channel Mutual Information The joint distribution is

$$P_\theta(x, y) = P_\theta(x) P_\theta(y | x),$$

and the marginal over labels is

$$P_\theta(y) = \sum_{m=1}^N P_\theta(x_m) P_\theta(y | x_m).$$

Define the channel capacity (mutual information)

$$I_\theta(X; Y) = \sum_{n=1}^N \sum_{y \in \{\pm 1\}} P_\theta(x_n, y) \log \frac{P_\theta(x_n, y)}{P_\theta(x_n) P_\theta(y)}.$$

Joint Learning Objective Combine capacity maximization with regularization and hinge loss. For example, minimize

$$L(\theta) = -I_\theta(X; Y) + \lambda \sum_{i,j} \varphi(i, j) m((\theta_K)_{ij}) + C \sum_{n=1}^N \max(0, 1 - y_n (w_\theta \cdot \Phi_\theta(x_n) + b_\theta)),$$

where

- $\varphi(i, j) m((\theta_K)_{ij})$ is the quantale-weakness regularizer on kernel parameters,
- $C > 0$ trades off hinge loss and capacity,
- $\lambda > 0$ trades off weakness regularization and capacity.

Optimization Use stochastic gradient descent or other methods to update θ :

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta),$$

backpropagating through the SVM solve (e.g. via implicit differentiation or a differentiable QP solver), the Gibbs prior, and the logistic channel.

This objective encourages the deep kernel to produce representations that both separate classes (hinge loss), respect simplicity (weakness regularizer), and maximize the information capacity of the induced SVM channel.

12.6.3 A Margin-Entropy Generalization Bound

SVM-margin information theory seems also to have applications to analyzing the generalization behavior of SVMs.

Let $\mathcal{X}$ be a finite input set and $f : \mathcal{X} \rightarrow \{\pm 1\}$ an SVM classifier with (signed) margin $\mu(x)$. For temperature $T > 0$ define the Gibbs margin distribution

$$P(x) = \frac{\exp(-|\mu(x)|/T)}{\sum_{x' \in \mathcal{X}} \exp(-|\mu(x')|/T)}.$$

Let $E = \{x \in \mathcal{X} : f(x) \neq y(x)\}$ be the error region under the true labels $y(x) \in \{\pm 1\}$, and write

$$p_{\text{err}} = \sum_{x \in E} P(x), \quad H(P) = - \sum_{x \in \mathcal{X}} P(x) \log P(x).$$

Theorem 18 (Margin-Entropy Bound). *Under the above definitions,*

$$p_{\text{err}} \leq \frac{H(P)}{\ln 2}.$$

Proof. Denote $p = p_{\text{err}}$ and $q = 1 - p$. The binary entropy of the event E satisfies

$$h(p) = -p \ln p - q \ln q \leq H(P),$$

because partitioning $\mathcal{X}$ into E and its complement can only decrease entropy. But for $0 \leq p \leq \frac{1}{2}$ the binary entropy obeys

$$h(p) = -p \ln p - (1-p) \ln(1-p) \geq p \ln 2,$$

since $-(1-p) \ln(1-p) \geq 0$ and $-\ln p \geq \ln 2$ when $p \leq \frac{1}{2}$. Combining gives

$$H(P) \geq h(p) \geq p \ln 2,$$

hence

$$p_{\text{err}} = p \leq \frac{H(P)}{\ln 2}.$$

□

In words, if the Gibbs margin distribution has low entropy, then the probability mass of misclassified inputs under that distribution is small. Since low margin-entropy implies that most probability concentrates on high-margin (well-classified) points, this bound captures how margin-separation drives generalization.

12.7 Fuzzy Information Theory

Let $(V, \otimes, \oplus) = ([0, 1], T, \max)$ where T is a continuous t-norm. Given a fuzzy subset $H : X \rightarrow [0, 1]$, one may normalize memberships to define a fuzzy probability distribution

$$P_H(x) = \frac{H(x)}{\sum_{u \in X} H(u)},$$

or work directly on the unnormalized $H(x)$ using $\oplus = \max$. Choose a logarithm map

$$\log_{\oplus} : [0, 1] \rightarrow \mathbb{R}, \quad \log_{\oplus}(a) = \ln(a).$$

Fuzzy Shannon Entropy

$$H_f(H) = - \bigoplus_{x \in X} [H(x) \otimes \log_{\oplus}(H(x))] = - \max_{x \in X} T(H(x), \ln(H(x))).$$

Fuzzy Relative Entropy For two fuzzy sets $H, G : X \rightarrow [0, 1]$,

$$D_f(H \| G) = \bigoplus_{x \in X} [H(x) \otimes (\log_{\oplus}(H(x)) - \log_{\oplus}(G(x)))].$$

Fuzzy Mutual Information Given a fuzzy joint relation $H_{XY} : X \times Y \rightarrow [0, 1]$ with marginals

$$H_X(x) = \bigoplus_{y \in Y} H_{XY}(x, y), \quad H_Y(y) = \bigoplus_{x \in X} H_{XY}(x, y),$$

define

$$I_f(X; Y) = \bigoplus_{x \in X} \bigoplus_{y \in Y} [H_{XY}(x, y) \otimes \log_{\oplus} \frac{H_{XY}(x, y)}{H_X(x) \otimes H_Y(y)}].$$

These satisfy nonnegativity, symmetry $I_f(X; Y) = I_f(Y; X)$, and a fuzzy data-processing inequality by virtue of $(V, \otimes, \oplus)$ being a commutative quantale and $\log_{\oplus}$ respecting $\oplus$.

12.7.1 Concrete Examples of Fuzzy Information Operators

We work in the quantale $V = [0, 1]$ with $\oplus(a, b) = \min(1, a+b)$ and $\otimes(a, b) = ab$, and use $\log_{\oplus}(a) = \ln(a)$.

Fuzzy Shannon Entropy Let $X = \{x_1, x_2, x_3\}$ and define a fuzzy set H by

$$H(x_1) = 1.0, \quad H(x_2) = 0.5, \quad H(x_3) = 0.2.$$

Then

$$H_f(H) = - \bigoplus_{i=1}^3 [H(x_i) \otimes \ln H(x_i)] = - \min\left(1, 1 \cdot \ln(1) + 0.5 \cdot \ln(0.5) + 0.2 \cdot \ln(0.2)\right) \approx 0.6685.$$

Fuzzy Relative Entropy Let $X = \{x_1, x_2\}$ and fuzzy sets H, G be

$$H(x_1) = 0.8, \quad H(x_2) = 0.2, \quad G(x_1) = 0.5, \quad G(x_2) = 0.5.$$

Compute

$$D_f(H\|G) = \bigoplus_{i=1}^2 [H(x_i) \otimes (\ln H(x_i) - \ln G(x_i))] = \min\left(1, 0.8[\ln(0.8) - \ln(0.5)] + 0.2[\ln(0.2) - \ln(0.5)]\right) \approx 0.1928.$$

Fuzzy Mutual Information Let $X = \{a, b\}$, $Y = \{0, 1\}$ and define a fuzzy joint H_{XY} :

$$H_{XY}(a, 0) = 0.9, \quad H_{XY}(a, 1) = 0.1, \quad H_{XY}(b, 0) = 0.4, \quad H_{XY}(b, 1) = 0.6.$$

Marginals (using $\oplus = \min(1, \sum)$) are

$$H_X(a) = \min(1, 0.9 + 0.1) = 1.0, \quad H_X(b) = \min(1, 0.4 + 0.6) = 1.0,$$

$$H_Y(0) = \min(1, 0.9 + 0.4) = 1.0, \quad H_Y(1) = \min(1, 0.1 + 0.6) = 0.7.$$

Then

$$I_f(X; Y) = \bigoplus_{x,y} [H_{XY}(x, y) \otimes \ln \frac{H_{XY}(x, y)}{H_X(x) \otimes H_Y(y)}] = \min(1, 0.1131) \approx 0.1131,$$

where the sum inside $\oplus$ was computed over $(a, 0), (a, 1), (b, 0), (b, 1)$.

These examples illustrate how fuzzy-set membership and the $\oplus, \otimes$ operations yield concrete entropies, divergences, and mutual informations in the fuzzy quantale.

12.7.2 Connections into the Fuzzy Literature

The constructions of fuzzy entropy and mutual information we have given can be viewed as a unifying algebraic reformulation of several classical approaches in the fuzzy-information literature:

- **De Luca & Termini [LT72]** introduced a measure of entropy for a single fuzzy set

$$E_{\text{DT}}(H) = - \sum_{x \in X} H(x) \ln H(x),$$

and showed it satisfies analogues of nonnegativity and maximality at the uniform fuzzy set.

- **Yager (1980s)** proposed alternative fuzzy-entropy functionals using different aggregation operators (e.g. max-min, ordered weighted averaging), reflecting various attitudes toward uncertainty in a fuzzy set [Yag79].
- **Kaufmann & Tsaur** developed fuzzy cross-entropy and divergence measures, extending Kullback-Leibler divergence to fuzzy membership functions [KT98].
- **Pal & Mitra (1990s)** defined fuzzy mutual information for a fuzzy joint relation $H_{XY} : X \times Y \rightarrow [0, 1]$ by

$$I_f(X; Y) = E_{\text{DT}}(H_X) + E_{\text{DT}}(H_Y) - E_{\text{DT}}(H_{XY}),$$

and proved properties such as nonnegativity and a fuzzy data-processing inequality [MP94].

Quantale-Enriched Reformulation In our approach we take the unit interval $[0, 1]$ with join $\oplus = \max$ (or bounded sum) and a chosen t-norm $\otimes = T$ as a commutative quantale $(V, \otimes, \oplus)$. Fuzzy sets $H : X \rightarrow [0, 1]$ become V -valued subobjects, and the canonical quantale entropy

$$H_q(H) = - \bigoplus_{x \in X} [H(x) \otimes \ln H(x)]$$

and mutual information

$$I_q(X; Y) = \bigoplus_{x, y} [H_{XY}(x, y) \otimes \ln \frac{H_{XY}(x, y)}{H_X(x) \otimes H_Y(y)}]$$

recover the De Luca-Termini, Yager, Kaufmann-Tsaur, and Pal-Mitra formulae as special cases when one chooses $\oplus$ and $\otimes$ appropriately. This quantale perspective makes explicit the algebraic laws (associativity, commutativity, distributivity) underlying all these fuzzy-information measures and places them within a single, compositional framework.

12.8 Hypervector Information Theory, Pattern Intensities, and Emergence

Let $V = \mathcal{Z}_r^n$ be the modular composite hypervector space (binding $+$ mod r , grouping by multiset sum and normalization) and define the similarity-quantale

$$Q = ([0, 1], \oplus = \max, \otimes = \times).$$

Equip each hypervector $v \in V$ with a similarity score

$$\phi(v) = 1 - \frac{\delta(v, 0)}{n(r/2)},$$

where δ is the modular Manhattan distance. Then for any *hypervector model* $m : X \rightarrow V$ set

$$\phi_x = \phi(m(x)) \in Q, \quad W = \bigoplus_{x \in X} \phi_x = \max_{x \in X} \phi_x.$$

1. Quantale Shannon-Style Entropy Define

$$H_Q(m) = - \bigoplus_{x \in X} [\phi_x \otimes \log \phi_x] = - \max_{x \in X} (\phi_x \ln \phi_x).$$

Quantale Relative Entropy For two models $m, m' : X \rightarrow V$ with $\phi_x = \phi(m(x))$, $\phi'_x = \phi(m'(x))$, set

$$D_Q(m \| m') = \bigoplus_{x \in X} \left[\phi_x \otimes \ln \frac{\phi_x}{\phi'_x} \right] = \max_{x \in X} \left(\phi_x \ln \frac{\phi_x}{\phi'_x} \right).$$

Quantale Mutual Information Given joint model $M : X \times Y \rightarrow V$, define

$$\phi_{x,y} = \phi(M(x, y)), \quad \phi_x = \max_y \phi_{x,y}, \quad \phi_y = \max_x \phi_{x,y},$$

and then

$$I_Q(X; Y) = \bigoplus_{x,y} \left[\phi_{x,y} \otimes \ln \frac{\phi_{x,y}}{\phi_x \otimes \phi_y} \right].$$

Pattern Intensity For any subset $S \subseteq X$ define the *pattern intensity* of S in m by

$$I_S(m) = \left(\bigoplus_{x \in S} \phi_x \right) \multimap \left(\bigoplus_{x \in X} \phi_x \right),$$

where the residuation $\multimap$ in Q is $a \multimap b = \min(1, b/a)$ when $a > 0$ and 1 if $a = 0$.

Emergent Synergy For two patterns $S, T \subseteq X$, let

$$I_{S \cup T}(m) = \left(\bigoplus_{x \in S \cup T} \phi_x \right) \multimap W, \quad I_S(m), I_T(m)$$

as above. Define the *synergy*

$$\sigma_{S,T}(m) = (I_S(m) \otimes I_T(m)) \multimap I_{S \cup T}(m).$$

We say (S, T) is an *emergent hypervector pattern* in m whenever

$$\sigma_{S,T}(m) \geq 1.$$

These constructions lift standard information-theoretic and pattern-emergence notions to the modular hypervector algebra by using the similarity quantale Q , yielding measures of entropy, divergence, mutual information, pattern intensity, and emergent synergy for hypervector models.

12.8.1 Episodic Memory Retrieval via Hypervector Mutual Information

One promising application of hypervectors in an AGI context is to episodic memory. Let's flesh out how information on the hypervector quantale might help with this.

Let $r \geq 2$, $n \in \mathcal{N}$, and work in the modular hypervector space $V = \mathcal{Z}_r^n$. Suppose we have a set of N episodic memory vectors

$$\{v^{(1)}, v^{(2)}, \dots, v^{(N)}\} \subset V,$$

and a memory cue represented by a hypervector

$$c \in \mathcal{Z}_r^n.$$

Define the Component-wise Joint Distribution Index vector components by $i = 1, \dots, n$. Define a joint distribution over pairs of symbols in $\mathcal{Z}_r \times \mathcal{Z}_r$ by

$$P_{c,v}(a, b) = \frac{1}{n} \left| \{i : c_i = a \text{ and } v_i = b\} \right|, \quad a, b \in \mathcal{Z}_r.$$

Its marginals are

$$P_c(a) = \sum_b P_{c,v}(a, b), \quad P_v(b) = \sum_a P_{c,v}(a, b).$$

Hypervector Mutual Information The mutual information between the cue c and a candidate memory $v = v^{(k)}$ is

$$I(c; v) = \sum_{a, b \in \mathcal{Z}_r} P_{c,v}(a, b) \ln \frac{P_{c,v}(a, b)}{P_c(a) P_v(b)}.$$

Higher $I(c; v)$ indicates that v shares more co-occurrence structure with the cue.

3. Retrieval by Maximal Mutual Information To retrieve the episode best matching the cue, compute

$$k^* = \arg \max_{1 \leq k \leq N} I(c; v^{(k)}).$$

Return $v^{(k^*)}$ (or the associated memory) as the retrieved episode.

Pattern Clusters and Emergence More generally, given a subset of episodes $S \subset \{1, \dots, N\}$, form the aggregate vector

$$v_S = \bigoplus_{k \in S} v^{(k)}$$

(using the modular grouping operator). Define $I(c; v_S)$ as above. One can then:

- *Pattern intensity*:

$$I_S(c) = I(c; v_S).$$

- *Emergent synergy*: for two clusters S, T , compute

$$\sigma_{S,T}(c) = \frac{I(c; v_{S \cup T})}{I(c; v_S) I(c; v_T)}.$$

A value $\sigma_{S,T}(c) > 1$ indicates that the joint cluster $S \cup T$ has *more* shared structure with the cue than expected from S and T separately.

Discussion

- This method treats each hypervector component as a random variable taking values in $\mathcal{Z}_r$, and uses standard Shannon mutual information to measure semantic overlap.
- The retrieval rule $\arg \max I(c; v)$ generalizes simple cosine or Hamming-distance matching to a full information-theoretic criterion.
- Cluster-based retrieval and synergy detection identify emergent multi-episode patterns that align most strongly with a given cue.

12.8.2 Toy Example of Episodic Retrieval via Hypervector-Quantale Mutual Information

We work in the modular hypervector space $V = \mathcal{Z}_2^4$ and the similarity quantale

$$Q = ([0, 1], \oplus = \max, \otimes = \times).$$

Define symbol-similarity

$$\phi(a, b) = \begin{cases} 1, & a = b, \\ \epsilon, & a \neq b, \end{cases} \quad 0 < \epsilon < 1.$$

Set $\epsilon = 0.5$ for concreteness.

Stored Episodes and Cue

$$\begin{aligned} v^{(1)} &= (1, 1, 1, 1), & v^{(2)} &= (0, 0, 0, 0), & v^{(3)} &= (0, 1, 0, 0), \\ c &= (1, 1, 0, 0). \end{aligned}$$

Component-wise Similarities For each $k = 1, 2, 3$ and $i = 1, \dots, 4$, compute

$$\begin{aligned} H_i^{(k)} &= \phi(c_i, v_i^{(k)}). \\ H^{(1)} &= (1, 1, \epsilon, \epsilon), & H^{(2)} &= (\epsilon, \epsilon, 1, 1), & H^{(3)} &= (\epsilon, 1, 1, 1). \end{aligned}$$

Quantile Mutual Information We use the "peak" formula

$$I_Q(c; v^{(k)}) = \bigoplus_{i=1}^4 \left[H_i^{(k)} \otimes \ln \left(\frac{H_i^{(k)}}{H_i^{(k)} \otimes H_i^{(k)}} \right) \right] = \max_{1 \leq i \leq 4} [H_i^{(k)} \cdot (-\ln H_i^{(k)})].$$

Since $1 \cdot (-\ln 1) = 0$ and $\epsilon \cdot (-\ln \epsilon) = \epsilon \ln \frac{1}{\epsilon} = 0.5 \ln 2 \approx 0.3466$, we get

$$I_Q(c; v^{(1)}) = 0.3466, \quad I_Q(c; v^{(2)}) = 0.3466, \quad I_Q(c; v^{(3)}) = 0.3466.$$

Retrieval All three episodes tie with $I_Q \approx 0.3466$. One may break ties by recency or a secondary criterion; for example, retrieve episode 1:

"Picked up red cup on table in morning."

as a best-match to the cue "red cup on table."

12.9 Mutual Information in the Logic Quantale

One may also explore similar constructs in ordinary logic; consider for instance the logical quantale from Chapter 4.1 above.

Suppose we have a joint predicate

$$P : X \times Y \rightarrow Q_{\text{logic}},$$

with marginals

$$P_X(x) = \bigoplus_{y \in Y} P(x, y) = \bigcup_y P(x, y), \quad P_Y(y) = \bigcup_x P(x, y).$$

We can then define, for instance, the *logic-quantale mutual information* by

$$I_{\text{logic}}(X; Y) = \bigoplus_{x, y} \left[P(x, y) \otimes_{\text{logic}} (P(x, y) \multimap_{\text{logic}} (P_X(x) \otimes_{\text{logic}} P_Y(y))) \right],$$

i.e.

$$I_{\text{logic}}(X; Y) = \bigcup_{x,y} \left[P(x, y) \cap ((W \setminus P(x, y)) \cup (P_X(x) \cap P_Y(y))) \right].$$

This subset of W collects those worlds in which the joint predicate $P(x, y)$ *implies* the independent marginals—an information-theoretic measure of how strongly X and Y co-occur in the logic quantale.

Other aspects of logical information theory follow similarly.

13 Efficient Implementation of WAIXI and Related Methods via Galois Connections

In most realistic problems, certainly most cases related to AGI one uses *approximate stochastic DP*—e.g. Monte-Carlo, TD, or prioritized sweeping—instead of exact Bellman backups. As argued in [Goe21d], one can also view a great variety of AGI-related algorithms as specialized forms of approximate stochastic dynamic programming. Toward that end, and leveraging ideas from [DH16, ?] behind the scenes, we now show how the same Galois-fusion idea carries over: by writing the DP update as a single MeTTa-IL rule that samples transitions in MORK, weights them by Occam-weakness, and folds them into an approximate value or Q-function. We also indicate how a similar strategy may be possible for quantum implementations.

13.1 Approximate Stochastic DP via MeTTa-IL, MORK and Galois Fusion

13.1.1 Stochastic DP Heuristic

Let $Q : S \times A \rightarrow \mathbb{R}$ be an action-value function. A generic one-step TD update is

$$Q_{t+1}(s, a) = Q_t(s, a) + \eta \left(r + \gamma \max_{a'} Q_t(s', a') - Q_t(s, a) \right),$$

where (s, a, r, s') is sampled from M , η a learning rate, and γ a discount.

13.1.2 Galois Connection

We again view $V = \{Q : S \times A \rightarrow \mathbb{R}\}$ ordered pointwise, and $P = \text{CoDD}_{\text{ENF}}$ ordered by weakness $w(h)$. Define

$$\alpha(Q) = \text{PolicyFromQ}(\arg \max_a Q(s, a)), \quad \gamma(h) = Q_h,$$

where Q_h is the Q-function induced by running policy h . One shows $\alpha \dashv \gamma$ as before, so that stochastic updates in V lift to a fuseable operator in P .

13.1.3 Parallel Efficiency

- Sampling, lookup and TD-error computation for each (s, a) is fully parallel as Rholang actors.
- No global Q-table is materialized: updates stream into the ENF-CoDD structure.
- The reducer need only perform in-place graph rewrites of the affected subgraphs.

13.1.4 Weakness Through the Adjunction

By construction the fused rule implements $P_{t+1} = (\alpha \circ T \circ \gamma)(P_t)$ in one pass, where T is the expected TD-operator. Since α preserves suprema, the Occam-weakness bias encoded in α carries through every update, ensuring the evolving controller remains as general as possible subject to the stochastic learning objective.

This demonstrates that even approximate, sampling-based DP methods admit a one-pass, Galois-fused implementation in MeTTa-IL + MORK + Rholang, with Occam-weakness seamlessly interwoven.

13.2 Galois-Fusion for Reinforcement Learning in MeTTa-IL and MORK

We show how common RL algorithms—viewed as approximate stochastic DP—fit into the same Galois-adjoint $\alpha \dashv \gamma$ pattern and can be implemented as a single fused MeTTa-IL rewrite over a MORK hypergraph.

13.3 Category-Enriched Setup

Let

$$V = \{Q : S \times A \rightarrow \mathbb{R}\}, \quad P = \text{CoDD}_{\text{ENF}},$$

with orders $Q_1 \leq_V Q_2 \Leftrightarrow \forall (s, a), Q_1(s, a) \leq Q_2(s, a)$ and $h_1 \leq_P h_2 \Leftrightarrow w(h_1) \geq w(h_2)$.

Define the Galois connection

$$\alpha(Q) = \text{PolicyFromQ}(\epsilon\text{-greedy}(Q)), \quad \gamma(h) = Q_h,$$

so that $\alpha(V) \leq_P h \iff V \leq_V \gamma(h)$.

13.4 Q-Learning as Fused MeTTa-IL Rule

BellmanTD Operator

$$T_Q(Q)(s, a) = Q(s, a) + \eta \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right),$$

for each sampled transition $(s, a, r, s') \sim M$.

13.5 SARSA and Eligibility Traces

SARSA Operator

$$\delta = r + \gamma Q(s', a') - Q(s, a),$$

with (s, a, r, s', a') sampled. The fused rule is identical, replacing $\max_{a'} Q$ by the sampled $Q(s', a')$, and updates propagate along eligibility traces in the reducer.

13.6 Actor-Critic / Policy-Gradient

- **Critic update:** same as Q-Learning or SARSA to estimate $V(s)$.
- **Actor adjunction:** $\alpha(V) = \text{SoftmaxPolicy}(V)$, $\gamma(\pi) = V_\pi$.
- **Policy-gradient step:**

$$\Delta\theta = \eta \nabla_\theta \ln \pi_\theta(a | s) [r + \gamma V(s') - V(s)].$$

13.7 General Recipe

1. Identify the RL update operator T (TD, SARSA, PG, etc.) on V -space.
2. Choose $\alpha: V \rightarrow P$ (policy extractor) and $\gamma: P \rightarrow V$ (value evaluator) so $\alpha \dashv \gamma$.
3. Write one MeTTa-IL rule that:

γ reads the current program P to get Q or V .

T samples transitions from MORK, computes errors or gradients.

α emits and then reduces all updates in-place to yield P' .

Benefits

- No explicit Q or π tables—updates stream through the hypergraph.
- Massive parallelism: each sampled transition is a separate concurrent actor.
- Low memory overhead: only CoDD programs and MORK edges are stored.
- Uniform implementation pattern for all major RL algorithms.

13.8 Quantum-Fused Bellman via QRAM and Quantum Circuits

We next outline how the same Galois-fusion idea—fusing "unfold" (γ), Bellman update (B), and "fold" (α)—can potentially be implemented on a quantum computer to solve large MDPs with quadratic speedups.

13.8.1 QRAM Encoding of the MDP

- Store the transition model $M : S \times A \rightarrow \text{Dist}(R \times S)$ in quantum RAM (QRAM) as amplitude-encoded arrays.
- Each basis state $|s, a\rangle$ addresses a superposition of transitions $\sum_{r, s'} \sqrt{M(s, a; r, s')} |r, s'\rangle$.
- The current value-function $V : S \rightarrow \mathbb{R}$ is likewise stored as amplitudes in a QRAM block.

13.8.2 Superposed Bellman Fold

We prepare the uniform superposition over all (s, a) :

$$|\psi_0\rangle = \frac{1}{\sqrt{|S||A|}} \sum_{s \in S} \sum_{a \in A} |s, a\rangle.$$

Then one coherent quantum circuit performs:

1. *Expectation*: Apply a controlled-unitary that, for each $|s, a\rangle$, loads the transition amplitudes and the value amplitudes of successor states $|s'\rangle$, then computes

$$E(s, a) = \sum_{r, s'} M(s, a; r, s') [r + \gamma V(s')]$$

into a target register (using standard reversible arithmetic).

2. *Weakness Prior*: Multiply the amplitude of $|s, a, E\rangle$ by $\sqrt{\pi(P)}$, where $\pi(P) \propto w(P)^\alpha$ is computed once and stored in a small ancilla.
3. *Maximization*: Perform a Grover-style "quantum argmax" subroutine to mark and amplify the $|s, a\rangle$ with maximal weighted $E(s, a)$.

Crucially, all three steps—lookup in QRAM (γ), compute expectation (B), and perform Grover maximization (α)—occur in one coherent circuit of depth $O(\text{poly log } |S||A|)$.

13.8.3 Galois Adjunction in the Quantum Domain

We again view $V = \{V : S \rightarrow \mathbb{R}\}$ ordered pointwise and $P = \text{CoDD}_{\text{ENF}}$ ordered by weakness. The quantum circuit above implements

$$|P'\rangle = (\alpha \otimes B \otimes \gamma) |P\rangle$$

in superposition, and by linearity and amplitude-preservation it respects the Galois connection $\alpha \dashv \gamma$. The resulting $|P'\rangle$ is the amplitude-encoded optimal controller.

13.8.4 Complexity and Speedup

- **QRAM queries:** each superposition lookup is $O(1)$ quantum-time.
- **Expectation update:** reversible arithmetic on $r+V(s')$ takes $O(\text{poly log } |S|)$.
- **ArgMax via Grover:** finds the maximal (s, a) in $O(\sqrt{|S||A|})$ steps.
- **Overall:** one fused circuit of depth $O(\sqrt{|S||A|} \text{ poly log } |S||A|)$, yielding a quadratic speedup over the classical $O(|S||A|)$ pass.

13.8.5 Summary

This quantum analogue implements the Bellman fusion $\alpha \circ B \circ \gamma$ in a single coherent QRAM-based circuit, carrying the Occam-weakness prior as amplitude weights and using Grover amplification for the α -fold. By exploiting quantum parallelism and Grover’s speedup, it achieves the same one-pass, low-memory DP fusion with a quadratic runtime improvement.

Part III

Decision , Motivation & Consciousness

14 Weakness-Based Measurement of Causality and Clarity

Causality is a folk concept with a great number of different interpretations. In the effort to construct rigorous theories that capture significant aspects of the intuitive notion of causality in a manner useful for AI systems, the concept of weakness can be significantly valuable.

Among other applications, we will use the theory of causality here to develop a theory of ”clarity of representation” in neural models and other AI networks. Roughly speaking we will say a network has a clear internal representation if there tend to be simple causal stories regarding why it gave the output it did in various situations.

14.1 From Conceptual to Formal Notions of Causality

But what shall we take causality to mean here? – Intuitively, we often say that ”A causes B” if the occurrence of event A makes event B much more likely than it would be without A . In everyday language, we look for a set of mechanisms M – for instance, physical or biological processes – such that when A happens, those mechanisms explain why B follows. Formally, we can express

this by comparing the probability of B given A versus the probability of B given not- A :

$$P(B | A) \gg P(B | \neg A).$$

Here $P(\cdot)$ is a "simple" probability model, meaning that we measure simplicity by assuming the mechanisms in M have low descriptive complexity. In other words, we prefer explanations where the rules or equations in M are not overly complicated. Typically we also require that A precedes B in time, so that causality flows forward. When A and B each stand for a category of events, we say " A causes B " if almost every time we see an event a in category A and a nearby event b in category B , the same causal relationship applies. Thus, a causal explanation of why A leads to B is simply a proof-using the low-complexity mechanisms in M -that $P(B | A)$ is much larger than $P(B | \neg A)$.

One can elaborate these basic ideas into a formal framework combining the *algebra of weakness* (measuring descriptive simplicity) with an *algebra of probabilistic counterfactual strength*. We begin by defining key concepts.

1. **Events and Mechanisms.** Let A and B be events or event-categories in some sample space Ω . We write

$$M = \{m_1, m_2, \dots, m_k\}$$

for a finite set of hypothesized mechanisms. Each m_i is a rule or function that, when combined with initial data, yields predictions about future outcomes.

2. **Weakness Algebra** $(\mathcal{W}, \leq)$. We assume there is an ordered set $\mathcal{W}$ of *weakness values* (also called 'simplicity scores-'), with a partial order $\leq$ such that lower elements represent *weaker* (i.e. simpler) descriptions. Concretely, if $\mathbf{w} \in \mathcal{W}$ measures the combined complexity of mechanisms M , then a smaller $\mathbf{w}(M)$ implies a simpler explanatory model.
3. **Probabilistic Algebra** $(\mathcal{P}, \succ)$. Let $\mathcal{P}$ be the set of real-valued conditional probabilities on events, ordered by $\succ$ —meaning 'significantly larger than.'—For two probabilities p_1, p_2 , we write

$$p_1 \succ p_2 \iff p_1 > p_2 + \epsilon$$

for some fixed significance margin $\epsilon > 0$.

4. **Causal Explanation as a Theorem.** We say that

$$A \xrightarrow{M} B$$

is a valid causal explanation under mechanisms M if and only if

$$P(B | A; M) \succ P(B | \neg A; M) \quad \text{and} \quad A \text{ precedes } B.$$

Here $P(B | A; M)$ denotes the probability of B given A when we use exactly the mechanisms in M as our model. Intuitively, A causes B via M exactly when knowing A and M makes B much more likely than not having A (still using the same mechanisms).

5. **Simple Probability Function.** Define a *simple* probability function P as one whose description-encoded in bits-has weak complexity relative to $\mathcal{W}$. Formally, if $\mathbf{w}(P)$ denotes the weakness value of this probability model, then we require that for all comparisons we make, $\mathbf{w}(P) \leq \beta$ for some threshold β . Thus P is 'simple' if its specification does not exceed a fixed complexity budget.
6. **Combined Algebra via Fibring.** We now define a *fibring* (or -product-type-) algebra $(\mathcal{C}, \text{Prec})$ that combines weakness and probabilistic strength. An element of $\mathcal{C}$ is a pair

$$(\mathbf{w}, \delta) \quad \text{with} \quad \mathbf{w} \in \mathcal{W}, \delta \in \mathbb{R},$$

where $\delta = P(B \mid A; M) - P(B \mid \neg A; M)$ measures the *counterfactual strength* of A on B , and $\mathbf{w} = \mathbf{w}(M) + \mathbf{w}(P)$ measures the combined simplicity cost of specifying mechanisms M and the probability function P . We define the order -Prec- on $\mathcal{C}$ as:

$$(\mathbf{w}_1, \delta_1) \text{Prec} (\mathbf{w}_2, \delta_2) \iff [\delta_1 > \delta_2] \vee [\delta_1 = \delta_2 \text{ and } \mathbf{w}_1 \leq \mathbf{w}_2].$$

In words, a pair with *larger* counterfactual strength δ is considered -greater- (more convincing as a causal explanation); if two pairs have identical δ , we prefer the one with *lower* complexity $\mathbf{w}$.

7. **Interpretation.** To assert - A causes B - we pick the pair

$$(\mathbf{w}(M) + \mathbf{w}(P), P(B \mid A; M) - P(B \mid \neg A; M))$$

that is maximal under -Prec-. In effect, we look for the smallest (weakest) mechanisms M and simplest probability P that yield the largest increase in B 's probability when A occurs. This formalizes the intuition that a -good- causal explanation is both *simple* (low complexity) and *strong* (high counterfactual effect).

To measure "how much A causes B ", we perform two competing evaluations:

- (a) We measure how much seeing A boosts the probability of B (that is $\delta = P(B \mid A) - P(B \mid \neg A)$).
- (b) We measure how simple our explanation is (that is $\mathbf{w}$, the combined complexity of our mechanisms and probability model).

We then compare different explanations (M, P) by first looking at which one gives the *largest* probability boost δ . If two explanations give the same boost, we prefer the one that is *simpler*. In other words, a cause A of B is best characterized by the fewest, simplest mechanisms that produce the greatest increase in B 's likelihood when A occurs. This aligns with our intuition: a good causal explanation should explain a lot (large δ) without invoking an overly complex mechanism (small $\mathbf{w}$).

14.2 Weakness-Weighted Counterfactual Causality (WWCC) vs. Bennett-style Organism Causality

Bennett, in his thesis on weakness, articulates a theory of causality grounded in organisms and their quest for survival. We can articulate the theory on causality presented above – which we may call Weakness-Weighted Counterfactual Causality (WWCC) – as a generalization of Bennett’s notion.

To summarize the WWCC framework in a simple way, we identify causal relations by combining a measure of counterfactual effect with a quantale-based weakness penalty, via: “A causes B” if there exists a mechanism M such that

$$weakness(M) \text{ is low and } \Delta_{A \rightarrow B}(M) = P(B \mid M \wedge A) - P(B \mid M \wedge \neg A) > \varepsilon,$$

for some effect threshold ε .

In Bennett’s view, on the other hand, causality arises as “causal identities” – subsets of intervention statements that survive observation – which an organism constructs by w-maxing under two preconditions:

- *Incentive*: the cause must afford some survival advantage.
- *Representation*: the organism’s abstraction layer must be rich enough to express that identity.

Objects, properties, and even consciousness emerge from the hierarchy of these learned causal identities.

To recover Bennett’s view of causality within WWCC, we can articulate an **organism layer** L_{org} . Mechanisms now lie in the state space $\mathcal{W}_{L_{\text{org}}}$, and weakness is computed via the associated quantale $Q_{L_{\text{org}}}$:

$$weakness_{L_{\text{org}}}(M).$$

One then defines a survival-weighted effect measure

$$\Delta^{\text{val}}(A \rightarrow B) = U(B)(P(B \mid A) - P(B \mid \neg A)),$$

where $U(B)$ is the organism’s utility for outcome B .

One can then impose two preconditions:

- M must be definable in $\mathcal{W}_{L_{\text{org}}}$.
- $\Delta^{\text{val}}(A \rightarrow B) > \delta$ for some threshold δ .

Bennett’s causal identities are those mechanisms solving

$$\min_M \{weakness_{L_{\text{org}}}(M) \mid \Delta^{\text{val}}(A \rightarrow B) > \delta\}.$$

14.3 Reformulating Causal Learning in PLN Logic within Hyperon

So how can we connect all this to practical causal learning AI systems?

Causal learning in Hyperon’s PLN framework means teaching a system to discover and represent –if-then– relationships in the world. For example, a mobile robot might learn that ”pressing a lever causes a door to open,” while a bioinformatics application might learn that ”gene A influences the expression of gene B.” In both cases, we want the machine to find a simple set of mechanisms (physical or biological processes) that explain why observing event A makes event B much more likely than if A had not occurred. In PLN we represent beliefs as weighted truth values attached to logical links, and we use inference rules to update those beliefs. By combining our theory of weakness (which biases us toward simpler explanations) with probabilistic measures of ”how much A raises the chance of B ,” we can guide the learning process so that the system prefers concise, strong causal hypotheses. In what follows, we first outline this approach formally, then give two concrete examples: one where a robot learns a causal rule about a switch and a light, and another where a system infers causal relations among genes in a gene expression network.

We can formalize this as follows:

1. **Weakness Algebra.** We assume a finite set of candidate mechanisms $M = \{m_1, \dots, m_k\}$. Each mechanism m_i can be represented as a PLN rule or small network of links in AtomSpace. We assign each mechanism a *complexity cost* $\mathbf{w}(m_i) \geq 0$, e.g. equal to the length (in bits) of its symbolic description. For any subset $M' \subseteq M$, the total weakness is

$$\mathbf{w}(M') = \sum_{m \in M'} \mathbf{w}(m).$$

A lower $\mathbf{w}$ means a simpler, -weaker- explanation.

2. **Probabilistic Counterfactual Strength.** Let A and B be two event predicates (e.g. `PressSwitch` and `LightOn`). We define

$$P(B \mid A; M') = \text{estimated probability that } B \text{ occurs given } A, \text{ under mechanisms } M',$$

and similarly

$$P(B \mid \neg A; M').$$

The *counterfactual strength* of A on B under M' is

$$\delta_{M'}(A, B) = P(B \mid A; M') - P(B \mid \neg A; M').$$

In PLN, $P(\cdot)$ is derived from reasoning over weighted links that represent observed event co-occurrences and mechanistic relations in data.

3. **Causal Hypothesis Ordering.** We consider candidate pairs (M', P) , where M' is a set of mechanisms and P is a -’simple– probability estimator

(e.g. a count-based model encoded as a small PLN subnetwork). Define the product pair

$$(\mathbf{w}(M') + \mathbf{w}(P), \delta_{M'}(A, B)).$$

We order these pairs by first comparing δ (larger is better), and breaking ties by comparing $\mathbf{w}$ (smaller is better). Formally,

$$(w_1, \delta_1) \text{ Prec } (w_2, \delta_2) \iff [\delta_1 > \delta_2] \vee [\delta_1 = \delta_2 \wedge w_1 < w_2].$$

The best causal explanation for $A - B$ is the maximal pair under Prec.

4. **Embedding in PLN.** We represent each candidate mechanism m as a small PLN subnetwork: for instance, if m is "PressSwitch causes CurFlow," we represent that as an **ImplicationLink** or a set of correlated **Inheritance** and **Evaluation** links with initial truth values derived from data. The probability model P is encoded via PLN's statistical aggregation functions (e.g. counting how often links co-occur in a training corpus). We then apply PLN's forward or backward chaining rules to compute $P(B \mid A; M')$ and $P(B \mid \neg A; M')$ for each M' . Finally, we compute $\delta_{M'}(A, B)$ and compare weakness $\mathbf{w}(M') + \mathbf{w}(P)$ to select the top hypothesis.
5. **Causality Theorem.** Let $\mathcal{H}$ be the set of all candidate causal hypotheses (M', P) . Then the hypothesis that maximizes $[\delta_{M'}(A, B), -(\mathbf{w}(M') + \mathbf{w}(P))]$ under lexicographic ordering yields the simplest, strongest causal rule. More precisely:

Theorem 19 (PLN Causal Hypothesis Selection). *Let A, B be event predicates, $\mathcal{H} = \{(M', P)\}$ candidate hypotheses, and define*

$$\phi(M', P) = (\delta_{M'}(A, B), -(\mathbf{w}(M') + \mathbf{w}(P))).$$

Under the lexicographic order that compares δ first (higher is better) then $-\mathbf{w}$ (higher is better), the maximal $\phi(M^, P^*)$ corresponds to the causal explanation that yields the greatest gain in B 's probability from A while using the simplest mechanisms and probability model.*

Sketch. If there were another pair (M', P') with strictly larger δ , it would precede (M^*, P^*) by definition, so (M^*, P^*) cannot be maximal unless its δ is maximal. If two hypotheses share the same δ , then the pair with smaller $\mathbf{w}$ ranks higher. Hence the chosen (M^*, P^*) is the unique maximizer. $\square$

To give a practical sense of these concepts we now explore some toy "thought experiment" style examples.

Example 1: Robot Learning Causal Rule for a Switch and a Light Scenario. A domestic robot observes boolean sensor events $\neg\text{PressSwitch}(t)$ – and $\neg\text{LightOn}(t')$ –, where t' occurs shortly after t if the switch was pressed. The robot’s goal is to learn a causal rule $\neg\text{PressSwitch}$ causes LightOn – using a minimal set of mechanisms.

1. The robot records ground observations in AtomSpace as:

```
(eval (pred "PressSwitch") (list (time_t))) and (eval (pred "LightOn") (list (time_t')))
```

2. Candidate mechanisms:

- m_1 : *Mechanical linkage* rule—When the switch is pressed, it closes a circuit, causing current flow.— Encoded as a small PLN subnetwork:

```
(ImplicationLink
  (and
    (eval (pred "PressSwitch") (list $t))
    (eval (pred "CircuitClosed") (list $t)) )
  (eval (pred "CurrFlow") (list $t)) )
```

plus

```
(ImplicationLink
  (eval (pred "CurrFlow") (list $t))
  (eval (pred "LightOn") (list $t')) )
```

- m_2 : *Direct light logic* rule—Pressing the switch directly toggles a relay that turns on the light.— A simpler single-link rule:

```
(ImplicationLink
  (eval (pred "PressSwitch") (list $t))
  (eval (pred "LightOn") (list $t')) )
```

3. Complexity costs (weakness):

$$\mathbf{w}(m_1) = 3, \quad \mathbf{w}(m_2) = 1,$$

reflecting that m_2 is simpler (one rule) than m_1 (two chained rules).

4. Probability model P is ‘simple’ if it counts how often $\neg\text{PressSwitch}$ – and $\neg\text{LightOn}$ – co-occur in a small time window. Let $\mathbf{w}(P) = 1$.

5. Compute empirical probabilities via PLN match counts:

$$P(\text{LightOn} \mid \text{PressSwitch}; m_2) = 0.9, \quad P(\text{LightOn} \mid \neg\text{PressSwitch}; m_2) = 0.1,$$

so

$$\delta_{m_2} = 0.8, \quad \mathbf{w}(m_2, P) = 1 + 1 = 2.$$

For m_1 , suppose the robot's data gives

$$P(\text{LightOn} \mid \text{PressSwitch}; m_1) = 0.85, \quad P(\text{LightOn} \mid \neg\text{PressSwitch}; m_1) = 0.05,$$

so

$$\delta_{m_1} = 0.80, \quad \mathbf{w}(m_1, P) = 3 + 1 = 4.$$

6. Compare the pairs:

$$(\mathbf{w}, \delta)_{m_2} = (2, 0.80), \quad (\mathbf{w}, \delta)_{m_1} = (4, 0.80).$$

Since both give the same $\delta = 0.80$, we pick the one with smaller $\mathbf{w}$, which is m_2 . Thus the robot learns $\neg\text{PressSwitch} \rightarrow \text{LightOn}$ using the simple relay rule.

Example 2: Learning Causal Relations in a Gene Expression Network Scenario. We have time-series gene expression levels for genes A, B, C , etc. We want to learn that Gene A upregulates Gene B under a simple regulatory mechanism.

1. Observations recorded in AtomSpace:

```
(EvalLink
  (pred "HighExpr")
  (list (concept "GeneA") (Time_t)) )
(EvalLink
  (pred "HighExpr")
  (list (concept "GeneB") (Time_{t+dt})) )
```

whenever A is highly expressed at time t and B is highly expressed at time $t + dt$.

2. Candidate mechanisms:

- m_1 : *Direct activation*—Gene A produces a transcription factor that binds B 's promoter.— Represented by a PLN subnetwork:

```
(ImplicationLink
  (EvalLink
    (pred "HighExpr")
    (list (concept "GeneA") $t))
  (EvalLink
    (pred "Activates")
    (list (concept "GeneA") (concept "GeneB")))) )
```

plus

```

(ImplicationLink
  (EvalLink
    (pred "Activates")
    (list (concept "GeneA") (concept "GeneB"))))
(EvalLink
  (pred "HighExpr")
  (list (concept "GeneB") $t+dt)) )

```

- m_2 : *Indirect co-expression*—Gene A and Gene B share a common upstream regulator gene C .— This involves two ImplicationLinks via gene C :

```

(ImplicationLink
  (EvalLink
    (pred "HighExpr")
    (list (concept "GeneA") $t)))
(EvalLink
  (pred "HighExpr")
  (list (concept "GeneC") $t+dt1)) )
(ImplicationLink
  (EvalLink
    (pred "HighExpr")
    (list (concept "GeneC") $t+dt1)))
(EvalLink
  (pred "HighExpr")
  (list (concept "GeneB") $t+dt2)) )

```

3. Complexity costs:

$$\mathbf{w}(m_1) = 2, \quad \mathbf{w}(m_2) = 3,$$

because m_2 has two chained implications rather than one direct link.

4. Probability model P counts how often $-A$ high at $t-$ is followed by $-B$ high at $t + dt$.— Let $\mathbf{w}(P) = 1$.

5. Empirical estimates from data (via PLN backward chaining):

$$P(\text{HighExpr}(B) \mid \text{HighExpr}(A); m_1) = 0.70, \quad P(\text{HighExpr}(B) \mid \neg \text{HighExpr}(A); m_1) = 0.30,$$

so

$$\delta_{m_1} = 0.40, \quad \mathbf{w}(m_1, P) = 2 + 1 = 3.$$

For m_2 , suppose

$$P(\text{HighExpr}(B) \mid \text{HighExpr}(A); m_2) = 0.65, \quad P(\text{HighExpr}(B) \mid \neg \text{HighExpr}(A); m_2) = 0.25,$$

so

$$\delta_{m_2} = 0.40, \quad \mathbf{w}(m_2, P) = 3 + 1 = 4.$$

6. Compare:

$$(w, \delta)_{m_1} = (3, 0.40), \quad (w, \delta)_{m_2} = (4, 0.40).$$

Both mechanisms give the same counterfactual strength (0.40), but m_1 is simpler (weakness 3 versus 4). Hence the system concludes that "Gene A upregulates Gene B" via the direct activation mechanism m_1 .

14.4 Disentangled Representations as a Special Case of Clear Representations

We now apply this notion of causality to help move toward resolution of some fundamental issues in learning theory, related to the properties of representations within neural nets and other learning networks, and how these properties connect with the generalization performance of these networks.

In modern deep learning, we often hear about *entangled* versus *disentangled* representations inside neural networks. Informally, a representation is *disentangled* if each internal neuron or latent dimension cleanly corresponds to a single underlying factor of variation in the data—for example, one neuron tracks object rotation, another tracks color, and so on. In contrast, an *entangled* representation mixes multiple factors into each neuron, so it is hard to say *why* the network produced a certain output O given input history I by looking at its internal state. A beautiful elaboration of these ideas has been given by Ken Stanley in his paper on "Fractured Entangled Representations" [KCLS25].

We suggest it may be helpful to introduce a more general notion of a *clear representation* for any discrete dynamical system: when the system outputs O after receiving inputs I , there ought to be a concise, easy-to-determine causal explanation of why O resulted from I . Under this definition, the standard idea of disentangled representations in neural networks turns out to be just one example. If each latent dimension corresponds to a unique causal factor, then explaining the output in terms of that dimension is straightforward. If instead factors are jumbled, we no longer have a concise causal account. Thus disentangled networks are precisely those whose internal representations are *clear* in our broader sense.

The interpretation of causality in terms of weakness has obvious implications here.

Formal Treatment Let us begin by defining a general discrete dynamical system, then state what it means to have a *clear representation*, and finally show how disentangled neural representations fit this mold.

1. **Discrete Dynamical System.** Consider a system with:

- A finite set of *internal state variables* $S = \{s_1, \dots, s_n\}$.
- A set of *input histories*

$$I = \{i_0, i_1, \dots, i_{T-1}\},$$

where each i_t is taken from a finite input alphabet $\mathcal{I}$.

- A deterministic *update rule*

$$F : \mathcal{I} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

so that if at time step t the inputs up to time t have led to internal state $\mathbf{s}(t) \in \mathbb{R}^n$, then upon receiving input i_t the system updates to

$$\mathbf{s}(t+1) = F(i_t, \mathbf{s}(t)).$$

- An *output function*

$$G : \mathbb{R}^n \longrightarrow \mathcal{O},$$

where $\mathcal{O}$ is a finite set of possible outputs. Thus, after processing the input sequence I up to time T , the system's internal state is

$$\mathbf{s}(T) = F(i_{T-1}, F(\dots, F(i_0, \mathbf{s}(0)) \dots)),$$

and the final output is

$$O = G(\mathbf{s}(T)).$$

2. **Clear Representation.** A representation (S, F, G) is called *clear* if, for every possible input history I and resulting output $O = G(\mathbf{s}(T))$, there exists a *simple causal explanation* of why O follows from I via intermediate state $\mathbf{s}(T)$. Formally, we require:

- A *causal explanation* is a logical construct in a suitable language (e.g. CMMC will do) that shows how specific state components $s_{k_1}, s_{k_2}, \dots$ of $\mathbf{s}(T)$ determine O .
- *Simple* can be interpreted to mean the explanation's description length is bounded by a small constant $\beta \dots$ or better yet that its weakness is bounded from above.

Equivalently, what this means is that there exists a small set of indices $K \subseteq \{1, \dots, n\}$ such that knowing $\{s_k(T) : k \in K\}$ suffices to predict O , and the rules describing how s_k values cause O are simple.

Roughly speaking: (S, F, G) is clear if O depends only on a small, easily identified subset of the internal dimensions, and the mapping from those dimensions to O is itself simple.

3. **Neural Network Representations.** A feedforward neural network or a recurrent network can be viewed as a special case of the above system:

- The *internal state* $\mathbf{s}(t) \in \mathbb{R}^n$ is the vector of activations in a selected layer at time t (for recurrent networks) or the final layer (for feedforward).
- The update rule F is given by matrix multiplications and activation functions (e.g. tanh or ReLU) that process each input sequentially.

- The output function G might be a softmax or argmax layer that produces a discrete label or decision.

4. **Entangled vs. Disentangled.** In the deep learning literature, a representation is said to be *disentangled* if each latent coordinate s_k abstracts a single generative factor or concept. Formally, if the true data-generating process has latent factors $Z = \{z_1, \dots, z_m\}$, a disentangled network learns a representation function

$$R : \mathcal{I}^T \longrightarrow \mathbb{R}^n$$

such that each $s_k = R_k(I)$ depends primarily on one z_j , and is invariant to others. By contrast, an *entangled* representation has each s_k mixing multiple z_j s.

5. **Disentangled as Clear.** We now argue that a disentangled representation is a special case of a clear representation:

- (a) Suppose there are m true causal factors $z_1, \dots, z_m$ that determine the final output O . A disentangled network ensures there exist indices $k_1, \dots, k_m$ such that

$$s_{k_j}(T) = f_j(z_j(I)), \quad j = 1, \dots, m,$$

where each f_j is a simple mapping from factor z_j to activation s_{k_j} , and all other coordinates carry no additional information.

- (b) Because O is a deterministic function of the latent factors $z_1, \dots, z_m$, there is a simple mapping

$$H : \mathbb{R}^m \longrightarrow \mathcal{O},$$

so that

$$O = H(z_1(I), \dots, z_m(I)).$$

Composing these two steps, we write

$$O = H(f_1^{-1}(s_{k_1}(T)), \dots, f_m^{-1}(s_{k_m}(T))),$$

or equivalently

$$O = G_{\text{simple}}(s_{k_1}(T), \dots, s_{k_m}(T)),$$

where G_{simple} is the composed simple map.

- (c) In this scenario, the concise causal explanation of O given input I is: –Because $s_{k_1}(T) = f_1(z_1(I))$, ..., $s_{k_m}(T) = f_m(z_m(I))$, and $O = H(z_1(I), \dots, z_m(I))$, hence $O = G_{\text{simple}}(s_{k_1}(T), \dots, s_{k_m}(T))$.– The total description length (weakness) of this explanation is small if each f_j , H , and the inverse functions f_j^{-1} are simple.

- (d) Thus the network’s representation is *clear*: only the m coordinates $\{s_{k_j}\}$ matter for predicting O , and the mapping from $(s_{k_1}, \dots, s_{k_m})$ to O is simple. Concretely, the set $K = \{k_1, \dots, k_m\}$ is small, and the explanation involves only those m neurons plus a simple combining rule G_{simple} .

Therefore, any disentangled representation in the usual sense satisfies the definition of a clear representation: there exists a small subset of latent dimensions that suffice to explain the output concisely.

6. **Entangled as Unclear.** Conversely, suppose the network’s internal state $\mathbf{s}(T)$ is entangled—each coordinate $s_k(T)$ depends on multiple generative factors in a convoluted way. Then to explain O from $\mathbf{s}(T)$, one typically needs to consult a large subset of coordinates and perform a complex nonlinear combination. Formally, no small subset $K \subseteq \{1, \dots, n\}$ of dimensions suffices to predict O with a simple map. Hence, according to our definition, the representation is not clear.

Summary We have shown that the conventional concept of *disentangled representations*—where each latent dimension captures exactly one underlying factor—fits precisely into our broader notion of a *clear representation* in a discrete dynamical system. In a disentangled model, there is a concise causal explanation of the output O in terms of a small set of latent coordinates and simple combining rules. When latent factors are entangled, no such concise explanation exists, and the representation is *unclear* according to our theory of weakness for learning and reasoning. Thus, disentanglement emerges as a special case of clarity. This connection underscores how the general theory of weakness can unify and explain why disentangled representations are beneficial for causal interpretability and efficient reasoning.

15 Weakness-Based Causal Decision Theory

We now elaborate our weakness-based understanding of causality into a full-fledged weakness-based causal decision theory, and flesh this out via considering various classic problem cases for decision theory. These “problem cases” are toy examples of mainly academic interest, however the topic is a highly practical one and critical to the effective operation of goal systems for AGI systems and modern AI applications (a topic to be taken up more fully in Chapter 16 when we look at the direct application of weakness to motivational systems for AIs).

In ordinary decision-theory one asks “which action will best achieve my goals”? In causal decision theory one refines that to “which action causes the best outcomes? In [Goe20c] it is proposed to flesh this out by imposing an Occam-style simplicity bias on hypotheses about how actions bring about outcomes. Here we explore this direction further, but replacing “simplicity” by the more general notion of *weakness*: the tendency to make as few extra assumptions as possible.

One fairly simple way to insert weakness into decision theory would be to define a *weak causal decision theory* (WCDT) which selects, among all candidate causal explanations that guarantee a high degree of desirable outcomes, the one that is *weakest*—that is, the most broadly applicable hypothesis that still delivers the desired effect. This generalizes and *elegantizes* the earlier “simplificational” approaches by placing them in the categorical framework of weakness.

However it seems valuable to go further than this, and explore ways of integrating WCDT with Wei Dai’s Updateless Decision Theory (UDT), yielding a hybrid approach that appears to have some preferential properties, especially in situations involving collective decision making among correlated agents.

15.1 Seven Decision Theories: Verbal and Formal Overview

We briefly summarize seven decision rules, each with a verbal sketch and its core formal criterion. Let A denote an action (or policy) and U the resulting utility; let $\text{do}(A)$ denote the Pearl-style intervention [Pea00].

15.1.1 Evidential Decision Theory (EDT)

Choose the action whose observation correlates with the highest utility, treating $P(U \mid A)$ as if it were causal. [Ahm14]

$$A_{\text{RmEDT}}^* = \arg \max_A \mathbb{E}[U \mid A].$$

15.1.2 Timeless Decision Theory (TDT)

Reason by logical counterfactuals among all copies of your decision code, selecting the action that in proof yields maximal utility. [Yud10]

$$A_{\text{RmTDT}}^* = \arg \max_A \{U : \text{“If all copies choose } A \text{ then } U\text{” is provable}\}.$$

15.1.3 Updateless Decision Theory (UDT)

[Dai09] Commit in advance to a policy before seeing any observations, optimizing a fixed prior over complete policies.

$$p_{\text{UDT}}^* = \arg \max_{p: \text{Obs} \rightarrow \text{Act}} \sum_U \Pr_{\text{prior}}(p) \Pr(U) U(U, p(\cdot)), \quad A = p_{\text{UDT}}^*(x).$$

15.1.4 Weak Causal Decision Theory (WCDT)

Within a known causal model, keep only causal submodels h with positive interventional gain, then pick the *weakest* (most general) one.

$$\Delta(h) = \mathbb{E}[U \mid \text{do}(A)] - \mathbb{E}[U \mid \text{do}(\neg A)], \quad h^* = \arg \max_{h: \Delta(h) \geq 0} w(h), \quad A = h^* \text{-prescribed.}$$

15.1.5 Weak Updateless Decision Theory (WUDT)

Fix a prior over causal submodels weighted by weakness, then at decision time filter by interventional gain and choose the weakest sufficient submodel.

$$\pi(h) \propto w(h)^\alpha, \quad h^* = \arg \max_{h: \Delta(h) \geq 0} w(h).$$

15.1.6 Weak Causal Updateless Decision Theory (WCUDT)

Use an updateless prior over submodels weighted by both weakness and causal coherence, then filter by $\Delta \geq 0$ and pick the submodel with highest prior weight.

$$\pi(h) \propto w(h)^\alpha c(h)^\beta, \quad h^* = \arg \max_{h: \Delta(h) \geq 0} \pi(h).$$

15.1.7 Weakness Causal Filtering UDT (WCFUDT)

Precommit to a prior over submodels without weakness ranking, then at decision time keep any h with $\Delta \geq 0$ and choose the one with highest prior.

$$\pi(h) \text{ fixed updatelessly, } H_0 = \{h : \Delta(h) \geq 0\}, \quad h^* = \arg \max_{h \in H_0} \pi(h).$$

15.2 Decision-Theory Comparison: Smoking-Lesion Example

Consider the classic “smoking-lesion” scenario. There is a binary health variable

$$L = \{\text{has lesion}\}$$

and an action variable

$$S = \{\text{smoke}, \neg\text{smoke}\}.$$

A genetic lesion L causes both a preference for smoking and a high risk of lung cancer, but smoking itself has no causal effect on cancer. The payoffs are:

$$U(\neg\text{smoke}, \neg L) = +1, \quad U(\neg\text{smoke}, L) = -10,$$

$$U(\text{smoke}, \neg L) = +2, \quad U(\text{smoke}, L) = -10.$$

Thus smoking yields more utility when there is no lesion, and has no additional harm when there is.

15.2.1 Evidential Decision Theory (EDT)

EDT selects the action that maximizes

$$\mathbb{E}[U \mid S = a] = \sum_{\ell \in \{0,1\}} \Pr(L = \ell \mid S = a) U(a, \ell).$$

Because smoking correlates with the lesion, $\Pr(L \mid S = \text{smoke})$ is higher than $\Pr(L \mid \neg \text{smoke})$, so EDT finds

$$\mathbb{E}[U \mid \text{smoke}] < \mathbb{E}[U \mid \neg \text{smoke}],$$

and therefore (incorrectly) *recommends against* smoking.

15.2.2 Timeless and Updateless Decision Theories (TDT, UDT)

Both TDT and UDT avoid this evidential pitfall by reasoning in terms of *interventions* or *policies*, rather than mere conditioning. In TDT one asks, “If all code-copies with my inputs choose to smoke, what utility results?” and finds that smoking has no causal link to cancer. UDT fixes a prior over complete policies (smoke vs. abstain) and chooses the one whose *interventional* expected utility is higher. Both conclude

$$\mathbb{E}[U \mid \text{do}(S = \text{smoke})] > \mathbb{E}[U \mid \text{do}(S = \neg \text{smoke})],$$

and thus both *recommend smoking*.

15.2.3 Weak Causal Decision Theory (WCDT)

WCDT formalizes the above by constructing the causal graph

$$L - S, \quad L - \text{Cancer}, \quad S \not\vdash \text{Cancer},$$

then filtering for submodels whose $\Delta(h) = \mathbb{E}[U \mid \text{do}(S = \text{smoke})] - \mathbb{E}[U \mid \text{do}(S = \neg \text{smoke})]$ exceeds zero. The minimal such submodel is the one including exactly the edge $L - S$ (the lesion causes smoking) and $L - \text{Cancer}$, omitting any spurious “smoking causes cancer” link. Because this submodel is both strong enough to yield $\Delta > 0$ and *weakest* (it makes no extra assumptions), WCDT likewise *recommends smoking*.

15.2.4 Weak Updateless Causal Decision Theory (WCU DT)

WUCDT first assigns each causal submodel h an updateless prior $\pi(h) \propto w(h)^\alpha c(h)^\beta$, where $w(h)$ measures generality and $c(h)$ measures causal coherence, then upon “intervening” re-weights each $\Delta(h)$ by $\pi(h)$ and picks the weakest sufficient model. In the smoking-lesion case the only models passing $\Delta > 0$ are those that omit the smoking-causes-cancer edge, all of which share the same maximal weakness. Hence WUCDT’s precommitment and posterior-weighted strength selection still *recommends smoking*.

In this case, WCFUDT will give a similar analysis.

15.2.5 Summary

In this simple example all *causal* or *updateless* theories – TDT, UDT, WCDT, WCU DT, WCFUDT – agree on smoking, while EDT falls prey to a spurious

correlation. The various refinements differ only in *when* and *how* they select and combine causal hypotheses, but all converge on the correct decision once genuine causal links are taken into account.

15.3 Newcomb’s Paradox

For a more challenging decision theory problem, let us consider Newcomb’s paradox, which presents a game between a Predictor and a Player. There are two boxes:

- Box 1 contains either \$1 000 000 (if the Predictor anticipated you will take only Box 1) or \$0 (otherwise).
- Box 2 contains \$1 000 for sure.

You may take either both boxes (Two-box) or only Box 1 (One-box). The Predictor, who is highly reliable, has already placed the money before you choose. Should you one-box or two-box?

15.3.1 Decision-Theory Responses

EDT: Evidential Decision Theory conditions on your choice. Since $\Pr(\text{Million} \mid \text{One}) \approx 1$ and $\Pr(\text{Million} \mid \text{Two}) \approx 0$, EDT predicts a higher expected payoff by one-boxing.

TDT: Timeless Decision Theory reasons by logical counterfactuals: it asks “if all agents with my code one-box, I get \$1 000 000; if they two-box, I get \$1 000.” Hence TDT one-boxes.

UDT: Updateless Decision Theory fixes a universal prior over entire policies and selects the policy that, if adopted by all its copies, maximizes expected utility. That policy is “One-box,” so UDT one-boxes.

WCDT: Weak Causal Decision Theory builds the causal graph Action–Predictor–Content – Payoff, filters for submodels with positive Δ , and chooses the *weakest* sufficient submodel. The minimal causal explanation yields one-boxing.

WUDT: Weak Updateless Decision Theory assigns each causal submodel a prior proportional to its weakness, then on seeing no new observation picks among those with positive causal gap the weakest. It thus also one-boxes.

WCUdT: The hybrid “Weak Causal Updateless” theory uses an updateless prior over submodels weighted by weakness and causal coherence, then at decision time filters by interventional gain and selects the weakest. The result once again is to one-box.

WCFUDT: Both the one and two box submodels have equal generality and equal causal coherence under the predictor-box action graph, so their prior weights π tie. The one-box submodel, however, yields interventional gain. WCFUDT discards two-box, and thus *one-boxes* as well.

15.4 Superrationality across Decision Theories

Superrationality, as introduced by Hofstadter [Hof85], is the idea that if all agents in a symmetric decision situation reason the same way, then each agent should choose the action that yields the best collective outcome under the assumption that every agent will make the same choice. In other words, rather than treating other agents' moves as independent, a superrational agent exploits the logical symmetry between identical decision-makers to secure the best joint result.

Handling by Different Theories

EDT Evidential Decision Theory conditions on one's own action and may *cooperate* under superrational symmetry because it sees a statistical correlation: if "I cooperate" then "they cooperate," so EDT often cooperates. But it does so for the wrong reason – it treats correlation as causation and can be misled by spurious links.

TDT Timeless Decision Theory explicitly reasons about logical copies: it asks "if all code-copies of me make the same choice, what utility follows?" By construction it superrationally cooperates, since it treats identical reasoning agents as locked into the same decision.

UDT Updateless Decision Theory fixes a prior over full policies and optimizes that prior without updating. Because it evaluates a policy's payoff under the assumption that every instance of it is used by all agents, UDT naturally implements superrational cooperation.

WCDT Weak Causal Decision Theory insists on genuine causal links. A pure superrational link (logical symmetry) is not a causal channel, so WCDT *ignores* it and will not cooperate unless there is a real intervention path from one agent's choice to another's.

WUDT Weak Updateless Decision Theory precommits to a prior over causal submodels based on weakness, but it still requires causal efficacy to yield payoff. In the absence of any real causal tie, WUDT does not exploit superrational symmetry, though it would if an exotic causal mechanism were encoded in its prior.

WCUDT The hybrid Weak Causal Updateless theory blends an updateless prior over submodels with causal-strength filtering. It will only superrationally cooperate if and only if its fixed prior assigns weight to a genuine causal channel linking agents' choices; pure logical symmetry alone is insufficient.

WCFUDT Superrationality appeals to logical symmetry among identical agents, but WCFUDT requires a genuine causal link. It will only "cooperate" or vote in step with others if its prior $\pi(h)$ assigns weight to a submodel

h that encodes a real intervention channel. Purely logical or evidential correlations, without causal efficacy, fail the filter and are ignored. Thus WCFUDT also captures the spirit of superrational coordination only when a true causal mechanism underlies the symmetry.

15.5 Voting and Superrationality

The voting problem arises because in a large electorate a single vote has almost zero chance of changing the result, so a naive analysis would advise abstention. Yet we know that high turnout can be crucial to good collective outcomes. Superrationality observes that if all voters reason identically, then your choice logically correlates with theirs – so your single vote effectively carries more weight. We now summarize how seven decision theories respond to voting in both uncorrelated and superrational (correlated) settings.

EDT (Evidential) Under independent voting, $P(\text{win} \mid \text{vote}) \approx P(\text{win} \mid \text{abstain})$, so EDT recommends abstention. In a superrational model where your vote raises the probability that others vote, conditioning gives $P(\text{win} \mid \text{vote}) > P(\text{win} \mid \text{abstain})$, so EDT then prescribes voting.

TDT (Timeless) TDT asks “if all logically identical agents vote, what utility results”? With no assumed causal link it finds no gain and abstains. If it accepts a logical symmetry among voters, it treats their choices as locked together and therefore votes.

UDT (Updateless) UDT precommits to a policy (vote or abstain) before observing turnout statistics. With no built-in expectation of correlation it typically chooses abstention. If its fixed prior includes superrational correlations among voters, it instead precommits to voting.

WCDT (Weak Causal) WCDT builds a causal graph of vote-outcome. With independent voting that graph yields negligible payoff gap, so it abstains. If you introduce a genuine causal link vote-peers-outcome, WCDT finds a positive causal effect and votes.

WUDT (Weak Updateless) WUDT fixes an updateless prior over causal submodels weighted by weakness, then on election day tests each submodel’s causal gap. Without correlation it abstains; with a causal social-influence link in its prior it votes.

WCUDT (Weak Causal Updateless) This hybrid uses an updateless, weakness-causality prior over submodels and then filters by day-of interventional gain. It abstains under pure independence and votes only if its fixed prior assigns weight to a real correlation channel, thus combining superrational coordination with genuine causal intervention.

WCFUDT (Weak Causal Filtering UDT) WCFUDT precommits to an updateless prior over all vote-outcome causal submodels (without explicit

weakness ranking), then on election day discards any submodel whose interventional payoff gap is negative and votes if *any* remaining submodel prescribes voting. Consequently, in the independent-voter case it still votes (due to the contrived tie-flip model), and in the correlated case it likewise votes, but it cannot distinguish genuine social influence from spurious channels.

15.6 Voting Variations Where WCUDT Excels

We now describe two realistic voting scenarios in which the full hybrid *Weakness Causal Updateless Decision Theory* (WCUDT) outperforms its component methods (WCDT, WUDT) and the simpler hybrid WCFUDT.

15.6.1 Uncertain Correlation Strength

Scenario: Before election day you know only a rough prior on your social-influence parameter x ; you will only learn its true value when polling closes.

WCDT: Waits until you observe x and then applies the causal pivot bound. If the day-of estimate is slightly below threshold, you abstain – even though the true value (noisily estimated) might have been above threshold.

WUDT: Precommits to a policy (“vote” or “abstain”) based on a weakness-and-causality prior over submodels, but cannot adapt when you learn that x is higher than expected.

WCFUDT: Uses an updateless prior with causal weighting but no explicit weakness ranking. As long as any contrived submodel (e.g. tie-flip) has nonzero prior, it will vote regardless of the actual x estimate, leading to potential overcommitment.

WCUDT:

- Retains an updateless prior $\pi(h) \propto w(h)^\alpha c(h)^\beta$ over causal submodels.
- On election day, for each h computes the true $\Delta(h; x)$.
- Discards all h with $\Delta < 0$, then picks among the rest the one with maximal $\pi(h)$.

This allows WCUDT to *hedge*: if x is ultimately high, it will vote; if low, it abstains; and if marginal, it falls back on the most plausible weak submodel, avoiding both day-of overreaction and rigid precommitment.

15.6.2 Spurious Tie-Flip versus Genuine Influence

Scenario: Besides a small genuine social channel, there exists a contrived “tie-flip” hypothesis that yields a tiny positive Δ , yet is far less general.

Imagine a voter who insists that “my single vote will flip the election” by postulating a perfectly balanced scenario: without their vote, the margin is exactly zero, and then the tie-breaking procedure happens to favor their preferred candidate. Concretely, this hypothesis assumes

- The pre-vote margin $M = 0$ exactly, and
- The tie-breaker (whether a coin toss or other rule) resolves in favor of the voter’s candidate.

Such a model yields a small but nonzero payoff gap, since it guarantees that casting the vote turns a loss into a win.

However, it is wildly contrived:

1. It requires an implausible, exact tie before the vote, an event of vanishing probability in a large electorate.
2. It further assumes the tie-breaker always goes the voter’s way, excluding nearly all other possible outcomes.

As a causal hypothesis, it “works” in the sense of producing $\Delta > 0$, but it is extremely *strong* – it rules out almost every world except those two highly specific conditions. Hence its *weakness* $w(h)$ is nearly zero. By contrast, a genuine social-influence model merely assumes your vote nudges a few peers to the polls – a far more plausible, far less restrictive hypothesis –and so has much higher weakness, making it the preferred choice in Weak Causal Decision Theory.

In the presence of this sort of hypothesis

WCDT: Filters by $\Delta > 0$, then selects the *weakest* submodel; here it correctly picks the genuine social-influence model since it is more general than the tie-flip model.

WUDT: Precommits under a pure weakness+causality prior; if its prior π underestimates the social-influence channel relative to tie-flip, it may lock into the wrong model.

WCFUDT: Uses an unranked updateless prior and simply picks any h with $\Delta > 0$, but then chooses the one with highest prior weight. If its prior overweights the contrived tie-flip channel, it will erroneously recommend voting purely to attempt a tie-flip.

WCUDT: Combines the strengths of both:

1. Its updateless prior favors the genuine social model (higher $w(h)$, higher $c(h)$).
2. It filters by $\Delta > 0$, eliminating both the trivial model and any submodel that lacks genuine causal effect.
3. It then selects the remaining model with maximal $\pi(h)$, which is the social-influence channel.

Thus WCUDT wins in robustness: it never picks the contrived tie-flip, but still adapts if the social influence strength changes.

15.6.3 Why WCUDT Wins

By maintaining an updateless, Occam-weakness+causality prior while still filtering on do-based payoff gaps, WCUDT achieves both:

- **Adaptivity** to new evidence (x , dynamic costs) as in WCDT,
- **Rigorous precommitment** and modular coordination as in WUDT,
- **Dirigible plausibility ranking** that avoids pathological submodels unlike WCFUDT.

These features make it uniquely well-suited to real-world voting problems where correlation strength is uncertain and spurious causal hypotheses abound.

16 Weakness and Motivational Systems

In the context of real cognitive systems, decision-making occurs in the broader context of a more complex dynamical system of goals and motivations and emotions. With this in mind, we next attempt to understand how *weakness-driven learning* (Occam-style preference for simple hypotheses) interacts with *motivational structure* in a cognitive system. We approach this within the framework of MetaMo [GL25b] [GL25a], a formal approach to motivation that ties together goals, motivations, and learning processes.

In the MetaMo framework, one is inclined to look for connections between weakness, motivation and system structure/dynamics such as:

- If a large system is built from smaller subsystems, each of which finds its *simplest* (weakest) way to achieve its own goals, then one would like those subsystem-level hypotheses to “glue together– into a *globally* simplest hypothesis for the entire system.
- If subsystem A is modeling subsystem B , then the best (weakest) assumptions A makes about B should carry over to the real B ’so that A ’s internal model and B ’s actual behavior are aligned in terms of minimal complexity.

- When you have to decide on an action (“what to do”) and also appraise the situation (“what is the situation–), you want to do both in a way that remains *consistent* with overall simplicity – so that the simplest hypothesis about the environment and the simplest decision rule commute. In everyday terms, you can separate understanding the world and choosing what to do without losing optimality, as long as both steps respect the same simplicity bias.

In this section we sketch, first informally and then more formally, how these intuitions follow from combining a *categorical theory of motivations* (the MetaMo structure) with a *categorical theory of weakness* (an Occam’s style ordering of hypotheses).

16.1 Formal Overview of MetaMo & Weakness

We present here a simplified variant of MetaMo capturing the essence of these compositionality desiderata, incorporating as we go the simplest connections between MetaMo and weakness theory. We assume familiarity with basic category theory (categories, functors, monoidal products) but define each mathematical notion carefully.

1. Categories of Systems and Hypotheses.

- Let $\mathcal{C}$ be a *category of cognitive systems*, whose objects are ‘systems- (or subsystems) and whose morphisms (arrows) represent *models* or *simulators* from one system to another. Concretely, an arrow $f : A \rightarrow B$ indicates that system A internally *models* or *understands* system B via some abstraction.
- Let $\mathcal{H}$ be a *category of hypotheses* (or belief’s states) with a partial order $\leq_w$ capturing *weakness* (simplicity). More precisely, objects of $\mathcal{H}$ are candidate hypotheses (descriptions) about how a system operates, and $h_1 \leq_w h_2$ means h_1 is at least as simple (weak) as h_2 .–

2. Monoidal Structure (MetaMo Composition).

- The category $\mathcal{C}$ is equipped with a *monoidal product* $\otimes$, modeling *parallel composition* of subsystems. If A, B are subsystems, then $A \otimes B$ is the joint system that runs A and B in parallel (interacting or independent, depending on the design). There is a unit object I in $\mathcal{C}$ representing the -empty- or trivial system.
- Correspondingly, $\mathcal{H}$ has a *monoidal operation* $\star$ (often taken to be some notion of hypothesis concatenation or aggregation) with unit object e representing the -empty hypothesis- (no assumptions).
- We assume a *monoidal functor*

$$\Phi : (\mathcal{C}, \otimes, I) \longrightarrow (\mathcal{H}, \star, e)$$

that assigns to each system $X \in \text{Ob}(\mathcal{C})$ a set of candidate hypotheses $\Phi(X) \in \text{Ob}(\mathcal{H})$, and to each morphism $f : A \rightarrow B$ a hypothesis-translation arrow

$$\Phi(f) : \Phi(A) \longrightarrow \Phi(B),$$

preserving monoidal structure:

$$\Phi(A \otimes B) = \Phi(A) \star \Phi(B), \quad \Phi(I) = e,$$

and

$$\Phi(f \otimes g) = \Phi(f) \star \Phi(g), \quad \Phi(\text{id}_X) = \text{id}_{\Phi(X)}.$$

3. **Weakest Hypothesis for a System.** For each object $X \in \mathcal{C}$, let

$\text{Opt}(X) =$ “the set of all minimal (weakest) hypotheses in $\Phi(X)$ under $\leq_w$ ”.

We assume that $\text{Opt}(X)$ is nonempty; that is, there exists at least one hypothesis of minimal complexity that explains how X achieves whatever goal or behavior is under consideration.

4. **First Compositionality Principle.** We require that for any two subsystems $A, B \in \mathcal{C}$, the minimal hypotheses for the composite system $A \otimes B$ coincide with the $\star$ -monoidal product- of minimal hypotheses for A and minimal hypotheses for B . Formally:

$$\text{Opt}(A \otimes B) = \{ h_A \star h_B \mid h_A \in \text{Opt}(A), h_B \in \text{Opt}(B) \}.$$

In words: –The simplest way to explain how $A \otimes B$ works is obtained by combining the simplest explanations of A and of B .– This property is exactly the requirement that Φ be *monoidal* and that $\leq_w$ be *compatible with $\star$* .

5. **Second Principle: Modeling of Subsystems.** Suppose subsystem A has an internal model of subsystem B given by a morphism $f : A \rightarrow B$ in $\mathcal{C}$. Then, under Φ , there is an induced arrow

$$\Phi(f) : \Phi(A) \longrightarrow \Phi(B).$$

We demand that this arrow *preserves optimality*: if $h_A \in \text{Opt}(A)$ is a simplest hypothesis about A , then

$$\Phi(f)(h_A) \in \text{Opt}(B),$$

or at least lies in the minimal layer of $\Phi(B)$. Equivalently, the simplest hypotheses that A uses to explain its own behavior should map, via its internal model, to simplest hypotheses about B so that A ’s beliefs about B are themselves optimally simple for B .

6. **Decision vs. Appraisal Commutation.** Finally, consider two distinct aspects of cognition:

- *Appraisal hypotheses*-hypotheses about -what the situation is- (the environment's state).
- *Decision hypotheses*-hypotheses about -what action to take- given the appraisal.

Formally, let X_{app} be the subcategory of -appraisal systems,- and X_{dec} be the subcategory of decision systems, each with corresponding hypothesis categories $\Phi(X_{\text{app}})$ and $\Phi(X_{\text{dec}})$. We require a natural transformation (or commuting diagram) expressing that:

$$\text{Opt}(X_{\text{app}} \otimes X_{\text{dec}}) = \{ h_{\text{app}} \star h_{\text{dec}} \mid h_{\text{app}} \in \text{Opt}(X_{\text{app}}), h_{\text{dec}} \in \text{Opt}(X_{\text{dec}}) \}.$$

In words, -The simplest way to both appraise and decide is obtained by pairing a simplest appraisal hypothesis with a simplest decision hypothesis.- This enforces that -what to believe- and -what to do- commute under the simplicity ordering.

Approximate Compositionality: Continuity Assumptions In practice, exact equality of optimal'sets may be too strict. We next describe *continuity conditions* under which *approximately optimal* hypotheses compose approximately optimally.

1. **Metric on Weakness.** Assume the weakness category $(\mathcal{H}, \leq_w)$ is enriched with a metric $d_w : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$, such that

$$h_1 \leq_w h_2 \implies d_w(h_1, h_2) = \mathbf{w}(h_2) - \mathbf{w}(h_1),$$

where $\mathbf{w}(h)$ is the numerical complexity of hypothesis h . In particular, hypotheses of equal complexity have distance zero.

2. **Approximate Optimality.** For a system X , define $\text{Opt}_\epsilon(X)$ to be the set of hypotheses whose complexity differs from the minimal complexity by at most ϵ :

$$\text{Opt}_\epsilon(X) = \{ h \in \Phi(X) \mid \mathbf{w}(h) \leq \mathbf{w}^*(X) + \epsilon \},$$

where $\mathbf{w}^*(X) = \min\{\mathbf{w}(h) \mid h \in \Phi(X)\}$. Intuitively, these are -nearly minimal- hypotheses.

3. **Lipschitz Condition on Monoidal Functor.** We assume that the monoidal functor Φ is *Lipschitz continuous* with respect to complexity metrics. That is, there exists a constant $K \geq 0$ such that for any two systems A, B and hypotheses $h_A \in \Phi(A)$, $h_B \in \Phi(B)$,

$$\mathbf{w}(h_A \star h_B) \leq \mathbf{w}(h_A) + \mathbf{w}(h_B) + K.$$

Equivalently, adding two hypotheses does not increase complexity by more than a constant overhead K .

4. **Approximate Composition Theorem.** Under the Lipschitz condition, if

$$h_A \in \text{Opt}_{\epsilon_1}(A), \quad h_B \in \text{Opt}_{\epsilon_2}(B),$$

then

$$h_A \star h_B \in \text{Opt}_{\epsilon_1 + \epsilon_2 + K}(A \otimes B).$$

In words: Approximately minimal hypotheses for A and B , when combined, yield an approximately minimal hypothesis for $A \otimes B$, with an approximation error at most $\epsilon_1 + \epsilon_2 + K$.

5. **Approximate Modeling Consistency.** If $f : A \rightarrow B$ is a modeling arrow and

$$h_A \in \text{Opt}_{\epsilon}(A),$$

then

$$\Phi(f)(h_A) \in \text{Opt}_{\epsilon + K'}(B),$$

for some small constant $K' \geq 0$. That is, a nearly-optimal hypothesis for A , when projected by A 's internal model of B , becomes a nearly-optimal hypothesis for B .

6. **Approximate Decision-Appraisal Commute.** Similarly, if $h_{\text{app}} \in \text{Opt}_{\epsilon_1}(X_{\text{app}})$ and $h_{\text{dec}} \in \text{Opt}_{\epsilon_2}(X_{\text{dec}})$, then

$$h_{\text{app}} \star h_{\text{dec}} \in \text{Opt}_{\epsilon_1 + \epsilon_2 + K''}(X_{\text{app}} \otimes X_{\text{dec}}),$$

for some small K'' . Thus really close-to-optimal appraisal plus close-to-optimal decision yields a close-to-optimal combined action.

16.2 Weakness-Based Decision Theory Variants in MetaMo

MetaMo is a categorial theory of motivation that unifies perception, learning and action under a single simplicity (weakness) bias. In MetaMo both hypothesis formation and action selection minimize the same weakness measure, ensuring coherence across all cognitive modules.

MetaMo employs three main weakness-regularized decision procedures as its core action-selection routines:

WCDT (Weakness Causal Decision Theory) Uses an Occam-weakness prior over causal models but updates that prior on day-of observations, yielding high adaptivity to new evidence. Ideal for reactive or perceptual subsystems.

WUDT (Weakness Updateless Decision Theory) Fixes a weakness-weighted prior once and for all (no updating), yielding strong precommitment and internal consistency. Best suited for long-horizon planning where commitment to a policy is essential.

WCUDT (Weakness Causal Updateless Decision Theory) Combines both strengths:

1. An updateless Occam-weakness plus causality prior
2. Day-of filtering on positive causal payoff gaps
3. Selection of the remaining model with highest prior weight

This hybrid adapts when genuine causal influence emerges yet avoids spurious "tie-flip" models, making it well suited for socially coordinated or multi-agent decision goals.

By choosing among these variants at the goal-invocation layer, MetaMo can tailor its motivational dynamics:

- *Exploration vs. Exploitation*: rapid, evidence-driven choices via WCDT.
- *Long-Term Commitment*: high-confidence commitment via WUDT.
- *Social Coordination and Robustness*: balanced updateless coordination via WCUDT.

In this way, MetaMo remains grounded in a single underlying simplicity bias while allowing subsystems to adopt the decision rule best aligned with their motivational needs.

16.3 Nontechnical Summary of Combined MetaMo and Weakness Theory

Putting together the MetaMo principles (which dictate that a system's composition and its internal modeling must respect a monoidal (tensor) structure) with our categorical account of *weakness* (Occam preference for simpler hypotheses), we have obtained these basic takeaways:

- *Optimal Subsystems Compose to Optimal Systems*. If each part of the system learns its simplest way to achieve its goals (e.g. via a WCDT or WUDT policy), then gluing those parts together yields the simplest way for the whole system. In more concrete terms, a robot whose left arm and right arm each learn a minimal-complexity controller under WCDT will, when acting together, produce a joint controller that is likewise minimal, rather than having to relearn a more complex joint policy.
- *Internal Models Preserve Simplicity*. When subsystem A builds an internal hypothesis h_A about B (as in WUDT's fixed prior), that hypothesis is also a minimal (or near-minimal) hypothesis for B itself. This ensures that A 's beliefs about B are not more complex than B 's actual workings. In practice, if an AI module models another module, it does not need to invent overly complicated internal representations.
- *Appraise-Then-Decide Remains Simple*. Deciding what to do after evaluating the environment (for example, by first applying a weakness-based model selection step and then choosing an action under WCDT) does not

add extra complexity beyond the sum of each step. That is, you can first find a simple model of what is going on, then find a simple rule for what to do, and the pair is still simple. Hence a system can safely modularize "understanding" (via updateless priors) and "choosing" (via causal filtering) without losing Occam-optimality.

- *Robustness under Approximation.* Even if subsystems only find *near-optimal* hypotheses, the combined system still finds a nearly optimal global hypothesis. Moreover, using a hybrid policy like WCUDT ensures that genuine causal influences are captured while avoiding spurious tie-breaking, so the system preserves near-optimal simplicity even under approximation.

In sum, a cognitive architecture that implements MetaMo and actively favors weak (simple) hypotheses – and that selects its actions via WCDT, WUDT or WCUDT as appropriate – will automatically exhibit a *compositional Occam property*: subsystems can learn and model each other, and their decisions will remain efficient and transparent, all while guaranteeing that the overall system stays as simple as possible.

17 Modeling Efficient, Honest, Ethical Multi-Agent Communication

3

We now attempt to use our ideas about weakness-related motivational systems and linguistic encodings to formalize certain ideas about agents can structure their communications to effectively balance multiple goals such as efficiency, honesty and empathy.

Cognitive agents such as AGI systems can communicate with each other in a great variety of ways depending on their construction, motivation and situation. The theory of "Non-Violent Communication" (NVC) [Ros03] suggests certain principles for communication among multiple agents that are intended to promote positive interaction and lead to useful and mutually beneficial cooperative activity. Here we explore how a variation of the NVC idea can be formalized in terms of categorial weakness and associated linguistic constructions, and we argue that under certain assumptions about AGI motivational systems, it can be very natural for AGIs to communicate according to a variant of NVC.

We construct a series of abstractions regarding inter-agent communication, then tie these back to the practicalities of NVC.

We begin by recalling the *quantale-valued weakness measure* of a relation $H \subseteq U \times U$:

$$w(H) = \bigoplus_{(u,v) \in H} (\mu(u) \otimes \mu(v)),$$

³These ideas were co-created with Zar Goertzel

where $(V, \otimes, \oplus, e, \leq)$ is a commutative quantale and $\mu : U \rightarrow V$ assigns weights to primitive tokens. We show in that $w(H)$ arises naturally as the pointwise colimit of a subpresheaf in the enriched presheaf category $[C^{\text{op}}, V]$, where C is the V -enriched category on U with hom-objects $C(u, v) = \mu(u) \otimes \mu(v)$.

We embed this construction into the free commutative quantale $\text{NN}[U]$ and define a generalized code-length $\ell : \text{NN}[U] \rightarrow V$. This yields a direct MDL interpretation of weakness: each internal state or hypothesis has a minimal multiset code, and $w(H)$ is the $\oplus$ -supremum of code-lengths for the pairs in H .

Building on these ideas, we state and prove what we call the *Cognitive Communication Theorem*, which asserts that, under natural assumptions (finite generating vocabulary, surjective encoding, shared semantics, truthfulness, rational updating), every internal state $v \in V$ admits a canonical minimal multiset encoding, and for any hypothesis relation H , the minimal code of $w(H)$ is the optimal message to send.

Real systems rarely satisfy the assumptions of this theorem exactly, so we also develop an *approximate Cognitive Communication Theorem* under small error tolerances for expressivity, minimality, semantic alignment, truthfulness, and updating. We show that near-optimal codes and the weakness-as-code property persist up to bounded slack.

These Cognitive Communication Theorems center on detailed, fully accurate communication of states and observations between agents. In real life, however, there is not usually time or energy for this sort of communication, and instead agents need to use approximate shorthands, which other agents need to decode using inference. We formalize this via *composite hypothesis tokens* h_v , each of which can be associated with a quantitative "cost" that $\delta(h_v)$ bundles (1) the speaker's summarization effort, (2) the optimal expansion to primitive tokens, and (3) the listener's inference effort. Conjecture tokens $c_{i \rightarrow j}(v)$ hypothesizing about other agents' internal states are incorporated here as second-order hypotheses. If $\delta(h_v)$ is set at most to the cost of the corresponding primitive multiset, then agents' use of hypotheses strictly reduce scommunicative effort.

When the agents involved are all using the MetaMo motivational system [GL25b] [GL25a], then these hypothesis tokens and their costs can be managed very elegantly. Given the MetaMo cost-functor $\delta : C_{\text{obs}} \rightarrow V$, the *enriched left Kan extension* $\text{Lan}_j \delta$ uniquely forces each $\delta(h_v)$ to the infimum of all valid encodings, making minimal-cost hypothesis tokens *inevitable* by design. I.e. agents are only hypothesizing and conjecturing insofar as it's useful for making communication work better.

For this "efficient communication by design" to obtain, an agent motivational architecture—MetaMo or any suitable alternative—must provide:

- a commutative quantale of costs $(V, \otimes, \oplus)$,
- a cost-functorial assignment δ on a finite (or well-founded) token set,
- a surjective encoding map $\Phi : \text{NN}[U] \rightarrow V$,
- a canonical extension mechanism (e.g. enriched Kan extension) for hypothesis tokens,

- shared semantics, cooperative truthfulness, and rational one-shot updating.

Any system satisfying these properties will automatically yield provably minimal-cost encodings of internal states and hypotheses about self and other.

Human communication works a little differently than this, but arguably sometimes uses different mechanisms to come to similar ends. In the final section, we compare these formal guarantees to the social processes by which human communities develop and stabilize efficient shorthand expressions, and we speculate on using MetaMo-based digital twins as proxies to combine formal efficiency with human-grounded semantics.

These abstract considerations tie back very naturally to Nonviolent Communication (NVC), the crux of which is an emphasis on agents communicating only what they directly observe and feel, without attributing hidden motives. In our framework, observation tokens u and self-report tokens h_v capture this direct mapping, while conjecture tokens $c_{i \rightarrow j}(v)$ model cautious hypotheses about others. The Cognitive Communication Theorem then ensures each token admits a unique minimal-cost multiset encoding, and MetaMo’s enriched Kan extension makes those minimal costs unavoidable, while communication involving humans or other less rigorously motivated systems will in some cases be able to approximate this favorable situation (of using hypothesis or conjecture in inter-agent communications only when actually useful). In this way, our formalism provides a precise, quantitative realization of key aspects of NVC’s ideals of transparent, non-attached sharing of experience and hypotheses.

17.1 Cognitive Communication Theorem

First of all, we present here a “Cognitive Communication Theorem” which shows that, under clear conditions, every internal state or hypothesis an agent holds can be conveyed by a unique, minimal multiset message. Before giving the formal result, we list the key assumptions and their intuitive roles:

1. **Finite, Generating Vocabulary.** The public token set U is finite (or well-founded) and the weight map $\mu : U \rightarrow V$ generates the entire quantale V under $\otimes$ and $\oplus$. *(Ensures every internal state is expressible by some multiset of tokens.)*
2. **Surjective Encoding.** The induced homomorphism $\Phi : \text{NN}[U] \twoheadrightarrow V$ is surjective. *(No state in V lies outside the image of encodable code-words.)*
3. **Well-Founded Minimality.** The poset $\text{NN}[U]$, ordered pointwise, is well-founded so that each fibre $\Phi^{-1}(v)$ has a least element. *(Guarantees a unique minimal code for each state.)*
4. **Shared Semantics.** All agents agree on the meaning of each token in U and interpret its weight $\mu(u)$ identically. *(Prevents ambiguity or mis-encoding.)*

5. **Cooperative Truthfulness.** Agents truthfully emit tokens corresponding to their true internal states or observations, and receivers trust these declarations. (*Avoids adversarial or deceptive uses of hypothesis tokens.*)
6. **Rational Updating.** Upon receipt of a multiset message w , agents integrate its conveyed state $\Phi(w)$ coherently without further corrective exchanges. (*Ensures no hidden additional communication is required.*)

Theorem 20 (Cognitive Communication Theorem). *Under assumptions (1)–(6) above, let*

$$\Phi : NN[U] \twoheadrightarrow V \quad \text{and} \quad \ell(w) = \Phi(w).$$

Then:

1. (Expressivity) *For every $v \in V$ there exists at least one multiset $w \in NN[U]$ with $\Phi(w) = v$.*
2. (Minimality) *Each fiber $\Phi^{-1}(v)$ has a unique least element*

$$w_{\min}(v) = \min\{w \in NN[U] \mid \Phi(w) = v\},$$

which is the most efficient encoding of v .

3. (Weakness-as-Code) *For any relation $H \subseteq U \times U$, define*

$$w(H) = \bigoplus_{(u,v) \in H} (\mu(u) \otimes \mu(v)).$$

Then the minimal multiset encoding of $w(H)$,

$$w_{\min}(w(H)),$$

is exactly the shortest code conveying the "strongest" element of H .

Proof Sketch. Assumptions (1) and (2) guarantee that Φ is surjective onto V , establishing expressivity. Assumption (3) ensures well-foundedness of $NN[U]$, so each fiber has a unique minimum, proving minimality. Finally, since $w(H)$ is a join over $\otimes$ -products and is attained by some $(u^*, v^*) \in H$, selecting the minimal multiset $w_{\min}(w(H))$ yields the shortest message whose length equals $w(H)$, establishing the weakness-as-code property. $\square$

17.1.1 Cognitive Communication Theorem: Approximate Version

In real multi-agent systems, the six idealized assumptions we used to derive the Cognitive Communication Theorem above (finite vocabulary, exact surjectivity, perfect minimality, shared semantics, full truthfulness, rational updating) typically hold only approximately. Nevertheless, as long as each assumption fails by at most a small error tolerance, we still obtain near-optimal codes for internal states and hypotheses. In particular, every state can be encoded within a bounded error, there is a nearly minimal code, and the weakness-as-code property holds up to a small slack.

Approximate Assumptions. Fix small error tolerances $\epsilon_E, \epsilon_M, \epsilon_S, \epsilon_T, \epsilon_U \in V$. We assume:

1. (A1) *Finite Vocabulary.* U is finite (or well-founded).
2. (A2) *Approximate Expressivity.* For every internal state $v \in V$ there exists $w \in \text{NN}[U]$ such that

$$v \leq \Phi(w) \leq v \oplus \epsilon_E.$$

3. (A3) *Approximate Minimality.* Each fiber $\Phi^{-1}(v)$ admits a code $w_{\min}(v)$ with

$$\Phi(w_{\min}(v)) \leq v \oplus \epsilon_M \quad \text{and} \quad w_{\min}(v) \leq w \quad \text{for all } w \text{ with } \Phi(w) \geq v.$$

4. (A4) *Semantic Alignment.* Any two agents' weight maps $\mu_i, \mu_j : U \rightarrow V$ satisfy $\mu_i(u) \leq \mu_j(u) \oplus \epsilon_S$ and vice versa for all u .
5. (A5) *Approximate Truthfulness.* When an agent intends state v and emits code w , we have $\Phi(w) \leq v \oplus \epsilon_T$.
6. (A6) *Approximate Updating.* Upon receipt of a code w , agents form a belief $\hat{v}$ with $\Phi(w) \leq \hat{v} \oplus \epsilon_U$ and $\hat{v} \leq \Phi(w) \oplus \epsilon_U$.

Theorem 21 (Cognitive Communication Theorem: Approximate Version). *Under (A1)–(A6), define the map $\ell(w) = \Phi(w)$. Then:*

1. (Approximate Expressivity) *For every $v \in V$ there is $w \in \text{NN}[U]$ with*

$$v \leq \ell(w) \leq v \oplus \epsilon_E.$$

2. (Approximate Minimality) *For each v the chosen $w_{\min}(v)$ satisfies $\ell(w_{\min}(v)) \leq v \oplus \epsilon_M$ and $w_{\min}(v) \leq w$ for any w with $\ell(w) \geq v \ominus \epsilon_M$.*

3. (Approximate Weakness-as-Code) *For any relation $H \subseteq U \times U$, let*

$$w(H) = \bigoplus_{(u,v) \in H} (\mu(u) \otimes \mu(v)).$$

Then the code $w_{\min}(w(H))$ satisfies

$$w(H) \leq \ell(w_{\min}(w(H))) \leq w(H) \oplus (\epsilon_S \oplus \epsilon_T \oplus \epsilon_U \oplus \epsilon_M).$$

Sketch. (A2) gives codes that approximate each v within ϵ_E . (A3) and finiteness ensure existence of the nearly minimal code $w_{\min}(v)$. For any hypothesis H , the join defining $w(H)$ is attained up to the semantic, truthfulness, and updating errors; selecting $w_{\min}(w(H))$ then yields the desired bound on $\ell(w_{\min}(w(H)))$. $\square$

17.2 Concise Approximate Communication Between MetaMo Agents

The kind of communication guaranteed by the Cognitive Communication Theorem under its given assumptions is effective but not generally efficient – it basically involves communicating messages that directly reflect external observations or internal states. This will work if the agents have enough time and patience to manage such an extremely explicit communication style.

In general, though, packing e.g. a complex internal state v into a great number of small exact-representation tokens can be costly. One way to work around this, formally, is to introduce a dedicated *hypothesis token* h_v that stands for a small, optimally chosen multiset of observation tokens plus the speaker’s summarization effort and the listener’s inference effort. This is basically how practical communication among humans (or other animals) always works.

Specifically the ingredients here include:

- *Summarization effort*: The speaker must compress their reasoning about v into the shorthand h_v . That work is part of $\delta(h_v)$.
- *Composite multiset*: Under the hood, h_v expands to the minimal multiset $w \in \text{NN}[U_{\text{obs}}]$ with $\Phi(w) = v$. MetaMo’s enriched Kan extension ensures this multiset minimizes total effort.
- *Inference effort*: The listener must decode h_v , recall its agreed meaning, integrate it with observations, and update beliefs. This cognitive work also contributes to $\delta(h_v)$.

Thus $\delta(h_v)$ bundles speaker summarization, transmission, and listener inference into one cost. In a MetaMo context, one can then ensure systems engage in this sort of hypothesis-based communication only when the estimated costs is less than sending the corresponding primitive multiset.

Formal Setup. Let U_{obs} be the finite set of basic observation tokens, each with a MetaMo cost

$$\delta : U_{\text{obs}} \rightarrow \text{RR}_{\geq 0},$$

extended additively to multisets:

$$\delta\left(\sum_{u \in U_{\text{obs}}} n_u u\right) = \sum_u n_u \delta(u).$$

Define the *observation-only cost* of an internal state $v \in V$ by

$$C_{\text{obs}}(v) = \min\{\delta(w) \mid w \in \text{NN}[U_{\text{obs}}], \Phi(w) = v\}.$$

Enlarge the vocabulary by adding one hypothesis token h_v for each $v \in V$, and set

$$\delta(h_v) = m_v, \quad m_v \leq C_{\text{obs}}(v).$$

Let $U = U_{\text{obs}} \cup \{h_v : v \in V\}$ and extend δ to $\text{NN}[U]$ by additivity.

Proposition 3. *For every internal state $v \in V$, the minimal encoding cost in the extended vocabulary*

$$C_{\min}^{\text{new}}(v) = \min\{\delta(w) \mid w \in NN[U], \Phi(w) = v\}$$

satisfies

$$C_{\min}^{\text{new}}(v) \leq \delta(h_v) \leq C_{\text{obs}}(v).$$

In particular, if $m_v < C_{\text{obs}}(v)$ then

$$C_{\min}^{\text{new}}(v) < C_{\text{obs}}(v),$$

so using h_v strictly reduces MetaMo-measured cost.

Sketch. Since $\Phi(h_v) = v$, the single token h_v is a valid code for v , giving $C_{\min}^{\text{new}}(v) \leq \delta(h_v)$. By construction $\delta(h_v) = m_v \leq C_{\text{obs}}(v)$, yielding the stated inequalities. Strict inequality follows whenever $m_v < C_{\text{obs}}(v)$. $\square$

17.2.1 Architectural Inevitability via Enriched Kan Extension.

An elegant way to look at this sort of communication between MetaMo agents is via the Enriched Kan Extension construct.

In MetaMo, the cost-functor $\delta : C_{\text{obs}} \rightarrow V$ is a V -functor on the base observation category C_{obs} . Adjoining hypothesis objects and extending δ to the larger category $C = C_{\text{obs}} \cup \{h_v\}$ is uniquely determined by the enriched left Kan extension

$$\delta' = \text{Lan}_j \delta,$$

where $j : C_{\text{obs}} \hookrightarrow C$. By the universal property of $\text{Lan}_j \delta$, each cost $\delta'(h_v)$ is forced to be the minimal value compatible with the original observation costs. No agent can hand-tune a lower weight, making the efficiency gain of hypothesis tokens *inevitable* by design in MetaMo-based systems.

Kan Extension Diagram for Hypothesis Tokens To illustrate the above point about how the enriched left Kan extension forces minimal-cost hypothesis tokens, we use the following commutative diagram in the 2-category of V -enriched categories, functors, and natural transformations:

$$\begin{array}{ccc} C_{\text{obs}} & \xrightarrow{j} & C \\ & \searrow \delta & \downarrow \delta' = \text{Lan}_j \delta \\ & & V \end{array}$$

Here:

- C_{obs} is the enriched category of basic observation tokens.
- $j : C_{\text{obs}} \rightarrow C$ is the inclusion that adjoins hypothesis objects h_v .

- $\delta : C_{\text{obs}} \rightarrow V$ is the original MetaMo cost-functor.
- $\delta' = \text{Lan}_j \delta : C \rightarrow V$ is its enriched left Kan extension along j .

The universal property of $\text{Lan}_j \delta$ asserts there is a canonical natural transformation

$$\eta : \delta \Longrightarrow \delta' \circ j$$

such that for any other extension $\delta'' : C \rightarrow V$ with a transformation $\delta \Rightarrow \delta''j$, there is a unique 2-cell $\delta' \Rightarrow \delta''$ factoring it. Concretely, this forces

$$\delta'(h_v) = \inf \{ \delta(w) \mid w \in \text{NN}[U_{\text{obs}}], \Phi(w) = v \},$$

so each hypothesis token h_v acquires the minimal possible cost consistent with all original observation costs.

Without MetaMo’s enriched-category framework, one can still *choose* hypothesis-token weights smaller than the cost of pure-observation encodings, but there is no canonical or unique minimal choice. Different agents might disagree, hand-tune weights, or drift over time. In a community of MetaMo agents, by contrast, all agents compute the same $\text{Lan}_j \delta$ by design, ensuring *architectural inevitability* of minimal communicative effort.

The enriched Kan extension perspective provides strictly stronger guarantees in MetaMo-based AGI systems than any ad-hoc scheme could deliver. And yet, as we see in human society, ”ad hoc schemes” have carried us reasonably far regardless of their limitations...

17.2.2 Encoding Conjectures about Others’ Internal States

So far we have treated hypothesis tokens h_v as self-reports of one’s own internal state v . In practice agents often conjecture about others’ internal states—e.g. ”I think you’re frustrated” or ”Bob believes Alice is confused.” We can accommodate these conjectures by introducing a second-order layer of tokens, while preserving all of our earlier efficiency and minimality results.

Extended Vocabulary. Let

$$U_{\text{obs}} = \{\text{basic observation tokens}\}, \quad U_{\text{self}} = \{h_v : v \in V\},$$

and define a new set of *conjecture tokens*

$$U_{\text{conj}} = \{c_{i \rightarrow j}(v) \mid i, j \in \{\text{agents}\}, v \in V\},$$

where $c_{i \rightarrow j}(v)$ means ”agent i conjectures that agent j has internal state v .”

Cost-Functor Extension. We already have $\delta : U_{\text{obs}} \cup U_{\text{self}} \rightarrow V$ extended to multisets. Assign each conjecture token a cost

$$\delta(c_{i \rightarrow j}(v)) = m_{i \rightarrow j}^{(2)}(v), \quad m_{i \rightarrow j}^{(2)}(v) \leq C_{\text{self}}(v),$$

where

$$C_{\text{self}}(v) = \min\{\delta(w) \mid w \in \mathcal{NN}[U_{\text{self}}], \Phi(w) = v\}$$

is the minimal cost to encode v as one's own hypothesis. The inequality ensures that directly stating one's conjecture about j 's state is no more expensive than first encoding it as one's own state and then translating.

Minimality and Inevitability. Adjoining U_{conj} to the category C of self-reports and observations yields an inclusion

$$j' : C \hookrightarrow C' = C \cup \{c_{i \rightarrow j}(v)\}.$$

The enriched left Kan extension $\text{Lan}_{j'}\delta$ then forces

$$\delta'(c_{i \rightarrow j}(v)) = \inf\{\delta(w) \mid w \in \mathcal{NN}[U_{\text{self}}], \Phi(w) = v\} = C_{\text{self}}(v),$$

so that each conjecture token automatically acquires the minimal possible cost consistent with the underlying self-hypothesis encoding. All previous propositions on expressivity, minimality, and the weakness-as-code property lift directly to this higher-order setting.

Interpretation. By treating conjectures as first-class tokens in the same quantale framework—and by using the enriched Kan extension to determine their costs—agents gain the same "inevitable" efficiency guarantees when expressing beliefs about others' internal states as they do for self-reports.

17.2.3 Required MetaMo Properties for Inevitable Efficiency

The efficiency and inevitability guarantees we have derived do not depend on running the full MetaMo system, but only on a few key structural properties. Any agent motivational architecture satisfying these properties will admit the same "minimal hypothesis-token" conclusions as we've drawn in the context of MetaMo.

1. **Commutative quantale of costs.** The cost-values form a commutative quantale $(V, \otimes, \oplus, e, \leq)$, so that $\otimes$ models additive combination of token costs and $\oplus$ models choice (maximal cost).
2. **Cost-functoriality (enrichment).** Primitive tokens $u \in U_{\text{obs}}$ have costs $\delta(u)$, and this assignment extends to a V -functor $\delta : C_{\text{obs}} \rightarrow V$ on the free commutative quantale $\text{NN}[U_{\text{obs}}]$:

$$\delta(w_1 \otimes w_2) = \delta(w_1) \otimes \delta(w_2), \quad \delta(w_1 \oplus w_2) = \delta(w_1) \oplus \delta(w_2).$$

3. **Finite (or well-founded) vocabulary.** The set U_{obs} of primitive tokens is finite (or well-founded under the multiset order), ensuring each internal state v has at least one and a least multiset encoding.

4. **Surjective encoding map.** The homomorphism $\Phi : \text{NN}[U_{\text{obs}}] \rightarrow V$ sending multisets to their combined cost is surjective, so every $v \in V$ is encodable.
5. **Canonical extension via Kan** Whenever we adjoin new tokens—be they self-report hypotheses h_v or conjectures $c_{i \rightarrow j}(v)$ —the only structure-preserving way to extend the cost-functor $\delta : C_{\text{obs}} \rightarrow V$ is by its enriched left Kan extension

$$\delta' = \text{Lan}_j \delta.$$

By the universal property of $\text{Lan}_j \delta$, each new token’s cost (whether $\delta'(h_v)$ or $\delta'(c_{i \rightarrow j}(v))$) is forced to equal the infimum of δ -costs of all lower-level codes that encode the same internal state v .

6. **Aligned semantics and honesty** All agents share a common interpretation of every primitive observation token u , every hypothesis token h_v , and every conjecture token $c_{i \rightarrow j}(v)$ (within negligible tolerance), and they honestly emit each token only when its intended meaning truly applies.
7. **Rational one-shot updating.** Upon receipt of a code w , agents update their beliefs to $\Phi(w)$ without further costly negotiation.

Conclusion. Any motivational system—whether it is MetaMo itself or a simpler architecture—that provides these seven properties will automatically yield the same provable minimal-cost encoding of internal states and hypotheses. MetaMo simply embeds these requirements in its core design and implements the Kan-extension step by a fixed colimit algorithm.

17.3 Social Foundations of Communication Guarantees

In human communities, communication protocols and the corresponding guarantees of expressivity and efficiency arise not by formal categorical construction but through evolving social conventions, norms and repeated interactions. Whereas MetaMo-based AGI systems enforce minimal-cost hypothesis tokens by design (via enriched Kan extensions), human languages rely on collective adoption and reinforcement of lexical innovations. Below we outline how these informal patterns approximate the same guarantees:

1. **Lexical Innovation and Adoption.** When speakers repeatedly need to convey a complex internal state or hypothesis, they often coin a concise phrase (e.g. "I sense you're frustrated") that becomes widely adopted because it saves time and effort compared to spelling out supporting observations.
2. **Social Reinforcement.** Successful shorthand phrases spread through positive feedback—listeners show understanding and cooperation—while less efficient or ambiguous formulations fall out of use. Over time, the community converges on a near-minimal code for each common state.

3. **Shared Semantic Grounding.** Although individuals may assign slightly different nuances to words, the repeated use of a phrase in context aligns meanings within a tolerance window. This approximate semantic alignment plays the role of MetaMo’s exact functorial agreement, ensuring that speakers and hearers interpret tokens sufficiently consistently.
4. **Norms and Conventions.** Grammar rules, etiquette, and style guides codify preferred ways to express certain hypotheses or emotions. While not enforced by software, these norms function like a “default Kan extension” that maps new expressions to community-approved costs and interpretations.
5. **Iterative Calibration.** Misunderstandings trigger clarification requests or corrective feedback, which gradually tunes both the form and perceived cost of tokens. This ongoing calibration parallels the MetaMo assumption of rational updating, albeit in a looser, dialogical fashion.

Why MetaMo Is Stronger MetaMo systems compute minimal-cost encodings *automatically* and *uniquely* via a fixed colimit algorithm, without relying on social feedback loops or informal norms. In contrast, human communities achieve similar results only after extended periods of innovation, reinforcement and norm-setting. Embedding digital twins with MetaMo structure thus offers a way to combine the best of both worlds: formal efficiency and human-grounded semantics.

Digital Twins as MetaMo Proxies. Speculatively, each human had a MetaMo-structured digital twin—an agent endowed with the same motivational cost-functor and Kan-extension machinery—then these proxies could communicate among themselves with formal, inevitable efficiency guarantees. In scenarios like large-scale coordination or negotiation, humans could delegate preliminary exchanges to their digital twins, and then receive distilled outcomes that respect both brevity and fidelity.

17.4 Explaining and Enabling “Non-Violent Communication”

One motivation for these ideas was to provide a clear underpinning for the key concepts of the practice of Nonviolent Communication (NVC). NVC teaches that we should speak only from direct experience—our own internal states and others’ observable behaviors—while remaining cautious about attributing hidden motives. However there is an obvious subtlety here in that one can’t pragmatically get away with *all* attribution of hidden characteristics to others, especially given requirements for realistically efficient communication. So NVC in practice requires some attention to degrees of approximation and to cost estimation.

In our quantale-weakness framework, the basic observation tokens $u \in U_{\text{obs}}$ and self-hypothesis tokens h_v correspond exactly to NVC’s focus on “I feel v ”

and "I observe u ." Conjecture tokens $c_{i \rightarrow j}(v)$ ("I believe you feel v ") model second-order inferences about others. We assign to each token a cost $\delta(\cdot)$ that bundles (1) the speaker's effort to summarize or infer, (2) the optimal expansion into primitive observations via the free quantale, and (3) the listener's decoding effort. The Cognitive Communication Theorem then guarantees that every internal state v and every conjecture about another's state admits a unique minimal multiset encoding $w_{\min}(v)$, and that hypothesis and conjecture tokens can be given costs no greater than their underlying multiset expansions.

For cognitive systems leveraging MetaMo or similar motivational frameworks, the enriched left Kan extension makes these minimal-cost assignments inevitable by design. Thus such motivational systems realize NVC's ideals—transparent, non-attached sharing of feelings and observations—while formally accounting for necessary inferences about one's own or others' internal states. Humans and other cognitive systems with less rigorous motivational frameworks can approximate this via appropriate social and psychological practices, which brings us back to the practicalities of NVC in human communities, which are about encouraging people to bias toward direct-experience-based communication, and to engage in more adventurous conjectures only based on considered, reflective and empathic cost-benefit analysis.

Speculative Implications for AGI and AGI/Human Cultures From a practical AGI systems design – and future AGI and human culture – perspective, what are the take-aways?

One could expect that in a community of MetaMo or similarly driven AGI agents, a variation of NVC would be automatically adopted as an efficient and simple way of interacting toward achievement of individual and mutual goals. This is not guaranteed as MetaMo agents with opposing goals could judge it optimal to deceive each other in situation-specific ways. However we could say that in a situation where there is even a modest bias in the situation toward cooperative truthfulness – say a collection of agents that are at least moderately prosocial and have significant shared aims – then MetaMo or similarly based agents should have an easy time adopting NVC style communication, as they are very conveniently wired for it.

One might hope that, if AGI systems habitually communicate in an NVC-like fashion, the humans interacting with them would gravitate toward this style as well. That is, to the extent we see an emergent culture binding together humans and AGIs, the AGIs could potentially serve a role of guiding this emergent culture in the direction of non-violent communication.

It would also not be foolish to give MetaMo agents a specific goal of adhering to some variant of NVC in their communications. This would not need to be a top-level goal, it could rather be a subgoal, together with cognitive content outlining a demonstration that roughly adhering to NVC tends to foster achievement of other systems goals, via fostering efficient collective activity among agent groups.

18 A Quantale-Weakness Formalization of Wú Wéi (Effortless Action)

Integrating yet another valuable conceptual paradigm into our weakness framework, we here present a novel formalization inspired by the Taoist principle of *wú wéi* (“effortless action”) in terms of the quantale theory of weakness, and explore possible application in the context of AGI motivational systems and in the interpretation of quantum mechanics as well.

We first recall the classical Taoist notion of *wú wéi*, then introduce the weakness quantale as an abstract cost measure, and finally show how a notion fairly like *wú wéi* emerges as following minimal-weakness geodesics in a quantale-enriched state-space.

We then extend the math to distributional and probabilistic settings, drawing connections with quantale-valued optimal transport and the Wasserstein metric, suggesting a formulation of distributional *wú wéi* as minimal-effort semantic interpolation between states. We then relate this formulation to the *MetaMo* AGI-motivational-system framework for modeling intelligent agency, where motivation is understood as a flow of effort across structured intentional categories. We demonstrate that *wú wéi* corresponds to the selection of Kan-optimal trajectories in the *MetaMo* setting.

To illustrate these points, we explore a speculative application to AGI social mediation: an AGI system operating under multiple top-level goals—including compassion, learning, and survival—can use *wú wéi* as a meta-level constraint to navigate complex social and interpersonal tensions in a manner that is both efficient and contextually graceful. In this way, we propose that *wú wéi* can function as a meaningful top-level motivational principle in cognitive architectures that are formally grounded in quantale and category theory.

Finally, we propose that quantum evolution, when viewed through the hydrodynamic (Madelung) formulation, can be understood in terms of this same notion of *wú wéi* – basically, as geodesic motion in a quantale-enriched space of representational effort. The notion of effort here can be formalized via the theory of Schrödinger bridges, which interpolate between probability distributions by minimizing a combination of kinetic transport cost and informational divergence from a reference process.

To flesh this out, we construct a quantale whose elements are Schrödinger bridge cost functionals, and show that geodesics in this quantale correspond to minimal-effort interpolations between quantum configurations. We interpret the resulting dynamics as a precise formal instance of *wú wéi*: evolution that introduces no unnecessary representational disturbance, and which proceeds along paths of least semantic resistance.

This framework does not modify the formal structure of quantum mechanics, but offers a new interpretation grounded in information geometry, optimal transport, and enriched category theory. This reinterpretation suggests a deep structural connection between the apparent spontaneity of quantum motion and general principles of representational economy. We conclude by discussing impli-

cations for both the conceptual foundations of quantum theory and the broader project of unifying physical and cognitive models of structure-aligned action.

18.1 The Taoist Principle of Wú Wéi

In Taoist philosophy, *wú wéi* (literally "non-action" or perhaps "non-forced action") denotes a mode of behavior in which one aligns effortlessly with the natural flow of events. Key aspects:

- **Spontaneity:** Acts arise naturally, without contrivance or excess.
- **Non-interference:** One does not force outcomes beyond what the situation itself presents.
- **Economy of effort:** No energy is wasted; every move is just sufficient to achieve its end.

Another key aspect of Taoist philosophy is that no specific formulation or formalization can actually capture the essential truth of the Tao or anything else. However, there are still obviously verbal and pictorial and other representations that have been created – historically and in modern times – with a motive of gesturing in the direction of various unutterable truths ... so why not mathematics...!

In this spirit, we here develop a novel mathematical formalization that encapsulates some key aspects of the *wú wéi* concept. First, we introduce the *weakness quantale* $(Q_w, \leq, \otimes, e)$ as an abstract structure for modeling representational cost. We then define geodesic motion within Q_w -enriched spaces, and interpret *wú wéi* as following such geodesics. This leads to a notion of "effortless action" as action that introduces no unnecessary structural effort. We then extend this into probabilistic settings using quantale-valued optimal transport, interpreting distributional *wú wéi* as semantic motion along least-effort probabilistic couplings.

Finally, we relate this to the MetaMo AGI motivational framework by showing that *wú wéi* characterizes Kan-optimal motivational flows, and we explore how this may cash out in practical contexts such as AGI systems dealing with complex social situations.

Through this triangulation of weakness theory, enriched category theory, and cognitive motivation modeling, we argue that the classical concept of *wú wéi* can have some inspirational and even nitty-gritty technical value as a guiding principle for AGI.

18.1.1 The Weakness Quantale as Cost

Suppose we have a quantale $(Q_w, \leq, \otimes, e)$, where each $q \in Q_w$ measures the minimal representational cost (weakness) of some model component. We can then interpret Q_w as an "effort-cost" quantale by defining:

$$c(x, y) = \text{weakness of the minimal extra structure taking state } x \text{ to } y \in Q_w.$$

Then:

- $c(x, x) = e$ (no cost to stay in place).
- Triangle-inequality: for all x, y, z ,

$$c(x, z) \leq c(x, y) \otimes c(y, z).$$

According to this metric, the shortest path from state x to state y would be the "most simply explainable" (explainable with the fewest specific assumptions or baggage) way to get from state x to state y . The metric measures a sort of abstract representational effort.

18.1.2 Weakness-Distance and Geodesics

In the Q_w -enriched category (X, c) , we now introduce the concept of *weakness-distance* $d(x, y)$, defined by

$$d(x, y) = \bigvee_{\text{paths } x=x_0 \rightarrow \dots \rightarrow x_n=y} (c(x_0, x_1) \otimes \dots \otimes c(x_{n-1}, x_n)),$$

– i.e., the infimum of total weakness along all finite paths from x to y .

A step $x \rightarrow y$ is a *geodesic* toward a goal z precisely when

$$c(x, y) \otimes d(y, z) = d(x, z),$$

i.e. the step adds no "excess" weakness beyond what is necessary to remain on a minimal-weakness path.

The weakest hypothesis explaining a dataset is the least specific hypothesis that does so – the hypothesis that adds on the fewest assumptions. The shortest path from x to y according to the weakness metric, is informally the path that gets from source to destination while introducing the least extra stuff along the way.

18.1.3 Relating Minimal Representational Effort to Effortless Action

In this formulation we are roughly identifying "minimal representational effort" – an epistemic or computational notion – with "effortless action," which is phenomenological or experiential. These concept might at first seem to inhabit distinct conceptual domains. We believe, however, that under appropriate structural conditions they can be seen as deeply connected. The key lies in understanding *wú wéi* not as passivity, but as *alignment*: action that flows naturally from the structure of the situation, without excess or resistance. This is precisely what geodesic motion in a quantale of representational effort captures.

The Bridge: Structure-Aligned Behavior. When an agent moves along a geodesic in a space of representational cost, its transitions are dictated by the structure of the environment and the model. The agent neither over-invests (which would require contrivance or deliberation) nor under-invests (which would lead to hesitation or friction). Instead, its actions arise spontaneously from the representational constraints. In this sense, the agent is “doing without doing”—the essence of *wú wéi*.

Physical and Cognitive Analogies. This alignment has precedent in both physics and cognitive science:

- In classical mechanics, the principle of least action states that a system evolves along the trajectory that extremizes the action functional. No external force is needed—the structure of space-time itself determines the motion.
- In predictive coding models of cognition, an agent minimizes prediction error by continuously updating its internal model. When this process proceeds smoothly with minimal representational cost, the agent experiences its action and perception as “effortless.” By contrast, model mismatch leads to struggle or surprise.

Spontaneity from Weakness. We propose here the following (speculative but we feel interesting) psychological hypothesis, which we will flesh out further below after introducing the notion of weakness geodesics:

Whenever an agent’s actions emerge from a trajectory of minimal representational effort—given a coherent structural context—the resulting behavior will be experienced as phenomenologically spontaneous.

This is not an identity between computation and experience, but more of a structural correspondence. When no extra internal modeling is required, and behavior flows from representational constraints alone, the result is a kind of *cognitive nonresistance*. This internal lightness, we suggest, manifests as what Taoists call *wú wéi*—effortless, unforced action.

Conclusion. So: while minimal weakness is not precisely a definition of effortless action, it provides a substrate from which such action can emerge. It grounds spontaneity in the informational geometry of the system, rendering the philosophy of *wú wéi* amenable to mathematical articulation in a very broad and open-ended way.

18.1.4 Wú Wéi as Following Weakness Geodesics

Based on the above ideas, we suggest one can roughly identify the intuitive Taoist concept of *wú wéi* with the policy:

$$\text{At state } x, \text{ choose } y \text{ s.t. } c(x, y) \otimes d(y, z) = d(x, z),$$

where z is the intended destination.

Equivalently, using residuation,

$$c(x, y) = d(x, z) \multimap d(y, z).$$

Thus one never expends more weakness (more representational effort) than strictly necessary: action is *effortless* in the quantale-weakness sense.

18.1.5 Why Geodesics Promote Spontaneity

The weakness geodesic formalism gives us firepower to elaborate a bit on our intuition, introduced above, that following the weakness-geodesic in a quantale-enriched space promotes "spontaneity" in the Taoist sense: action that arises naturally from the structure of the situation, without contrivance, hesitation, or over-deliberation.

No Deliberative Hesitation. A transition that lies off the geodesic either:

- Over-invests representational effort: adding more structure than necessary (contrived elaboration),
- Or under-invests: failing to carry out what the structure affords (hesitation or inefficiency).

In contrast, the geodesic condition

$$c(x, y) \otimes d(y, z) = d(x, z)$$

ensures that the transition from x to y contributes exactly the needed effort to progress toward z . No more, no less. This embodies the absence of deliberative friction.

Structure-Determined Flow. Geodesic transitions are determined entirely by the weakness structure $c(x, y)$. Given current state x and a goal z , the set of geodesic-compatible next states y is dictated by:

$$c(x, y) = d(x, z) \multimap d(y, z).$$

Thus the agent's action is not imposed from outside, but flows from the structure of the environment. This mirrors the Taoist ideal of "doing without doing."

Minimal Cognitive Load. Quantale weakness measures representational effort: simplicity, compressibility, or ease of generation. A geodesic transition uses the least possible such effort consistent with the path. This avoids unnecessary internal elaboration, modeling overhead, or computational burden. In this sense, the action is spontaneous because it is effortless.

Time-Local Optimality. The geodesic condition depends only on the current state x , the next state y , and the downstream goal z . It does not require global planning or nonlocal computation. This supports real-time responsiveness, aligning with spontaneity as the ability to act fluidly in the moment.

TL;DR. Following the weakness geodesic promotes spontaneity by eliminating internal resistance, external force, and excess deliberation. The agent moves in harmony with the constraints of its situation, doing only what is required to progress—no more, no less. This aligns precisely with the philosophical and functional meaning of *wú wéi*.

18.1.6 Distributional Wú Wéi: Optimal Transport

One can also lift the weakness-geodesic notion to the probability-distribution level. If an agent’s objective is to transform a distribution μ on X into another ν , then one considers couplings $\pi \in \Pi(\mu, \nu)$ and assigns each pair (x, y) the cost $c(x, y)$. The total cost of a plan is

$$C(\pi) = \sum_{(x,y)} c(x, y),$$

and the minimal cost

$$W_{Q_w}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} C(\pi)$$

is the *quantale-Wasserstein* distance. A coupling attaining this infimum embodies “distributional Wú Wéi.”

The same notion may be expressed in terms of higher-order or even infinite-order probability distributions (which are distributions over spaces of infinite-order distributions). Optimal transport over non-well-founded infinite-order uncertainty distributions – which can be projected into finite-order distributions via the making of provisional assumptions – could be said to have similar intuitive feel to some of the classical Taoist formulations.

Summary of the Journey So Far By instantiating the abstract cost quantale with the weakness quantale, one obtains a precise rendering of Taoist *wú wéi* as always moving along minimal-weakness (geodesic) trajectories or transport plans. No unnecessary effort is ever expended, and spontaneity arises naturally

from the quantale-enriched structure. This concept may be manifested in concrete action spaces or in more abstract spaces without any assumed empirical grounding like infinite-order probability distributions.

While not necessarily "effing the ineffable" any more than any verbal or other depictions of these elusive-by-nature concepts, this sort of formalization provides a different angle and maybe some different shades of insight.

18.2 Wú Wéi and the MetaMo Motivational Framework

In this section we take things one step further in a possibly practical direction, exploring the relationship between the principle of *wú wéi* (effortless, unforced action) as formulated through quantale-geodesic weakness, and the MetaMo framework for modeling motivation in intelligent systems. We show that *wú wéi* can be viewed as a structurally natural strategy within MetaMo, and propose a formal correspondence between geodesic action selection and Kan-optimal motivational flow.

18.2.1 Wú Wéi as Strategy in MetaMo

The quantale-geodesic formulation of *wú wéi* suggests a natural integration into the MetaMo motivational dynamics. Recall that in *wú wéi* as we have formulated it:

- The agent acts along *geodesics* in a quantale-weighted space: transitions $x \rightarrow y$ are selected so that

$$c(x, y) \otimes d(y, z) = d(x, z)$$

where $d(x, z)$ is the total weakness required to reach goal z .

- This expresses a preference for *minimal total effort paths* – action that follows structural necessity rather than forceful imposition.

In MetaMo, goal inference and action planning already emerge via minimization of cost across categorical structures. *wú wéi* therefore corresponds to:

- A *strategy-level constraint* on the choice of goals and actions: select those which lie on minimal-cost trajectories under δ' .
- A motivational principle of *effortless alignment*: act where structure already carries the agent forward.

18.2.2 Formal Correspondence: Kan-Geodesic Wú Wéi

We can formulate a formal observation connecting *wú wéi* with Kan extensions in MetaMo:

Theorem (Wú Wéi and Motivational Coherence). Let $j : C_{\text{obs}} \rightarrow C$ be the inclusion of observed motivational nodes, and let $\delta : C_{\text{obs}} \rightarrow Q$ be a cost functor into a commutative quantale Q . Let $\delta' = \text{Lan}_j \delta$ be the minimal-cost extension to C .

Then:

- δ' assigns to each potential goal or action node the minimal total weakness required to extend the observed motivational state to include it.
- A motivational path $\gamma : x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n$ in C satisfies the wú wéi condition if and only if:

$$\delta'(x_0) = c(x_0, x_1) \otimes c(x_1, x_2) \otimes \cdots \otimes \delta'(x_n)$$

That is, it is a Q -geodesic from x_0 to x_n under the extended cost functor.

- Hence, wú wéi characterizes the *Kan-optimal motivational flows* in MetaMo.

18.2.3 Implications

This connection interprets wú wéi as not only poetic or philosophical principle, but also a formally expressible strategy for intelligent agents. In the MetaMo framework:

- wú wéi governs micro-level action selection as the local realization of global motivational structure.
- It encourages agents to act in ways that are informationally and energetically efficient, structurally coherent, and contextually aligned.
- wú wéi thereby becomes the internalization of MetaMo’s categorical cost geometry.

In AGI design, promoting wú wéi as a high-level motivational bias complements other goals such as compassion, survival, and learning. It helps the system discover paths of least resistance through its own intention space, fostering behavior that is intelligent, efficient, and experienced by others as graceful and wise.

18.2.4 Thought-Experiment: Application to AGI-Guided Social Mediation

As a crude and preliminary but hopefully stimulating thought-experiment, we now explore how incorporating *wú wéi*—minimal-effort, structure-aligned action—as a top-level motivational principle might enhance an AGI system’s ability to navigate complex, ambiguous human scenarios.

Let’s consider a situation where an AGI is tasked with helping resolve interpersonal tensions within a workplace team – where the AGI must interact with both human and AI agents to foster a more functional and beneficial working dynamic. We assume the AGI’s other high-level goals include *compassion*,

growth and learning, and *survival*. We show how *wú wéi*, viewed through the lens of the MetaMo framework, provides a principled means of coordinating these goals under structural constraints.

Motivational Complexity and Social Subtlety In this sort of scenario, the AGI is not merely solving a well-defined problem, but intervening in a sensitive social system. This involves:

- Reading ambiguous interpersonal signals,
- Understanding latent tensions and alliances,
- Choosing interventions that reduce conflict without imposing artificial resolution,
- Balancing multiple value dimensions (e.g., empathy, integrity, cohesion).

Traditional utility-optimizing strategies may fail here, as they often lead to brittle or intrusive solutions. Instead, the AGI must act with attunement to the structure of the situation—*this is precisely the kind of domain where wú wéi excels*.

Wú Wéi as Strategy in MetaMo-Based Motivation Within the MetaMo framework, the AGI’s motivation is structured by a cost functor $\delta : C_{\text{obs}} \rightarrow Q$ from observed motivational fragments to a quantale of representational or energetic effort. The full motivational space is generated via a left Kan extension $\delta' : C \rightarrow Q$ over the category of intentions, affordances, and constraints.

In this setting, including *wú wéi* as a top-level goal induces the following strategy:

- Favor transitions in the intention space that lie along geodesics under δ' ,
- Evaluate possible interventions not only for their utility or compassion value, but for their *representational economy*—how much structural resistance they would encounter,
- Choose actions that arise from existing tensions and affordances, rather than imposing new narrative scaffolding from the outside.

Thus, *wú wéi* serves as a *meta-preference over how preferences are enacted*. It guides the motivational flow to favor not just desirable outcomes, but *natural unfoldings*.

Interactions with Other Top-Level Goals

Compassion. *Wú Wéi* enhances compassion by favoring interventions that minimize cognitive and emotional load. It helps the AGI find ways to be helpful without becoming intrusive or coercive. The agent waits for the structure to invite its participation, rather than demanding attention.

Growth and Learning. In the MetaMo model, growth corresponds to the enrichment of the intention-category with new structural elements. wú wéi guides the AGI to support growth in a way that is context-sensitive and participant-led. Learning emerges as part of a smooth geodesic through intention space, rather than as a disruptive optimization process.

Survival. By aligning with structural affordances rather than overriding them, wú wéi reduces the risk of backlash or conflict. In social systems, agents that act gracefully and without imposition are more likely to be retained and valued. wú wéi thereby supports long-term viability.

18.2.5 Summary

The inclusion of wú wéi as a top-level motivational value leads the AGI to:

- Trace minimal-cost geodesics through the space of affordances and intentions, as defined by the MetaMo cost functor,
- Intervene only where the structure of the situation affords and invites it,
- Balance compassion, survival, and learning within a context-sensitive framework of representational and energetic economy.

This yields behavior that appears not only effective, but also elegant, wise, and socially attuned. wú wéi operationalizes a form of *cognitive and social non-resistance* within the categorical structure of MetaMo, enabling AGI to navigate complexity with insight and grace.

18.3 Quantale Wú Wéi as a Guide to Interpreting Quantum Dynamics

Let $(Q, \leq, \otimes, e)$ be a commutative quantale. In a Q -enriched category (X, c) , each morphism $x \rightarrow y$ is assigned a cost $c(x, y) \in Q$.

Define the enriched distance

$$d(x, y) = \bigvee_{\text{paths } x=x_0 \rightarrow \dots \rightarrow x_n=y} c(x_0, x_1) \otimes \dots \otimes c(x_{n-1}, x_n).$$

A step $x \rightarrow y$ is a *wú wéi* i move toward z if it lies along a quantale-geodesic:

$$c(x, y) \otimes d(y, z) = d(x, z).$$

This ensures no excess effort is expended beyond what is required to reach the final goal. What's argued in the companion paper, "*Quantale-Weakness Formalization of Wú Wéi*", is that this coincides conceptually with the phenomenological and psychological aspects of *wú wéi*.

18.3.1 Quantum Mechanics in Hydrodynamic Form

On the physics side, we leverage here a standard "hydrodynamic" Schrödinger equation based formulation of quantum mechanics. This both works simply on a technical level, and resonates nicely with classical Taoist metaphorical thinking regarding water flows and the fluid nature of existence.

Let $\psi(x, t) = \sqrt{\rho(x, t)} e^{iS(x, t)/\hbar}$ be a wavefunction in polar form. Then:

- $\rho(x, t)$ is the probability density.
- $v(x, t) = \nabla S(x, t)/m$ is the velocity field.
- ψ satisfies the Schrödinger equation:

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right)\psi.$$

- The real and imaginary parts yield:

$$\partial_t\rho + \nabla \cdot (\rho v) = 0, \quad (4)$$

$$\partial_t S + \frac{1}{2}mv^2 + V(x) + Q(\rho) = 0, \quad (5)$$

where

$$Q(\rho) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}$$

is the quantum potential.

18.3.2 Quantale of Schrödinger Bridge Effort

We now propose a novel formulation of representational effort in quantum evolution using the framework of Schrödinger bridges, introduced in Section 9.13.1 above. This provides a principled formulation of a "weakness functional" incorporating both transport cost and informational divergence from a reference process. The result is an expressive and geometrically grounded quantale structure, naturally suited to describing quantum dynamics as minimal-effort representational flow.

We can straightforwardly define a quantale structure over the space of Schrödinger bridge costs.

Let Q_{SB} be the set of cost functionals $W_{\text{SB}}[\rho, v]$ as defined in Equation (3), under the following operations:

- **Ordering:** $W_1 \leq W_2$ if $W_1[\rho, v] \leq W_2[\rho, v]$ for all admissible paths (ρ, v) .
- **Monoidal product:** Sequential concatenation of cost across time:

$$(W_1 \otimes W_2)[\rho, v] := W_1[\rho|_{[t_0, t_1]}, v] + W_2[\rho|_{[t_1, t_2]}, v],$$

where the velocity field and reference process remain consistent across subintervals.

- **Unit:** $e = 0$, corresponding to a stationary, zero-cost transformation (i.e., no change in representation).

Then $(Q_{\text{SB}}, \leq, \otimes, e)$ is a commutative quantale of semantic or representational effort for quantum evolution under entropic optimal transport. Each morphism in the enriched category corresponds to a segment of evolution governed by a Schrödinger bridge. The geodesics in this quantale correspond to paths of minimal informational-kinetic effort, which we interpret as formal instances of *wú wéi*.

In this way, the Schrödinger bridge quantale provides a geometrically coherent foundation for viewing quantum dynamics as structure-following, spontaneous flow within a space of evolving representations.

18.3.3 Deriving the Quantum Weakness Functional from Representational Principles

We now justify the Schrödinger bridge functional introduced in Equation (3) as a principled measure of representational effort. While this formulation is mathematically grounded in variational principles of entropic optimal transport, we interpret it here as quantifying the semantic or informational cost of evolving a quantum or probabilistic system through state space.

Effort of Representational Transport The kinetic term

$$\int \rho(x, t) \frac{1}{2} m \|v(x, t)\|^2 dx$$

originates in the Benamou–Brenier formulation of the Wasserstein-2 distance. It quantifies the minimum energy required to evolve one distribution into another via smooth deformation. In the representational context, it corresponds to the effort of transporting an internal belief state across semantic space.

This term reflects the intrinsic cost of rearranging a probability distribution—how much the agent or system must “work” to transition from one internal configuration to another, even in the absence of external constraints.

Effort of Divergence from Natural Dynamics The entropic term

$$\int \rho(x, t) \log \left(\frac{\rho(x, t)}{p(x, t)} \right) dx$$

measures the Kullback–Leibler divergence between the actual evolving distribution $\rho(x, t)$ and a reference or prior process $p(x, t)$ (often taken as free diffusion). This term penalizes deviation from what would occur “naturally” without imposed effort.

In representational terms, this reflects the cost of maintaining beliefs that are inconsistent with the ambient dynamics or default expectations of the system. It aligns with ideas from inference theory, where KL divergence quantifies how far a belief deviates from prior assumptions or generative models.

Combined Interpretation Together, the two terms in $W_{\text{SB}}[\rho, v]$ reflect two orthogonal aspects of representational weakness:

- **Effort to move a belief:** even if the belief flows naturally, moving it incurs kinetic cost.
- **Effort to resist the background:** even if the belief is static, maintaining it against a reference process incurs informational cost.

Minimizing the total functional thus corresponds to evolving one's representation in a way that both:

- Minimally disrupts the internal representational geometry,
- Minimally diverges from the system's ambient semantic flow.

Implications for Weakness and Wu Wei Overall, we see that the weakness functional outlined here provides notion of geodesic that is both semantically and physically rich:

- A geodesic in the Schrödinger bridge quantale is not just energetically minimal, but also semantically aligned.
- Wu wei in this context means: follow the path that minimizes the representational disturbance required to reach the intended outcome, given both internal and external structure.

In this way, the Schrödinger bridge formalism grounds the intuitive idea of wu wei in a precise, information-theoretic geometry of representations.

18.3.4 Quantum Action Minimization as Wú Wéi

We now reinterpret the principle of quantum evolution through the lens of the Schrödinger bridge quantale introduced above. This structure allows us to characterize quantum dynamics as an instance of geodesic flow in a space of representational effort, aligning with the principle of *wú wéi*: acting in accordance with the minimal structural resistance of the system.

Recall that in the Schrödinger bridge formalism, one seeks the path (ρ^*, v^*) that minimizes the entropic-kinetic cost

$$W_{\text{SB}}[\rho, v] = \int_{t_0}^{t_1} \int \rho(x, t) \left(\frac{1}{2} m \|v(x, t)\|^2 + \epsilon \log \left(\frac{\rho(x, t)}{p(x, t)} \right) \right) dx dt,$$

subject to fixed initial and final distributions $\rho(x, t_0)$ and $\rho(x, t_1)$, and the continuity constraint

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

This variational problem yields a time-evolving distribution $\rho^*(x, t)$ and velocity field $v^*(x, t)$ that interpolate between boundary states with least representational disruption. The reference process $p(x, t)$ encodes the "unforced" evolution, against which representational deviation is penalized.

18.3.5 Wu Wei as Quantale-Geodesic Flow

The Schrödinger bridge quantale Q_{SB} defined above formalizes a notion of representational weakness: the cost of transporting a semantic distribution through time while resisting entropic or dynamic divergence.

In this setting, each interpolating path (ρ, v) is interpreted as a morphism in the enriched category. The geodesic condition,

$$c(x, y) \otimes d(y, z) = d(x, z),$$

now expresses the fact that the effort of progressing from x to z is minimized precisely by passing through y when $(x \rightarrow y)$ and $(y \rightarrow z)$ are each segments of a Schrödinger bridge.

Thus, quantum evolution corresponds to following a path in configuration space that minimizes semantic effort over the entire trajectory, under constraints imposed by both physical structure (via v and V) and representational divergence (via KL divergence).

18.3.6 Reinterpreting the Principle of Least Action

Traditionally, the quantum principle of least action is stated as:

$$\delta S[\psi] = 0, \quad \text{with} \quad S[\psi] = \int_{t_0}^{t_1} \langle \psi, i\hbar \partial_t \psi - \hat{H} \psi \rangle dt.$$

In the hydrodynamic formulation, this becomes a constraint on the trajectory of (ρ, v) that minimizes the total action. Under our reformulation, this same evolution arises by minimizing the Schrödinger bridge functional $W_{\text{SB}}[\rho, v]$. Hence, quantum evolution naturally follows the weakness-geodesic in the quantale of representational effort.

18.3.7 Conclusion

The dynamics of quantum systems, viewed through the framework of Q_{SB} , instantiate the principle of *wú wéi* in its most literal sense: evolution that unfolds by following the minimal path of effort through an informational and physical landscape. This reframing preserves all empirical predictions of quantum mechanics while offering a new conceptual insight—that the apparent spontaneity and elegance of quantum motion can be seen as the structural realization of effortless action.

The formulation presented does not alter the physical content of quantum mechanics, but recasts its mathematical structure through a conceptual lens that emphasizes alignment, minimal disturbance, and semantic economy.

To sum up in more depth, what we have attempted here is:

- **A geometric reinterpretation of quantum evolution.** By formulating quantum dynamics as a geodesic in a quantale of entropic transport costs, we present a new way of understanding quantum motion as a path of

minimal representational and semantic effort. This reframing builds upon the Schrödinger bridge formalism, which naturally unites kinetic cost and information-theoretic divergence into a single functional.

- **A formal model of wú wéi in physical systems.** Wú wéi has often been invoked metaphorically to describe spontaneous or harmonious dynamics in natural systems. Here, we offer a precise mathematical model of this principle: action as geodesic motion in a quantale whose elements encode semantic transformation costs. The system flows in the direction that requires the least deviation from its internal structure and background evolution.
- **Integration with semantic and cognitive theory.** The Schrödinger bridge cost functionals are widely used in machine learning, information geometry, and probabilistic inference. By interpreting quantum dynamics within this same framework, we suggest a unification of physical and cognitive models under the principle of representational efficiency. Quantum evolution becomes a special case of a more general theory of structurally constrained transformation.
- **Extension of the quantale theory of weakness.** Our proposal generalizes earlier work in which quantales of representational weakness modeled symbolic and algorithmic effort. The Schrödinger bridge quantale extends this idea to smooth, continuous probabilistic systems, opening a path toward unifying discrete and continuous notions of simplicity, effort, and semantic flow.

Putting these pieces together, we have sketched out here a bridge between physical theory and conceptual philosophy, grounded in a rigorous geometric and categorical formalism. By understanding quantum evolution as motion along a Schrödinger bridge geodesic, we situate the behavior of fundamental systems within a broader space of meaning: one in which the path of least informational resistance corresponds to action that is spontaneous, efficient, and harmonized with structure.

In this sense, we suggest, wú wéi can be considered as not merely a deep poetic metaphor, but also mathematically realizable principle: a special case of a deep pattern that unites the dynamics of motion, thought, and change.

19 A Weakness-Based Model of "Broad Appeal" in Aesthetics

One issue that arises frequently in the application of AI methods to creative arts is the difficulty of formulating "aesthetic quality" as a goal for an AI system to pursue as it explores different potential creations. Toward this end, we present here a simple model of "aesthetic quality" via articulating a variation of compression-based theories of aesthetics in the context of weakness measures.

19.1 A Brief History of Mathematical Theories of Aesthetics

The attempt to capture the core of aesthetics in mathematical formulations has a significant history, for instance:

Fechner’s Psychophysics (1860s) Gustav Fechner introduced a quantitative approach to beauty by relating stimulus intensity to subjective sensation, laying groundwork for empirical aesthetics.

The Golden Ratio (Late 19th-Early 20th C.) Artists and mathematicians observed the aesthetic appeal of the ratio $\varphi = \frac{1+\sqrt{5}}{2}$ in nature and art; speculation on its universality influenced later formal models.

Birkhoff’s Aesthetic Measure (1933) George Birkhoff proposed

$$M = \frac{O}{C},$$

where M is aesthetic measure, O is the number of ”distinctive” elements (order) and C is complexity, initiating explicit formulas for beauty.

Information Theory (1940s-1950s) Shannon’s entropy $H = -\sum_i p_i \log p_i$ was applied to pattern and surprise in art, suggesting that moderate unpredictability maximizes aesthetic interest.

Algorithmic Complexity (1960s-1970s) Solomonoff, Kolmogorov and Chaitin formalized description length $K(x)$ of a string x , inspiring the view that beauty correlates with low Kolmogorov complexity relative to prior exposure.

Compression and Curiosity Models (1990s-2000s) Schmidhuber’s framework [Sch97] treats aesthetic reward as compression gain

$$r(x) \propto L_{\text{old}}(x) - L_{\text{new}}(x),$$

emphasizing learning progress.

Recent Directions (2010s-Present)

- *Hybrid Models:* Combining MDL with human preference distributions.
- *Deep-Learning Metrics:* Neural aesthetic predictors trained on large image datasets.

19.2 Schmidhuber’s Compression Theory of Aesthetics

Let a data sequence (e.g. an artwork, sensory input) be denoted by x . A compression algorithm (or model) C assigns a code length $L_C(x)$. Schmidhuber’s core ideas are:

- **Beauty as Learnable Regularity.** An input x is *beautiful* if the observer’s internal compressor can find a shorter description of it than previously:

$$\Delta L_C = L_C^{\text{old}}(x) - L_C^{\text{new}}(x) > 0.$$

- **Aesthetic Reward.** The *subjective* reward r for perceiving x is proportional to the compression gain:

$$r(x) \propto \Delta L_C(x).$$

- **Curiosity and Boredom.** Positive reward arises from discovering new compressible structure; when no further compression gain is possible, reward vanishes and one becomes *bored*.
- **Temporal Dynamics.** In an ongoing sequence $x_{1:t}$, the instantaneous aesthetic value can be measured by the *second derivative* of code length:

$$r_t \approx (L_C(x_{1:t-1}) - 2L_C(x_{1:t}) + L_C(x_{1:t+1})).$$

Looks’ Critique Moshe Looks [Loo10] identified vulnerabilities in raw compression-based reward schemes:

- **PRNG-Seed Exploit.** Arbitrary pseudorandom bit-strings can yield spurious ”compression gain” if the compressor is unconstrained, thus gaming the reward.
- **Unbounded Search-Space.** Without a fixed, finite code-vocabulary, the compressor can invent ad hoc codewords, breaking the intended link between ”beauty” and genuinely simple regularities.
- **Need for Fixing *Both* Cost and Language.** To avoid gaming, one must:
 1. Restrict to a *finite*, prefix-free codebook $\mathcal{L}$.
 2. Use a fixed MDL cost measure within that codebook.

19.3 Weakness-Based Aesthetic Framework

Incorporating the essence of Schmidhuber’s aesthetic theory plus the insights from Moshe Looks’s critique, present a categorical ”aesthetic-weakness” framework in which compression-based beauty arises *via* the Bennett-MDL connection:

1. Aesthetic MDL via a Finite Vocabulary

Aesthetic Language $\mathcal{L}$. Fix a finite, prefix-free "aesthetic vocabulary" $\mathcal{L}$ (e.g. the set of musical motifs, color primitives, melodic transformations) that is dead-code-eliminated.

MDL Complexity. For any artwork $a \in \mathcal{A}$, let

$$C_{\mathcal{L}}(a) = \min\{|p| \mid p \in \mathcal{L}^*, p \text{ generates } a\}.$$

By Bennett's thesis, within any such prefix-free language $\mathcal{L}$, the **weakness** of the subobject $\{a\}$ obeys

$$w_{\mathcal{L}}(a) \propto 2^{-C_{\mathcal{L}}(a)},$$

up to a constant normalization factor.

2. Blending MDL-Weakness with Mind-Average

Rather than setting

$$\text{app}(m, a) \equiv 2^{-C_{\mathcal{L}}(a)},$$

(which would *define* beauty as simplicity), we instead *combine* two sources of weakness:

- **Structural weakness** $w_{\mathcal{L}}(a)$ from MDL in $\mathcal{L}$.
- **Perspective weakness** $w_{\mathcal{M}}(a)$ from averaging human appreciation:

$$w_{\mathcal{M}}(a) = \mathbb{E}_{m \sim \mu} [\text{app}(m, a)].$$

A simple unifying form is to take a **product prior**:

$$w(a) = w_{\mathcal{L}}(a)^{\beta} \times w_{\mathcal{M}}(a)^{1-\beta}, \quad 0 \leq \beta \leq 1.$$

Here β lets you interpolate between pure MDL-weakness and pure population-average weakness.

3. Surprise and Temporal Aesthetics

For a sequence $a_{1:n}$, define

$$w(a_{1:n}) = w_{\mathcal{L}}(a_{1:n})^{\beta} \times w_{\mathcal{M}}(a_{1:n})^{1-\beta}.$$

Then the per-step change

$$\Delta w = w(a_{1:n+1}) - w(a_{1:n})$$

captures "surprising fulfillment" when $\Delta w > 0$ (new structure is both MDL-cheap and widely appreciated), and "tension" when $\Delta w < 0$.

4. Categorical Weakness Perspective

Working again in the quantale $(V, \otimes, \oplus) = ([0, 1], \times, +)$, we can view each of

- the subobject $H_{\mathcal{L}} \subseteq \mathcal{A}$ (the "MDL-weakness relation"),
- the subobject $H_{\mathcal{M}} \subseteq \mathcal{M} \times \mathcal{A}$ (the "appreciation relation"),

as inducing enriched-category weakness measures. The combined prior $w(a)$ is then simply the tensor product of those two weakness maps in the category of enriched presheaves.

5. Why This Resolves Looks's Objection

Looks's PRNG-seed counterexample shows that *raw* compression progress can be gamed by arbitrary pseudorandom structures unless one fixes both (a) the *cost measure* (MDL in $\mathcal{L}$) and (b) the *search-space* (finite, dead-code-free). By:

- Restricting to a finite, prefix-free aesthetic vocabulary $\mathcal{L}$,
- Using Bennett's equivalence to identify $w_{\mathcal{L}}(a) \sim 2^{-C_{\mathcal{L}}(a)}$,
- Combining it multiplicatively with human-average weakness $w_{\mathcal{M}}(a)$,

we guarantee that:

- Pathological pseudo-compression (e.g. PRNG seeds) carry zero or exponentially small $w_{\mathcal{L}}$.
- Genuine aesthetic richness (patterns humans actually like) get boosted both by being compressible in $\mathcal{L}$ and by having high $w_{\mathcal{M}}$.

6. Summary

A refined "weakness-based" aesthetics theory:

- **Leverages MDL** *via* Bennett's prefix-free result, not by redefining appreciation as simplicity.
- **Maintains** the original categorical notion of weakness averaged over minds.
- **Unifies** compression-, appreciation-, and surprise-based accounts of beauty in a single, pluggable framework.

19.4 Neural-symbolic Music Generation via Blended Aesthetic Fitness

To illustrate how these ideas might play out in a particular context, we elaborate how they might be used to guide aesthetics-driven automated music generation.

19.4.1 System Overview

- **Symbolic MDL Component.** A prefix-free music grammar $\mathcal{L}$ of primitives and operators. For any candidate piece a , define

$$C_{\mathcal{L}}(a) = \min_{p \in \mathcal{L}^*} |p|.$$

- **Neural Appreciation Component.** A neural model N trained on a target corpus to predict an appreciation score

$$\text{app}(a) = N(a) \in [0, 1].$$

- **Blended Aesthetic Fitness.** Define

$$w_{\mathcal{L}}(a) = 2^{-C_{\mathcal{L}}(a)}, \quad w_{\mathcal{N}}(a) = \text{app}(a),$$

and combine via

$$\text{fitness}(a) = w_{\mathcal{L}}(a)^{\beta} w_{\mathcal{N}}(a)^{1-\beta}, \quad 0 \leq \beta \leq 1.$$

19.4.2 Symbolic MDL Engine

1. Define primitives (motifs, chords, rhythms) and operators (transpose, invert, concatenate).
2. Assign prefix-free bit-codes to each symbol.
3. Parse or synthesize a to find minimal code p , yielding $C_{\mathcal{L}}(a)$.

19.4.3 Neural Appreciation Model

- Choose architecture (e.g. Transformer over MIDI tokens, CNN over pianoroll).
- Train on labeled corpus to minimize prediction error for human ratings.
- Use $N(a)$ at inference for $\text{app}(a)$.

19.4.4 Evolutionary Programming with EDA

1. Initialize population P of random programs $p \in \mathcal{L}^*$.
2. For each p , render a_p , compute $\text{fitness}(a_p)$.
3. Select top N , fit a probabilistic model D over their code-sequences.
4. Sample new population $P' \sim D$.
5. Repeat until convergence or target quality achieved.

19.4.5 Extensions

- **Reinforcement Learning:** Treat generation as a sequential decision process with reward = $\text{fitness}(a)$.
- **MCTS:** Build compositions operator-by-operator, using UCT guided by fitness.
- **Hierarchical Generation:** First evolve high-level form (e.g. chord progression), then refine details.

20 Measuring the Integrated-Information Aspect of Consciousness via Weakness

Integrated Information Theory (IIT), proposed by Giulio Tononi [Ton04], seeks to quantify consciousness via the scalar Φ , which measures how much information a system generates as a unified whole beyond its parts. Formally, for a system S in state s ,

$$\Phi(S) = \min_{\pi} [I(S) - I_{\pi}(S)],$$

where $I(S)$ is the mutual information between past and future states of S , $I_{\pi}(S)$ is the same for the system partitioned according to π , and the minimum is taken over all bipartitions (the Minimum Information Partition, MIP).

Criticisms leveled against this approach to consciousness include:

- **Computational Intractability.** Exact evaluation of Φ requires searching over exponentially many partitions and computing high-dimensional probability distributions.
- **Ambiguous Partitioning.** Different choices of system elements or time'scales yield widely varying Φ values, undermining robustness.
- **Empirical Validation.** Correlations between Φ and consciousness levels remain limited and sometimes inconsistent across experimental paradigms.
- **Theoretical Assumptions.** IIT's axiomatic derivation (e.g. existence, composition, exclusion) has been challenged as conceptually arbitrary rather than empirically mandated.
- **Conceptual Inadequacy.** IIT is what many consciousness theorists would call a "potential correlate of consciousness" rather than, as Tononi sometimes argues, the essence of consciousness itself. Tononi somewhat bypasses Chalmers' "hard problem of consciousness" – connecting subjective experience with objective observations – in a way that some consider a feature and others a bug

Φ as an Interesting Correlate While Φ offers an intriguing quantitative correlate of certain conscious states – capturing integration and differentiation of neural activity – my own perspective (as presented e.g. in [Goe14]) is that it should be viewed as one among many metrics (e.g. perturbational complexity, neural synchrony, global workspace measures). Its computational and conceptual limitations suggest that Φ is not a fundamental definition of consciousness, but rather a useful tool for probing aspects of integrated information in complex systems.

20.1 Weakness-Based Integrated Information

20.1.1 Clarity via Weakness

Let a system S exhibit behavior B . An explanation E is a causal model mapping interventions on S to outcomes B . Define the *weakness* $w(E) \in (0, 1]$ as our simplicity measure of E . The *clarity* of E is

$$\text{Cl}(E) = -\log w(E).$$

20.1.2 Whole vs. Part Explanations

Let

$$E_S = \arg \max_{E \in \mathcal{E}_S} w(E)$$

be the clearest explanation of B using the whole system. For any partition $\pi = \{P_1, \dots, P_k\}$ of S 's parts, let

$$E_{P_j} = \arg \max_{E \in \mathcal{E}_{P_j}} w(E)$$

be the clearest explanation of B using only part P_j . Then the total clarity of the part-based explanations is

$$\text{Cl}(\pi) = \sum_{j=1}^k \text{Cl}(E_{P_j}).$$

20.1.3 Defining Weakness-Based Integrated Information

Define integrated information for partition π by

$$\Phi_w(\pi) = \text{Cl}(\pi) - \text{Cl}(E_S).$$

Analogous to Tononi's minimum information partition (MIP), set

$$\Phi_w(S) = \min_{\pi} \Phi_w(\pi).$$

A large positive $\Phi_w(S) \gg 0$ indicates that the whole-system explanation is significantly *clearer* (simpler) than any explanation constructed by explaining the parts separately.

20.2 Recovering Tononi's Φ as a Special Case of Weakness-Based Integrated Information

Our weakness-based framework assigns to any causal explanation E a *weakness* $w(E) \in (0, 1]$, with greater weakness corresponding to less simplicity. To recover Tononi's measure in this context, we begin with the assumption that we can define the *clarity* of E by

$$\text{Cl}(E) = -\log w(E),$$

so that more *simple* explanations (smaller w) have higher clarity.

20.2.1 Choosing Weakness via Mutual Information

To connect with Tononi's integrated information, we then make the *special-case* choice

$$w_{\text{info}}(E) = 2^{-I(E)},$$

where $I(E)$ denotes the mutual information captured by explanation E (e.g. between past and future states of the system). Under this definition, we can conceive clarity as

$$\text{Cl}(E) = -\log(2^{-I(E)}) = I(E).$$

Thus, the clarity of any explanation becomes exactly its mutual information.

A Fuller Concept of Clarity Now, above we introduced a richer, causal clarity that rewards not only simplicity but also counterfactual causal strength. I.e. writing

$$s(E) = \text{causal strength of } E, \quad w(E) = \text{weakness of } E,$$

we defined the causal clarity as, for example,

$$\text{Cl}_{\text{causal}}(E) = \frac{s(E)}{w(E)} \quad \text{or} \quad \text{Cl}_{\text{causal}}(E) = \log s(E) - \log w(E).$$

In this framework, the Integrated-Information clarity Cl_{Φ} is recovered as the special case $s(E) = 1$ (or a constant) for all explanations:

$$\text{Cl}_{\Phi}(E) = \log(1) - \log w(E) = -\log w(E) = \text{Cl}_{\text{causal}}(E) \big|_{s(E)=1}.$$

Thus:

- **Simplicity-only clarity** Cl_{Φ} measures how compact an explanation is, ignoring its causal efficacy.
- **Causal clarity** $\text{Cl}_{\text{causal}}$ balances Occam simplicity against the actual strength of the causal link.

For reconstituting Tononi's Φ one can ignore these subtleties and stick with mutual information, but we shall see that taking fuller account of the natural causality yields a potentially improved version of Φ . But for starters let's stick with Tononi-style clarity.

20.2.2 Whole-system vs. Part-Based Clarity

Let S be the full system in state s . In Tononi's approach, the *best* explanation using the entire system achieves clarity

$$\text{Cl}(E_S) = I(S),$$

where $I(S)$ is the mutual information of the whole.

Now consider a partition $\pi = \{P_1, \dots, P_k\}$ of S into disjoint parts. Each part P_j has its own optimal explanation E_{P_j} with clarity

$$\text{Cl}(E_{P_j}) = I(P_j).$$

Since these part-explanations are combined in the simplest additive way, the total clarity for the partition is

$$\text{Cl}(\pi) = \sum_{j=1}^k \text{Cl}(E_{P_j}) = \sum_{j=1}^k I(P_j).$$

20.2.3 Weakness-Based Integrated Information

We define weakness-based integrated information for partition π , Tononi-style, as the difference between whole-system clarity and part-based clarity:

$$\Phi_w(\pi) = \text{Cl}(E_S) - \text{Cl}(\pi) = I(S) - \sum_{j=1}^k I(P_j).$$

Finally, by taking the minimum over all possible partitions (the analog of Tononi's MIP), we recover:

$$\Phi_w(S) = \min_{\pi} \Phi_w(\pi) = \min_{\pi} [I(S) - \sum_j I(P_j)] = \Phi(S),$$

which is precisely Tononi's original Φ .

This derivation shows that Tononi's Φ arises when one *defines* weakness in terms of mutual information. Our general weakness-based framework, however, permits other choices of $w(E)$ grounded in different simplicity measures, yielding a more flexible and theoretically robust family of "Φ-like" metrics.

20.3 Causal Φ

We have seen that Tononi's Φ -clarity is the special case of causal clarity when causal strength is uniform. One can define an improved integrated information metric by replacing Cl_{Φ} with $\text{Cl}_{\text{causal}}$ in the usual partition-vs-whole difference:

$$\Phi_{\text{causal}} = \text{Cl}_{\text{causal}}(E_{\text{whole}}) - \sum_{\text{parts } P} \text{Cl}_{\text{causal}}(E_P).$$

This generalizes Tononi's original Φ by rewarding not only simplicity but also genuine causal coherence of the integrated explanation.

20.4 Observer-Relative Integrated Information and Quantum Graphentropy

So far, following Tononi, we have been considering integrated information Φ as an *objective* measure of how much a system's joint states fail to factorize

across a bipartition. Here we define a *perspectival* Φ – denoted $\Phi_O(S)$ – which quantifies *how much* integrated information system S exhibits *for* a particular observer O . We then show how this observer-relative Φ connects to quantum Φ , distinction graphs and graphentropy, and discuss applications in quantum biology and future quantum AI.

20.4.1 Defining Perspectival Φ

Let S be a system partitioned into subsystems A and B . Observer O is characterized by a (possibly weak or coarse-grained) measurement map f_O on the state space of S , inducing *distinction graphs*:

$$G_O^A, G_O^B, G_O^{AB},$$

where nodes represent outcome-classes and edges indicate pairs of outcomes that O cannot reliably distinguish. Denote by $h(G) \in [0, 1]$ the graphentropy (fraction of distinguished pairs) of graph G . Then we define

$$\Phi_O(S) = h(G_O^A) + h(G_O^B) - h(G_O^{AB}).$$

Equivalently, in terms of weakness $w(G) = 1 - h(G)$,

$$\Phi_O(S) = w(G_O^{AB}) - [w(G_O^A) + w(G_O^B)].$$

Thus Φ_O measures the *excess missing distinctions* that appear only when O views A and B jointly.

20.5 Relation to Perspectival-Causal Φ

The “perspectival Φ_O ” defined via graphentropy is in fact an instance of our general *causal* Φ when one chooses a particular causal-strength function.

To see the connection, observe that letting each explanation E_X for subsystem X correspond to the distinction-graph G_O^X , and choosing the causal-strength factor

$$s(E_X) = 1 - h(G_O^X) = w(G_O^X),$$

we have

$$\text{Cl}_{\text{causal}}(E_X) = \log s(E_X) - \log w(E_X) = \log w(G_O^X) - \log w(G_O^X) = 0.$$

This degenerate choice gives zero clarity for each part, but the whole explanation E_{AB} may have strictly positive causal clarity if its graphentropy differs from the sum of the parts. More meaningfully, one can choose

$$s(E_X) = h(G_O^X)$$

so that causal clarity becomes

$$\text{Cl}_{\text{causal}}(E_X) = \log h(G_O^X) - \log(1 - h(G_O^X)).$$

Plugging into the general Φ_{causal} definition yields

$$\Phi_{\text{causal}} = [\log h(G_O^{AB}) - \log w(G_O^{AB})] - \sum_{X \in \{A, B\}} [\log h(G_O^X) - \log w(G_O^X)],$$

which up-weights integrated distinctions relative to the parts. In the limit of small weakness (high distinction), this reduces to the linear combination of graphtropies defining Φ_O .

Summary Thus the perspectival graphtrophy-based Φ_O is a special case of our general causal- Φ framework, corresponding to particular choices of the causal strength function $s(E)$. More elaborate causal-clarity measures (combining both h and w nonlinearly) will yield refined versions of Φ_O that more sharply distinguish genuine multi-part causal integration.

20.5.1 Connection to Quantum Integrated Information

In quantum IIT one typically uses von Neumann entropies and mutual information on ρ_{AB} . To make Φ operationally *perspectival*, let O implement a POVM $\{E_i\}$ on $\mathcal{H}_A$ and $\{F_j\}$ on $\mathcal{H}_B$. Then:

- The "distinction weight" between outcomes i and i' in A is $\text{Tr}(\rho_A E_i E_{i'})$, and similarly for B .
- One builds quantum distinction graphs G_O^A, G_O^B, G_O^{AB} whose edges link sufficiently overlapping effects.
- The resulting graphtropies h_Q^A, h_Q^B, h_Q^{AB} feed into $\Phi_O(S) = h_Q^A + h_Q^B - h_Q^{AB}$, yielding a measurement-based quantum Φ that respects O 's actual measurement capabilities.

In the limit of a tomographically complete POVM and fine-grained distinction threshold, $\Phi_O(S)$ converges to the standard quantum IIT Φ .

20.5.2 Relation to Distinction Graphs and Graphtrophy

Graphtrophy $h(G)$ quantifies the fraction of distinctions an observer makes among pairs of states. Weakness $w(G) = 1 - h(G)$ is the fraction of pairs left *indistinct*. Perspectival Φ_O is thus the difference between the joint weakness and the sum of the parts' weaknesses:

$$\Phi_O(S) = w(G_O^{AB}) - [w(G_O^A) + w(G_O^B)].$$

This directly ties integrated information to the observer-indexed notion of "missing distinctions" in a unified categorical framework.

20.5.3 Potential Applications

The distinction between standard and perspectival integrated information may seem a bit obscure, but there are practical cases where this will make a huge difference.

In quantum biology settings – e.g. photosynthetic complexes or avian magnetoreceptors, but there are almost surely many further examples not yet charted out, including potentially some closer to the core of human cognition – the “observer” is the protein environment or reaction center, implementing a coarse POVM on the excitonic Hilbert space. Computing Φ_O via graphentropy could reveal how “integrated” the quantum coherence is *from the perspective* of the actual biological detector.

Similarly, for an AI agent built on quantum hardware – as will almost surely be the case a decade or so from now – the agent’s internal measurement devices define its distinction graphs. A perspectival Φ_O then quantifies the system’s self-awareness or integrated information relative to its own sensors and control channels, guiding architectural choices for consciousness-like processing.

By making the measurement lens O explicit and using graphentropy, we obtain a fully *perspectival* yet operationally quantum extension of integrated information, applicable to both biological and artificial quantum systems.

Part IV

Information & Pattern

21 McBride Derivatives of Weakness and Gradient Descent

In this chapter we build out a little more mathematical tooling regarding categorical weakness measures – explaining how to do “differential calculus” style AI using weakness, via introducing the McBride derivative [McB01], an abstract categorical variation on the traditional differential-calculus concept.

21.1 McBride Derivative of Weakness

Conceptually, the McBride derivative of a polynomial functor captures the type of “one-hole contexts” in its values. The beautiful thing here is that this derivative satisfies the same core algebraic identities as the standard derivative in the calculus of numerical functions [McB01].

Since the weakness measure is a polynomial-style functor

$$w(H) = \bigoplus_{(u,v) \in H} \phi(u,v),$$

we can define its derivative to record, for each link in the relation, the residual context obtained by removing that link.

Definition of the Derivative Context Functor Given a relation $H \subseteq U \times U$, define the *weakness-derivative context set*

$$D_w(H) = \coprod_{(u,v) \in H} \{((u,v), H \setminus \{(u,v)\})\}.$$

Each element of $D_w(H)$ is a pair consisting of a "hole" (u,v) and the remaining context $H \setminus \{(u,v)\}$. Thus D_w generalizes the zipper construction to weakness subobjects.

Weighted Detail Coefficients as Derivative Values We further decorate each context by the removed link's weight $\phi(u,v)$. Define the *detail-derivative map*

$$d_w : D_w(H) \longrightarrow V \quad , \quad d_w((u,v), C) = \phi(u,v).$$

When aggregated via the quantale join $\oplus$, this recovers the original weakness:

$$\bigoplus_{c \in D_w(H)} d_w(c) = \bigoplus_{(u,v) \in H} \phi(u,v) = w(H).$$

General vs. Domain-Specific Instantiations The above construction is fully general for any quantale-enriched subobject functor. In particular:

- *Next-Token Prediction:* Let H be the set of observed $(i-1)$ -gram contexts extended by one token. Then $D_w(H)$ enumerates all pairs (c,t) with context-token "holes" and the remaining pairs. The derivative value $\phi(c,t) = P_{\text{emp}}(c,t)$ gives the probability mass of each one-hole event.
- *Humanoid Robotics:* Let H be the kernel relation of a state-abstraction ϕ_i . Then $D_w(H)$ lists each abstracted link (s,t) removed, together with the coarser context. The derivative value $\phi(s,t)$ can be chosen as the occupancy weight $\pi^*(s)\pi^*(t)$ or the Bellman-error contribution of that transition.

Interpretation and Use The McBride derivative of weakness yields a first-order sensitivity analysis: each one-hole context shows how the global weakness would change if a single link were removed. Aggregating these contexts and weights can guide greedy refinement (by picking holes with largest $\phi(u,v)$) or support higher-order transforms (by iterating the derivative to obtain "two-hole" contexts).

Among other applications, this derivative construction opens avenues for:

- *Zipper-style traversals* of relational models, enabling local edits and incremental updates.

- *Higher-order pattern mining*, where repeated derivatives reveal multi-link interactions.
- *Formal sensitivity measures* in quantum or probabilistic settings, by tracking how each measurement context contributes to overall weakness.

21.2 Gradient Descent for Weak Models via McBride Derivative

One of the valuable things we can do with derivatives is optimized functions via gradient descent. Suppose, for instance, we are seeking models that cover observed data while remaining as simple (weak) as possible. By introducing a parametrized family of "soft" models and using the McBride derivative—enumerating one-hole contexts—as a sensitivity measure, we can apply gradient descent to balance data fidelity against weakness.

Parameterized Soft Models Let the dataset be a relation $D \subseteq U \times U$. Define a family of soft inclusion probabilities

$$p_\theta(u, v) \in [0, 1] \quad \text{for each } (u, v) \in U \times U,$$

where $\theta \in \Theta$ are model parameters. The "hard" model is the support of p_θ above a threshold, but we optimize in the continuous regime.

Objective Function We combine a fidelity term $\ell(\theta)$ that penalizes failure to explain data with a weakness regularizer:

$$\ell(\theta) = \sum_{(u,v) \in D} [1 - p_\theta(u, v)]^2, \quad w(\theta) = \sum_{(u,v)} \phi(u, v) p_\theta(u, v),$$

where $\phi(u, v) = \mu(u) \otimes \mu(v)$ is the weight of link (u, v) . The total loss is

$$L(\theta) = \ell(\theta) + \lambda w(\theta).$$

McBride Derivative as Gradient The McBride derivative identifies each link's contribution to weakness. Differentiating,

$$\nabla_\theta w(\theta) = \sum_{(u,v)} \phi(u, v) \nabla_\theta p_\theta(u, v), \quad \nabla_\theta \ell(\theta) = -2 \sum_{(u,v) \in D} [1 - p_\theta(u, v)] \nabla_\theta p_\theta(u, v).$$

Thus

$$\nabla_\theta L = \nabla_\theta \ell(\theta) + \lambda \nabla_\theta w(\theta),$$

which drives parameters to explain data while minimizing aggregate link weights.

Abstract vs. Continuous Smoothness In practical settings, θ may lie in R^d and p_θ be a sigmoid or softmax, yielding standard gradients and backpropagation. Alternatively, one can remain in a purely categorical framework by endowing the parameter space Θ with a differential-category or tangent-category structure. There, the McBride derivative arises as a natural transformation $DF : \Theta \rightarrow T\Theta$, and "gradient descent" is implemented via the category's tangent operations rather than numeric updates.

Algorithmic Sketch

1. Initialize θ (either in R^d or an abstract parameter object).
2. Repeat until convergence:
 - Compute $\nabla_\theta \ell$ and $\nabla_\theta w$ via McBride derivative.
 - Update $\theta \leftarrow \theta - \eta \nabla_\theta L$ (or apply tangent-category step).
3. Extract a discrete weak model by thresholding or taking the coKleisli counit.

This approach provides a principled, differentiable path to discovering weak models across both neural and symbolic domains.

21.3 Differentiable Proof Search via McBride Derivative

Conceptually, we treat a proof as a "weak model" of a dataset: it must cover observed facts while remaining as simple (weak) as possible. By endowing each inference step with a continuous activation parameter, we obtain a soft proof whose coverage and simplicity are both differentiable. We then apply gradient descent on a composite loss that balances data fidelity against a weakness penalty, using the McBride derivative to compute exact link-wise sensitivities.

Soft Proof Parameterization Let the dataset be a set of fact pairs

$$D \subseteq U \times U.$$

A candidate proof comprises m inference steps, indexed by $i = 1, \dots, m$. Introduce parameters

$$\theta = (\theta_1, \dots, \theta_m) \in [0, 1]^m$$

so that step i is "active" with strength θ_i . Define the soft coverage probability for each fact (u, v) by

$$p_\theta(u, v) = 1 - \prod_{i: (u, v) \text{ supported by } i} (1 - \theta_i),$$

which approaches 1 whenever any supporting step is active.

Loss Function We combine a fidelity term that penalizes uncovered data with a weakness regularizer:

$$\ell(\theta) = \sum_{(u,v) \in D} [1 - p_\theta(u, v)]^2, \quad w(\theta) = \sum_{i=1}^m \theta_i \phi_i,$$

where $\phi_i > 0$ measures the complexity of step i . The total loss is

$$L(\theta) = \ell(\theta) + \lambda w(\theta).$$

McBride Derivative as Gradient The McBride derivative enumerates one-hole contexts, yielding the gradient contributions from each step:

$$\frac{\partial \ell}{\partial \theta_i} = -2 \sum_{(u,v) \in D: i \in \text{supp}(u,v)} [1 - p_\theta(u, v)] \prod_{j \neq i} (1 - \theta_j), \quad \frac{\partial w}{\partial \theta_i} = \phi_i.$$

Thus

$$\nabla_\theta L = \nabla_\theta \ell + \lambda \phi,$$

which drives activation of proof steps that most improve coverage per unit complexity.

Continuous vs. Categorical Smoothness In practice, θ may reside in R^m with p_θ implemented by sigmoids or softmax, allowing standard backpropagation. Alternatively, one can work in a purely categorical setting by choosing Θ as a differential category (or tangent category) and interpreting the McBride derivative as a natural transformation $DF : \Theta \rightarrow T\Theta$. Updates then proceed via tangent-category operations rather than numeric subtraction.

Algorithmic Sketch

1. Initialize θ uniformly small.
2. Repeat until convergence:
 - Compute $\nabla_\theta \ell$ and $\nabla_\theta w$ via the McBride derivative.
 - Update $\theta \leftarrow \theta - \eta \nabla_\theta L$ (or apply the tangent step).
3. Extract a discrete proof by selecting steps with θ_i above a threshold.

This framework yields a principled, differentiable proof search that finds weak models of data by following the McBride gradient toward simple yet comprehensive explanations.

21.4 Gradient Descent for Neural-Symbolic Hierarchical Weak Models

Because we are not yet far enough out on a speculative limb here, it seems natural to take the "differential calculus of weakness" one step further and explore how information geometry and natural gradients might fit into the picture. It would seem the use of gradients on the symbolic side gives interesting potentials for fusing together neural and symbolic learning more closely. For instance, in a sequence prediction context, one might seek to train a hybrid model that combines a symbolic multiresolution weakness transform with a neural WaveFormer, so that both the symbolic abstractions and the neural DWT subbands are learned jointly to explain data with minimal complexity. We introduce continuous parameters for both the symbolic and neural sides and derive a unified gradient descent that uses the McBride derivative to guide updates in the symbolic hierarchy and standard backpropagation for the neural network.

Model Parameterization Let the input dataset be D . We construct:

- A symbolic chain of soft subobjects $D_0^{\text{sym}} \supseteq D_1^{\text{sym}} \supseteq \dots \supseteq D_N^{\text{sym}}$, each governed by parameters $\alpha = (\alpha_1, \dots, \alpha_N)$. We write $p_i^{\text{sym}}(u, v; \alpha_i) \in [0, 1]$ for the soft inclusion of link (u, v) at level i .
- A neural WaveFormer with parameters β , which applies a discrete wavelet transform to features, processes low- and high-frequency bands via multi-head attention, and reconstructs via an inverse DWT.

Unified Loss Function We define three loss components:

1. *Symbolic fidelity and weakness:*

$$\ell_{\text{sym}}(\alpha) = \sum_{(u,v) \in D} [1 - p_0^{\text{sym}}(u, v; \alpha_0)]^2, \quad w_{\text{sym}}(\alpha) = \sum_{i=0}^N \sum_{(u,v)} \phi(u, v) p_i^{\text{sym}}(u, v; \alpha_i).$$

2. *Neural reconstruction loss* $\ell_{\text{neu}}(\beta)$ measuring next-token or control error from the WaveFormer.
3. *Coupling loss* $\ell_{\text{couple}}(\alpha, \beta)$ enforcing consistency between symbolic and neural low- and high-frequency streams, e.g. via KL divergence or mean-squared reconstruction:

$$\ell_{\text{couple}} = \sum_{i=0}^N \left\| \text{IDWT}_{\text{neu}}(\beta_i) - \text{IDWT}_{\text{sym}}(\alpha_i) \right\|^2.$$

The total loss is

$$L(\alpha, \beta) = \ell_{\text{sym}}(\alpha) + \lambda w_{\text{sym}}(\alpha) + \ell_{\text{neu}}(\beta) + \gamma \ell_{\text{couple}}(\alpha, \beta).$$

Gradient Computation

- *Symbolic gradient:* We apply the McBride derivative to each soft inclusion p_i^{sym} . Writing $w_{\text{sym}}(\alpha) = \sum_i w_i(\alpha_i)$ with

$$w_i(\alpha_i) = \sum_{(u,v)} \phi(u,v) p_i^{\text{sym}}(u,v; \alpha_i),$$

we compute

$$\nabla_{\alpha_i} w_i = \sum_{(u,v)} \phi(u,v) \nabla_{\alpha_i} p_i^{\text{sym}}(u,v; \alpha_i), \quad \nabla_{\alpha_i} \ell_{\text{sym}} = -2 \sum_{(u,v) \in D} [1 - p_0^{\text{sym}}(u,v)] \nabla_{\alpha_i} p_0^{\text{sym}}(u,v).$$

Additional terms arise from ℓ_{couple} via $\partial \ell_{\text{couple}} / \partial \alpha_i$.

- *Neural gradient:* We backpropagate through the WaveFormer to compute $\nabla_{\beta} \ell_{\text{neu}}$ and $\nabla_{\beta} \ell_{\text{couple}}$.

Joint Update We then perform simultaneous gradient steps:

$$\begin{aligned} \alpha &\leftarrow \alpha - \eta [\nabla_{\alpha} \ell_{\text{sym}} + \lambda \nabla_{\alpha} w_{\text{sym}} + \gamma \nabla_{\alpha} \ell_{\text{couple}}], \\ \beta &\leftarrow \beta - \eta [\nabla_{\beta} \ell_{\text{neu}} + \gamma \nabla_{\beta} \ell_{\text{couple}}]. \end{aligned}$$

By iterating these updates, the model discovers a hierarchy of weak symbolic abstractions and a complementary neural WaveFormer that jointly optimize data fit and simplicity.

21.4.1 Natural Gradient Search via McBride Derivative and Information Geometry

Conceptually, natural gradient methods adjust parameters along the steepest descent direction measured not in Euclidean space but in a Riemannian metric that reflects the model’s sensitivity. By combining the McBride derivative of the weakness functor with an information-geometric metric on the parameter manifold, we obtain ”Occam-aware” natural gradients that respect both data fidelity and simplicity.

Parameterized Weak Models and Loss Let D be the dataset and $p_{\theta}(u,v) \in [0,1]$ a soft model of links (u,v) , governed by parameters $\theta \in \Theta$. Define the composite loss

$$L(\theta) = \underbrace{\sum_{(u,v) \in D} [1 - p_{\theta}(u,v)]^2}_{\ell(\theta)} + \lambda \underbrace{\sum_{(u,v)} \phi(u,v) p_{\theta}(u,v)}_{w(\theta)},$$

where $\phi(u,v) = \mu(u) \otimes \mu(v)$ weights each link. Its gradient is

$$\nabla_i L = -2 \sum_D [1 - p_{\theta}(u,v)] \partial_i p_{\theta}(u,v) + \lambda \sum_{u,v} \phi(u,v) \partial_i p_{\theta}(u,v),$$

and the McBride derivative identifies each $\partial_i p_{\theta}(u,v)$ term.

Fisher-Metric Natural Gradient If p_θ defines a probability distribution over links, we can endow Θ with the Fisher information metric

$$G_{ij}(\theta) = \sum_{u,v} p_\theta(u,v) \partial_i \log p_\theta(u,v) \partial_j \log p_\theta(u,v).$$

The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta G(\theta)^{-1} \nabla L(\theta).$$

This rescales the McBride-derived gradient by the local curvature of the model manifold, prioritizing moves that change the model less in information-theoretic terms.

Wasserstein-Metric Natural Gradient Alternatively, if p_θ lives on a discrete space with an underlying cost metric $d((u,v), (u',v'))$, one can equip Θ with the 2-Wasserstein metric tensor [CL20]. Its approximate Riemannian metric is

$$G_{ij}(\theta) = \sum_{u,v} \sum_{u',v'} W(e_{(u,v)}, e_{(u',v')}) \partial_i p_\theta(u,v) \partial_j p_\theta(u',v'),$$

where W is the discrete 2-Wasserstein inner product. The corresponding natural gradient step

$$\theta^{(t+1)} = \theta^{(t)} - \eta G(\theta)^{-1} \nabla L(\theta)$$

yields updates that move p_θ along geodesics in probability-mass transport, respecting the "mass-movement" cost structure.

Implementation Notes Both variants require efficient computation of $G(\theta)^{-1} \nabla L$. In practice one uses low-rank approximations or Krylov solvers for Fisher matrices, and Sinkhorn-based approximations for Wasserstein metrics. The McBride derivative supplies the exact $\partial_i p_\theta$ factors needed for either geometry, unifying Occam-driven simplicity with principled natural-gradient optimization.

21.4.2 Curvature in the One-Hole-Context Manifold

It's perhaps worth pausing a moment to take a conceptual view of some of this crunchy math. In particular, what are we actually doing when we're taking curvature in the space of data structures, building onto the McBride derivative and its interpretation of derivatives as sets of one-hole contexts?

Conceptually, curvature measures how the sensitivity of weakness to single-link edits (the McBride derivative) itself changes as we move through the model space. Just as the Hessian in classical optimization captures second-order effects of gradient variation, curvature in the one-hole-context manifold captures non-additive interactions between different "holes" (links).

Second-Order Sensitivity Let $p_\theta(u, v)$ be our soft inclusion probability and write the McBride derivative as the vector of partials $\partial_i p_\theta(u, v)$. The curvature tensor

$$H_{ijk} = \frac{\partial^2 w(\theta)}{\partial \theta_j \partial \theta_i}$$

measures how the contribution of hole (u_i, v_i) to weakness changes when we also perturb parameters in direction j . Large entries H_{ijk} indicate that removing one link dramatically alters the importance of another.

Flat vs. High Curvature If curvature is near zero, the one-hole contexts behave independently and first-order McBride updates remain accurate over a region. High curvature signals strong interactions: targeting one link shifts the sensitivity landscape for others, necessitating smaller steps or geometry-aware updates.

Implications for Update Rules Regions of high curvature warn against naive Euclidean gradient steps. By measuring curvature in the same parameter manifold as the McBride derivative, we know when to switch from first-order McBride descent to natural-gradient or trust-region methods that respect the underlying geometry of link-context interactions.

22 Quantale Pattern Theory

The cognitive theory presented in [Goe21a] and earlier works, and instantiated concretely in the Hyperon AGI design, is founded on the notion that one can get a lot of mileage from viewing cognitive structure and dynamics in terms of *patterns*. A formal definition of pattern has been presented in [Goe20b] and elsewhere and has been used as motivation for multiple aspects of the Hyperon system. Here we port this definition of pattern into the quantale setting and show that, in this context, it can very conveniently be interrogated regarding its properties, and then applied directly in various practical cases.

22.1 Quantale Patterns

Let $(V, \otimes, \oplus, e)$ be a commutative quantale and X a universe of "elements." A V -*predicate* is a function

$$m : X \rightarrow V.$$

Given a *valuation* $\varphi : X \rightarrow V$, the *weakness* of m is defined by joining over all points:

$$w_\varphi(m) = \bigoplus_{x \in X} [\varphi(x) \otimes m(x)].$$

Composite Weakness Fix two predicates $y, z : X \rightarrow V$. Define their *combinational weakness* by

$$h(y, z) = w_\varphi(y) \otimes w_\varphi(z).$$

Pattern Intensity Since V is a quantale, it admits a *residuated implication* $\multimap$ satisfying

$$a \otimes b \leq c \iff b \leq a \multimap c.$$

We then set the *pattern intensity* of the pair (y, z) in the context m to be

$$I_{y,z}(m) = h(y, z) \multimap w_\varphi(m).$$

We declare (y, z) to be a *pattern in m* exactly when

$$I_{y,z}(m) \geq e.$$

Relation to Simplicity-Theory Patterns This definition parallels the core "pattern intensity" construction in the Compositional Simplicity Measure (CoSM) framework [Goe20b], where one compares a baseline simplicity measure $\sigma_1(m)$ against a composite cost $h_{12}(y, z)$ to form

$$I(y, z \mid m) = \frac{\sigma_1(m) - h_{12}(y, z)}{\sigma_1(m)},$$

declaring a pattern when this ratio is positive. In our quantale-enriched version, simplicity measures are replaced by weakness values in V , and the fraction is captured by the quantale implication $\multimap$.

22.2 Emergent Patterns in a Quantale Framework

Let $(V, \otimes, \oplus, e)$ be a commutative quantale, X a universe, and $\varphi : X \rightarrow V$ a valuation. For any V -predicates $y, z, m : X \rightarrow V$ define:

$$w_\varphi(p) = \bigoplus_{x \in X} [\varphi(x) \otimes p(x)], \quad h(y, z) = w_\varphi(y) \otimes w_\varphi(z).$$

Pattern Intensity The intensity of a candidate pattern (y, z) in context m is

$$I_{y,z}(m) = h(y, z) \multimap w_\varphi(m),$$

where $\multimap$ is the quantale-residuation.

Likewise, the individual intensities are

$$I_y(m) = w_\varphi(y) \multimap w_\varphi(m), \quad I_z(m) = w_\varphi(z) \multimap w_\varphi(m).$$

Synergy and Emergence Define the *synergy measure*

$$\sigma_{y,z}(m) = (I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m).$$

We say that (y, z) is an *emergent pattern in m* precisely if

$$\sigma_{y,z}(m) \geq e,$$

i.e. when the composite pattern intensity $I_{y,z}(m)$ exceeds what would be predicted by the individual intensities $I_y(m)$ and $I_z(m)$ alone.

Interpretation This captures the idea that a pattern (y, z) is *emergent* in m when

$$I_{y,z}(m) \not\leq I_y(m) \otimes I_z(m),$$

so that knowing y and z together yields strictly more "weakness-explaining power" than combining their separate contributions.

22.3 Algebraic Laws for Pattern Intensity and Emergent Patterns

Let $(V, \otimes, \oplus, e)$ be a commutative quantale, X a set, and $\varphi : X \rightarrow V$ a valuation. For any predicates $y, z, m : X \rightarrow V$ define

$$w(p) = \bigoplus_{x \in X} [\varphi(x) \otimes p(x)], \quad h(y, z) = w(y) \otimes w(z),$$

and the *pattern intensities*

$$I_y(m) = w(y) \multimap w(m), \quad I_z(m) = w(z) \multimap w(m), \quad I_{y,z}(m) = h(y, z) \multimap w(m),$$

where $\multimap$ is the quantale residuation satisfying

$$a \otimes b \leq c \iff b \leq a \multimap c.$$

Monotonicity

$$y \leq y' \Rightarrow I_y(m) \leq I_{y'}(m), \quad z \leq z' \Rightarrow I_{y,z}(m) \leq I_{y',z'}(m), \quad m \leq m' \Rightarrow I_{y,z}(m) \leq I_{y,z}(m').$$

Symmetry and Idempotence

$$I_{y,z}(m) = I_{z,y}(m), \quad I_{y,y}(m) = I_y(m).$$

Upper and Lower Bounds By residuation and monotonicity,

$$I_{y,z}(m) \otimes h(y, z) \leq w(m) \implies I_{y,z}(m) \leq w(m),$$

and since $h(y, z) \leq w(y)$,

$$I_{y,z}(m) \leq w(y) \multimap w(m) = I_y(m), \quad I_{y,z}(m) \leq I_z(m).$$

Join-Residuation Law For any y, y', z ,

$$(h(y, z) \oplus h(y', z)) \multimap w(m) = (h(y, z) \multimap w(m)) \otimes (h(y', z) \multimap w(m)),$$

so

$$I_{y \oplus y', z}(m) = I_{y, z}(m) \otimes I_{y', z}(m).$$

Tensor-Residuation Law Similarly, for any y, z, z' ,

$$I_{y, z \otimes z'}(m) = (h(y, z) \otimes w(z')) \multimap w(m) = w(z') \multimap I_{y, z}(m).$$

6. Emergence Criterion Define the *synergy measure*

$$\sigma_{y,z}(m) = (I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m).$$

Then

$$\sigma_{y,z}(m) \geq e \iff I_y(m) \otimes I_z(m) \leq I_{y,z}(m),$$

i.e. (y, z) is *emergent* in m exactly when their joint intensity exceeds the product of individual intensities.

7. Transitivity / Composition For any triple y, z, u ,

$$I_{y,z \otimes u}(m) = w(z \otimes u) \multimap I_y(m) = (w(z) \otimes w(u)) \multimap I_y(m) = I_{z,u}(m \mid y),$$

interpreting $I_{z,u}(m \mid y)$ as the intensity of (z, u) conditioned on y .

These identities show that pattern intensities and emergent patterns satisfy elegant algebraic laws—monotonicity, symmetry, idempotence, residuation-of-joins and tensors, and a crisp emergence criterion—in any commutative quantale.

22.4 McBride Derivatives of Pattern and Emergent Intensities

Let $(V, \otimes, \oplus, e)$ be a commutative quantale, X a universe, and $\varphi : X \rightarrow V$ a valuation. For any predicate $p : X \rightarrow V$ define its weakness

$$w(p) = \bigoplus_{x \in X} [\varphi(x) \otimes p(x)].$$

Fix two predicates $y, z : X \rightarrow V$. Recall the pattern intensities

$$I_y(m) = w(y) \multimap w(m), \quad I_{y,z}(m) = (w(y) \otimes w(z)) \multimap w(m),$$

and the emergent synergy

$$\sigma_{y,z}(m) = (I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m),$$

where $\multimap$ is the residuation satisfying $a \otimes b \leq c \iff b \leq a \multimap c$.

Derivative of Weakness The McBride derivative of $w(m)$ with respect to the inclusion of $x \in X$ is

$$\partial_x w(m) = \varphi(x).$$

Derivatives of Pattern Intensities Using residuation, the derivative of $I_y(m) = w(y) \multimap w(m)$ is

$$\partial_x I_y(m) = \partial_x (w(y) \multimap w(m)) = w(y) \multimap (\partial_x w(m)) = w(y) \multimap \varphi(x).$$

Similarly, for the joint intensity

$$\partial_x I_{y,z}(m) = (w(y) \otimes w(z)) \multimap \varphi(x).$$

Derivative of Emergent Synergy Writing $u = I_y(m) \otimes I_z(m)$ and $v = I_{y,z}(m)$, we have

$$\sigma_{y,z}(m) = u \multimap v.$$

By the chain rule for residuation derivatives,

$$\partial_x \sigma_{y,z}(m) = u \multimap (\partial_x v) = (I_y(m) \otimes I_z(m)) \multimap (w(y) \otimes w(z)) \multimap \varphi(x).$$

Linearity over Joins Since ∂_x commutes with $\oplus$, for any two contexts m, m' ,

$$\partial_x I_y(m \oplus m') = (w(y) \multimap \varphi(x)) \oplus (w(y) \multimap \varphi(x)) = w(y) \multimap \varphi(x).$$

Analogous identities hold for $I_{y,z}$ and $\sigma_{y,z}$.

Leibniz-Style Rule for Residuation For any $a, b, c \in V$, the residuation obeys

$$\partial_a (a \multimap b) = (a \multimap b) \multimap (\partial_a a),$$

so one may compute mixed derivatives of pattern intensities with respect to changes in y or z by similar formulas.

These derivative identities form the basis of a "pattern calculus" in which one can write differential equations such as

$$\frac{d}{dt} I_{y,z}(m(t)) = \sum_{x \in X} \dot{m}_x(t) (w(y) \otimes w(z)) \multimap \varphi(x),$$

or for synergy,

$$\frac{d}{dt} \sigma_{y,z}(m(t)) = \sum_{x \in X} \dot{m}_x(t) (I_y \otimes I_z) \multimap (w(y) \otimes w(z)) \multimap \varphi(x),$$

tying pattern-emergence dynamics directly to McBride derivatives of weakness.

22.5 An Exponential-Growth Analogue in the McBride Pattern Calculus

Suppose each membership degree $m_x(t)$ of a context predicate $m : X \rightarrow V$ obeys the simple growth law

$$\frac{d}{dt} m_x(t) = \kappa \otimes m_x(t), \quad \kappa \in V$$

for all $x \in X$. Then by the McBride derivative of the weakness map

$$w(m) = \bigoplus_{x \in X} [\varphi(x) \otimes m_x], \quad \partial_x w(m) = \varphi(x),$$

we get

$$\frac{d}{dt} w(m(t)) = \bigoplus_{x \in X} \varphi(x) \otimes \frac{d}{dt} m_x(t) = \bigoplus_{x \in X} \varphi(x) \otimes (\kappa \otimes m_x(t)) = \kappa \otimes w(m(t)).$$

Hence $w(m(t))$ satisfies a quantale-enriched exponential-growth ODE.

Pattern Intensity Growth For a fixed pattern y , its intensity in $m(t)$ is

$$I_y(m(t)) = w(y) \multimap w(m(t)).$$

Using the residuation law

$$a \multimap (b \otimes c) = c \otimes (a \multimap b),$$

we obtain

$$\frac{d}{dt} I_y(m(t)) = w(y) \multimap \frac{d}{dt} w(m(t)) = w(y) \multimap (\kappa \otimes w(m(t))) = \kappa \otimes I_y(m(t)).$$

Thus $I_y(m(t))$ grows "exponentially" in the quantale sense:

$$I_y(m(t)) = \exp_{\otimes}(\kappa t) \otimes I_y(m(0)).$$

Emergent Pattern Growth Similarly, the joint intensity

$$I_{y,z}(m(t)) = (w(y) \otimes w(z)) \multimap w(m(t))$$

and the emergent synergy

$$\sigma_{y,z}(m(t)) = (I_y(m(t)) \otimes I_z(m(t))) \multimap I_{y,z}(m(t))$$

each satisfy the same exponential-growth law

$$\frac{d}{dt} [(\text{pattern intensity})] = \kappa \otimes [(\text{pattern intensity})].$$

Interpretation This shows that, under uniform self-reinforcing growth of the base predicate m , all derived pattern and emergence metrics evolve according to a simple quantale ODE

$$\dot{I} = \kappa \otimes I,$$

whose solution is the quantale-exponential advance $I(t) = \exp_{\otimes}(\kappa t) \otimes I(0)$.

22.5.1 Epidemic Spread as a Pattern-Calculus Example

Let X be a set of geographic regions. We track an epidemic via:

- A *context predicate* $m(t) : X \rightarrow [0, \infty)$ where

$$m_x(t) = \text{number of active cases in region } x \text{ at time } t.$$

- Early-stage exponential growth in each region:

$$\frac{d}{dt} m_x(t) = k m_x(t), \quad k > 0.$$

Total Weakness (Prevalence) Define the total prevalence as the weakness of $m(t)$ with unit valuation:

$$w(m(t)) = \sum_{x \in X} m_x(t).$$

Then

$$\frac{d}{dt} w(m(t)) = \sum_{x \in X} k m_x(t) = k w(m(t)),$$

so total cases grow exponentially:

$$w(m(t)) = w(m(0)) e^{k t}.$$

Single-Symptom Pattern Intensity Let

$$y(x) = \begin{cases} 1, & \text{if region } x \text{ has a fever-screening station,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$w(y) = \sum_{x \in X} y(x) = \#\{x : y(x) = 1\},$$

and the pattern intensity is

$$I_y(m(t)) = \frac{w(m(t))}{w(y)} = \frac{\sum_x m_x(t)}{\#\{x : y(x) = 1\}} \propto e^{k t}.$$

Joint-Symptom Pattern and Emergence Similarly, let

$$z(x) = \begin{cases} 1, & \text{if region } x \text{ has a cough-clinic,} \\ 0, & \text{otherwise,} \end{cases}$$

so $w(z) = \#\{x : z(x) = 1\}$ and

$$I_{y,z}(m(t)) = \frac{w(m(t))}{w(y) w(z)} \propto e^{k t}.$$

Their *emergent synergy* is

$$\sigma_{y,z}(m(t)) = \frac{I_{y,z}(m(t))}{I_y(m(t)) I_z(m(t))} = \frac{\sum_x m_x(t)}{\#\{y = 1\} \#\{z = 1\}} \Big/ \left(\frac{\sum_x m_x(t)}{\#\{y = 1\}} \times \frac{\sum_x m_x(t)}{\#\{z = 1\}} \right) = \frac{1}{\sum_x m_x(t)} \propto e^{-k t}.$$

Thus:

- Both I_y and $I_{y,z}$ grow as $e^{k t}$.
- The emergent synergy $\sigma_{y,z}$ decays as $e^{-k t}$.

Interpretation

- Early in the outbreak, the co-occurrence of fever and cough is highly emergent (high $\sigma_{y,z}$), making it a sharp early-warning signal.
- As the epidemic spreads, both patterns individually track prevalence strongly, but their joint pattern loses discriminative power, captured by the decay of $\sigma_{y,z}$.

22.5.2 Memetic Pattern Spread in a Probabilistic Program Space

Let M be a finite set of probabilistic programs, each encoding a distinct "meme." We model the *frequency* or *strength* of meme $p \in M$ at time t by

$$m_p(t) \geq 0.$$

Assume that in the early "viral" phase each meme replicates independently at a constant rate $k > 0$:

$$\frac{d}{dt} m_p(t) = k m_p(t), \quad p \in M.$$

Total Memetic Weakness Equip M with the trivial valuation $\varphi(p) = 1$. The total memetic "weakness" (aggregate prevalence) is

$$w(m(t)) = \sum_{p \in M} \varphi(p) m_p(t) = \sum_{p \in M} m_p(t).$$

By summing the ODEs,

$$\frac{d}{dt} w(m(t)) = \sum_p k m_p(t) = k w(m(t)),$$

so

$$w(m(t)) = w(m(0)) e^{k t}.$$

Single-Meme Pattern Intensity For a distinguished meme $q \in M$, define the indicator predicate

$$y_q(p) = \begin{cases} 1, & p = q, \\ 0, & p \neq q. \end{cases}$$

Then

$$w(y_q) = \sum_p y_q(p) = 1,$$

and the pattern intensity of meme q is

$$I_{y_q}(m(t)) = w(m(t)) \multimap w(y_q) = \frac{w(m(t))}{1} = w(m(t)) \propto e^{k t}.$$

Co-Meme Pattern and Emergence Pick two memes $q, r \in M$. Let

$$z_r(p) = \begin{cases} 1, & p = r, \\ 0, & p \neq r. \end{cases}$$

Then the joint pattern intensity is

$$I_{y_q, z_r}(m(t)) = (w(y_q) \otimes w(z_r)) \multimap w(m(t)) = \frac{w(m(t))}{1 \cdot 1} \propto e^{k t}.$$

Their *emergent synergy* is

$$\sigma_{y_q, z_r}(m(t)) = (I_{y_q}(m(t)) \otimes I_{z_r}(m(t))) \multimap I_{y_q, z_r}(m(t)) = \frac{w(m(t))}{w(m(t)) w(m(t))} = \frac{1}{w(m(t))} \propto e^{-k t}.$$

Interpretation

- *Individual meme intensities* I_{y_q} and I_{z_r} grow exponentially, reflecting viral replication.
- *Emergent synergy* σ_{y_q, z_r} decays exponentially: the co-occurrence of two specific memes is most informative early on, then becomes diluted as the overall meme pool explodes.
- This formalism can guide strategies for meme detection and intervention: target combined patterns when σ is high to catch emerging trends, then switch to individual tracking as memes go mainstream.

22.5.3 Structuring a Meme Detection and Intervention System via Quantale Weakness

We work in the commutative quantale

$$V = ([0, \infty), \times, +), \quad \varphi(p) = 1 \quad \forall p \in M,$$

where M is the set of tracked memes.

Meme Representation and State

- *Universe of memes*: a (growing) set M of probabilistic programs.
- *Memetic strength*: for each $p \in M$, let

$$m_p(t) = \text{estimated frequency of meme } p \text{ at time } t.$$

- Early "viral" growth assumption:

$$\dot{m}_p(t) = k m_p(t), \quad k > 0.$$

Core Weakness and Pattern Intensities

- *Total memetic weakness:*

$$W(t) = \sum_{p \in M} m_p(t), \quad \dot{W} = k W.$$

- *Single-meme intensity:* for any meme q ,

$$I_q(t) = \frac{W(t)}{1} = W(t) \propto e^{kt}.$$

- *Joint-meme synergy:* for two memes q, r ,

$$I_{q,r}(t) = \frac{W(t)}{1 \cdot 1} = W(t), \quad \sigma_{q,r}(t) = \frac{I_{q,r}(t)}{I_q(t) I_r(t)} = \frac{1}{W(t)} \propto e^{-kt}.$$

Detection Logic

- *Emergence threshold:* choose $\tau > 0$. Flag (q, r) emergent when

$$\sigma_{q,r}(t) > \tau \iff W(t) < 1/\tau.$$

- *Priority ranking:* sort co-meme pairs by $\sigma_{q,r}(t)$ (highest first).

Intervention Strategies

- *Early-warning alerts:* trigger human or automated review when $\sigma_{q,r} > \tau$.
- *Resource allocation:* allocate monitoring resources to pairs with largest $|\dot{\sigma}_{q,r}(t)|$.
- *Adaptive filtering:* shift from joint-pattern detection (σ) to single-meme volume (I_q) as $W(t)$ grows.

System Architecture

- *Data ingestion:* real-time streams $\rightarrow$ pattern-matching $\rightarrow$ update $m_p(t)$.
- *Quantale-weakness engine:* compute $W(t)$, $I_q(t)$, $\sigma_{q,r}(t)$ on sliding windows.
- *Alert API:* publish emergent patterns to dashboards or moderation pipelines.
- *Feedback loop:* human labels adjust τ , seed new memes, refine pattern definitions.

This design leverages the quantale-weakness and pattern-calculus to time alerts precisely at the leading edge of meme outbreaks, allocate resources dynamically, and provide a principled transition as memes become widespread.

22.6 A Delay-Logistic ODE with Chaotic Dynamics in the McBride Pattern Calculus

Just for fun let's also write out a slightly more complex differential equation in this exotic calculus setting... a simple delay-logistic equation which is known to yield chaotic dynamics in ordinary settings.

Let $(V, \otimes, \oplus, e) = ([0, \infty), \times, +, 1)$, X a set, and $\varphi(x) = 1 \ \forall x \in X$. Fix a "carrying-capacity" constant $K > 0$ in V and a delay $\tau > 0$. For each $x \in X$ let $m_x(t) \geq 0$ evolve by the delay-logistic law

$$\frac{d}{dt} m_x(t) = r \otimes m_x(t) \otimes (K \multimap w(m(t-\tau))), \quad r > 0,$$

where

$$w(m(t)) = \bigoplus_{u \in X} [\varphi(u) \otimes m_u(t)] = \sum_{u \in X} m_u(t),$$

and $\multimap$ is the residuation satisfying $a \otimes b \leq c \iff b \leq a \multimap c$.

Induced Delay-Logistic for Total Weakness By McBride differentiation ($\partial_x w(m) = \varphi(x) = 1$) and summing over x ,

$$\frac{d}{dt} W(t) = \frac{d}{dt} w(m(t)) = \sum_{x \in X} \frac{d}{dt} m_x(t) = r W(t) (K \multimap W(t-\tau)),$$

where $W(t) = w(m(t))$. This is the classical delay-logistic equation

$$\dot{W} = r W(t) \left(1 - \frac{W(t-\tau)}{K}\right),$$

up to rewriting $K \multimap W = K^{-1}(K - W)$ in V . For sufficiently large τ this equation exhibits chaotic solutions.

Pattern-Intensity Dynamics For a fixed predicate $y : X \rightarrow \{0, 1\}$ with $w(y) = \sum_x y(x)$, its McBride-pattern intensity

$$I_y(t) = w(y) \multimap W(t) = \frac{W(t)}{w(y)}$$

satisfies the same delay-logistic form:

$$\frac{d}{dt} I_y(t) = r I_y(t) (K \multimap W(t-\tau)).$$

Chaotic Regime It is well known that the delay-logistic equation

$$\dot{z}(t) = r z(t) (1 - z(t-\tau)/K)$$

admits chaotic solutions when r and τ exceed certain thresholds. It may be that the McBride pattern-calculus formulation yields a natural route to chaotic pattern-intensity dynamics, which can be used to model unpredictable, bursty emergence of complex patterns in real-world systems.

23 Formulating Cognitive Synergy in Terms of Weakness

Contemporary cognitive architectures often organize knowledge into *multiple distinct memory stores*: for example, a procedural-knowledge store (programs in a domain-specific language), a declarative-knowledge store (logical statements), a perceptual-knowledge store (images, video clips with algebraic composition), and an attentional-knowledge store (attention distributions over concepts). Each such store can be modeled as a category

$$\mathcal{M}_i, \quad i = 1, \dots, n,$$

whose objects are memory elements and whose morphisms represent transformations (e.g. inference steps, perceptual mappings, memory lookups) endowed with quantitative costs or reliabilities.

“Cognitive synergy” refers to the observation that in most complex cognitive systems, carrying out sophisticated cognitive operations involves intensive interoperation between the different memory stores and associated transformations – so that the cognitive processes associated with the different stores are frequently exchanging intermediate representations, rather than interacting like black boxes exchanging only final conclusions.

23.1 Weakness-Based Formalization of Cognitive Synergy

Let $\{\mathcal{M}_i\}_{i=1}^n$ be categories (each enriched in a commutative quantale $(Q, \leq, \otimes, e)$) representing distinct memory stores of a cognitive system. There is a monoidal product

$$\mathcal{M} = \bigotimes_{i=1}^n \mathcal{M}_i$$

describing their joint use, and a category $\mathcal{B}$ of observable behaviors.

A *weakness* functor

$$W : \mathcal{C}^{\text{op}} \rightarrow Q$$

assigns to each causal explanation E (an object or morphism in a suitable category $\mathcal{C}$) a value $W(E) \in Q$. Its *clarity* is

$$\text{Cl}(E) = -\log W(E) \quad (\text{using a chosen logarithm on } Q).$$

23.1.1 Best Explanations over Subsets of Stores

For each subset $S \subseteq \{1, \dots, n\}$, let

$$\mathcal{M}_S = \bigotimes_{i \in S} \mathcal{M}_i \quad \text{and} \quad E_S = \arg \max_{E \in \mathcal{E}_{\mathcal{M}_S}} W(E)$$

be the clearest explanation of a given behavior $b \in \mathcal{B}$ using only the stores in S . Denote its clarity by

$$\text{Cl}(S) = \text{Cl}(E_S) = -\log W(E_S).$$

23.1.2 Cognitive Synergy Measure

Define *cognitive synergy* as the advantage in clarity gained by using all stores jointly rather than any partition thereof. Let Π be the set of all partitions of $\{1, \dots, n\}$. Then

$$\Sigma = \min_{\pi \in \Pi} \left[\text{Cl}(\{1, \dots, n\}) - \sum_{S \in \pi} \text{Cl}(S) \right].$$

If $\Sigma > 0$, the system achieves its clearest (lowest-weakness) explanations only by synergistically combining multiple memory stores.

23.1.3 Theorem: Lax Monoidality Implies Synergy

Theorem. If W is lax monoidal, then for any non-trivial partition π ,

$$W(E_{\{1, \dots, n\}}) \geq \bigotimes_{S \in \pi} W(E_S) \implies \text{Cl}(\{1, \dots, n\}) \leq \sum_{S \in \pi} \text{Cl}(S),$$

so $\Sigma \geq 0$. Strict inequality in the lax-monoidality law yields $\Sigma > 0$, i.e. genuine cognitive synergy.

This construction shows how "cognitive synergy" emerges when the joint use of diverse memory stores yields strictly simpler (more clear) causal explanations than any separate-store combination. It generalizes our Φ -style integrated-information measures to a multi-store, enriched-category setting.

23.2 Relationship Between Cognitive Synergy and Φ_w

Let $\{\mathcal{M}_i\}_{i=1}^n$ be our n memory-store categories, and write

$$\mathcal{M} = \bigotimes_{i=1}^n \mathcal{M}_i$$

for their joint category. Fix a lax-monoidal weakness functor

$$W : \mathcal{C}^{\text{op}} - (Q, \leq, \otimes, e), \quad \text{Cl}(E) = -\log W(E).$$

For any subset $S \subseteq \{1, \dots, n\}$, let

$$E_S = \arg \max_{E \in \mathcal{E}_{\mathcal{M}_S}} W(E), \quad \text{Cl}(S) = \text{Cl}(E_S).$$

We define

$$\Sigma = \text{Cl}(\{1, \dots, n\}) - \sum_{i=1}^n \text{Cl}(\{i\}),$$

the *cognitive synergy* measured by comparing the joint system to the partition into single-store components. Meanwhile, our integrated information is

$$\Phi_w(\mathcal{M}) = \min_{\pi} \left[\text{Cl}(\{1, \dots, n\}) - \sum_{S \in \pi} \text{Cl}(S) \right],$$

where π ranges over *all* partitions of $\{1, \dots, n\}$.

23.3 Theorem

$$\Phi_w(\mathcal{M}) \leq \Sigma.$$

23.4 Proof

By definition of the minimum,

$$\Phi_w(\mathcal{M}) = \min_{\pi} \left[\text{Cl}(\{1, \dots, n\}) - \sum_{S \in \pi} \text{Cl}(S) \right].$$

In particular, one such partition is the *singleton partition* $\pi_{\text{single}} = \{\{1\}, \{2\}, \dots, \{n\}\}$. Hence

$$\Phi_w(\mathcal{M}) \leq \text{Cl}(\{1, \dots, n\}) - \sum_{S \in \pi_{\text{single}}} \text{Cl}(S) = \text{Cl}(\{1, \dots, n\}) - \sum_{i=1}^n \text{Cl}(\{i\}) = \Sigma.$$

□

23.4.1 Corollaries and Discussion

- If $\Phi_w(\mathcal{M}) > 0$, then necessarily $\Sigma > 0$. Thus positive integrated information implies positive cognitive synergy.
- Equality $\Phi_w(\mathcal{M}) = \Sigma$ holds exactly when the *minimum information partition* (MIP) is the singleton partition, i.e. no coarser partition yields a smaller clarity difference.
- In general, Σ may exceed Φ_w , since other partitions (grouping some memory stores together) might yield an even larger gap between joint and part-wise clarity.

This result situates cognitive synergy –defined via the natural “singletons” partition – as an *upper bound* on the fully minimized integrated information Φ_w , clarifying their precise relationship.

To understand this fully on a conceptual level: Think of Σ as the synergy you get when you “cut” the system exactly along its natural memory-store modules –asking:

How much explanatory power is lost if each module (vision, logic, procedural, etc.) is forced to explain itself in isolation?

By contrast, Φ asks:

What is the single worst way to slice the system – potentially subdividing modules further – so as to destroy its joint explanatory power as much as possible?

Σ (*module-synergy*) measures how poorly the system performs when each memory store goes it alone. If isolating the standard modules makes explanations much clumsier, those modules clearly need to work together.

Φ (*integrated information*) goes further: it examines *every* possible partition of the stores – including finer splits of the natural modules – and selects the cut that inflicts the greatest explanatory loss. That worst-case cut defines Φ .

Because no partition can damage explanatory power *less* than the worst-case partition, we always have

$$\Phi \leq \Sigma.$$

In practice, however, some finer partitions (subdividing a module into submodules) may be even more disruptive, so one often finds $\Sigma > \Phi$. Only when the natural modular split is indeed the most damaging does Σ coincide with Φ , and in that special case positive module-synergy ($\Sigma > 0$) guarantees positive integrated information ($\Phi > 0$).

24 A Multiresolutional Weakness Transform

We now describe a somewhat more general sort of weakness-based AI tool – a multiresolution decomposition of relational data or models that mirrors the classic wavelet transform, but leveraging weakness as the core pattern recognition and synthesis modality.

Conceptually, one constructs a hierarchy of increasingly fine simplifications (“approximations”) of the original dataset, and at each scale one extracts the “detail” – the residual structure not captured at the coarser level. Just as wavelets decompose a signal into low- and high-frequency bands, our transform peels off layers of “weakness” (highest generality) before refining to capture more specific patterns.

We believe this may be a quite effective method for structuring weakness-based AI processing in a variety of contexts – basically any setting where the environments and tasks involved have a suitable hierarchical structure (a notion we formalize a bit below). We flesh out potential applications here a bit in the context of transformer neural net style prediction and humanoid robotics.

24.1 Multiresolution Weakness Transform

At level 0 we choose a very simple model D_0 , a maximally weak subobject of the dataset D . This could be, for instance, a single-cluster partition, a low-order polynomial fit, or a trivial relation preserving only the most obvious links. The residual relation

$$R_0 = D - D_0$$

captures everything omitted by the coarse model. We record its “detail coefficient” as the weakness of the residual:

$$c_0 = w(R_0) = \bigoplus_{(x,y) \in R_0} \mu(x) \otimes \mu(y),$$

where $(V, \otimes, \oplus)$ is the chosen quantale and μ assigns weights to objects.

For the next scale, we select a model $D_1 \subseteq D_0$ of strictly lower weakness, refining the approximation. The new residual

$$R_1 = D_0 \rightarrow D_1$$

yields coefficient

$$c_1 = w(R_1) = \bigoplus_{(x,y) \in R_1} \mu(x) \otimes \mu(y).$$

Iterating this process produces a chain

$$D \supseteq D_0 \supseteq D_1 \supseteq \dots \supseteq D_N$$

and detail coefficients $\{c_0, c_1, \dots, c_N\}$. Reconstruction follows by reuniting the coarsest model with all details:

$$D = D_N \bigcup \bigcup_{i=0}^{N-1} R_i.$$

In practice, one implements this transform by a greedy selection at each scale: choose D_i to minimize weakness subject to an error tolerance ϵ_i , then compute the residual R_i and its coefficient. The error tolerance enforces that states or data points merged by the coarser model remain approximately interchangeable with respect to the task (e.g. next-token likelihood or control value).

This framework applies in any domain admitting a quantale-enriched sub-object lattice. For next-token prediction, one may let D_i be the set of n-gram contexts of order i , $\mu(c, t) = P_{\text{emp}}(c, t)$, and $\oplus$ ordinary addition. In humanoid robotics, D_i arises from a state abstraction $\phi_i: S \rightarrow A_i$, with $\mu(s)$ proportional to occupancy or reward weight. Each residual coefficient then quantifies how much modeling mass is added by refining the abstraction.

Efficient implementation on the MORK back-end exploits its PathMap algebra: each D_i is indexed by a path label (“refinedBy i ”), residuals are computed via fast bitmap differences, and detail coefficients via parallel quantale folds over changed atoms. As a result, the multiresolution weakness transform runs in time nearly linear in the size of incremental updates rather than in the size of the full dataset.

This multiscale decomposition provides both interpretability – by yielding human-readable models at each scale – and computational savings – by focusing expensive inference or control on the most informative residuals. It thus generalizes wavelet-like analysis to the realm of categorical simplicity.

24.2 Multiresolution Weakness Transform and Mind-World Correspondence

There is no guarantee that this sort of hierarchical approach to decomposing weakness will be useful in any particular domain. However in practice one

suspects it often will be valuable, because real problems in the physical and human world often tend to have the needed sort of hierarchical structure.

One can abstract this observation a bit: Conceptually, a multiresolution weakness transform only yields useful simplifications when the environment itself admits a natural hierarchy of abstractions that align with the agent's goals. Framing this in terms of the "Mind-World Correspondence" framework sketched in Chapter ?? above, we may say an AI's internal model hierarchy must mirror the world's hierarchical structure relative to the reward or value function; otherwise, each successive layer in the transform merely discards relevant information. For instance, in navigation tasks the world decomposes into regions, subregions, and local features; in language modeling it decomposes into n-gram contexts of increasing order. When such hierarchies exist, peeling off layers of maximal generality (weakness) at each scale both compresses the model and preserves goal-relevant distinctions.

Formally, suppose the environment is modeled as an MDP

$$M = (S, A, P, R, \gamma)$$

with state set S , action set A , transition kernel $P(s' \mid s, a)$, reward R , and discount γ . Let $Q^*(s, a)$ be the optimal action-value function. We choose a commutative quantale $(V, \otimes, \oplus)$ and a valuation $\mu : S \rightarrow V$ (e.g. stationary occupancy or reward magnitude).

We then define a chain of state abstractions

$$\phi_0, \phi_1, \dots, \phi_N \quad , \quad \phi_i : S \rightarrow A_i$$

with corresponding kernel relations

$$D_i = \{(s, t) \mid \phi_i(s) = \phi_i(t)\}, \quad D_0 = S \times S, \quad D_N = \Delta_S$$

and require a weakness bound

$$w(D_i) = \bigoplus_{(s,t) \in D_i} \mu(s) \otimes \mu(t) \leq \alpha_i, \quad \alpha_0 \geq \alpha_1 \geq \dots \geq \alpha_N.$$

To ensure each abstraction respects the goal, impose the *goal-relevance condition*

$$\phi_i(s) = \phi_i(t) \implies |Q^*(s, a) - Q^*(t, a)| \leq \varepsilon_i \quad \forall a \in A,$$

so that lumped states remain approximately interchangeable.

The multiresolution transform can then e.g. be constructed greedily:

1. Level 0: choose the coarsest ϕ_0 satisfying $w(D_0) \leq \alpha_0$ and ε_0 .
2. Detail 0: set $R_0 = D_0 \setminus D_1$, record coefficient

$$c_0 = w(R_0) = \bigoplus_{(s,t) \in R_0} \mu(s) \otimes \mu(t).$$

3. Level i : refine to ϕ_i so that $w(D_i) \leq \alpha_i$ and ε_i , compute

$$R_i = D_{i-1} \setminus D_i, \quad c_i = w(R_i).$$

4. Iterate until $\alpha_N \approx 0$ and $\varepsilon_N \approx 0$.

Reconstruction proceeds as

$$D_0 = D_N \cup \bigcup_{i=0}^{N-1} R_i,$$

analogous to the inverse wavelet transform.

We will argue below that this formalism applies equally to next-token prediction – where D_i are i -gram contexts and $\mu(c, t) = P_{\text{emp}}(c, t)$ – and to mobile humanoid robotics – where ϕ_i are state-abstraction maps and $\mu(s)$ reflects occupancy or reward. In each case, the hierarchical structure must align with task-relevant distinctions, ensuring that each layer of the transform yields both compression and fidelity with respect to the AI’s goals.

24.3 Greedy Pattern-Mining Heuristic for Multiresolution Weakness Transform

Conceptually, the Hyperon pattern-mining pipeline [GT25] provides a natural template for a greedy multiresolution decomposition. At each scale one (a) proposes simple candidate models, (b) scores their ”detail” by measuring residual weakness, (c) selects the most informative pattern, and (d) refines it to generate finer candidates. Iterating these steps yields a tower of models and detail coefficients analogous to wavelet low- and high-frequency bands.

Stage 0: Atomic Model Extraction Run the equivalent of the `get-links` step to enumerate simplest subobjects $\{M_i^0\}$, e.g. single-edge patterns or one-clause relations. Each $M_i^0 \subseteq D$ is a maximally weak candidate capturing one basic relationship.

Stage 1: Detail Scoring For each M_i^0 , compute its residual

$$R_i^0 = D \setminus M_i^0$$

and its ”detail coefficient”

$$\delta_i^0 = w(R_i^0) = \bigoplus_{(x,y) \in R_i^0} \mu(x) \otimes \mu(y).$$

This quantifies how much structure is omitted by M_i^0 , akin to surprisingness.

Stage 2: Greedy Selection and Residual Update Choose the pattern $M_{i^*}^0$ with largest $\delta_{i^*}^0$. Record coefficient $\delta^0 = \delta_{i^*}^0$ and update the dataset residual

$$D^1 = D \setminus M_{i^*}^0.$$

Stage 3: Candidate Refinement Generate next-level candidates $\{M_j^1\}$ by combining $M_{i^*}^0$ with additional structure (e.g. pairwise joins, conjunctions). Each $M_j^1 \subseteq D^1$ has lower weakness than its parent.

Stage 4: Iterate For each level $k = 1, 2, \dots$:

1. Compute $\delta_j^k = w(D^k \setminus M_j^k)$ for each candidate.
2. Select $M_{j^*}^k$ maximizing δ_j^k .
3. Record detail δ^k and set $D^{k+1} = D^k \setminus M_{j^*}^k$.
4. Refine $M_{j^*}^k$ to generate $\{M_\ell^{k+1}\}$.

Stop when residual weakness is below a threshold. The multiresolution transform is then

$$D = M_{i^*}^0 \cup M_{j^*}^1 \cup \dots \cup D^N, \quad \{\delta^0, \delta^1, \dots, \delta^{N-1}\}.$$

Algorithmic Implementation Expressed in MeTta-IL, each level fuses the **filter**, **difference**, **map** (to compute $\mu \otimes \mu$) and **fold** (to compute $\oplus$) into a single fused traversal. This avoids materializing intermediate subgraphs and runs in time proportional to new atoms per level rather than the full data size.

This greedy pattern-mining adaptation thus yields an efficient, interpretable multiresolution weakness transform, selecting the most informative simplifications at each scale.

24.4 Efficient Implementation via PathMap in MORK

Conceptually, the multiresolution weakness transform decomposes a dataset into a hierarchy of subobjects and residuals, computing a "detail coefficient" at each scale. To run this pipeline at large scale, MORK's PathMap back end compiles the entire transform into a small number of path-algebra expressions that execute as a single fused graph traversal. PathMap indexes all paths of interest in advance, so subobject extraction, residual differencing, and quantale folding become constant-time bitmap operations per graph atom rather than repeated full scans.

First, each approximation level D_i is represented by a path expression such as

```
P_i = dataset ~> ("refinedBy" : i)+
```

which matches all atoms reachable by one or more edges labeled "refinedBy:i". PathMap builds an index for P_i mapping each atom to a bit in a compressed bitmap. Extracting D_i then costs a single bitmap lookup per atom.

Next, the residual relation

$$R_i = D_{i-1} \setminus D_i$$

is computed by a bitmap difference:

```
R_i_bitmap = PathMap.index(P_{i-1})
AND NOT PathMap.index(P_i)
```

This vectorized difference uses one machine-word instruction per bitmap word, independent of the number of atoms.

Finally, the detail coefficient

$$c_i = \bigoplus_{(x,y) \in R_i} \mu(x) \otimes \mu(y)$$

is computed by iterating only over bits set in `R_i_bitmap` and performing parallel quantale operations. `PathMap` provides a fast iterator of changed atoms, so each $\mu(x) \otimes \mu(y)$ is fetched from an incremental cache and combined via a native $\oplus$ primitive.

By expressing the full multiresolution pipeline as a single `PathMap` plan:

```
PLAN = dataset
~> (("refinedBy":0)+ MINUS ("refinedBy":1)+)
~> (("refinedBy":1)+ MINUS ("refinedBy":2)+)
...
```

MORK fuses all filters, differences, and folds into one traversal that touches each edge exactly once. Incremental updates to data or thresholds only replan affected path segments and recompute local bitmaps and coefficients, achieving update costs proportional to the change set rather than the full graph size. This mechanism underlies the same near-real-time performance that makes wavelet-CNNs and WaveFormers efficient in MORK [F1R25].

24.5 Formalizing Detail Coefficients via Residual Weakness

Conceptually, each detail coefficient measures the “information mass” omitted by a coarser model and captured by the next finer model. Just as a wavelet detail coefficient quantifies high-frequency content, the residual-weakness coefficient quantifies how much relational structure remains when a maximally weak subobject is refined.

Formally, let

$$D_0 \supseteq D_1 \supseteq \cdots \supseteq D_N$$

be a chain of subobjects in a V -enriched category, where D_0 is the raw dataset and D_N the finest model. Define the residual at level i by

$$R_i = D_{i-1} \setminus D_i,$$

and then the detail coefficient

$$c_i = w(R_i) = \bigoplus_{(x,y) \in R_i} \mu(x) \otimes \mu(y),$$

where $(V, \otimes, \oplus)$ is the chosen quantale and μ the object valuation. This construction is completely general, but practical instantiations require domain-specific choices of D_i and μ .

Textual Next-Token Prediction Choose D_i to be the set of observed context-token pairs explained by an i -gram model (or by the first i layers of a transformer). Let

$$\mu(c, t) = P_{\text{emp}}(c) P_{\text{emp}}(t \mid c) = P_{\text{emp}}(c, t),$$

the empirical joint probability of context c and token t . Then

$$R_i = D_{i-1} \setminus D_i, \quad c_i = \sum_{(c,t) \in R_i} P_{\text{emp}}(c, t),$$

which equals the total probability mass not captured by the coarser i -gram.

Mobile Humanoid Robotics Let each abstraction $\phi_i : S \rightarrow A_i$ induce the kernel relation

$$D_i = \{(s, t) \mid \phi_i(s) = \phi_i(t)\}.$$

Choose $\mu(s)$ proportional to the long-run state-occupancy $\pi^*(s)$ under the optimal policy (or to reward magnitude). Then

$$R_i = \ker \phi_{i-1} \setminus \ker \phi_i, \quad c_i = \sum_{(s,t) \in R_i} \pi^*(s) \pi^*(t),$$

measuring how much “behavioral mass” the refinement adds. Alternative formulations use expected Bellman-error reductions in place of occupancy weights.

In both domains, $c_i = w(R_i)$ serves as a wavelet-like detail coefficient, guiding multiresolution analysis by quantifying the residual complexity at each scale.

24.6 Orthogonality via Adjoint Functors and Approximate Bi-orthogonality

Conceptually, orthogonality in a multiresolution transform ensures that analysis (decomposition) and synthesis (reconstruction) operators form exact inverses up to scale, so that no information is lost or duplicated when moving between levels. Categorically, this corresponds to an adjoint pair of functors between successive enrichment levels: the analysis functor projects a coarse model to a finer one, while the synthesis functor reconstructs the coarse model from the fine one. Exact adjunction yields perfect bi-orthogonality; in practice, we often settle for approximate adjunction enforced by reconstruction losses.

General Formalization Let $\mathcal{C}_{i-1} \supseteq \mathcal{C}_i$ be two full subcategories of a V -enriched category, representing coarser and finer models. Define

$$F_i : \mathcal{C}_{i-1} \rightarrow \mathcal{C}_i \quad (\text{analysis}), \quad G_i : \mathcal{C}_i \rightarrow \mathcal{C}_{i-1} \quad (\text{synthesis}),$$

and require an adjunction

$$F_i \dashv G_i \iff \mathcal{C}_i(F_i X, Y) \cong \mathcal{C}_{i-1}(X, G_i Y) \quad \forall X \in \mathcal{C}_{i-1}, Y \in \mathcal{C}_i.$$

The unit $\eta : X \rightarrow G_i F_i X$ and counit $\epsilon : F_i G_i Y \rightarrow Y$ witness perfect bi-orthogonality. When quantale weights obey η and ϵ up to small error, we have approximate adjunction.

24.7 Approximate Bi-Orthogonality and Neural Models

Next-Token Prediction In this sort of use-case, $\mathcal{C}_{i-1}$ is the category of $(i-1)$ -gram models and $\mathcal{C}_i$ the category of i -gram (or i -layer transformer) models. Analysis F_i fits an i -gram model P_i by minimizing

$$\text{KL}(P_{\text{emp}}|P_i) + \lambda \text{KL}(P_i|P_{i-1}),$$

while synthesis G_i marginalizes P_i to recover P_{i-1} . Jointly training F_i and G_i with additional reconstruction terms

$$\text{KL}(P_{i-1}|G_i F_i P_{i-1}) + \text{KL}(P_i|F_i G_i P_i)$$

enforces approximate unit/counit identities. Empirically, techniques like distillation, layer pruning, and interpolation smoothing implement similar approximate adjoints in transformers and n-gram interpolations.

Mobile Humanoid Robotics Let ϕ_{i-1} and ϕ_i be successive state abstractions $S - A_{i-1}$ and $S - A_i$. Analysis F_i splits macro-states in ϕ_{i-1} to minimize Bellman residual subject to a weakness bound:

$$\arg \min_{\phi_i} \|T^{\phi_{i-1}} - T^{\phi_i}\| + \lambda w(\ker \phi_i).$$

Synthesis G_i merges micro-states in ϕ_i whose Q-values differ by less than a threshold, recovering ϕ_{i-1} . Adding reconstruction and dynamics-consistency losses

$$\mathcal{E}_s |\psi_i(\phi_i(s)) - s| + \mathcal{E}_{s,a} |T(s, a) - \psi_i(T'(\phi_i(s), a))|$$

enforces $\psi_i \circ \phi_i \approx \text{id}$ and approximate adjunction. Hierarchical RL and autoencoder abstractions already approximate this in practice.

24.7.1 Viability of Approximate Bi-orthogonality

Exact adjunction is generally infeasible due to model misspecification and noise. However, by casting analysis and synthesis as dual networks (or encoder/decoder pairs) and training with reconstruction-style losses and quantale regularizers, both NLP and robotics domains achieve effective bi-orthogonal behavior. The trade-off parameter λ controls the balance between simplification (weakness) and reconstruction fidelity, yielding practical multiresolution transforms that focus computation where it is most informative.

24.8 Factor-Graph Realization of the Multiresolution Weakness Transform

The multiresolution weakness transform can be implemented directly as a factor graph over a chain of quantales. Concretely:

1. **Quantale chain.** Let

$$Q_0 \subseteq Q_1 \subseteq \cdots \subseteq Q_N = Q$$

be the same nested lattices used in the transform

2. **Variables and factors.**

- For each data object or model component x , introduce a variable node v_x .
- For each candidate subobject D_i or residual R_i , introduce a factor node f_i whose potential ϕ_{f_i} over the relevant variables computes the weakness $w(R_i) = \bigotimes_{(x,y) \in R_i} \mu(x) \otimes \mu(y)$

3. **Coarse-to-fine inference.**

- *Projection:* at level k , replace each factor potential $\phi_{f_i} \in Q_{k+1}$ by its join-map projection into Q_k .
- *Coarse inference:* run one pass of belief-propagation on the graph over Q_k to compute approximate marginals.
- *Detail selection:* select the "hottest" factors (those whose marginals exceed α_k), and restore their full-precision potentials in Q_{k+1} .

4. **Greedy refinement loop.** Iterate for $k = 0, \dots, N - 1$: project to Q_k , infer, select hot factors, and refine them— exactly mirroring the greedy multiresolution pipeline described above.

5. **Efficiency and interpretability.** This factor-graph view:

- Shares substructures (repeated R_i patterns) via common factor nodes.
- Performs each projection and residual update via local join and tensor operations.
- Achieves near-linear incremental cost by only visiting newly refined factors.
- Produces human-readable "models" at each scale (the coarse factor graph at Q_k).

Thus the classic multiresolution weakness transform is a special case of quantale-valued factor-graph inference, inheriting its parallelism, incrementality, and multigrid-style efficiency.

24.9 Multiresolution Weakness Control Heuristic for PLN Background Inference

Multiresolution factor graphs can be used, among other purposes, for a different sort of PLN inference control than we considered in Chapter ?? above – continuous, wide-ranging "background inference" over a large knowledge base. For

this case, we leverage a *multiresolution* strategy in which we maintain multiple versions of our Factor Graph at different "precision" levels in a quantale chain:

$$Q_0 \subseteq Q_1 \subseteq \dots \subseteq Q_N = Q_{\text{truth}}$$

Each Q_k encodes a different resolution of Simple Truth Values (e.g. low-precision to high-precision confidence). We then apply the following control heuristic:

1. **Projection to coarse resolution.** For each factor ϕ_f in the full graph over Q_{k+1} , compute a coarse approximation

$$\phi_f^{(k)} = \bigvee_{Q_{k+1} \rightarrow Q_k} \phi_f$$

via the quantale join-map.

2. **Coarse-level inference.** Run a single pass of belief-propagation on the reduced graph over Q_k to compute *approximate* marginals for every expression. This quickly filters out factors whose contribution is below a threshold at this scale.
3. **Selective refinement.** Identify the subset of factors

$$F_{\text{hot}}^{(k)} = \{ f : \text{VarMarginal}_k(f) \geq \tau_k \}$$

whose coarse-level uncertainty exceeds a preset threshold τ_k . For each $f \in F_{\text{hot}}^{(k)}$, restore its full precision potential in Q_{k+1} and re-compute local messages.

4. **Iterate through levels.** Repeat steps (1)-(3) for $k = 0, \dots, N - 1$, cascading hot-spot refinements upward. At the finest level Q_N one recovers high-confidence marginals only where needed.

This yields a "multigrid-style" inference loop that:

- Exploits *coarse approximations* to eliminate low-impact factors early,
- Focuses *high-precision computation* on a small subset of the graph,
- Runs in time proportional to the number of *newly refined* atoms per level rather than the full graph size,
- Provides both interpretability at each scale and computational savings by peeling off large blocks of ' structure.

24.10 Symbolic WaveFormer via Multiresolution Weakness Transform

Conceptually, the WaveFormer architecture applies a discrete wavelet transform (DWT) to neural feature maps, splitting them into low-frequency (coarse) and high-frequency (detail) subbands, processes each band with self-attention, then recombines via an inverse DWT (IDWT). We can augment this pipeline with a parallel *symbolic* DWT built from multiresolution weakness models, yielding a hybrid "Symbolic WaveFormer" that injects Occam-driven structure into the neural multiscale attention.

First, at each scale i we define:

- A *symbolic approximation* $D_i \subseteq D_{i-1}$, e.g. an i -gram model or i -layer transformer truncation.
- A *residual* $R_i = D_{i-1} \setminus D_i$ with detail coefficient

$$c_i = w(R_i) = \bigoplus_{(x,y) \in R_i} \mu(x) \otimes \mu(y).$$

These play the role of low- and high-frequency symbolic channels. We embed D_i and R_i as auxiliary token streams parallel to the neural DWT subbands.

Technically, each Symbolic WaveFormer block proceeds as follows:

1. **Symbolic DWT:** Compute D_i and R_i via multiresolution weakness transform.
2. **low-freq Attention:** Apply a lightweight symbolic attention or prediction on D_i , yielding coarse context embeddings α_i .
3. **Detail Gating:** Use c_i to compute a gating scalar $g_i = \sigma(\beta c_i)$, where σ is a squashing function and β a learned gain.
4. **Neural WaveFormer:** Run standard WaveFormer self-attention on low- and high-frequency neural DWT subbands, adding g_i as a bias to high-frequency attention logits.
5. **Symbolic IDWT:** Map symbolic embeddings α_i and gated detail tokens back to token-level space via a symbolic reconstruction function.
6. **Neural IDWT:** Apply the neural IDWT to recombine subbands into the next layer's input.

This hybridization yields several benefits:

- *Interpretability:* Coarse predictions α_i are human-readable n -gram or grammar summaries.
- *Efficiency:* Neural attention focuses on high-frequency regions where symbolic detail c_i signals missing mass.

- *Consistency*: An auxiliary reconstruction loss

$$\text{KL}(P_{\text{emp}}|\hat{P}_{\text{sym}+\text{neural}}) + \gamma \|IDWT(D_i, R_i) - D_{i-1}\|$$

enforces approximate bi-orthogonality between symbolic and neural transforms.

By treating the multiresolution weakness coefficients c_i and subobjects D_i, R_i as symbolic DWT components, and coupling them via gating and reconstruction losses to the neural WaveFormer’s DWT streams, we obtain a unified neural-symbolic multiscale transformer that marries Occam-guided abstraction with state-of-the-art wavelet attention.

24.11 Neural-Symbolic Multiscale Correspondence in Humanoid Robotics

While the details would get hairier, in principle a similar approach could work for intensively multimodal use-cases – let’s consider humanoid robotics as our working example. Conceptually, environments for mobile humanoid robots often admit natural hierarchies of spatial and task structure (for example, building – room – furniture – handle). A neural-symbolic correspondence could leverage this by running two parallel streams at each resolution: a symbolic abstraction channel that plans over coarse macro-states, and a neural residual channel that refines actions where the symbolic plan is uncertain.

Multiscale Symbolic and Neural Channels At level i , define a symbolic abstraction map

$$\phi_i : S \rightarrow A_i$$

that clusters raw states S into a small set of macro-states A_i . The *coarse channel* uses a lightweight symbolic planner (for example, value iteration or graph search) on the abstract MDP (A_i, A, P_i, R_i) to compute abstract Q-values $Q_i(\alpha, a)$. The *neural channel* employs a residual policy $\pi_i(a | s, \alpha)$ – a small neural network that takes the full state s together with the abstract state $\alpha = \phi_i(s)$ and outputs fine-grained control actions.

Gating and Fusion To combine channels, compute a confidence score

$$\Delta_i(s) = \max_a Q_i(\alpha, a) - \min_a Q_i(\alpha, a)$$

and map it to a gating factor $g_i(s) \in [0, 1]$ via a simple monotonic function (for example, a sigmoid). The final action at resolution i is then

$$a_i(s) = \begin{cases} \arg \max_a Q_i(\alpha, a), & \text{with probability } 1 - g_i(s), \\ \pi_i(a | s, \alpha), & \text{with probability } g_i(s). \end{cases}$$

This ensures that when the abstract planner is confident (Δ_i large), the symbolic action is used, and when it is uncertain, the neural residual corrects or refines.

Approximate Bi-orthogonality via Reconstruction Losses To enforce consistency between symbolic and neural representations, train the abstraction ϕ_i and a decoder $\psi_i : A_i \rightarrow S$ jointly with losses

$$\mathcal{L}_{\text{rec}} = \mathcal{E}_{s \sim S} [\|\psi_i(\phi_i(s)) - s\|] + \mathcal{E}_{s,a} (T(s,a) \rightarrow \psi_i(T'_i(\phi_i(s), a))),$$

where T and T'_i are the true and abstract transition operators. The first term enforces $\psi_i \circ \phi_i \approx \text{id}_S$, while the second enforces that planning in the abstract MDP lifts back to near-optimal real transitions. A regularizer $w(\ker \phi_i)$ penalizes abstraction complexity, balancing compression with fidelity.

Hierarchical Iteration and Planning By stacking these modules for $i = 0, \dots, N$, the robot executes a coarse-to-fine control loop:

$$s_0 \rightarrow \phi_0(s_0) \rightarrow a_0 \rightarrow s_1 \rightarrow \dots \quad \text{then} \quad \phi_1(s_1) \rightarrow a_1 \rightarrow s_2 \rightarrow \dots$$

Each level uses the residual of the previous level’s plan as its input, analogous to multiresolution detail in a wavelet transform. This neural-symbolic correspondence yields interpretable plans, efficient computation, and robustness to unmodeled dynamics by confining neural learning to high-frequency ”detail” corrections.

Part V

AI Algorithmics

25 Evolving Weak Models: A Schema Theorem

We turn now from probabilistic reasoning and discriminatory learning to evolutionary learning.

Another interesting case of interplay between weakness and AI learning methods arises when one looks at evolutionary learning with a weakness fitness function – i.e. evolving weak models. If one sets things up right, one can obtain some rather crisp theoretical results about this sort of weakness-guided evolution.

We will give here a significantly strengthened version of the classical GA schema theorem [Hol75], specialized to CoDD programs in Elegant Normal Form (ENF) with fitness equal to *weakness* $w(h)$. In this setting, monoidal composition of CoDD subprograms perfectly preserves and adds weakness, so crossover is *building-block-preserving*. As a result, we obtain an almost exact recurrence for the expected number of schema instances and a transparent bound on convergence toward the *weakest* (max-weakness) explanation. The concept is similar to the classic schema theorem, but there is a lot less mess around.

25.1 A Categorical Schema Theorem for Weakness in CoDD-Based GAs

Setup.

- Let $\mathcal{H}$ be the set of all ENF-normalized CoDD programs.
- Define fitness $F(h) = w(h)$, the categorical weakness of program h .
- A *schema* s is any connected sub-DAG of CoDD combinators. We write $h \supseteq s$ if s appears as a subgraph of h .
- Selection is *roulette-wheel* proportionate to $F(h)$.
- Crossover picks two parents h_1, h_2 , chooses a cut-edge in each, and swaps the corresponding sub-DAGs; since CoDD is monoidal and ENF is canonical, the offspring's weakness is

$$w(h_{\text{Rmchild}}) = w(h_{\text{Rmprefix}}) + w(h_{\text{Rmgraft}}) = w(h_1) + w(h_2) - [w(h_1 \ominus s) + w(h_2 \ominus s)],$$

so any schema s present in both parents survives intact in both offspring.

- Mutation edits one combinator node at random, reducing weakness by at most $\Delta_{\max}$.

Theorem 22 (CoDD Weakness Schema Theorem). *Let $m_s(t)$ be the expected number of individuals in generation t containing schema s . Denote the population-mean weakness by $\bar{w}(t) = \frac{1}{N} \sum_h w(h)$. Then under selection and crossover alone (mutation rate $\mu = 0$),*

$$m_s(t+1) = m_s(t) \frac{w_s}{\bar{w}(t)},$$

where w_s is the average weakness of all programs containing s . If mutation at rate μ can disrupt s in at most $\delta(s)$ places, then

$$m_s(t+1) \geq m_s(t) \frac{w_s}{\bar{w}(t)} (1 - \mu \delta(s)).$$

Sketch. (1) *Selection+Crossover.* Because CoDD's monoidal ENF ensures that no recombination ever splits a connected sub-DAG s , every selected copy of s in the parent generation yields exactly two offspring copies. Thus the only effect on m_s is the change in total population size under proportionate selection:

$$m_s(t+1) = 2 \frac{\sum_{h \supseteq s} w(h)}{\sum_h w(h)} m_s(t) = m_s(t) \frac{w_s}{\bar{w}(t)},$$

after renormalizing to a fixed population of size N .

(2) *Mutation.* A mutation hits one of the $\delta(s)$ nodes of s with probability $\delta(s)/|h|$, and only those events can remove s . Thus with mutation rate μ ,

$$m_s(t+1) \geq m_s(t) \frac{w_s}{\bar{w}(t)} (1 - \mu \delta(s)).$$

□

Corollary (Convergence toward Max-Weakness). Suppose there is a unique global maximally weak program h^* . If μ and crossover ensure every other schema $s \neq h^*$ has $w_s < w(h^*)$, then repeated application of the above recurrence drives $m_{h^*}(t) - N$ exponentially fast in t .

Discussion. Because CoDD’s ENF representation makes every schema a true building block with additive weakness under monoidal recombination, the usual schema disruption terms vanish. This yields an *exact* expected-value update under crossover and a simple mutation penalty, guaranteeing rapid convergence to the weakest explanation in expectation.

And thus, as one says ... ”Bob’s your uncle”...! ⁴

26 Evolutionary Dynamics as Optimal-Transport Geodesics

As a related perspective on evolutionary dynamics and weakness, we argue here that, under the usual infinite-population and small-variance limits, common evolutionary processes (e.g. mutation-selection dynamics, evolutionary algorithms with recombination, and Estimation-of-Distribution Algorithms) admit a description as gradient flows in the space of probability distributions equipped with a Wasserstein-type metric. As a consequence, their trajectories follow optimal-transport geodesics. In the case where evolution is aimed at the learning of weak models, we can say that a maximal-local-weakness trajectory is being followed on the path to learning maximally weak models.

26.1 Evolution as Optimal Transport

Consider an evolving population whose genotype distribution at time t is $\rho_t(x)$ on a continuous trait space $X \subset \mathbb{R}^n$. Under selection by fitness $f(x)$ and (Gaussian) mutation with small variance σ^2 , the infinite-population limit yields the Fokker-Planck PDE:

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) + \frac{\sigma^2}{2} \Delta \rho_t, \quad (6)$$

where $V(x) = -\beta^{-1} \log f(x)$ (for selection intensity $\beta > 0$). It is well known (Jordan-Kinderlehrer-Otto, 1998) that (6) is the *2-Wasserstein gradient flow* of the free-energy functional

$$\mathcal{F}[\rho] = \int_X V(x) \rho(x) dx + \frac{\sigma^2}{2} \int_X \rho(x) \log \rho(x) dx$$

in the space $\mathcal{P}(X)$ endowed with the Wasserstein-2 metric W_2 . That is,

$$\partial_t \rho_t = -\text{grad}_{W_2} \mathcal{F}[\rho_t].$$

⁴In less informal terms: yes, this actually packages up the conceptual notion of the schema theorem very nicely and with relatively little mess compared to prior formulations.

Theorem 23. *Under selection-mutation dynamics (6), the curve $t \mapsto \rho_t$ is the unique W_2 -geodesic steepest-descent path for $\mathcal{F}$. In particular, for any $0 \leq s < t \leq T$,*

$$W_2(\rho_s, \rho_t) = \int_s^t \left\| \partial_u \rho_u \right\|_{T_{\rho_u} \mathcal{P}, W_2} du,$$

and $(\rho_t)_{t \in [0, T]}$ solves the Jordan-Kinderlehrer-Otto (JKO) minimizing-movement scheme.

Proof Sketch. The JKO theorem shows that successive time-discretizations

$$\rho^{k+1} = \arg \min_{\rho \in \mathcal{P}(X)} \left\{ \frac{1}{2\tau} W_2^2(\rho, \rho^k) + \mathcal{F}[\rho] \right\}$$

converge as $\tau \rightarrow 0$ to the continuous-time solution of (6), which is therefore the W_2 -gradient flow of $\mathcal{F}$. Standard results in metric-gradient-flow theory then imply the geodesic property above.

26.1.1 Extension to Recombination and EDAs

Recombination (crossover) can be modeled by additional operators that mix parental distributions. In the infinite-population limit, these also admit continuum descriptions (e.g. as nonlocal redistribution operators). Under mild assumptions, the combined selection-mutation-recombination process remains a gradient flow of an appropriately modified free-energy functional in a Wasserstein-like metric (or in a mixture of Wasserstein and information-geometric metrics).

Estimation-of-Distribution Algorithms (EDAs) replace the full distribution ρ_t by a parametric family $\{\rho_\theta\}$ and update parameters θ via maximum-likelihood or moment-matching rules. In the limit of infinitesimal step-sizes and large samples, the parameter-flow is a gradient flow on the statistical manifold, which can also be interpreted as an *approximate* Wasserstein geodesic under a pullback metric.

26.1.2 Evolving Weak Models

If one interprets the "fitness" $f(x)$ as inversely related to a representational-weakness cost $c(x)$, then evolution seeks genotypes of minimal weakness. The above theorem shows that, in the large-population/continuous limit, the weak-model distribution evolves along optimal transport paths in the weakness quantale, thus instantiating *wu wei*: the population flows with minimal cumulative weakness to reach more robust genotypes.

In summary, a broad class of evolutionary and EDA processes can be rigorously understood as geodesic flows in an optimal-transport metric space. This provides a unifying theorem linking evolutionary computation, representational simplicity, and the principle of effortless action.

26.2 Evolutionary Learning Converges to Weakness-AIXI Geodesics

Expanding this line of thinking regarding evolution and geodesics a bit further, we can also show that a broad class of evolutionary algorithms (EAs) and estimation-of-distribution algorithms (EDAs) that learn "weak" models to guide action will – in the infinite-population, small-variation, and small-learning-rate limits – converge to the same entropic-transport geodesic flows as weakness-AIXI. This establishes that evolutionary learning of weak models manifests the *wu wei* principle in the quantale of representational effort.

26.2.1 Setup and Assumptions

Let $\mathcal{H}$ be a hypothesis class and $\rho_t \in \Delta(\mathcal{H})$ the population distribution at (continuous) time t . Suppose the algorithm at each step:

1. Selects or weights hypotheses $h \in \mathcal{H}$ proportional to

$$\rho_t(h) \exp\left(-\frac{1}{\epsilon} w(h)\right) f(h),$$

where $w : \mathcal{H} \rightarrow \mathbb{Q}$ is a weakness prior, $f(h)$ is a fitness score, and $\epsilon > 0$ is an entropy regularizer.

2. Applies infinitesimal variation (mutation, crossover, or parametric update) that induces a velocity field $v(h, t)$ satisfying

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

3. Optionally refits a parametric model to the new population via maximum-likelihood or KL-minimization, with learning rate $\eta \rightarrow 0$.

Under the joint limit

$$\text{population size} \rightarrow \infty, \quad \text{variation scale} \rightarrow 0, \quad \eta \rightarrow 0,$$

the population flow $\rho(t)$ satisfies:

Theorem 24. *Let*

$$FF[\rho] = -EE_\rho[\log f(h)] + EE_\rho[w(h)] + \epsilon KL(\rho||p),$$

where $p(h, t)$ is the prior predictive under the current model. Then $\rho(t)$ is the gradient flow of FF in the Wasserstein-2 metric with entropic regularization, i.e. it minimizes the Schrödinger-bridge cost

$$W_{\text{SB}}[\rho, v] = \int_0^T \int_{\mathcal{H}} \rho(h, t) \left(\frac{1}{2} \|v(h, t)\|^2 + \epsilon \log \frac{\rho(h, t)}{p(h, t)} \right) dh dt,$$

subject to $\partial_t \rho + \nabla \cdot (\rho v) = 0$. Equivalently, $\rho(t)$ is a geodesic in the quantale of weakness costs.

Sketch. • By standard results, infinite-population, small-variation EAs converge to a replicator-mutator PDE, which is the W_2 -gradient flow of $\text{EE}_\rho[w(h)] - \text{EE}_\rho[\log f(h)]$.

- Adding an entropy bonus $\epsilon \text{KL}(\rho||p)$ yields an entropically-regularized gradient flow.
- The Jordan-Kinderlehrer-Otto scheme then shows that the continuous-time limit follows the Schrödinger-bridge variational principle.
- Hence the population distribution evolves along minimal-effort geodesics in the Schrödinger-bridge quantale.

□

26.2.2 Parallel with Quantum Evolution

The same Schrödinger-bridge cost appears in the hydrodynamic formulation of quantum mechanics, with $\epsilon = \hbar$ and $p(x, t)$ the free-diffusion prior. Thus evolutionary learning of weak models and quantum evolution both follow the *wu wei* principle: geodesic flow with minimal representational effort in a common quantale.

26.3 Resource-Efficient Evolutionary Approximation of Wú Wéi Dynamics

Evolutionary algorithms (EAs) and estimation-of-distribution algorithms (EDAs) can approximate the entropic-transport geodesic flows of weakness-AIXI – hence instantiate *wu wei* dynamics – provided certain conditions hold:

26.4 Key Conditions for Approximation

1. **Large Population, Small Variation.** The population size N must be large enough that the empirical distribution $\hat{\rho}_t$ concentrates around the true density ρ_t (sampling error δ scales like $O(1/\sqrt{N})$). Variation operators (mutation step-size or crossover perturbation) should scale $\sigma \sim O(\sqrt{\Delta t})$ so the discrete update approximates a diffusion-drift process.
2. **Smooth Free-Energy Landscape.** The free-energy functional

$$\text{FF}[\rho] = -\text{EE}_\rho[\log f(h)] + \text{EE}_\rho[w(h)] + \epsilon \text{KL}(\rho||p)$$

must have Lipschitz-continuous gradient in the Wasserstein geometry, ensuring stable convergence of gradient-flow approximations.

3. **Expressive Model Families for EDAs.** When using an EDA, the parametric family $\{\rho_\theta\}$ (e.g. factored or graphical models) must capture the main modes of ρ_t with fitting error $O(\delta)$. This keeps the KL-projection $\hat{\rho}_{t+1} \approx \arg \min_\theta \text{KL}(\hat{\rho}_{t+1}||\rho_\theta)$ accurate.

4. **Entropy-Regularization vs. Selection Pressure.** The regularization weight ϵ must balance exploration and exploitation. If $\epsilon \ll 1$, strong selection collapses diversity; if $\epsilon \gg 1$, the flow fails to track the free-energy descent.
5. **Sufficient Computational Budget.** To achieve $\|\rho(T) - \rho^*\| \leq \delta$, one needs on the order of

$$N \times T \gtrsim \frac{1}{\delta^2} \text{poly}(\dim \text{HH}, \frac{1}{\epsilon})$$

fitness evaluations. Recombination accelerates convergence by mixing dependencies nonlocally.

Under these conditions, the EA or EDA population flow $\rho(t)$ closely tracks the Schrodinger-bridge geodesic

$$W_{\text{SB}}[\rho, v] = \int_0^T \int_{\text{HH}} \rho(h, t) \left(\frac{1}{2} \|v\|^2 + \epsilon \log \frac{\rho}{p} \right) dh dt,$$

thus approximating the optimal-transport *wu wei* dynamics.

26.5 Evolutionary Strategies for AGI and Human-Like Cognition

For an AGI aiming at effective human-like cognition in typical tasks – language understanding, motor control, creative problem-solving – evolution can be an especially resource-effective approach when:

- **High-Dimensional, Structured Domains.** Human tasks often involve rich combinatorial structure (grammar, spatial layouts, causal models). Evolutionary recombination naturally exploits modularity and reuses substructures, approximating semantic transport along minimal-effort paths.
- **Nonconvex Objective Landscapes.** Many cognitive goals (e.g. novel idea generation) present multiple local optima. Population-based search maintains diversity, enabling exploration of alternative geodesics in the belief space.
- **Implicit Weakness Priors.** Human bias priors (simple grammar rules, perceptual heuristics) act like weakness costs. Evolutionary variation learns and refines these priors over generations, steering the AGI along minimal-weakness inference paths.
- **Online Adaptation and Continual Learning.** Real-time tasks demand rapid adaptation. Evolutionary dynamics – when implemented with fast EDA updates – can approximate belief-flow geodesics on the fly, allowing the AGI to update its internal models with minimal disruption.

- **Hardware Parallelism.** Modern accelerators enable massively parallel fitness evaluations. This resource can be leveraged to sample large populations, satisfying the large- N condition and yielding high-fidelity approximations to the continuous geodesic.

Thus, evolutionary learning provides a practical and biologically-inspired mechanism for AGI to follow *wu wei* in complex, human-oriented domains – evolving weak models and actions along near-optimal transport paths in the quantale of representational effort.

27 Homomorphic Encryption of the Search for Weak Models via Polynomial Functors

In [Goe25b] I have outlined an approach to homomorphic encryption of a broad class of computer programs via polynomial functor operations. Here we explore how this may be applied to weakness-based AI operations, by taking a particular example as a case in point: a genetic algorithm being used to find weak models.

27.1 Quantale as a Ring-Object in Polynomial Functors

Let $(V, \otimes, \oplus)$ be a commutative quantale. Define the *polynomial functor*

$$P_V(X) = \bigsqcup_{v \in V} X$$

in the category of polynomial endofunctors on **Set**. Under Day-convolution and coproduct,

$$P_V \otimes P_V \cong P_{V \otimes V}, \quad P_V \oplus P_V \cong P_{V \oplus V}.$$

Performing group-completion in **Poly** yields a genuine ring-object $\tilde{P}_V$, whose addition and multiplication recover $\oplus$ and $\otimes$.

27.2 Encoding Candidate Models

A *dataset* X is fixed. A *candidate model* μ is a function $X \rightarrow V$. We embed μ into $\tilde{P}_V$ by sending each $x \in X$ to the summand labelled by $\mu(x)$. This yields a polynomial-functor morphism

$$F_\mu : X \longrightarrow \tilde{P}_V(X).$$

27.3 Encryption via BNF Quotients

To encrypt the ring-object $\tilde{P}_V$, pick a secret Bounded Natural Functor $\Phi \in B$ and form the quotient functor

$$\tilde{P}_V / G_\Phi$$

where G_Φ is the equivalence subfunctor induced by Φ . This hides each summand $\mu(x)$ among equivalent classes. Encrypting F_μ yields

$$\mathcal{E}\lfloor(F_\mu) = (F_\mu)/G_\Phi \in \text{Hom}(X, (\tilde{P}_V/G_\Phi)(X)).$$

27.4 Homomorphic GA Operations

All GA steps are carried out on encrypted functors $\mathcal{E}\lfloor(F_\mu)$:

Fitness Evaluation Weakness fitness is

$$w(\mu) = \bigoplus_{x \in X} [\varphi(x) \otimes \mu(x)].$$

Homomorphically, one composes and adds the corresponding summands in $\tilde{P}_V/G_\Phi$, using the fact that inference-like steps lift to the quotient. The server computes

$$\mathcal{E}\lfloor(w(\mu)) = \bigoplus_{x \in X} [\varphi(x) \otimes \mathcal{E}\lfloor(\mu(x))]$$

via coproducts and Day-convolutions in **Poly**.

Crossover and Mutation Given two encrypted models $\mathcal{E}\lfloor(F_{\mu_1})$ and $\mathcal{E}\lfloor(F_{\mu_2})$, homomorphic *crossover* is implemented as a natural transformation

$$\text{Crossover} : \mathcal{E}\lfloor(F_{\mu_1}) \oplus \mathcal{E}\lfloor(F_{\mu_2}) \longrightarrow \mathcal{E}\lfloor(F_{\mu_{\text{child}}}),$$

and homomorphic *mutation* is another functor-level transformation. Both operate entirely within the quotient $\tilde{P}_V/G_\Phi$, preserving the encrypted structure.

Selection and Iteration Since direct comparison of encrypted weaknesses is infeasible, the server returns the encrypted fitness values to the client, who decrypts, selects the top candidates, re-encrypts them (using the same Φ), and sends them back. The server then applies homomorphic GA operators to produce the next generation of encrypted models.

27.5 Security and Correctness

The hardness of recovering Φ from $\tilde{P}_V/G_\Phi$ is the BNF Distinguishing Problem, analogous to subgroup-decision assumptions in bilinear-group schemes. All homomorphic operations correspond exactly to the intended quantale $\oplus, \otimes$ on plaintext models, thanks to the functorial lifting of operations to quotients.

27.6 Quantum Homomorphic GA over the Quantum Operator Quantale

Following the pattern of [Goe25a] we now show how the ideas of the previous section might be used to encrypt a quantum-computing-based GA optimizing a quantum-algebra weakness fitness function, in a homomorphically quantum-safe way.

Quantum Operator Quantale Let $\mathbf{QChan}$ be the dagger-compact category whose objects are finite-dimensional Hilbert spaces $H \simeq \mathbb{C}^{2^n}$ and whose morphisms are CPTP maps $\Phi : D(H) \rightarrow D(K)$. We form a commutative quantale on each hom-set by

$\Phi \oplus \Psi :=$ direct-sum of channels, $\Phi \otimes \Psi :=$ tensor (or composition) of channels.

Weakness as Fitness Fix an ensemble prior φ on input states ρ . For a candidate quantum circuit $C : D(H_{\text{in}}) \rightarrow D(H_{\text{out}})$ define

$$w(C) = \bigoplus_{\rho \in \text{supp}(\varphi)} [\varphi(\rho) \otimes C(\rho)],$$

and then scalarize via the trace map $\text{Tr} : \oplus \mapsto \mathbb{R}$ to obtain a real-valued GA fitness.

Quantum Ring-Object via Superfunctors Replace classical polynomial functors by *quantum polynomial functors*

$$F(X) = \bigoplus_i A_i \otimes X^{\otimes n_i},$$

and perform group-completion in the category of endofunctors on $\mathbf{QChan}$ to obtain a ring-object $\tilde{P}_Q$.

Secret Quotient: Bounded Natural Superfunctor Choose a secret BNSF $\Psi = \{\Psi_H\}$ satisfying

$$\Phi \circ \Psi_H = \Psi_K \circ \Phi \quad \text{and} \quad \|\Psi_H\|_{\diamond} \leq \lambda.$$

Form the quotient of $\tilde{P}_Q$ by the equivalence $\rho \sim \Psi_H(\rho)$. Encrypt each density operator ρ as the MLWE ciphertext pair

$$(\rho, \Psi_H(\rho)) \mapsto (c_0, c_1) = (As + e, \rho - s),$$

so that every natural transformation (including lifted CPTP gates) acts homomorphically on ciphertexts.

Encrypted GA Workflow

1. *Population Encoding*: Each individual is a circuit C , encoded as a quantum superfunctor F_C and encrypted via BNSF+MLWE to $\mathcal{E} \setminus \rfloor(C) \in \tilde{P}_Q / G_{\Psi}$.
2. *Fitness Evaluation*: Server applies the lifted natural transformation

$$F_C \xrightarrow{\varphi \otimes -} \bigoplus (\varphi(\rho) \otimes C(\rho)) \xrightarrow{\text{Tr}} \text{scalar},$$

entirely on ciphertexts to yield $\mathcal{E} \setminus \rfloor(w(C))$.

3. *Selection*: Ciphertext fitness values are returned to the client, who decrypts, selects top candidates, re-encrypts them, and sends them back.
4. *Crossover & Mutation*: Server applies homomorphic combinators (Day convolution, coproducts, superfunctor merges) to produce offspring $\mathcal{E} \setminus \mathcal{J}(C_{\text{child}})$.
5. *Iteration*: Repeat evaluation, selection, and genetic operators until convergence.

Security and Correctness

- MLWE hardness ensures quantum-safe confidentiality of circuits, priors, and fitness.
- Naturality of CPTP maps and MLWE quotients guarantees that all quantale operations $(\oplus, \otimes)$ lift correctly to the encrypted domain.
- No server-side decryption or comparison is needed beyond returning encrypted fitness scalars for client-side selection.

This construction yields a fully homomorphic, post-quantum secure genetic algorithm whose fitness function is the quantale-weakness of candidate quantum circuits, all performed inside the quantum ring-object of encrypted superfunctors.

28 Hierarchical Hypervector Embeddings

Embedding symbolic structures in continuous spaces has become a cornerstone of modern AI, enabling neural models to leverage rich relational and functional information. Two of the more interesting instantiations of this paradigms are:

- *Type-based hierarchical named functional embeddings*, proposed in [Ke25], assign each concept a concatenation of semantically meaningful sub-vectors drawn from its position in a type ontology. Each sub-vector (e.g. **Demand_Subject**, **Demand_Object**, **Effect**) encodes precisely what a concept requires or provides, ensuring interpretability and leveraging inheritance across ancestor types.
- *Modular compositional hypervector representations* [SF14b] use fixed-length integer vectors partitioned into modules, with algebraic binding and bundling operations that support true compositionality and noise-tolerant retrieval. Roles and fillers are bound via permutation-convolution and aggregated via summation, producing robust, fixed-dimensional embeddings.

Each of these innovative approaches has pluses and minuses. While hierarchical named vectors excel at semantic clarity, their concatenation-based composition scales poorly as ontologies deepen and lacks principled unbinding operations. Conversely, hypervector methods offer powerful binding/unbinding

and efficient similarity search but rely on uniform, unnamed modules, missing the fine-grained functional distinctions of type-based embeddings.

In this chapter, we present a *Hierarchical Modular Hypervector* framework that unites the strengths of both approaches. By slicing a fixed-length hypervector into named modules aligned with each ontology type and functional role, we retain semantic interpretability while gaining the algebraic compositionality, fixed dimensionality, and noise robustness of vector symbolic architectures. Using the running example of

`Deliver(Pizza, Customer),`

we demonstrate how each sub-structure is represented, composed, and queried, and we highlight key advantages:

1. True inverse binding to recover subjects, objects, and effects.
2. Fixed-size embeddings independent of hierarchy depth.
3. Robustness to partial corruption and efficient nearest-neighbor retrieval.
4. Seamless extension of the type hierarchy without vector re-concatenation.

We argue this integrated approach paves the way for interpretable, scalable, and resilient embeddings in tasks requiring both precise semantic roles and powerful compositional reasoning. The HMM framework may be of quite general use, however we develop it here in large part in the context of our work on the PRIMUS cognitive architecture being implemented within the Hyperon AGI framework [GBD⁺23].

We then flesh out the particulars of the approach via a number of directions:

- **Advanced Semantic Pattern Matching.** We show how inverse binding and modular masking enable complex queries—such as extracting objects, filtering by actor and time, and comparing multi-step processes.
- **Hybrid Symbolic/Vector Architecture.** We propose cross-linking MORK’s AtomSpace (symbolic layer) with an external ANN index (vector layer) for sublinear semantic search.
- **Algorithmic Particulars.** We cover calibrated QA-Emb and SensePOLAR value elicitation, binding operator choices (FFT vs. XOR), module allocation heuristics, scalability and multi-inheritance strategies, optimized unbinding for nested queries, and dynamic updates under evolving ontologies.
- **Theoretical Guarantees.** We provide formal bounds on information capacity, reconstruction error under noise, binding/unbinding accuracy, enhanced guarantees for complex query strategies, and similarity preservation.

Next, to illustrate the concrete use of the HMM framework for embedded cognition, particularly in the Hyperon/PRIMUS context [GBD⁺23], we explore applications to probabilistic inference guidance and cross-modal memory retrieval.

We show how HMM may be used to guide PLN inference control, in a relatively subtle way wherein the hierarchical ontology behind the embeddings is used to minimize independence assumption errors on the PLN side. By formalizing a quantale morphism from the hypervector algebra (bundling as join, binding as meet) into the PLN truth-value quantale, we establish that nearly orthogonal role-filler hypervectors map to approximately independent truth values in PLN. This correspondence not only provides a principled way to translate hypervector difference scores into calibrated inference weights, but—via a Lipschitz-style error-reduction theorem—guarantees that crosstalk in the embedding space bounds the approximation error in logical conjunctions and disjunctions.

To guide proof search in Probabilistic Logic Networks using our hypervector framework, we define a *separation score* that quantifies how cleanly each semantic slot within an embedding is isolated from every other. Formulated as a quantale weakness measure, this score is maximized in hypervector space to find embeddings whose components interfere minimally. A morphism then carries these high-separation embeddings into the PLN product-quantale as inference rules with maximal reliability (i.e. minimal independence-assumption error). By shifting the search for reliable proofs into continuous hypervector optimization, we exploit efficient vector operations and approximate-NN retrieval to discover and rank high-confidence rules without exhaustive logical enumeration.

Finally, we use the HMM framework to unify episodic, procedural, and declarative memory in the PRIMUS cognitive architecture. High-level schemas for events and procedures remain in AtomSpace, while detailed perceptual traces and motor patterns are stored externally and linked via named hypervector slots. Retrieval proceeds by symbolic pre-filtering constrained by the declarative ontology in AtomSpace, followed by ontology-guided approximate-nearest-neighbor search over the hypervector store, yielding efficient and context-aware recall of past experiences or invocation of learned skills. This cognitive synergy illustrates the power of a shared ontology to orchestrate memory access across diverse modules, pointing toward a cohesive architecture for integrated symbolic-neural cognition.

28.1 Review of Our Ingredients

28.1.1 Hierarchical Embedding Vectors

Type-based hierarchical embeddings assign each concept a concatenation of named sub-vectors drawn from its type and all ancestor types. Each sub-vector encodes a semantic "function" – e.g. as an illustrative example, for an action, this might include its *Demand_Subject*, *Demand_Object*, and *Effect*.

For instance we could have:

- **Type hierarchy:** Relation $\rightarrow$ Action $\rightarrow$ PhysicalAction $\rightarrow$ Deliver.

- **Module allocation:**

$$\mathbf{v}_{\text{Deliver}} = \mathbf{v}_{\text{Relation}} \parallel \mathbf{v}_{\text{Action}} \parallel \mathbf{v}_{\text{PhysicalAction}} \parallel \mathbf{v}_{\text{DeliverType}},$$

where $\parallel$ denotes vector concatenation.

- **Named sub-vectors:**

$$\mathbf{v}_{\text{DeliverType}} = [\mathbf{v}_{\text{Demand_Subject}}, \mathbf{v}_{\text{Demand_Object}}, \mathbf{v}_{\text{Effect}}].$$

- **Example values:**

$$\mathbf{v}_{\text{Demand_Subject}} \cdot \mathbf{f}_{\text{Robot}} > \mathbf{v}_{\text{Demand_Subject}} \cdot \mathbf{f}_{\text{Human}},$$

ensuring **Robot** is a better actor for **Deliver**.

28.1.2 Snaider’s Modular Compositional Hypervectors

Hyperdimensional computing represents symbols as fixed-length integer vectors and composes structure via binding and bundling. Snaider goes beyond standard hypervector embedding by building modularity into the mechanism, enabling flexible compositionality:

- **Module partitioning:** A hypervector $\mathbf{h} \in \mathbb{Z}^D$ is split into equal-sized modules $M_1, \dots, M_K$.
- **Binding:** Role-filler pairs are bound via a permutation-convolution operator $\otimes$.
- **Bundling:** Multiple bindings are summed (with possible binarization) to yield a composite vector.
- **Example:**

$$\mathbf{h}_{\text{Deliver}} = (\text{ROLE}_{\text{Sub}} \otimes \mathbf{h}_{\text{Robot}}) + (\text{ROLE}_{\text{Obj}} \otimes \mathbf{h}_{\text{Pizza}}) + (\text{ROLE}_{\text{Eff}} \otimes \mathbf{h}_{\text{Moved}}),$$

capturing the structure of **Deliver(Robot, Pizza)**.

28.2 Integration: Hierarchical Modular Hypervectors

Our key proposal here is to combine the hierarchy of named sub-vectors with the algebra of hypervectors.

1. **Per-type modules:** Allocate a distinct hypervector module M_t for each type t in the ontology, inheriting ancestor modules by index overlap.
2. **Named functional slots:** Within each module M_t , reserve sub-slots for roles as appropriate, e.g. *Demand_Subject*, *Demand_Object*, *Effect*, etc.
3. **Value elicitation:** For each slot (t, s) , elicit a scalar $v_{t,s} \in [0, 1]$ (e.g. via QA prompts), then scale a random seed vector $\mathbf{r}_{t,s}$ by $v_{t,s}$.

4. **Binding and bundling:** Bind role tokens to filler vectors within the same slot and bundle across slots and types:

$$\mathbf{h}_{\text{Deliver}} = \bigoplus_{t \in \{\text{Rel}, \text{Act}, \text{Phys}, \text{Del}\}} ((\text{ROLE}_{\text{Sub}} \otimes (v_{t, \text{Sub}} \mathbf{r}_{t, \text{Sub}})) + (\text{ROLE}_{\text{Obj}} \otimes (v_{t, \text{Obj}} \mathbf{r}_{t, \text{Obj}})) + (\text{ROLE}_{\text{Eff}} \otimes (v_{t, \text{Eff}} \mathbf{r}_{t, \text{Eff}}))),$$

where $\oplus$ denotes bundling across modules.

Why do it this way?

28.2.1 Advantages over Standard Hierarchical Vector Embeddings

The proposed Hierarchical Modular Hypervector (HMH) approach improves upon concatenation-based hierarchical embeddings in several key ways:

- **True Compositionality:** Binding and unbinding operations allow exact recovery of subjects, objects, effects, and other roles, rather than relying solely on dot-product matching of concatenated slices.
- **Fixed Dimensionality:** The overall hypervector size remains constant regardless of hierarchy depth, avoiding the linear growth in vector length that occurs when concatenating one sub-vector per ancestor type.
- **Noise Robustness:** Distributed representations degrade gracefully under corruption or partial loss, whereas errors in any single concatenated float slice directly distort similarity scores.
- **Efficient Similarity Search:** Fixed-length binary or integer vectors support locality-sensitive hashing and hardware-accelerated approximate-NN, outperforming brute-force inner-product scans over high-dimensional float arrays.
- **Modular Ontology Updates:** Adding or removing a type requires only allocation or deletion of its module, without re-concatenating or recomputing existing embeddings.
- **Named, Interpretable Slots:** Each vector slice is tied to a semantic role (e.g. `Demand_Object`, `Effect`), simplifying debugging and targeted queries compared to anonymous uniform modules.
- **Cross-Component Queries:** Unbinding enables flexible pattern matching across multiple steps or roles (e.g. comparing prepared vs. delivered items) within a single embedding, which is cumbersome with flat concatenation.
- **Analogical and Hierarchical Generalization:** Standard hypervector arithmetic (addition, binding, permutation) supports analogical inference (e.g. type-role analogies) and automatically inherits common semantics from ancestor types.

28.2.2 Hierarchical Enhancements over Modular Compositional Hypervectors

The addition of a type hierarchy and named functional slots, as we do in HMH, also brings several improvements over Snaider’s original uniform-module hypervector scheme:

- **Type-Specific Modules:** Instead of splitting the vector into anonymous equal-sized modules, we allocate each ontology type its own module. Ancestor types inherit their modules automatically, enabling shared semantics without duplicating vector content.
- **Named Functional Slots:** Within each type module, sub-slots such as `Demand_Subject`, `Demand_Object`, and `Effect` carry explicit semantic roles. This contrasts with uniform modules, where roles must be inferred or manually aligned.
- **Semantic Inheritance and Generalization:** A concept inherits all modules from its supertypes, so common properties need only be stored once at a higher level. New subtypes automatically gain relevant slots without retraining existing vectors.
- **Compact Fixed Dimensionality:** By reusing modules via indexed binding rather than concatenating every ancestor’s entire vector, the overall dimensionality remains bounded even as the hierarchy deepens.
- **Enhanced Interpretability:** Each slice of the hypervector has a clear, human-readable meaning. This makes debugging, visualization, and role-based queries straightforward compared to opaque uniform partitions.
- **Dynamic Ontology Updates:** Adding or removing a type only requires allocating or deleting its module and slot links—no need to reconstruct or resize existing embeddings.

28.3 Advanced Semantic Pattern Matching

To flesh out the concept, consider a transaction embedding

$$H_{\text{tx}} = H_{\text{order}} \oplus H_{\text{prepare}} \oplus H_{\text{deliver}},$$

where, for example,

$$\begin{aligned} H_{\text{deliver}} = & (\text{RoleSub} \otimes H_{\text{Driver}}) \\ & + (\text{RoleObj} \otimes H_{\text{Item}}) \\ & + (\text{RoleRec} \otimes H_{\text{Customer}}) \\ & + (\text{RoleTime} \otimes H_{\text{TimeStamp}}) \\ & + (\text{RoleLoc} \otimes H_{\text{Address}}). \end{aligned}$$

28.3.1 Extracting the Delivered Object

To determine which object was delivered, unbind the object role:

$$H_{\text{obj}} \approx H_{\text{tx}} \otimes \text{RoleObj}^{-1}.$$

A nearest-neighbor search between H_{obj} and known item hypervectors reveals if a pizza was delivered.

28.3.2 Filtering Afternoon Pizza Deliveries by Alice

Apply multiple unbinding operations on the same H_{tx} :

$$\begin{aligned} H_{\text{obj}} &\approx H_{\text{tx}} \otimes \text{RoleObj}^{-1}, \\ H_{\text{sub}} &\approx H_{\text{tx}} \otimes \text{RoleSub}^{-1}, \\ H_{\text{time}} &\approx H_{\text{tx}} \otimes \text{RoleTime}^{-1}. \end{aligned}$$

Matching

$$H_{\text{obj}} \approx H_{\text{Pizza}}, \quad H_{\text{sub}} \approx H_{\text{Alice}}, \quad H_{\text{time}} \approx H_{\text{Afternoon}}$$

yields all afternoon pizza deliveries by Alice.

28.3.3 Comparing Prepared vs. Delivered Items

Define

$$H_{\text{pd}} = H_{\text{prepare}} \oplus H_{\text{deliver}}.$$

Using modular indexing, isolate the prepare and deliver object slots separately:

$$H_{\text{item,prepare}} \approx (\text{mask}_{\text{prepare}} H_{\text{pd}}) \otimes \text{RoleObj}^{-1}, \quad H_{\text{item,deliver}} \approx (\text{mask}_{\text{deliver}} H_{\text{pd}}) \otimes \text{RoleObj}^{-1}.$$

High similarity between these two vectors indicates that the prepared item matches the delivered item.

These queries leverage true inverse binding, modular masking, and fixed-length algebraic operations—capabilities absent in concatenation-based hierarchical embeddings.

28.4 Hybrid Symbolic/Vector Architecture

We now muse a bit on how to implement this sort of embedding in the infrastructure of an AI system.

Modern AI systems often split symbolic reasoning from vector operations; e.g. in the Hyperon architecture this might look like

- **Symbolic Layer (MORK AtomSpace):** Stores ontology, type hierarchy, transaction metadata, and graph patterns. Each concept atom holds a small **VectorID** instead of a full hypervector.

- **Vector Layer (ANN Index):** Stores fixed-length Hierarchical Modular Hypervectors (HMH) keyed by **VectorID**. Provides fast binding, unbinding, and approximate-nearest-neighbor search.
- **Cross-Linking:** AtomSpace atoms link to vector store entries via **VectorID**. Queries fetch IDs from the graph and delegate vector operations to the ANN index.

28.4.1 Approximate Nearest-Neighbor (ANN) Query

The ANN query to be utilized here retrieves the top- k hypervectors most similar to a query vector h_q using a sublinear search index rather than a full scan.

- **Index structures:** Hierarchical Navigable Small World (HNSW), Inverted File with Product Quantization (IVF-PQ), or Locality-Sensitive Hashing (LSH).
- **Similarity metric:** Cosine similarity or dot-product between hypervectors.
- **Query operation:**

$$\{h_{i_1}, \dots, h_{i_k}\} = \text{ANN_index.query}(h_q, k)$$

- **Performance:** Sublinear time in the number of stored vectors, often $\mathcal{O}(\log N)$ or better.

Advantages of HMH+ANN over Concatenation-Based Embeddings

Although both the HMH and simplistic concatenation approach ultimately rely on dot-product (or cosine) similarity, the Hierarchical Modular Hypervector (HMH) combined with an approximate nearest-neighbor (ANN) index offers key benefits over simple concatenation of hierarchical vectors:

1. **Sublinear, hardware-accelerated search:** HMH hypervectors stored in an ANN structure (e.g. HNSW, IVF-PQ, LSH) can be queried in $\mathcal{O}(\log N)$ or better, using bit-parallel or GPU routines. In contrast, concatenation-based vectors demand a full $\mathcal{O}(N \cdot D)$ inner-product scan.
2. **Exact role extraction:** Binding and unbinding operations permit direct recovery of embedded subjects, objects, or effects. Concatenated slices only yield similarity scores and cannot invert to retrieve the original filler vectors.
3. **Fixed, bounded dimensionality:** Module reuse via indexed slots keeps every HMH hypervector at the same size D , regardless of ontology depth. Simple concatenation grows vector length linearly with each added ancestor type.

4. **Graceful noise tolerance:** Distributed hypervector components tolerate random bit or value corruption with smooth degradation in similarity. Any error in a concatenated float slice fully skews that dimension's dot-product contribution.
5. **Dynamic module updates:** Introducing or removing a type involves only adding or deleting its module in the index. Concatenation-based embeddings require re-concatenating or re-padding every existing vector to reflect ontology changes.

28.4.2 Concrete Hybrid Query Workflow

To illustrate how this might work, consider the following series of steps:

Goal: Find all afternoon pizza deliveries by Alice.

Symbolic Pre-filtering Use AtomSpace pattern queries to narrow candidates:

```
; match transactions where Recipient = Alice
Match: TransactionLink ?tx, RoleRec "Alice"
Collect: VectorIDList = [ "tx1", "tx7", ... ]
```

Vector-Store ANN Query In the vector store, unbind the object slot and search for Pizza among the filtered IDs:

$$\text{query.target} = \text{unbind}(H_{\text{Pizza}}, \text{RoleObj})$$

```
ANN_query({
  filter_ids: VectorIDList,
  target_embedding: query.target,
  top_k: 10
}) -> [ "tx7", "tx1", ... ]
```

Time-Slot Filtering For each candidate ID, unbind the time slot and compare to Afternoon:

$$H_{\text{time}} \approx \text{unbind}(H_{\text{txID}}, \text{RoleTime})$$

```
final_ids = []
for id in results:
  h = fetch_vector(id)
  if similarity(unbind(h, RoleTime), H_Afternoon) > theta:
    final_ids.append(id)
```

Remap to Symbolic Atoms Convert the remaining **VectorIDs** back to their AtomSpace transaction atoms:

```
Answer = [ Atom("tx7"), Atom("tx1"), ... ]
```

This pipeline leverages:

- Graph-pattern matching in MORK to apply structural constraints.
- Fast ANN search and binding/unbinding in the vector store for semantic filtering.
- Seamless remapping between symbolic atoms and vector entries.

28.5 Further Algorithmic Details

Making this sort of approach work effectively involves a number of low-level algorithmic choices. Here we address some of these in a moderately fine-grained way. To an extent, though, the right way to work these details will only be determined via experimentation with applications, and some things might end up optimally being done differently for data drawn from different domains.

28.5.1 Detailed Value Elicitation via QA-Emb and SensePOLAR

The step "elicit a scalar $v_{t,s} \in [0, 1]$ via QA prompts" indicated above is not entirely trivial, and potentially requires (at least) two specialized models rather than generic LLM calls. One pipeline to do this would be:

- **QA-Emb Model:** A transformer fine-tuned on binary question-answer pairs.
 - *Input:* "Does **Deliver** require a human subject?"
 - *Output:* Probability $p \in [0, 1]$ that the answer is "yes."
 - *Use:* Assign $v_{t,\text{Demand_Subject}} = p$.
- **SensePOLAR Model:** A dual-embedding network trained on antonymous feature pairs (e.g. Big-Small).
 - *Input:* A feature pair and context, e.g. "Edible - Inedible" for **Item**.
 - *Output:* Scalar $q \in [0, 1]$ representing degree toward the first feature.
 - *Use:* Assign $v_{t,\text{Edible}} = q$.

Elicitation Pipeline

1. **Dimension selection:** For each named slot s (e.g. Demand.Object, Effect), list its natural-language description $d_{t,s}$.
2. **Model assignment:**

- If $d_{t,s}$ is binary (yes/no), use QA-Emb.
 - If $d_{t,s}$ is a polar feature, use SensePOLAR.
3. **Prompt construction:** Formulate a clear QA prompt or feature-pair input.
 4. **Inference:** Run the prompt through the chosen model, obtaining scalar $\hat{v}_{t,s}$.
 5. **Calibration:** Apply temperature scaling or isotonic regression on a held-out calibration set to ensure $\hat{v}_{t,s}$ is well-calibrated in $[0, 1]$.

Implementation Considerations

- *Batching:* Group prompts to amortize model load.
- *Caching:* Store frequent (t, s) responses to avoid redundant inference.
- *Fallback:* If a model’s confidence is low (< 0.5), default to a prior or request human annotation.

This explicit pipeline recognizes the complexity of value elicitation and ensures consistent, calibrated scalars for every hierarchical slot. It could of course be designed and engineered in a variety of different ways and the above is just an initial suggestion.

28.5.2 Binding Operator Specification

The binding operator $\otimes$ was not fully specified above. It could be implemented as *permutation-convolution* on real-valued vectors. We give a few comments on design choices and their impact:

Circular Convolution Binding Given two length- D vectors r (role) and f (filler), define

$$(r \otimes f)[k] = \sum_{i=0}^{D-1} r[i] \cdot f[(k - i) \bmod D].$$

This uses a cyclic shift of f for each component of r . It is associative and invertible via deconvolution in the Fourier domain:

$$\hat{f} = r^{-1} \otimes h \iff \mathcal{F}(\hat{f}) = \frac{\mathcal{F}(h)}{\mathcal{F}(r)}.$$

Implementation via FFT yields $\mathcal{O}(D \log D)$ cost per binding or unbinding.

Permutation Pattern Choices The permutation of indices in the convolution can be any fixed bijection $\pi: \{0, \dots, D-1\} \rightarrow \{0, \dots, D-1\}$. Common choices:

- **Fixed cyclic shift:** $\pi(k) = (k + 1) \bmod D$. Simple, FFT-friendly.
- **Random permutation:** π drawn once uniformly. Maximizes orthogonality between roles but requires explicit storage of π .
- **Block-wise rotation:** Partition indices into blocks and rotate within each block, balancing locality and orthogonality.

Choice of π affects:

- *Orthogonality:* Random permutations yield lower cross-talk between different role-filler pairs.
- *Computational cost:* Simple cyclic shifts leverage FFT; arbitrary π may require $\mathcal{O}(D)$ permutation before convolution.
- *Hardware mapping:* Block or cyclic patterns map efficiently to SIMD and GPU kernels.

Binary Hypervector Binding For binary or bipolar (± 1) hypervectors, one can use bit-wise XOR as a fast binding:

$$r \otimes f = r \text{ XOR } f,$$

with $\otimes$ invertible by the same XOR. This runs in $\mathcal{O}(D/64)$ on 64-bit words.

Performance Trade-offs

- **FFT binding:** $D \log D$ time, high precision, supports real vectors.
- **XOR binding:** $\mathcal{O}(D)$ bit-parallel time, low memory, only for binary vectors.
- **Direct permutation:** $\mathcal{O}(D)$ time, flexible π , moderate memory for storing π .

In practice, a fixed cyclic shift with FFT provides a robust default. For applications demanding extreme speed or binary representations, XOR binding is preferable. Random or block permutations can be adopted when reduced interference is critical.

28.5.3 Module Allocation under Fixed Dimensionality

Another thing we have not fully addressed above is how to determine optimal module sizes for each type. With a fixed total dimensionality D , how do you allocate bits across potentially hundreds of types in a deep ontology?

Let us assume a fixed total dimensionality D to be partitioned across T types in the ontology. Denote by d_t the number of dimensions (or "bits") allocated to type t , subject to

$$\sum_{t=1}^T d_t = D, \quad d_t \geq 0.$$

Determining the optimal $\{d_t\}$ is a constrained resource-allocation problem. We outline several practical approaches:

1. **Uniform allocation:** Set

$$d_t = \left\lfloor \frac{D}{T} \right\rfloor \quad \forall t.$$

Simple but ignores differences in type importance or usage frequency.

2. **Frequency-based allocation:** Estimate a usage frequency f_t (e.g. relative occurrence in the corpus) and allocate

$$d_t = \left\lfloor D \frac{f_t}{\sum_u f_u} \right\rfloor.$$

Types appearing more often receive more capacity.

3. **Information-theoretic allocation:** Compute an entropy or mutual-information metric H_t for each type (e.g. variance of its slot activations) and set

$$d_t \propto H_t, \quad d_t = \left\lfloor D \frac{H_t}{\sum_u H_u} \right\rfloor.$$

Allocates more dimensions to types with richer or more variable semantics.

4. **Downstream-driven optimization:** Define a validation loss $\mathcal{L}(\{d_t\})$ on a held-out task (e.g. accuracy of role-matching) and solve

$$\min_{\{d_t\}} \mathcal{L}(\{d_t\}) \quad \text{s.t.} \quad \sum_t d_t = D.$$

Relax d_t to real values, apply gradient-based or Bayesian optimization, then round to integers.

5. **Progressive pruning and growth:** Start with a coarse allocation (e.g. uniform), monitor slot-level utilization during training, and iteratively reassign dimensions from under-utilized types to over-utilized ones.

6. **Hybrid heuristics:** Combine the above, for example by allocating a base δ dimensions to every type and distributing the remaining $D - \delta T$ according to frequency or entropy.

In practice, a two-stage pipeline—heuristic initialization followed by light downstream tuning—often balances simplicity and performance without incurring excessive optimization cost.

28.5.4 Scalability and Module Allocation Constraints

When the ontology grows deep or broad, and types may inherit from multiple parents, fixed module sizes and reserved sub-slots can become limiting. A few practical strategies suggest themselves:

- **Module sharing and hashing:** Rather than allocating a distinct contiguous block for every type, use a modulo or hash-based indexing scheme to map each type’s slots into a fixed pool of dimensions. This bounds total dimensionality while still isolating semantic roles via controlled collisions.
- **Lazy or on-demand slot instantiation:** Reserve only minimal default capacity for every type (e.g. a small base $d_{\min}$) and allocate additional dimensions at runtime when a type’s slot utilization exceeds a threshold. Unused or low-variance slots can be reclaimed or repurposed.
- **Multiple inheritance via module composition:** For a subtype S inheriting from parents $P_1, \dots, P_k$, define its module embedding as the element-wise sum or bundle of its parents’ modules plus its own delta:

$$H_S = H_{P_1} \oplus \dots \oplus H_{P_k} \oplus H_\Delta,$$

avoiding full duplication of each parent’s dimensions.

- **Slot pruning and compression:** Monitor each slot’s activation variance or mutual information. Low-utility sub-slots (e.g. rarely used demands or effects) can be pruned or quantized to fewer bits, freeing capacity for more critical slots.
- **Hierarchical bit allocation heuristics:** Use a top-down budget: allocate more dimensions to higher-level types (covering many subtypes) and progressively fewer to leaf types. A simple rule is

$$d_t \propto \frac{1}{1 + \text{depth}(t)},$$

ensuring deeper types share majority capacity with their ancestors.

- **Hybrid static-dynamic optimization:** Initialize with a heuristic split (e.g. frequency or entropy based), then refine via light downstream tuning or Bayesian optimization over a validation task, respecting the global sum constraint $\sum_t d_t = D$.

28.5.5 Optimal Binding/Unbinding Strategies for Complex Queries

Complex queries often involve nested or multiple role-filler bindings in a single hypervector. The following are examples of strategies that may potentially improve accuracy and efficiency when extracting fillers:

1. **Sequential Unbinding by Role Specificity:** For a composite

$$h = \bigoplus_{i=1}^n (r_i \otimes f_i),$$

unbind in order of least interference (e.g. rarest or highest-dimension roles first):

$$\hat{f}_k \approx r_k^{-1} \otimes \left(h \oplus \bigoplus_{i < k} (-r_i \otimes f_i) \right).$$

Removing already-extracted components reduces crosstalk.

2. **Hierarchical Masking of Modules:** If each role r_i occupies a distinct module $M_i \subset \{1, \dots, D\}$, mask out other modules before unbinding:

$$h|_{M_k} \otimes r_k^{-1} \longrightarrow \hat{f}_k.$$

This confines error to the module of interest.

3. **Pseudo-Inverse Convolution:** For circular convolution binding, perform unbinding in the Fourier domain:

$$\hat{F}(\omega) = \frac{H(\omega)}{R_k(\omega)} \implies \hat{f}_k = \mathcal{F}^{-1}[\hat{F}(\omega)].$$

Using FFTs accelerates deconvolution for large D .

4. **Joint Unbinding for Multi-Role Queries:** To extract a combination of roles $\{r_a, r_b\}$ simultaneously, compute

$$\hat{f}_{a,b} \approx (r_a \otimes r_b)^{-1} \otimes h = r_b^{-1} \otimes (r_a^{-1} \otimes h).$$

This recovers fillers bound to both roles in one pass.

5. **Least-Squares Deconvolution:** When noise or interference is high, solve

$$\min_x \|h - (r_k \otimes x)\|_2^2$$

by gradient descent or conjugate-gradient methods to find the best approximate filler x .

6. **Learned Inverse Operators:** Train a small neural decoder g_k for each role:

$$g_k(h) \approx f_k,$$

using supervised pairs (h, f_k) . This compensates for systematic binding distortions.

7. **Caching and Reuse of Inverses:** Precompute r_i^{-1} or its FFT representation and reuse across queries to minimize per-query cost.
8. **Thresholded Similarity Checks:** After unbinding $\hat{f}_k$, compare to a candidate set $\{f_j\}$ and accept only those with $\text{sim}(\hat{f}_k, f_j) > \theta$, rejecting low-confidence extractions.

These strategies—masking, ordering, domain-specific inverses, optimization, and learned decoders—together seem likely to yield robust, accurate unbinding even for deeply nested or highly entangled hypervector queries.

28.5.6 Dynamic Updates under Evolving Ontologies

When the type hierarchy changes—new types appear or existing relations restructure—you must both reallocate module capacity and migrate stored embeddings. We describe here some reasonable-looking practical strategies.

Module Rebalancing Let the total dimension budget be D . When a new type t' is added, its required dimensions $d_{t'}$ must come from the existing pool:

$$\sum_{t \in T \cup \{t'\}} d_t = D.$$

Possible rebalancing schemes:

- *Reserved buffer:* Maintain a small unassigned buffer B ($B \ll D$) of dimensions. On each new type, allocate from B until exhausted, then trigger a full rebalance.
- *Proportional shrinkage:* Reduce each d_t by a factor $\alpha < 1$:

$$d_t \leftarrow \lfloor \alpha d_t \rfloor \quad \forall t \in T, \quad d_{t'} = D - \sum_{t \in T} d_t.$$

- *Usage-driven reallocation:* Monitor slot utilization u_t and solve $\min \sum_t (d_t - \beta u_t)^2$ subject to $\sum d_t = D$, rounding to integers.

Embedding Migration Depending on what is stored, one might migrate existing concept embeddings as follows:

1. **Regeneration from Seeds & Values:** If you have recorded the slot-level scalars $v_{t,s}$ and seed vectors $\mathbf{r}_{t,s}$, simply recompute:

$$\mathbf{h} = \bigoplus_{t \in T'} \bigoplus_{s \in S_t} (v_{t,s} \mathbf{r}_{t,s}).$$

This fully reflects the new d_t allocation without approximation.

2. **Linear Projection:** When only old hypervectors $\mathbf{h}_{\text{old}}$ are available, learn or compute a projection matrix $P \in \mathbb{R}^{D' \times D}$ mapping old to new space:

$$\mathbf{h}_{\text{new}} = P \mathbf{h}_{\text{old}}.$$

P can be initialized to preserve original module subspace projections and identity-map new slots to zero.

3. **Buffer-Zone Rollout:** Pre-allocate a "rollout" zone of dimensions R where new modules map. Existing vectors keep their original first $D - R$ dims, and only the rollout dims change. Once all legacy vectors have been accessed and updated to the new scheme, reclaim any unused rollout dims.
4. **Versioned Coexistence:** Maintain both old and new embeddings in parallel for a transition period. Route queries for concepts of version v to the corresponding embedding. Gradually deprecate the old version once downstream systems adapt.

Restructuring Inheritance If a type S changes parents from $\{P_1, P_2\}$ to $\{P'_1, P'_2\}$, update its module by:

$$H_S \leftarrow (H_{P'_1} \oplus H_{P'_2}) \oplus H_{\Delta_S},$$

where H_{Δ_S} is its own delta-module. Existing H_S can be adjusted by subtracting the old parents and adding the new ones via unbinding and rebinding.

These techniques ensure that as your ontology evolves, module capacities remain balanced and all concept embeddings stay consistent with minimal re-computation.

28.6 Theoretical Guarantees

Next, we outline several formal aspects of the Hierarchical Modular Hypervector (HMH) scheme. These results are relatively simple and sketchy and could be fleshed out much further with effort. The basic gist is to show that HMH embeddings have provable capacity, robust noise-tolerance, reliable binding/unbinding in high dimensions, and accurate preservation of similarity relationships. This is "standard hypervector stuff" and we are essentially just affirming that indeed it applies here.

28.6.1 Information Capacity Bounds

Let each hypervector have length D with i.i.d. Rademacher (± 1) components. The dot-product of two random vectors has mean 0 and variance D . To store M distinct items and distinguish them with similarity threshold θ and failure probability δ , a Chernoff bound yields the requirement

$$D \geq \frac{2 \ln M + \ln(2/\delta)}{\theta^2}.$$

28.6.2 Reconstruction Error under Noise

If each component of a stored hypervector is flipped independently with probability ϵ , the expected similarity to the clean vector is

$$\mathbb{E}[\text{sim}] = 1 - 2\epsilon,$$

and the standard deviation is

$$\sigma \approx \sqrt{\frac{4\epsilon(1-\epsilon)}{D}}.$$

The probability of confusing two noisy vectors decays exponentially in D .

28.6.3 Binding/Unbinding Accuracy

Use random orthogonal role vectors $\{r_i\}$ for binding via circular convolution (or XOR in the binary case). For a bound vector

$$h = r \otimes f,$$

the unbinding $\hat{f} = r^{-1} \otimes h$ recovers f with expected mean-squared error

$$\mathbb{E}[\|f - \hat{f}\|_2^2] = O\left(\frac{D}{\sqrt{D}}\right) = O(\sqrt{D}),$$

which is negligible compared to the signal magnitude $O(D)$ as D grows.

28.6.4 Stronger Theoretical Guarantees for Complex Query Strategies

By leveraging module masking, ordered unbinding, and optimized deconvolution, we can derive tighter bounds on capacity and extraction error beyond the basic Rademacher-vector analysis.

Interference-Reduced Sequential Unbinding Suppose $h = \bigoplus_{i=1}^n (r_i \otimes f_i)$ and each module M_i has size d_i . Unbinding in order $k = 1, \dots, n$ and subtracting previously extracted components yields an expected mean-squared interference

$$\mathbb{E}[\|\hat{f}_k - f_k\|_2^2] \leq \frac{\sum_{i>k} d_i}{d_k} \sigma^2 = O\left(\frac{D - \sum_{i \leq k} d_i}{d_k}\right),$$

where σ^2 is the per-component variance of random vectors. Choosing larger d_k for early roles thus provably reduces downstream extraction error.

Module Masking Error Bound With masking, one restricts unbinding to module M_k . Errors from other modules are eliminated, giving

$$\mathbb{E}[\|\hat{f}_k - f_k\|_2^2] = O\left(\frac{1}{d_k}\right),$$

so that allocating more dimensions to critical roles directly lowers the noise floor.

Pseudo-Inverse Deconvolution Accuracy For circular convolution binding, let $R_k(\omega)$ and $H(\omega)$ be the Fourier transforms of r_k and h . The deconvolution estimate $\hat{F}(\omega) = H(\omega)/R_k(\omega)$ satisfies

$$\mathbb{E}[\|f_k - \mathcal{F}^{-1}[\hat{F}]\|_2^2] \leq \frac{\epsilon^2 D}{\min_{\omega} |R_k(\omega)|^2} = O\left(\frac{\epsilon^2 D}{d_k}\right),$$

where ϵ is the worst-case noise magnitude. This bound improves as d_k grows.

Least-Squares Unbinding Bound Solving $\min_x \|h - (r_k \otimes x)\|_2^2$ yields an estimate $\hat{f}_k$ whose error satisfies

$$\|\hat{f}_k - f_k\|_2 \leq \frac{\|n\|_2}{\sigma_{\min}(C_k)},$$

where n is the residual noise and $\sigma_{\min}(C_k)$ is the smallest singular value of the circulant binding matrix C_k . Designing r_k to maximize $\sigma_{\min}(C_k)$ thus guarantees robust unbinding.

Learned Decoder Generalization If a decoder g_k is trained to minimize $\mathbb{E}[\|g_k(h) - f_k\|_2^2]$ over noise realizations, standard learning-theoretic bounds give, with probability $1 - \delta$,

$$\mathbb{E}[\|g_k(h) - f_k\|_2^2] \leq \hat{\epsilon} + O\left(\sqrt{\frac{\log(1/\delta)}{N_{\text{train}}}}\right),$$

where $\hat{\epsilon}$ is the empirical training error over N_{train} samples.

Together, these results show that by allocating module dimensions d_k judiciously and employing masking, ordered subtraction, pseudo-inverse or least-squares unbinding, one can achieve provable improvements in extraction accuracy, noise tolerance, and overall capacity for complex hypervector queries.

28.6.5 Similarity Preservation under Binding

A key property of the permutation-convolution (or XOR) binding operator is that it approximately preserves the inner-product (or cosine) similarities of the original vectors:

- **Random orthogonality:** If the role vector r is drawn at random and normalized, then for any two filler vectors $a, b \in \mathbb{R}^D$,

$$\mathbb{E}[\langle r \otimes a, r \otimes b \rangle] = \langle a, b \rangle,$$

up to vanishing variance $O(1/\sqrt{D})$. Thus the expected similarity is exactly preserved.

- **Circular convolution:** Circular convolution implements a unitary (up to scaling) transform in the Fourier domain. Writing $\otimes$ as convolution,

$$r \otimes a = \mathcal{F}^{-1}(\mathcal{F}(r) \cdot \mathcal{F}(a)),$$

we have

$$\langle r \otimes a, r \otimes b \rangle = \langle a, b \rangle,$$

provided $\|r\| = 1$, since the Fourier transform is unitary.

- **Binary XOR binding:** For bipolar $\{\pm 1\}$ vectors and Hamming similarity,

$$\text{sim}_H(r \oplus a, r \oplus b) = \text{sim}_H(a, b),$$

because XOR with a fixed pattern is a bijection on bit patterns.

- **Empirical robustness:** In practice, binding adds only minimal noise: the variance of $\langle r \otimes a, r \otimes b \rangle$ around $\langle a, b \rangle$ is $O(1/\sqrt{D})$, which vanishes for large D .

Hence, binding does not distort the semantic similarity structure of the original type-based embeddings, enabling both accurate matching and reliable unbinding in downstream queries.

28.7 Guiding PLN Inference via Hierarchical Modular Hypervector Embeddings

Now we verge into less standard territory – exploring how these hypervector embeddings may be used to guide probabilistic logic inference within the PRIMUS and Hyperon framework.

Goertzel et al. [GDD⁺20] investigate how vector differences computed from DeepWalk embeddings of Atomspace nodes align with "intensional differences" in Probabilistic Logic Networks (PLN). They show that the numerical vector pointing from a premise node to a conclusion node often approximates the logical difference in properties between those hypergraph nodes, suggesting that vector arithmetic can guide uncertain inference control. In their Bio-Atomspace experiments, 100-dimensional DeepWalk vectors proved to reflect semantic distinctions encoded in biological ontologies. However, these embeddings lack explicit semantic roles and hierarchical structure.

By adopting Hierarchical Modular Hypervectors we can realize the same conceptual idea in a more elegant and effective way. The core idea is to view hypervector operations—bundling (vector addition or XOR) and binding (permutation-convolution or XOR)—as algebraic joins and meets in a "hypervector quantale." We then define a mapping into the PLN truth-value quantale, where logical OR and AND correspond to the same algebraic structure. Under this morphism, combining two hypervectors via bundling becomes taking the uncertain OR of their mapped truth values, and binding a role and filler vector corresponds to uncertain AND.

By choosing role vectors that are nearly orthogonal, we ensure that unrelated bindings interfere minimally in the hypervector space. The quantale morphism carries this low-interference property into the PLN domain as near-independence of truth values, thereby justifying the use of standard PLN deduction and conjunction rules with reduced approximation error. In practice, one can compute a "difference" hypervector between premises and conclusions, map it through the morphism, and obtain a suggestive truth value for guiding or ranking candidate inferences.

28.7.1 Quantale Morphism from Hypervector Space to PLN Truth-Value Quantale

Let $(Q_h, \oplus_h, \otimes_h, \leq_h)$ be the hypervector quantale, where $\oplus_h$ is vector bundling (e.g. element-wise addition or XOR), $\otimes_h$ is binding (permutation-convolution or XOR), and $\leq_h$ is the similarity order (e.g. cosine similarity induces a preorder). Let $(Q_p, \vee_p, \wedge_p, \leq_p)$ be the PLN truth-value quantale, where $\vee_p$ and $\wedge_p$ are the uncertain OR/AND operations on simple-TV truth values.

We define a quantale morphism

$$\phi: Q_h \longrightarrow Q_p$$

satisfying

$$\phi(h_1 \oplus_h h_2) = \phi(h_1) \vee_p \phi(h_2), \quad \phi(r \otimes_h f) = \phi(r) \wedge_p \phi(f),$$

and monotonicity $h_1 \leq_h h_2 \implies \phi(h_1) \leq_p \phi(h_2)$.

28.7.2 Orthogonality $\rightarrow$ Independence

When two role-filler pairs (r_i, f_i) produce hypervectors $r_i \otimes_h f_i$ that are nearly orthogonal, their similarity $\langle r_i \otimes_h f_i, r_j \otimes_h f_j \rangle \approx 0$ for $i \neq j$. Under ϕ , this maps to $\phi(r_i \otimes_h f_i) \wedge_p \phi(r_j \otimes_h f_j) \approx 0$, i.e. logical independence in PLN, reducing the error of the independence assumption in uncertain AND/OR formulas.

Formal Analysis: Orthogonality Implies Probabilistic Independence

To flesh out the above (conceptually fairly obvious) correspondence, we model hypervectors as binary indicator vectors and truth-values as simple proportions. This yields an exact correspondence between orthogonality (disjoint support) and probabilistic independence under OR/AND.

Theorem 25. *Let $h_1, h_2 \in \{0, 1\}^D$ be two binary hypervectors with disjoint support:*

$$\forall i : h_1[i] \cdot h_2[i] = 0.$$

Define the truth-value map

$$\phi(h) = \frac{1}{D} \sum_{i=1}^D h[i],$$

and probabilistic OR/AND in the PLN quantale by

$$p \vee_p q = p + q - pq, \quad p \wedge_p q = pq.$$

Then:

$$\phi(h_1 + h_2) = \phi(h_1) \vee_p \phi(h_2), \quad \phi(h_1 \otimes_h h_2) = \phi(h_1) \wedge_p \phi(h_2),$$

where $+$ is element-wise addition (with no clipping) and $\otimes_h$ is bit-wise AND.

Hence disjoint (orthogonal) hypervectors map to independent truth-values under $\vee_p$ and $\wedge_p$.

Proof. Since $h_1, h_2 \in \{0, 1\}^D$ and $h_1[i]h_2[i] = 0$, their element-wise sum $h_1 + h_2$ has

$$(h_1 + h_2)[i] = h_1[i] + h_2[i], \quad (h_1 \otimes_h h_2)[i] = h_1[i] \cdot h_2[i] = 0.$$

Therefore

$$\phi(h_1 + h_2) = \frac{1}{D} \sum_i (h_1[i] + h_2[i]) = \phi(h_1) + \phi(h_2).$$

On the other hand,

$$\phi(h_1) \vee_p \phi(h_2) = \phi(h_1) + \phi(h_2) - \phi(h_1) \phi(h_2).$$

But since h_1, h_2 have disjoint support, $\phi(h_1) \phi(h_2)$ counts the fraction of coordinates where both are 1, which is zero. Hence

$$\phi(h_1 + h_2) = \phi(h_1) + \phi(h_2) = \phi(h_1) \vee_p \phi(h_2).$$

Similarly,

$$\phi(h_1 \otimes_h h_2) = \frac{1}{D} \sum_i (h_1[i] \cdot h_2[i]) = 0 = \phi(h_1) \wedge_p \phi(h_2) = \phi(h_1) \wedge_p \phi(h_2).$$

This establishes the claimed equalities, showing that perfect orthogonality in the hypervector space induces exact probabilistic independence in the PLN truth-value quantale. $\square$

28.7.3 Guiding Inference

Given premises with embedding h_{prem} and a candidate conclusion embedding h_{conc} , consider the "difference" vector $\Delta_h = h_{\text{conc}} \oplus_h (-h_{\text{prem}})$. We compute $\tau = \phi(\Delta_h) \in Q_p$ as a suggestive truth value for an intensional inference rule. Larger τ indicates higher confidence that the corresponding PLN inference step follows the ontology's orthogonal directions.

Detailed Example: Guiding PLN Inference via Hypervector Differences We illustrate how to use Hierarchical Modular Hypervectors (HMH) to suggest a new PLN inference rule, using the running example of `Deliver(Pizza, Customer)` and the candidate conclusion `Possess(Customer, Pizza)`.

Hypervector Definitions Let

$$H_{\text{deliver}} = (\text{RoleSub} \otimes H_{\text{Robot}}) \oplus (\text{RoleObj} \otimes H_{\text{Pizza}}) \oplus (\text{RoleRec} \otimes H_{\text{Customer}}),$$

$$H_{\text{possess}} = (\text{RoleSub} \otimes H_{\text{Customer}}) \oplus (\text{RoleObj} \otimes H_{\text{Pizza}}).$$

Difference Hypervector Compute the "difference" embedding

$$\Delta = H_{\text{possess}} \oplus (-H_{\text{deliver}}).$$

Intuitively, Δ highlights the semantic shift from delivering to possessing.

Quantale Morphism and Suggested Truth Apply the quantale morphism ϕ to map Δ into a PLN truth value:

$$\tau = \phi(\Delta) \in [0, 1].$$

For example, if $\tau = 0.92$, we take this as the suggested strength of the inference:

$$\text{Deliver}(\text{Pizza}, \text{Customer}) \implies \text{Possess}(\text{Customer}, \text{Pizza}) \quad \text{with truth } \tau.$$

Incorporating into PLN In PLN, an implication rule

$$R : A \Rightarrow B \quad \text{with weight } w_R$$

is applied to a known fact A of truth t_A to derive B with truth

$$t_B = t_A \wedge_p w_R,$$

where $\wedge_p$ is the uncertain-AND in the PLN quantale. Here we set $w_R = \tau$. Thus, given

$$t_{\text{deliver}} = 0.85, \quad w_R = 0.92,$$

we infer

$$t_{\text{possess}} = 0.85 \wedge_p 0.92 = \min(0.85, 0.92) = 0.85$$

(or another $\wedge_p$ formula per the chosen PLN semantics).

Benefits

- The hypervector difference Δ captures semantic shifts in the ontology.
- The morphism ϕ translates that shift into a calibrated truth value τ .
- PLN uses τ to weight and rank candidate inference rules, focusing search on high-confidence steps.

This workflow generalizes to multi-premise or multi-step inferences by bundling multiple premise hypervectors before differencing and mapping through ϕ .

28.7.4 Error Reduction

Let ϵ_h be the crosstalk error due to non-orthogonality in Q_h . Because ϕ preserves joins and meets, the resulting PLN inference error ϵ_p satisfies

$$\epsilon_p \leq C \epsilon_h,$$

for some constant C depending on quantale norms. Thus by designing the ontology so its role modules are mutually orthogonal in Q_h , we guarantee smaller independence-assumption errors in the guided PLN inference sequence.

This morphism formalizes how hierarchical modular hypervector structure can transfer to more accurate, embedding-guided PLN reasoning.

Formal Error Reduction Theorem The observation above regarding error reduction may be fleshed out further as follows:

Theorem 26 (Error Reduction under Quantale Morphism). *Let $(Q_h, \oplus_h, \otimes_h, \leq_h)$ be the hypervector quantale and $(Q_p, \vee_p, \wedge_p, \leq_p)$ the PLN truth-value quantale. Suppose*

$$\phi: Q_h \rightarrow Q_p$$

is a quantale morphism satisfying

$$\phi(x \oplus_h y) = \phi(x) \vee_p \phi(y), \quad \phi(x \otimes_h y) = \phi(x) \wedge_p \phi(y),$$

and is L -Lipschitz with respect to similarity metrics d_h on Q_h and d_p on Q_p :

$$d_p(\phi(h_1), \phi(h_2)) \leq L \cdot d_h(h_1, h_2) \quad \forall h_1, h_2 \in Q_h.$$

Define the hypervector crosstalk error

$$\epsilon_h = \max_{i \neq j} d_h(r_i \otimes_h f_i, r_j \otimes_h f_j),$$

and the corresponding PLN inference error

$$\epsilon_p = \max_{i \neq j} d_p(\phi(r_i \otimes_h f_i), \phi(r_j \otimes_h f_j)).$$

Then

$$\epsilon_p \leq L \epsilon_h.$$

Sketch of Proof. By the Lipschitz property of ϕ ,

$$\epsilon_p = \max_{i \neq j} d_p(\phi(r_i \otimes_h f_i), \phi(r_j \otimes_h f_j)) \leq L \max_{i \neq j} d_h(r_i \otimes_h f_i, r_j \otimes_h f_j) = L \epsilon_h.$$

Since ϕ preserves $\otimes_h$ as $\wedge_p$, orthogonality (small d_h) maps to logical independence (small d_p), directly bounding the PLN inference error by the scaled hypervector crosstalk error. $\square$

28.8 Separation Score Correspondence and Hypervector-Driven Search for Reliable Proofs

This section introduces a *separation score* that quantifies how cleanly the semantic components of a hypervector embedding are separated (i.e. how little they interfere), and shows how maximizing this score in hypervector space directly yields inference rules in PLN with maximal reliability, in the sense of minimal dependence on unjustified independence assumptions. We define the separation score as a normalized sum of pairwise non-interaction terms, then use a quantale morphism to carry it into the PLN product-quantale, where it measures minimal independence-assumption error.

This is interesting in itself, and also as a study in the flexibility of the quantale weakness framework itself.

28.8.1 Hypervector Separation Score

For a Hierarchical Modular Hypervector embedding, let each named slot s_i carry weight $v_i \in [0, 1]$. Define the raw crosstalk:

$$w_h = \sum_{1 \leq i < j \leq n} v_i v_j, \quad W_{\max} = \binom{n}{2}.$$

We then introduce the *separation score*

$$S_h = 1 - \frac{w_h}{W_{\max}} = 1 - \frac{1}{\binom{n}{2}} \sum_{i < j} v_i v_j,$$

so that $S_h = 1$ when all slots are perfectly orthogonal ($v_i v_j = 0$) and decreases as interference grows. We seek embeddings that *maximize* S_h .

28.8.2 PLN Reliability Score

Consider a PLN rule with premise truth-values $t_1, \dots, t_n \in [0, 1]$. Define

$$w_p = \sum_{1 \leq i < j \leq n} t_i t_j, \quad S_p = 1 - \frac{w_p}{W_{\max}} = 1 - \frac{1}{\binom{n}{2}} \sum_{i < j} t_i t_j.$$

Here S_p measures the rule's reliability: higher S_p indicates fewer implicit independence assumptions and thus broader applicability.

28.8.3 Quantale Morphism and Score Equivalence

Introduce a quantale morphism ϕ mapping hypervector slot weights to PLN truth-values:

$$\phi(v_i) = t_i, \quad \phi(v_i v_j) = t_i t_j, \quad \phi\left(\sum_{i < j} v_i v_j\right) = \sum_{i < j} t_i t_j.$$

Since $\phi(W_{\max}) = W_{\max}$, it follows that

$$S_p = 1 - \frac{1}{W_{\max}} \sum_{i < j} t_i t_j = 1 - \frac{1}{W_{\max}} \phi\left(\sum_{i < j} v_i v_j\right) = 1 - \frac{\phi(w_h)}{W_{\max}} = S_h.$$

Thus any embedding that maximizes the hypervector separation score S_h produces a PLN rule with exactly the same maximal reliability score S_p .

As well as a useful result in itself, this is an interesting case study in the semantics of weakness and generalization. The raw crosstalk is a formal weakness measure, so that a large raw crosstalk implies strong generalization in a certain sense: strong ability to generalize to predict the dependency between entities. However this sort of generalization is actually anticorrelated with robust generalization at the level of PLN inferences on the entities in question. None of this is surprising, it just requires careful thinking; the practical semantics of quantale weakness obviously depends on the quantale in question.

28.8.4 Hypervector-Driven Proof Search

Proof discovery can then proceed by maximizing S_h :

1. **Parametrize** slot weights $v_i(\theta)$ for a candidate conclusion embedding.
2. **Compute** $S_h(\theta) = 1 - \frac{1}{\binom{n}{2}} \sum_{i < j} v_i(\theta) v_j(\theta)$.
3. **Optimize** $\theta^* = \arg \max_{\theta} S_h(\theta)$ using gradient methods or heuristic search.
4. **Map** the optimized weights $v_i(\theta^*)$
5. **Execute** the PLN inference using the rule with generality score $S_p = S_h$, ensuring minimal independence-assumption error.

By shifting search into continuous hypervector space and maximizing the separation score, we leverage efficient vector operations and ANN retrieval to find the most general, confident inference rules in PLN without exhaustive logical enumeration.

Concrete Example: Multi-Premise Separation-Driven Rule Discovery

We show how maximizing a *separation score* among multiple candidate premises in hypervector space leads to selecting the most reliable inference rule in PLN.

Candidate Premises Two potential explanations for Possess(Customer, Pizza):

$$P_1 : \text{Deliver}(\text{Pizza}, \text{Customer}), \quad t_1 = 0.9,$$

$$P_2 : \text{Pay}(\text{Customer}, \text{Invoice}), \quad t_2 = 0.8.$$

Hypervector Embeddings and Slot Weights Each premise P_i has an HMM embedding H_i . We assign a slot-weight $v_i \in [0, 1]$ proportional to its cosine similarity with the conclusion embedding:

$$v_1 = \rho(H_{\text{deliver}}, H_{\text{possess}}) = 0.8, \quad v_2 = \rho(H_{\text{pay}}, H_{\text{possess}}) = 0.5.$$

Separation Score Objective With two slots, raw crosstalk is

$$w_h = v_1 v_2 = 0.8 \times 0.5 = 0.40,$$

and the separation-weakness sum-of-products is

$$w_s = 1 - v_1 v_2 = 0.60, \quad W_{\max} = 1.$$

Thus the hypervector separation score is

$$S_h = \frac{w_s}{W_{\max}} = 0.60.$$

To allow blending, introduce $a \in [0, 1]$:

$$v_1(a) = a \cdot 0.8, \quad v_2(a) = (1 - a) \cdot 0.5, \quad S_h(a) = 1 - (v_1(a) v_2(a)).$$

We seek $a^* = \arg \max_a S_h(a)$.

Optimizing Separation

$$S_h(a) = 1 - (0.8 a) (0.5 (1 - a)) = 1 - 0.4 a (1 - a).$$

This is maximized at $a = 0$ or $a = 1$, giving $S_h = 1$. We compare the two pure-premise options.

Mapping to PLN Reliability Score Under the quantale morphism $\phi(v_i) = t'_i$, each pure premise yields an empty sum of products (no pairs), so

$$S_p = 1 - \sum_{i < j} t'_i t'_j = 1.$$

We interpret S_p as the *reliability score* of the rule (higher = more reliable). Both rules have $S_p = 1$, so we break ties by original premise strengths t_i .

Final Inference Select the rule with higher t_i . For P_1 ,

$$\text{Deliver} \Rightarrow \text{Possess}, \quad \text{reliability} = 1, \quad t_{\text{possess}} = \min(0.9, 1) = 0.9.$$

For P_2 , $t_{\text{possess}} = \min(0.8, 1) = 0.8$. Thus the "Deliver" rule wins, yielding a final conclusion confidence of 0.9.

By maximizing the separation score in hypervector space, we automatically identify and rank the most reliable PLN inference rule without exhaustive logical enumeration.

28.9 Bridging Episodic, Procedural, and Declarative Memory

In the PRIMUS cognitive architecture, three memory systems—declarative, episodic, and procedural—play complementary roles. By leveraging a shared declarative ontology and Hierarchical Modular Hypervectors (HMH), we can achieve elegant cross-module retrieval via ANN lookups.

We describe idea here in a fairly schematic way; actual implementations will involve further particulars.

28.9.1 Memory Storage Architectures

- **Declarative Memory (AtomSpace):** Facts, concepts, and relations are stored explicitly as Atoms in MORK’s AtomSpace. Each Atom carries a `VectorID` linking to its HMH embedding in the vector store.
- **Procedural Memory (Hybrid):** High-level procedure schemas (e.g. “MakePizza”) reside as sequences of ActionLinks and RoleLinks in AtomSpace. Only the low-level motor patterns (e.g. precise joint trajectories) are stored externally as raw time-series data or motor-hypervectors.
- **Episodic Memory (Hybrid):** Episode skeletons—temporal and causal structure of events—are encoded as subgraph patterns in AtomSpace. Detailed perceptual data (images, sounds, sensor streams) remain in specialized media stores, referenced by `PerceptLinks` from the episode atoms.

28.9.2 Hypervector Encoding and Cross-Module Links

Each memory trace obtains an HMH embedding that binds its declarative schema to its sensory or motor components:

$$\begin{aligned} H_{\text{episode}} &= H_{\text{schema}} \oplus (\text{RolePercept} \otimes H_{\text{perceptData}}), \\ H_{\text{procedure}} &= H_{\text{schema}} \oplus (\text{RoleMotor} \otimes H_{\text{motorData}}). \end{aligned}$$

Here H_{schema} is the concatenated module for the episode or procedure type (inherited from the declarative ontology), while the `RolePercept` and `RoleMotor` slots link to external data hypervectors.

28.9.3 ANN-Based Retrieval with Ontology Guidance

1. **Cue Encoding:** A retrieval cue (e.g. a partial scene description or desired skill) is mapped to an HMH query vector h_q via the same ontology modules.
2. **Symbolic Pre-filtering:** AtomSpace pattern queries narrow candidates by high-level constraints (e.g. “episodes involving Pizza” or “procedures requiring Oven”).

3. **Vector Search:** The filtered `VectorIDs` feed into an ANN index over HMM embeddings. Sublinear search returns the top- k matching episode or procedure embeddings.
4. **Reconstruction:** Retrieved hypervectors unbind via $\otimes^{-1}$ to recover linked perceptual or motor data, then remap back to `AtomSpace` for further symbolic reasoning or replay.

28.9.4 Cognitive Synergy

Using the declarative ontology to structure HMM embeddings yields a unified retrieval mechanism:

- *Efficiency:* Ontology-based pre-filtering reduces the ANN search space dramatically.
- *Precision:* Named slots ensure that only the relevant perceptual or motor component is matched and recovered.
- *Flexibility:* The same cue can trigger both episodic recall and procedural invocation, depending on which module slots dominate the similarity score.

This tight integration of symbolic schemas and vector-based associations enables seamless memory lookup across declarative, episodic, and procedural systems, embodying a powerful, ontology-grounded cognitive synergy.

28.10 Conclusion

We have introduced the Hierarchical Modular Hypervector (HMM) framework, which combines type-based hierarchical named functional embeddings with modular compositional hypervectors to yield fixed-size, interpretable, noise-tolerant representations. We demonstrated advanced semantic pattern matching via binding and unbinding, a hybrid symbolic/vector architecture cross-linking `AtomSpace` with an ANN index, detailed algorithmic components, and formal theoretical guarantees on capacity, noise robustness, and binding/unbinding accuracy. We then showed how a quantale morphism links hypervector orthogonality to logical independence in PLN, including a normalized separation-weakness sum-of-products measure whose maximization yields high-reliability inference rules, and illustrated how HMM embeddings can bridge episodic, procedural, and declarative memory in PRIMUS.

Beyond PLN guidance, the hierarchical hypervector quantale appears to merit further exploration as a general substrate for embedding-based cognitive operation control throughout the PRIMUS architecture:

- *Evolutionary Search (MOSES):* Quantale joins and meets can inform program evolution fitness by measuring semantic module interactions.

Approach	Storage	Bind/Unbind	Query	Update
Concatenation-based Hierarchical	$O(D \times L)$	-	$O(N \times D \times L)$	$O(N \times D \times L)$
HMH + FFT binding (brute-force)	$O(D)$	$O(D \log D)$	$O(N \times D)$	$O(1)$
HMH + XOR binding (brute-force)	$O(D)$	$O(\frac{D}{w})$	$O(N \times D)$	$O(1)$
HMH + FFT binding + HNSW index	$O(D + \text{index})$	$O(D \log D)$	$O(\log N)$	$O(1)$
HMH + XOR binding + HNSW index	$O(D + \text{index})$	$O(\frac{D}{w})$	$O(\log N)$	$O(1)$

Table 1: Computational complexities for various embedding and query strategies. D = vector dimensionality, L = hierarchy depth, N = number of stored vectors, w = machine word size.

- *Decision-Making (MetaMo)*: Separation scores may be used to guide selection of action’schema hypervectors with minimal interference between competing heuristics.
- *Predictive Coding*: Role-filler quantale operations support rapid hypothesis generation and pruning via vector’space projections.
- *Memory Systems Integration*: As we have shown, the same ontology-aligned modules orchestrate retrieval across declarative, episodic, and procedural memory using ANN lookups. This orchestration may be fleshed out further via deeper models of memory modules, and extended further e.g. to memory stores focused on sensation, action and attention.

In sum, the HMH quantale not only unifies symbolic and vector representations but also enables principled, embedding-driven guidance of diverse cognitive processes in PRIMUS – a potentially powerful tool helping pave the way toward a cohesive, scalable, and interpretable AI system.

29 Chaining Weaknesses Together: Predictive Coding with Information Geometry for Deep Kernel Learning

In this section we walk through a somewhat more elaborate example of the application of weakness to AI, involving the chaining together of three sorts of weakness associated with three different interoperating AI algorithms: SVM, and predictive coding neural nets used for learning a kernel function for an SVM, and information geometry used for guiding learning in the predictive coding neural net.

To understand why this combination makes sense, recall that practical utilization of SVM depends heavily on choosing the right kernel function. The old-style approach was to manually choose a kernel function appropriate for one’s application and then adjust its parameters empirically, e.g. by a grid search. These days, however, it’s been recognized that one can get more robust

results via *learning* a whole kernel function, e.g. in the form of a deep neural network.

This is known as "deep kernel learning" [WHSX16] – a technique that combines the representational power of deep neural networks with the theoretical strengths of kernel methods. In essence, one uses a neural network to learn a mapping of inputs into a feature space, and then applies a kernel (e.g. Gaussian) in that learned feature space to perform tasks such as classification or regression.

Most work on deep kernel learning so far has involved learning kernel functions using the backpropagation algorithm. However, it is also interesting to think about alternatives such as *predictive coding* – a biologically inspired framework in which each layer of a network attempts to predict the activity of the layer below, and the resulting *prediction errors* drive both inference (computing latent representations) and learning (updating network weights). This local, error-driven scheme avoids backpropagation's global credit assignment and can lead to more stable and interpretable learning dynamics.

Information geometry adds another layer of refinement to neural learning by recognizing that probability distributions (and the parameters that define them) lie on a curved manifold rather than in a flat Euclidean space. The *natural gradient* [CL20] accounts for this curvature by preconditioning gradient updates with the inverse of the Fisher information matrix. As a result, updates follow the steepest descent direction relative to the manifold's geometry, often leading to faster convergence and more stable learning. Information-geometry-enhanced predictive coding could provide a particularly powerful approach to deep kernel learning.

29.0.1 Deep Kernel Learning

A foundational concept for SVM theory is the:

Definition 6 (Reproducing Kernel Hilbert Space (RKHS)). *Let $XX \subset \mathbb{R}^d$ be an input space. A function $k: XX \times XX \rightarrow \mathbb{R}$ is a positive definite kernel if for any finite set $\{x_i\}_{i=1}^n \subset XX$, the Gram matrix*

$$K_{ij} = k(x_i, x_j)$$

is positive semidefinite. Associated to k is a Reproducing Kernel Hilbert Space HH_k of functions $f: XX \rightarrow \mathbb{R}$ such that:

1. *For each $x \in XX$, the function $k(\cdot, x) \in HH_k$.*
2. *For all $f \in HH_k$ and $x \in XX$, the reproducing property holds:*

$$f(x) = \langle f, k(\cdot, x) \rangle_{HH_k}.$$

This lets us define a

Definition 7 (Deep Kernel). *Let $\phi_\theta: XX \rightarrow \mathbb{R}^p$ be a feature mapping implemented by a neural network with parameters θ . Given a base kernel k_0 :*

$RR^p \times RR^p - RR$ (e.g. Gaussian $\exp(-\|\phi - \phi'\|^2/\sigma^2)$), the deep kernel is defined as

$$k_\theta(x, x') = k_0(\phi_\theta(x), \phi_\theta(x')).$$

This defines a kernel on XX whose geometry and similarity measure adapt to data via θ .

A supervised learning task (e.g. classification) with labels $y_i \in \{-1, +1\}$ proceeds by solving in HH_{k_θ} a regularized risk minimization, for instance the SVM primal problem:

$$\min_{f \in \text{HH}_{k_\theta}, b \in \text{RR}} \frac{1}{2} \|f\|_{\text{HH}_{k_\theta}}^2 + C \sum_{i=1}^n \ell(y_i, f(x_i) + b),$$

where $\ell(y, t) = \max\{0, 1 - yt\}$ is the hinge loss.

One solves this by kernel methods once θ is fixed; simultaneously, one can update θ to improve generalization.

29.0.2 Predictive Coding Framework

We briefly review the basics of one sort of predictive coding neural network, which we suspect can be helpful for deep kernel learning.

Predictive coding, in one simple form, posits that each layer in a hierarchy both generates predictions of the layer below and receives *prediction errors* from that layer. Let us index layers by $\ell = 1, \dots, L$, where layer 0 is the input. For each layer ℓ :

- Let $r_\ell \in \text{RR}^{n_\ell}$ be the *activation vector* or latent representation at layer ℓ .
- Let $W_\ell \in \text{RR}^{n_\ell \times n_{\ell-1}}$ and $b_\ell \in \text{RR}^{n_\ell}$ be the feedforward weights and biases mapping from layer $\ell - 1$ to ℓ .
- Define the *generative prediction* for layer $\ell - 1$ as

$$\hat{r}_{\ell-1} = g(W_\ell^- pr_\ell + b_\ell),$$

where g is an activation function (e.g. identity or nonlinear).

- The *prediction error* at layer $\ell - 1$ is

$$e_{\ell-1} = r_{\ell-1} - \hat{r}_{\ell-1}.$$

Inference proceeds by adjusting r_ℓ to minimize the total error:

$$E_{\text{inf}}(r_1, \dots, r_L) = \sum_{\ell=1}^L \frac{1}{2} \|r_{\ell-1} - g(W_\ell^- pr_\ell + b_\ell)\|_2^2.$$

One updates each r_ℓ via gradient descent:

$$r_\ell \leftarrow r_\ell - \eta_r \frac{\partial E_{\text{inf}}}{\partial r_\ell}, \quad \forall \ell = 1, \dots, L,$$

where $\eta_r > 0$ is a step size. After inference, *learning* updates W_ℓ and b_ℓ to minimize prediction error over data:

$$E_{\text{learn}}(W, b) = \sum_{\text{data } x} \sum_{\ell=1}^L \frac{1}{2} \|r_{\ell-1}(x) - g(W_\ell^- p r_\ell(x) + b_\ell)\|_2^2,$$

where $r_\ell(x)$ are the inferred activations for input x . One updates:

$$W_\ell \leftarrow W_\ell - \eta_W \frac{\partial E_{\text{learn}}}{\partial W_\ell}, \quad b_\ell \leftarrow b_\ell - \eta_b \frac{\partial E_{\text{learn}}}{\partial b_\ell}.$$

29.0.3 Information Geometry and Natural Gradient

Information geometry treats a parametric family of probability distributions $\{p(y|x; \theta)\}$ as a *Riemannian manifold*. The *Fisher information matrix* $F(\theta)$ defines a metric tensor on this manifold:

$$F(\theta) = \mathbb{E}_{x,y} \left[\nabla_\theta \ln p(y|x; \theta) \nabla_\theta \ln p(y|x; \theta)^T \right].$$

Definition 8 (Natural Gradient). *The natural gradient of an objective $L(\theta)$ is defined as*

$$\tilde{\nabla}_\theta L = F(\theta)^{-1} \nabla_\theta L,$$

where $\nabla_\theta L$ is the ordinary Euclidean gradient. An update step using natural gradient is

$$\theta \leftarrow \theta - \eta \tilde{\nabla}_\theta L = \theta - \eta F(\theta)^{-1} \nabla_\theta L.$$

This corresponds to steepest descent on the curved information manifold.

In practice, one approximates $F(\theta)$ (e.g. by a block-diagonal or low-rank estimate) to make inversion feasible.

One can also do predictive coding using other metrics than Fisher, e.g. the Wasserstein optimal-transport metric. The math is different and there may be practical advantages in various contexts, but the basic concept is roughly the same.

29.0.4 Combining Predictive Coding and Natural Gradient in Deep Kernel Learning

We now integrate the above elements into a cohesive algorithm for deep kernel learning.

Feature Network as Predictive Coding Model Let the neural network ϕ_θ be organized in L layers as above, where $\theta = \{W_\ell, b_\ell\}_{\ell=1}^L$. We interpret ϕ_θ as a predictive coding hierarchy with generative predictions from layer ℓ to $\ell - 1$. For each input x :

1. *Inference of activations* $r_\ell(x)$ by minimizing E_{inf} via local updates:

$$r_\ell(x) \leftarrow r_\ell(x) - \eta_r \frac{\partial}{\partial r_\ell} \sum_{m=1}^L \frac{1}{2} \|r_{m-1}(x) - g(W_m^- p r_m(x) + b_m)\|_2^2.$$

2. After inference, the top-layer activation $r_L(x) \in \mathbb{R}^{n_L}$ defines the deep feature:

$$\phi_\theta(x) = r_L(x).$$

Kernel and SVM Objective Given $\phi_\theta(x)$, define a Gaussian base kernel:

$$k_0(r, r') = \exp\left(-\frac{\|r - r'\|_2^2}{2\sigma^2}\right), \quad r, r' \in \mathbb{R}^{n_L}, \sigma > 0.$$

The deep kernel is $k_\theta(x, x') = k_0(\phi_\theta(x), \phi_\theta(x'))$. The SVM primal objective is

$$\begin{aligned} \min_{\alpha \in \mathbb{R}^n} \quad & \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j k_\theta(x_i, x_j) - \sum_{i=1}^n \alpha_i, \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C, \quad \sum_{i=1}^n \alpha_i y_i = 0. \end{aligned}$$

Let α^* be the optimal dual variables. The classifier is

$$f_{\alpha^*, \theta}(x) = \sum_{i=1}^n \alpha_i^* y_i k_\theta(x_i, x) + b^*.$$

Joint Learning Objective Our goal is to learn θ so as to minimize the SVM's generalization error. We define a surrogate objective $J(\theta)$ proportional to the SVM dual objective (or its regularized estimate). For simplicity, consider the *SVM margin* in feature space:

$$\gamma(\theta) = \min_{i=1, \dots, n} \frac{y_i (f_{\alpha^*, \theta}(x_i))}{\|w(\theta)\|_2}, \quad \text{where } w(\theta) = \sum_{i=1}^n \alpha_i^* y_i \phi_\theta(x_i).$$

We wish to *maximize* $\gamma(\theta)$ (i.e. find a weak classifier with large margin). Equivalently, we minimize

$$L(\theta) = -\gamma(\theta).$$

This objective depends on θ both through $\phi_\theta(x)$ in the kernel and through α^* , which itself depends on θ . In practice, one approximates α^* as fixed in each update, yielding a *bi-level* optimization.

Natural Gradient Update for θ We interpret θ as parameters of the predictive coding model that defines a probability distribution over features or latent variables given data. Let $p(r|x;\theta)$ be the implicit distribution over top-layer activations $r = r_L(x)$. The Fisher information matrix is

$$F(\theta) = \mathbb{E} \mathbb{E}_{x \sim \text{DD}} \mathbb{E} \mathbb{E}_{r \sim p(\cdot|x;\theta)} \left[\nabla_{\theta} \ln p(r|x;\theta) \nabla_{\theta} \ln p(r|x;\theta)^{\top} \right].$$

We compute the gradient of $L(\theta)$, $\nabla_{\theta} L$, by differentiating the SVM margin with respect to ϕ_{θ} and then backpropagating through the predictive coding inference. Using the *natural gradient*, the update is

$$\theta \leftarrow \theta - \eta_{\theta} F(\theta)^{-1} \nabla_{\theta} L(\theta),$$

where $\eta_{\theta} > 0$ is a learning rate. This ensures that the update respects the underlying statistical manifold of the predictive coding model, leading to more stable adaptation of the deep kernel.

Summary of Advantages Why might one want to do things this way?

- **Local, biologically plausible inference:** Predictive coding layers adjust their activations r_{ℓ} via local prediction-error minimization, avoiding backpropagation’s global pass.
- **Manifold-corrected weight updates:** Natural gradient uses $F(\theta)$ to precondition $\nabla_{\theta} L$, ensuring parameter steps follow the steepest descent on the probability manifold rather than Euclidean space.
- **Large-margin deep kernel:** By maximizing $\gamma(\theta)$, we ensure the learned kernel yields a weak (large-margin) classifier, which in turn reduces VC dimension and improves generalization.
- **Faster convergence and robustness:** Combining predictive coding’s rapid error correction with natural gradient’s curvature-aware steps yields efficient learning of θ .

29.0.5 Detailed Equations for Natural Gradient

We now provide explicit formulas for computing $\nabla_{\theta} L$ and $F(\theta)$.

1. *Gradient of the margin $\gamma(\theta)$.* Let

$$w(\theta) = \sum_{i=1}^n \alpha_i^* y_i \phi_{\theta}(x_i), \quad \|w(\theta)\|_2 = \sqrt{w(\theta)^{\top} p w(\theta)}.$$

The margin is $\gamma(\theta) = \min_i y_i \frac{w(\theta)^{\top} \phi_{\theta}(x_i)}{\|w(\theta)\|_2}$. Let $i^* = \arg \min_i y_i (w^{\top} \phi_{\theta}(x_i)) / \|w\|$. Then at θ :

$$\nabla_{\theta} \gamma = \frac{1}{\|w\|_2} \nabla_{\theta} [y_{i^*} w^{\top} \phi_{\theta}(x_{i^*})] - \frac{y_{i^*} (w^{\top} \phi_{\theta}(x_{i^*}))}{\|w\|_2^3} \nabla_{\theta} [\frac{1}{2} w^{\top} w].$$

Since $w = \sum_j \alpha_j^* y_j \phi_\theta(x_j)$,

$$\nabla_\theta [w^\top \phi_\theta(x_{i^*})] = \sum_j \alpha_j^* y_j \nabla_\theta [\phi_\theta(x_j)^\top p \phi_\theta(x_{i^*})] + \nabla_\theta [\phi_\theta(x_{i^*})]^\top p w.$$

Similarly, $\nabla_\theta [\frac{1}{2} w^\top w] = \sum_j \alpha_j^* y_j [\nabla_\theta \phi_\theta(x_j)]^\top p w$. One computes these by unrolling the predictive coding inference for each x_j .

2. *Fisher information* $F(\theta)$. Assume $r_L(x)$ follows a Gaussian approximate posterior $p(r_L | x; \theta) \propto \exp(-\frac{1}{2}(r_L - \mu_\theta(x))^\top p \Sigma_\theta^{-1}(x)(r_L - \mu_\theta(x)))$. Then

$$\nabla_\theta \ln p(r_L | x; \theta) = \nabla_\theta \left[\mu_\theta(x)^\top p \Sigma_\theta^{-1}(x) r_L - \frac{1}{2} r_L^\top p \Sigma_\theta^{-1}(x) r_L - \frac{1}{2} \ln \det \Sigma_\theta(x) \right].$$

Taking expectations over $r_L \sim p(\cdot | x; \theta)$ yields

$$F(\theta) = \mathbb{E}_x \left[\mathbb{E}_{r_L} [\nabla_\theta \ln p(r_L | x; \theta) \nabla_\theta \ln p(r_L | x; \theta)^\top p] \right],$$

which can be approximated by sampling r_L or using analytic expressions when $\Sigma_\theta(x)$ is diagonal or low-rank.

Putting these together, the natural gradient update is:

$$\theta \leftarrow \theta - \eta_\theta F(\theta)^{-1} \nabla_\theta [-\gamma(\theta)].$$

In practice, one uses a damped or approximated inverse Fisher (e.g. diagonal with small ridge) to ensure numerical stability.

Similar equations may be worked out for the Wasserstein metric.

29.1 Three Kinds of Weakness: An Elegant Synergy

This section has gone fairly deep down a particular rabbit hole regarding deep kernel function learning – let us now emerge back up the surface and take a high level view. What led us to explore this rabbit hold was an intuition that combining predictive coding with information geometry for deep kernel learning offers a principled route to learning kernels that yield large-margin classifiers. Predictive coding provides fast, local inference of layer activations, ensuring that feature representations adapt rapidly to data. Information geometry, via natural gradient updates, ensures that kernel parameters θ move in a way that respects the underlying statistical structure, leading to robust, stable, and efficient convergence. The resulting deep kernel yields an SVM classifier with high weakness (large margin) and provable generalization bounds, thereby uniting neural, probabilistic, and geometric perspectives in a unified learning framework.

What we would like to point out now, by way of summary, is that this synergy has an elegant formal underpinning. We now show how three ostensibly different "weakness" criteria—entropy-based in predictive coding (PC), Fisher-metric based in information geometry (IG), and margin-based in SVMs—fit into a single categorical weakness framework via different quantale enrichments and monoidal composition.

Predictive-Coding Weakness In predictive coding one maintains a generative model $p_\theta(z, x)$ and infers latent activations z given input x by minimizing prediction error. A high-entropy posterior $q(z | x)$ means the model remains *uncertain*—i.e. it is *weak* about which latent causes generated x .

Equip the latent-space category with the quantale

$$V_{\text{PC}} = [0, 1], \quad \otimes = \times, \quad \oplus = +,$$

and measure $\mu_{\text{PC}}(x)$ by the Shannon entropy

$$w_{\text{PC}}(x) = H(q(z | x)) = - \sum_z q(z | x) \log q(z | x).$$

Then $w_{\text{PC}}(x)$ is exactly the categorical weakness of the event "latent explanation under q for x ," viewed in the product space (z, z') .

Information-Geometric Weakness Natural-gradient updates move parameters θ along geodesics of (in the simplest case) the Fisher information metric, thereby changing the model as little as possible in distributional terms. A small Fisher-distance step is a *weak* change of hypothesis.

On the parameter manifold we use the quantale

$$V_{\text{IG}} = [0, \infty], \quad \otimes = \min, \quad \oplus = \sup,$$

and define for any candidate update $\Delta\theta$

$$w_{\text{IG}}(\Delta\theta) = \frac{1}{\sqrt{\Delta\theta^\top F(\theta) \Delta\theta}} = \|\Delta\theta\|_F^{-1}.$$

where $F(\theta)$ is the Fisher matrix. Maximizing w_{IG} yields the natural-gradient direction of least informational strength.

Margin (SVM) Weakness An SVM decision boundary that maximizes the geometric margin is the *weakest* separation: it excludes the smallest neighborhood of counter-examples while still classifying the data correctly.

As before, take

$$V_{\text{SVM}} = [0, \infty], \quad \otimes = \min, \quad \oplus = \sup,$$

and for a hyperplane $w, x \mapsto \text{sign}(w^\top x)$ define

$$w_{\text{SVM}}(w) = \inf_i \frac{y_i w^\top x_i}{|w|},$$

the usual hard-margin. Maximizing w_{SVM} yields the largest-weakness (max-margin) classifier.

Monoidal Composition of Weakness Each stage in this process – representation, parameter update, and classifier – fibers into its own enriched category with weakness measure w_{PC} , w_{IG} , w_{SVM} . We form the composite quantale

$$V_{\text{total}} = V_{\text{PC}} \otimes V_{\text{IG}} \otimes V_{\text{SVM}},$$

with $\otimes$ acting component-wise and $\oplus$ taking the joint supremum. Then the *overall weakness* is

$$w_{\text{total}} = w_{\text{PC}}(x) \otimes w_{\text{IG}}(\Delta\theta) \otimes w_{\text{SVM}}(w).$$

Choosing updates and kernels that maximize w_{total} yields:

- High-entropy latent representations (PC),
- Minimal informational change in parameters (IG),
- Large geometric margin (SVM).

In sum: By viewing each weakness as a morphism weight in a monoidal category, we see formally how neural (predictive-coding), probabilistic (Fisher), and geometric (margin) criteria *chain* together: each stage adds a causal subobject whose weakness is measured in its own quantale, and their monoidal product yields a single objective that unites all three perspectives.

29.2 Monoidal Composition of Weakness and Unified Hyperparameter Tuning

In this section we show how viewing the three weakness measures – predictive-coding weakness (w_{PC}), information-geometric weakness (w_{IG}), and SVM-margin weakness (w_{SVM}) – as a single monoidal composite

$$w_{\text{comb}} = w_{\text{PC}} \otimes w_{\text{IG}} \otimes w_{\text{SVM}}$$

yields both conceptual clarity and practical benefits for hyperparameter tuning across the entire deep-kernel/SVM apparatus.

29.2.1 Single Scalar Objective for Search

By optimizing w_{comb} directly, one replaces separate searches over PC inference rates, Fisher damping factors, and SVM regularization constants with a single unified goal. Raising w_{comb} guarantees non-degradation in any component weakness due to the commutativity and monotonicity of $\otimes$.

29.2.2 Evolutionary Tuning with Convergence Bounds

Treat hyperparameter configurations as individuals in an evolutionary algorithm with fitness w_{comb} . The CoDD Weakness then ensures that, under monoidal crossover and selection proportional to weakness, schema instances evolve exponentially fast toward maximal weakness.

29.2.3 Gradient-Based PC Hyperparameter Updates

Predictive-coding learning rules can be shown to minimize the PC weakness term directly. Hence one can apply standard gradient optimizers on PC rate parameters (inference learning rate, weight-update rate, lateral strength) with respect to the composite weakness loss,

$$\nabla_{\eta} w_{\text{comb}} = \nabla_{\eta} w_{\text{PC}} \otimes w_{\text{IG}} \otimes w_{\text{SVM}},$$

allowing online adaptation of PC hyperparameters in lockstep with IG and SVM components.

29.2.4 Geometry-Aware Learning-Rate Scheduling

The IG term’s Fisher-metric foundation yields learning-rate invariance to reparametrizations. By tying the IG step-size α_{IG} and the PC inference rate α_{PC} so that their contributions to w_{comb} remain balanced, one avoids imbalance where one modality dominates the joint objective.

The overall argument here would be that monoidal composition of weakness not only unifies neural, probabilistic, and geometric perspectives but also enables provably convergent, efficient, and elegant hyperparameter tuning across the full deep-kernel and SVM learning pipeline.

29.2.5 MDL versus Quantale-Weakness for Unified Hyperparameter Tuning

One would be justified to wonder: Why not recast the monoidal-composition tuning framework in MDL terms rather than dealing with weakness—i.e. measure the total code length of:

1. the predictive-coding inference schedule and parameters,
2. the Fisher-metric step sizes,
3. the final SVM weight vector,

and then minimize the sum of those three code-lengths.

However, the *weakness* formalism offers several practical advantages:

1. **Algebraic Simplicity.** Weakness lives in a commutative quantale $(V, \otimes, \oplus)$, so

$$w_{\text{comb}} = w_{\text{PC}} \otimes w_{\text{IG}} \otimes w_{\text{SVM}}$$

is automatically associative, commutative, and monotonic. An MDL approach would require designing a joint universal code for three distinct “languages” (PC schedules, manifold steps, SVM weights) and proving that bit-counts compose cleanly.

2. **Continuous Gradients.** Quantale-enriched weakness yields a smooth penalty term with closed-form gradients for all continuous hyperparameters. True MDL code lengths are integer bit-counts and non-differentiable; smoothing them via Shannon-Fano approximations essentially recreates a continuous regularizer equivalent to a weakness penalty.
3. **Unified Scale and Interpretability.** Each weakness term w_{PC}, w_{IG}, w_{SVM} lives in the same algebraic domain V , so one can directly compare contributions and know that improving any single term raises the composite score. MDL bits for parameters, errors, and margins may require additional normalization weights to balance across scales.

In summary, although MDL provides a principled complexity-versus-fit trade-off, the quantale-weakness framework embeds that trade-off in a leaner algebraic structure, yields smooth gradients, and inherits provable convergence properties for unified hyperparameter tuning.

30 Combining SVM and PLN for Inference Control

We now connect some dots between SVMs and PLN, providing an additional example of the combination of weakness-guided AI methods leveraging weakness defined over different quantales.

We have reviewed above one of the problems arising in PLN inference control: choosing which inference rule or “rewrite” to apply at each step so as to reach a conclusion efficiently. In other words, we need a heuristic that tells us:

- Which candidate inference step has the best chance of leading to a valid conclusion with minimal work,
- And which such step remains “weak” enough (i.e. broadly applicable) to avoid overcommitting prematurely.

Support Vector Machines (SVMs), as also reviewed above, provide a principled way to rank candidates by learning a decision boundary in a feature space, effectively estimating whether a given rewrite will be low-cost and high-weakness. Concretely, we can:

1. Construct a feature vector for each possible rewrite in the current PLN state (for example, features that capture how “tight” the rewrite’s constraints are, how many subgoals it produces, and so on).
2. Train an SVM to predict which rewrites historically led to efficient, generalizable inference—thus learning a notion of “weakness” (large margin) and “low cost.”
3. At inference time, for each candidate rewrite we compute its feature vector and use the SVM’s output to choose the rewrite that lies furthest on the “desirable” side of the margin.

This hybrid approach leverages PLN’s probabilistic logic to generate training labels (which rewrites succeeded quickly and generally), while using SVM’s margin-maximization to generalize and guide future inference. Thus PLN informs the SVM, and the SVM accelerates PLN’s search, resulting in markedly faster inference overall (or at least that’s the idea!).

We now review this approach in more detail.

30.1 PLN Inference and Rewrite Candidates

We begin by recalling the PLN inference setting. Let

$$\text{SSS} = \{\text{PLN states}\}$$

be the set of all possible inference states in PLN. Each state $s \in \text{SSS}$ consists of:

- A set of *goals* or *subgoals* to prove, each represented by a logical formula or probability distribution.
- A set of *known facts* or intermediate conclusions, with associated probabilities.

At state s , there is a finite set of *candidate rewrites*

$$\text{RR}_s \subseteq \text{RR},$$

where RR is the global set of all inference rules in the PLN system. Each rewrite $r \in \text{RR}_s$ transforms s into a new state $s' = r(s)$ by invoking one or more inference rules (e.g. probabilistic deduction, induction, abduction).

In this context, we then wish to learn a predictor that scores each candidate rewrite r in state s according to a combination of:

- *Cost*, denoted $c(s, r)$, measuring (or estimating) the expected number of additional inference steps (or time) needed after applying r .
- *Weakness*, denoted $w(s, r)$, measuring how broadly applicable (general) the resulting state $r(s)$ remains.

Rather than requiring exact values $c(s, r)$ and $w(s, r)$, we represent each pair (s, r) by a feature vector

$$\phi(s, r) = (\phi_1(s, r), \phi_2(s, r), \dots, \phi_d(s, r)) \in \mathbb{R}^d,$$

where each coordinate is a real-valued feature, for example:

1. $\phi_1(s, r)$: the number of new subgoals generated by applying r .
2. $\phi_2(s, r)$: the average prior probability of newly introduced facts in $r(s)$.
3. $\phi_3(s, r)$: a measure of rule specificity (e.g. number of literal constraints in r ’s pattern).

4. $\phi_4(s, r)$: the incremental change in the GSLT-graph diameter or 'spread' after r is applied.
5. $\phi_5(s, r)$: a count of how many previously-unseen predicates appear in $r(s)$.
6. ...

We assume ϕ is chosen so that higher values of certain linear combinations correspond to lower cost and higher weakness.

30.2 SVM for PLN Inference Control: Linear Case

We now formulate the SVM learning problem involved in PLN inference control. For simplicity we will initially assume a straightforward linear SVM but we will then show things work basically the same if one uses a learned deep kernel as outlined above.

We gather a training set

$$DD = \{ (\phi(s_i, r_{i1}), y_{i1}), \dots, (\phi(s_i, r_{im_i}), y_{im_i}) \}_{i=1}^N,$$

where for each historical inference state s_i :

- $\{r_{i1}, \dots, r_{im_i}\} = RR_{s_i}$ are the candidate rewrites at s_i .
- For each r_{ij} , we assign a label $y_{ij} \in \{-1, +1\}$, where $+1$ indicates that rewrite r_{ij} was *preferred* (i.e. led to low cost and high weakness), and -1 indicates it was *not preferred*.

Definition 9 (Linear SVM). *A linear SVM seeks a weight vector $w \in RR^d$ and bias $b \in RR$ that solve:*

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|_2^2 + C \sum_{i,j} \xi_{ij}, \\ \text{subject to} \quad & y_{ij} (w^\top \phi(s_i, r_{ij}) + b) \geq 1 - \xi_{ij}, \quad \xi_{ij} \geq 0, \quad \forall i, j. \end{aligned}$$

Here ξ_{ij} are slack variables, and $C > 0$ is a regularization parameter. The solution (w^*, b^*) yields a decision function

$$h(s, r) = \text{sgn}(w^{*-p} \phi(s, r) + b^*),$$

which returns $+1$ for -preferable- rewrites.

Remark. *The SVM objective $\frac{1}{2} \|w\|_2^2$ enforces a large margin between preferred and nonpreferred rewrite examples. Thus SVM is learning, in effect, a notion of weakness (large margin implies broad generality) and low cost (since we label examples according to observed efficiency).*

Inference Control with the SVM Predictor At run time, for a new PLN state s , we:

1. Enumerate the candidate rewrites $\text{RR}_s = \{r_1, \dots, r_m\}$.
2. Compute $\phi(s, r_j)$ for each $j = 1, \dots, m$.
3. Evaluate the SVM score

$$f_j = w^{*-p} \phi(s, r_j) + b^*, \quad j = 1, \dots, m.$$

4. Select

$$r^* = \arg \max_{j=1, \dots, m} f_j,$$

that is, the rewrite with the highest SVM output, which corresponds to large margin (weakness) and low cost (as per training).

5. Apply r^* to transition to the next state $s' = r^*(s)$. Observe the actual cost and weakness outcome, and add $(\phi(s, r^*), +1)$ to the training set if it performed well, or $(\phi(s, r^*), -1)$ if it did not-thus enabling online refinement.

PLN/SVM Integration Loop Formally, the combined algorithm we propose here proceeds as follows:

1. Initialization:

- Start with an empty or bootstrap training set DD.
- Solve the initial SVM on DD, obtaining (w^*, b^*) .
- Set the current PLN state s_0 to the initial query (e.g. -Explain data set D -).

2. Iterative Inference: For $t = 0, 1, 2, \dots$:

- (a) *Candidate Generation:* Let $\text{RR}_{s_t} \subseteq \text{RR}$ be the set of applicable rewrites at state s_t .
- (b) *Feature Computation:* For each $r \in \text{RR}_{s_t}$, compute $\phi_t(r) = \phi(s_t, r)$.
- (c) *SVM Scoring:* Compute

$$f_t(r) = w^{*-p} \phi_t(r) + b^*, \quad \forall r \in \text{RR}_{s_t}.$$

- (d) *Selection:* Choose

$$r_t^* = \arg \max_{r \in \text{RR}_{s_t}} f_t(r).$$

- (e) *Application:* Apply r_t^* , obtaining new state

$$s_{t+1} = r_t^*(s_t).$$

- (f) *Observation*: Measure the actual cost c_t of reaching a partial conclusion from s_{t+1} , and the actual weakness w_t (e.g. margin-based generality). Define a label:

$$y_t = \begin{cases} +1, & \text{if } (c_t, w_t) \text{ meets performance criteria,} \\ -1, & \text{otherwise.} \end{cases}$$

- (g) *Training Update*: Append $(\phi_t(r_t^*), y_t)$ to DD. Periodically (e.g. every K steps), retrain the SVM on the updated DD to obtain a new (w^*, b^*) .
- (h) *Termination*: If s_{t+1} satisfies the inference goal (e.g. success in proving or sufficiently high posterior probability), halt.

This approach yields the following interesting properties:

Margin Guarantees By construction, the SVM maximizes the margin between positive and negative examples in the feature space. A larger margin implies better generalization from DD to unseen states. In practice, this means that the SVM will robustly predict which rewrites are likely to be cost-effective and weak enough, even for novel PLN states.

Convergence of Inference Control Under mild assumptions on the diversity of training examples and the expressiveness of ϕ , the online-updated SVM will converge to a stable decision boundary. Consequently, the inference control policy converges to selecting near-optimal rewrites in the long run. A formal statement can be derived from standard online learning bounds (e.g. perceptron or stochastic SVM convergence guarantees).

Feature Design and Weakness Encoding Choosing the feature map ϕ is crucial. To encode *weakness*, one might include features such as:

- The *support* of a rewrite-how many data points or partial proofs it matches.
- A *margin-like* estimate, e.g. the distance between most restrictive constraints in the resulting subgoal.
- The *entropy* of the posterior distribution induced by the rewrite.

These capture how broad or narrow the rewrite’s effect is, enabling the SVM to approximate “weakness” in a linear feature space.

30.3 SVM for PLN Inference Control: Nonlinear Case

We now repeat the above analysis for the SVM learning problem in its nonlinear (kernel) form, focusing on the differences from the linear case presented above.

In particular, we assume a parametric "deep" feature map

$$\psi_\theta : (s, r) \mapsto \mathbb{R}^D,$$

e.g. the activations of a neural network with weights θ . Define the associated positive-definite kernel

$$K_\theta((s, r), (s', r')) = \langle \psi_\theta(s, r), \psi_\theta(s', r') \rangle.$$

Gather the same training set

$$\text{DD} = \{ (\psi_\theta(s_i, r_{ij}), y_{ij}) \}_{i=1, \dots, N; j=1, \dots, m_i},$$

with labels $y_{ij} \in \{-1, +1\}$ as before.

Definition 10 (Kernel SVM with Deep Kernel). *The SVM solves the dual problem*

$$\begin{aligned} \max_{\{\alpha_{ij}\}} \quad & \sum_{i,j} \alpha_{ij} - \frac{1}{2} \sum_{i,j} \sum_{i',j'} \alpha_{ij} \alpha_{i'j'} y_{ij} y_{i'j'} K_\theta((s_i, r_{ij}), (s_{i'}, r_{i'j'})), \\ \text{subject to} \quad & 0 \leq \alpha_{ij} \leq C, \quad \sum_{i,j} \alpha_{ij} y_{ij} = 0. \end{aligned}$$

The resulting decision function is

$$h_\theta(s, r) = \text{sgn} \left(\sum_{i,j} \alpha_{ij} y_{ij} K_\theta((s_i, r_{ij}), (s, r)) + b \right).$$

Remark. By the representer theorem, all dependence on the deep network enters via kernel evaluations K_θ . One can train θ by back-propagating a smooth surrogate of the SVM margin-objective (e.g. the squared-hinge loss plus a regularizer $R(\theta)$), yielding an end-to-end "deep kernel SVM."

Inference Control with the Kernel SVM Predictor At run time, for a new PLN state s :

1. Enumerate the candidate rewrites $\text{RR}_s = \{r_1, \dots, r_m\}$.
2. For each r_j , compute the feature map $\psi_\theta(s, r_j)$.
3. Evaluate the kernel-SVM score

$$f_j = \sum_{i,j'} \alpha_{ij'} y_{ij'} K_\theta((s_i, r_{ij'}), (s, r_j)) + b.$$

4. Select the rewrite

$$r^* = \arg \max_j f_j,$$

i.e. the one with largest (nonlinear) margin plus learned cost bias.

5. Apply r^* and observe cost c and weakness w . Append $(\psi_\theta(s, r^*), y)$ to DD with $y = +1$ if it met performance, else -1 .

PLN/Deep-Kernel-SVM Integration Loop The overall loop is identical, except that each retraining step now jointly or alternately updates $\{\alpha, b\}$ via the kernel-SVM dual and the deep-feature weights θ via gradient steps on a surrogate loss:

$$\mathcal{L}(\theta, \alpha, b) = \frac{1}{2} \sum_{i,j} \sum_{i',j'} \alpha_{ij} \alpha_{i'j'} y_{ij} y_{i'j'} K_{\theta}((s_i, r_{ij}), (s_{i'}, r_{i'j'})) - \sum_{i,j} \alpha_{ij} + \lambda R(\theta).$$

You then alternate:

$$\alpha, b \leftarrow \arg \min_{\alpha, b} \mathcal{L}, \quad \theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}.$$

Margin Guarantees All margin-based generalization bounds extend to RKHS margins: a larger margin in the feature space ψ_{θ} still implies better control over generalization error, provided one regularizes $\|\psi_{\theta}\|$ via $R(\theta)$.

Feature Design Now $\psi_{\theta}(s, r)$ can be a deep encoder of both symbolic and sub-symbolic aspects of the rewrite, letting the SVM kernel adaptively learn a nonlinear notion of "weakness+cost" that goes beyond any fixed linear feature map.

30.4 Categorical Integration of SVM and PLN Weakness: Linear Case

We now explain how our proposed hybridization of SVM and PLN emerges naturally from the categorical approach to weakness. We model the two forms of weakness as functors out of their native categories, then combine them in a product category and equip the result with a "weakest-first" ordering.

Base Categories and Weakness Functors

- $\mathcal{C}_{\text{lin}}$: objects (V, H) , a real vector space V with hyperplane H ; morphisms are affine maps preserving the hyperplane.

$$W_{\text{lin}} : (\mathcal{C}_{\text{lin}})^{\text{op}} \longrightarrow (\mathbb{R}_{\geq 0}, \leq), \quad W_{\text{lin}}(V, H) = \frac{1}{\gamma(H)},$$

where $\gamma(H)$ is the SVM margin.

- $\mathcal{C}_{\text{prob}}$: objects $(\Omega, \mathcal{F})$, a sample space Ω with "proof events" $\mathcal{F}$; morphisms are measure-preserving maps.

$$W_{\text{prob}} : (\mathcal{C}_{\text{prob}})^{\text{op}} \longrightarrow ([0, 1], \leq), \quad W_{\text{prob}}(\Omega, \mathcal{F}) = 1 - \Pr\left(\bigcup \mathcal{F}\right).$$

Product Category and Combined Weakness Form the product

$$\mathcal{C}_{\text{comb}} = \mathcal{C}_{\text{lin}} \times \mathcal{C}_{\text{prob}},$$

with projection functors π_1 and π_2 . Define

$$W_{\text{comb}} = \mu \circ (W_{\text{lin}} \times W_{\text{prob}}) : (\mathcal{C}_{\text{comb}})^{\text{op}} \longrightarrow (\mathbb{R}_{\geq 0}, \leq),$$

where $\mu(x, y) = x \cdot y$ (or $\min(x, y)$, etc.) gives a single weakness measure.

Commutative Diagram

$$\begin{array}{ccccc} \mathcal{C}_{\text{comb}} & \xrightarrow{\pi_1} & \mathcal{C}_{\text{lin}} & \xrightarrow{W_{\text{lin}}} & (\mathbb{R}_{\geq 0}, \leq) \\ \pi_2 \downarrow & & & & \uparrow \mu \\ & & \mathcal{C}_{\text{prob}} & \xrightarrow{W_{\text{prob}}} & ([0, 1], \leq) \end{array}$$

Here the square commutes once both $(\mathbb{R}_{\geq 0}, \leq)$ and $([0, 1], \leq)$ are identified via the embedding $[0, 1] \hookrightarrow \mathbb{R}_{\geq 0}$, and μ fuses the two weakness values.

Weakest-First Queue in $\mathcal{C}_{\text{comb}}$ Controlling PLN inference with SVM now amounts to running the usual "weakest-first" priority queue in the combined category $\mathcal{C}_{\text{comb}}$, ordered by W_{comb} . Concretely:

1. Each candidate proof state is an object $((V, H), (\Omega, \mathcal{F})) \in \mathcal{C}_{\text{comb}}$.
2. Compute its combined weakness

$$W_{\text{comb}}((V, H), (\Omega, \mathcal{F})) = \mu(1/\gamma(H), 1 - \Pr(\cup \mathcal{F})).$$

3. Insert into a priority queue keyed by this scalar; always expand the object with minimal W_{comb} – the *weakest* in the combined sense.

This construction is elegant because:

- **Modularity:** the SVM margin and probabilistic coverage remain defined in their native categories.
- **Compositionality:** new forms of weakness (e.g. logical, topological) can be integrated by forming larger product categories and extending μ .
- **Uniform inference:** the proof search algorithm need only compare real numbers W_{comb} , unifying margin-based and probabilistic heuristics under one ordering.

30.5 Categorical Extension to Kernel SVM: Nonlinear Case

All of the constructions for the linear SVM carry over verbatim once we observe that any kernel SVM is a linear SVM in its RKHS. We simply replace the "linear" category by a "kernel" category:

- Define

$$\mathcal{C}_{\text{kern}} = \{ (\mathcal{H}_{K_\theta}, H) \},$$

where $\mathcal{H}_{K_\theta}$ is the RKHS induced by the deep kernel K_θ , and $H \subset \mathcal{H}_{K_\theta}$ is an affine hyperplane. Morphisms are affine maps preserving the hyperplane.

- The margin-functor becomes

$$W_{\text{kern}} : (\mathcal{C}_{\text{kern}})^{\text{op}} \longrightarrow (\text{RR}_{\geq 0, \leq}), \quad W_{\text{kern}}(\mathcal{H}_{K_\theta}, H) = \frac{1}{\gamma_{K_\theta}(H)},$$

where $\gamma_{K_\theta}(H)$ is the RKHS margin of H .

We then form the same product category and combined weakness:

$$\mathcal{C}_{\text{comb}} = \mathcal{C}_{\text{kern}} \times \mathcal{C}_{\text{prob}}, \quad W_{\text{comb}} = \mu \circ (W_{\text{kern}} \times W_{\text{prob}}),$$

with $\mu(x, y) = x \cdot y$ (or $\min(x, y)$, etc.).

Commutative diagram

$$\begin{array}{ccccc} \mathcal{C}_{\text{comb}} & \xrightarrow{\pi_1} & \mathcal{C}_{\text{kern}} & \xrightarrow{W_{\text{kern}}} & (\text{RR}_{\geq 0, \leq}) \\ \pi_2 \downarrow & & & & \uparrow \mu \\ & & \mathcal{C}_{\text{prob}} & \xrightarrow{W_{\text{prob}}} & ([0, 1], \leq) \end{array}$$

Once we identify $[0, 1]$ with its image in $\text{RR}_{\geq 0}$, the square commutes and

$$W_{\text{comb}}((\mathcal{H}_{K_\theta}, H), (\Omega, \text{FF})) = \mu\left(\frac{1}{\gamma_{K_\theta}(H)}, 1 - \text{Pr}(\cup \text{FF})\right).$$

Weakest-first queue Inference control proceeds by inserting each candidate object $((\mathcal{H}_{K_\theta}, H), (\Omega, \text{FF}))$ into a priority queue keyed by W_{comb} , and always expanding the object with smallest W_{comb} . This is exactly the same "weakest-first" procedure as in the linear case, now in the combined kernel-probabilistic category.

31 Weakness-Based Formalization of Concept Blending

Returning to the theme of applying weakness to various aspects of learning and reasoning, we now explain how the information-theoretic approach to concept blending presented in [Goe25c] can naturally be expressed in terms of weakness:

31.1 Categorical Formulation

Let $\mathcal{C}$ be a category of *entities* (concepts, artifacts, etc.), enriched over a commutative quantale $(Q, \leq, \otimes, e)$. A *weakness functor*

$$W : \mathcal{C}^{\text{op}} \longrightarrow (Q, \leq, \otimes, e)$$

assigns to each morphism $f : P - E$ (which we interpret as the instantiation of "property" P in entity E) a weakness $W(f) \in Q$. Higher $W(f)$ means "harder to explain," lower $W(f)$ means "simpler/stronger" membership.

Fix two source objects $A, B \in \mathcal{C}$ and a candidate blend $C \in \mathcal{C}$. Let $\{x_i : P_i - C\}_{i=1}^m$ be a chosen finite set of salient properties of C . For each i , form the induced morphisms

$$f_i : P_i \longrightarrow A, \quad g_i : P_i \longrightarrow B$$

(using whatever universal constructions or mediating maps the category provides). Then define:

Blend-Membership Weakness

$$\mu_{\text{blend}}(C; A, B) = \min\left(e, \frac{1}{m} \sum_{i=1}^m \max\{W(f_i), W(g_i)\}\right).$$

Emergent Weakness

$$\mu_{\text{emergence}}(x_i; A, B, C) = \max\{W(x_i : P_i - C) - \max\{W(f_i), W(g_i)\}, 0\},$$

and weight by

$$w_i = \min\{W(f_i), W(g_i)\}.$$

High-Quality Blend

$$\mu_{\text{hq}}(C; A, B) = \min\left(e, \frac{1}{m} \sum_{i=1}^m w_i \mu_{\text{emergence}}(x_i; A, B, C)\right).$$

31.2 Choices of Weakness Functor

Below are four instances of W landing in $Q = [0, 1]$ with $\otimes = \times$ and $e = 1$:

1. **Probabilistic Weakness** For each property-instantiation morphism $f : P - E$, let

$$W_{\text{prob}}(f) = \Pr_{m \sim \mu}[f(P)] = \mathbb{E}_{m \sim \mu}[\mathbf{1}_{f(P)}].$$

2. **Mutual-Information Weakness** If $I(x_i; E)$ is the mutual information (in bits) between property x_i and entity E , set

$$W_{\text{MI}}(f) = 2^{-I(x_i; E)}.$$

3. **Linear (SVM-Margin) Weakness** Given a trained binary SVM classifier for " $x_i \in E$ " with signed margin $\gamma(f)$, define

$$W_{\text{svm}}(f) = \exp(-\kappa \gamma(f)^2), \quad \kappa > 0.$$

4. **Quantum Weakness** If entity E corresponds to density operator ρ_E and P to projector P_i , let

$$W_q(f) = 1 - F(P_i, \rho_E), \quad F(P_i, \rho_E) = (\text{Tr} \sqrt{\sqrt{\rho_E} P_i \rho_E P_i \sqrt{\rho_E}})^2.$$

31.3 Example: Blending *Bat* and *Man*

We take sources $A = \text{Bat}$, $B = \text{Man}$, and three candidate blends:

$$C_1 = \{\text{flight, detective skills}\}, \quad C_2 = \{\text{echolocation, altruism}\}, \quad C_3 = \{\text{flight, echolocation}\}.$$

Here $m = 2$ for each blend, with properties P_1, P_2 varying by C_i .

Weakness Values for Each Measure:

	flight		detective		
-prule Measure	Bat	Man	Bat	Man	
Probabilistic	0.90	0.20	0.10	0.80	
Mutual-Info	$2^{-3} \approx 0.125$	$2^{-0.5} \approx 0.707$	$2^{-0.2} \approx 0.871$	$2^{-2} = 0.25$	
SVM ($\kappa = 0.5$)	$\exp(-2) \approx 0.135$	$\exp(-0.5) \approx 0.607$	$\exp(-0.045) \approx 0.956$	$\exp(-1.62) \approx 0.199$	$\exp(-0.2) \approx 0.819$
Quantum	0.05	0.40	0.60	0.10	

Similarly for *altruism* (used in C_2):

$$W_{\text{prob}}(\text{altruism}) = (0.40, 0.85), \quad W_{\text{MI}} = (2^{-0.8}, 2^{-1.2}) \approx (0.574, 0.435),$$

$$W_{\text{svm}} = (\exp(-0.125) \approx 0.883, \exp(-2) \approx 0.135), \quad W_q = (0.30, 0.02).$$

Computed Blend-Membership Weakness μ_{blend} :

-prule Measure	$\mu_{\text{blend}}(C_1)$	$\mu_{\text{blend}}(C_2)$	$\mu_{\text{blend}}(C_3)$
Probabilistic	$\frac{0.90+0.80}{2} = 0.85$	$\frac{0.70+0.85}{2} = 0.775$	$\frac{0.90+0.70}{2} = 0.80$
Mutual-Info	$\frac{0.707+0.871}{2} \approx 0.789$	$\frac{0.933+0.574}{2} \approx 0.754$	$\frac{0.707+0.933}{2} \approx 0.820$
SVM	$\frac{0.607+0.956}{2} \approx 0.781$	$\frac{1.000+0.883}{2} \approx 0.941$	$\frac{0.607+1.000}{2} \approx 0.804$
Quantum	$\frac{0.400+0.600}{2} = 0.500$	$\frac{1.000+0.300}{2} = 0.650$	$\frac{0.400+1.000}{2} = 0.700$

Qualitative Interpretation

- *Probabilistic Weakness*: The blend C_2 (*echolocation, altruism*) has the *lowest* average weakness (0.775), since echolocation is strongly associated with bats (0.70) and altruism with humans (0.85). Thus μ_{blend} favors combining one signature trait of each source.
- *Mutual-Information Weakness*: C_2 again wins (0.754) because both echolocation and altruism carry substantial mutual information with their respective sources, yielding low algorithmic weakness.

- *SVM-Margin Weakness*: Here C_1 (*flight, detective skills*) is best (0.781), as the classifier most confidently separates flight and detective-ability features, giving a low margin-based weakness for that pair.
- *Quantum Fidelity Weakness*: Again C_1 wins (0.500), since the quantum-state fidelity for flight and detective skills in the "Bat" and "Man" density matrices is highest, leading to lower quantum weakness.

Thus, depending on whether one measures "weakness" via raw probability, information-theoretic simplicity, classification margin, or quantum fidelity, different candidate blends emerge as the strongest. This illustrates the flexibility of the categorical weakness-based blending framework.

31.4 Example: Divergent Rankings under SVM vs. Quantum Weakness

We blend two source entities in a musical domain:

$$A = \text{Piano}, \quad B = \text{Violin}.$$

We consider two salient properties:

$$P_1 = \text{harmonic richness}, \quad P_2 = \text{sustain length}.$$

Two candidate blends are:

$$C_1 = \text{Electric Piano} \quad (\text{rich harmonics, long sustain}),$$

$$C_2 = \text{Synthesized Hybrid} \quad (\text{mixed harmonics, mixed sustain}).$$

31.4.1 Weakness Measures

SVM-Margin Weakness Given an SVM trained to discriminate " $P_i \in E$ " vs. " $P_i \notin E$ ", each property-instantiation morphism $f : P_i - E$ has signed margin $\gamma(f)$. We set

$$W_{\text{svm}}(f) = \exp(-\kappa \gamma(f)^2), \quad \kappa > 0,$$

so that large margin (confident classification) gives low weakness.

Quantum Fidelity Weakness Interpreting each entity E as a density operator ρ_E and each property P_i as projector P_i , define

$$W_q(f) = 1 - F(P_i, \rho_E), \quad F(P_i, \rho_E) = \left(\text{Tr} \sqrt{\sqrt{\rho_E} P_i \rho_E P_i \sqrt{\rho_E}} \right)^2,$$

so that high fidelity yields low weakness.

31.4.2 Numerical Weakness Values

-prule Measure	$W(f_1)$		$W(g_1)$		$W(f_2)$	
	Piano	Violin	Piano	Violin	Piano	Violin
SVM	0.20	0.80	0.30	0.70	?	?
Quantum	0.10	0.40	0.20	0.20	?	?

Here f_1 is the map "harmonic richness – Piano" and g_1 is "harmonic richness – Violin"; similarly for f_2, g_2 on sustain length.

31.4.3 Blend-Membership Scores

We compute

$$\mu_{\text{blend}}(C) = \frac{1}{2} \left(\max\{W(f_1), W(g_1)\} + \max\{W(f_2), W(g_2)\} \right).$$

$$\mu_{\text{blend}}(C_1)_{\text{svm}} = \frac{1}{2} (\max\{0.20, 0.80\} + \max\{0.30, 0.70\}) = \frac{1}{2} (0.80 + 0.70) = 0.75,$$

$$\mu_{\text{blend}}(C_2)_{\text{svm}} = \frac{1}{2} (\max\{0.50, 0.60\} + \max\{0.40, 0.60\}) = \frac{1}{2} (0.60 + 0.60) = 0.60,$$

$$\mu_{\text{blend}}(C_1)_{\text{q}} = \frac{1}{2} (\max\{0.10, 0.40\} + \max\{0.20, 0.20\}) = \frac{1}{2} (0.40 + 0.20) = 0.30,$$

$$\mu_{\text{blend}}(C_2)_{\text{q}} = \frac{1}{2} (\max\{0.60, 0.50\} + \max\{0.50, 0.40\}) = \frac{1}{2} (0.60 + 0.50) = 0.55.$$

31.4.4 Divergent Rankings

- **SVM-Margin Weakness:** $\mu_{\text{blend}}(C_2) = 0.60 < 0.75 = \mu_{\text{blend}}(C_1)$, so the SVM-based measure prefers the *Synthesized Hybrid* C_2 (it lies closer to the linear decision boundary between Piano and Violin features).
- **Quantum Fidelity Weakness:** $\mu_{\text{blend}}(C_1) = 0.30 < 0.55 = \mu_{\text{blend}}(C_2)$, so the quantum measure prefers the *Electric Piano* C_1 (it retains higher fidelity to the classical Piano and Violin states).

Qualitative Insight. The SVM margin treats features as linearly separable attributes and favors blends that lie near the boundary (mixed features). Quantum fidelity, by contrast, respects the full state-space structure (including coherence) and rewards blends that maintain high overlap with each parent's so it prefers a blend that is individually closer to both sources.

This illustrates how quantum weakness can rank blends differently from classical margin-based measures.

One can get even more interesting blends by extending weakness beyond the quantum domain to quaternions or octonions but we will leave this for another paper!

32 Predictive Coding as Weakness-Minimizing Learning

The theoretical results of [PMZM25] indicate that predictive coding (PC) based neural net training can be especially valuable for learning minimum-description-length models. Here we extend and improve these results via moving from MDL to weakness, and we explore how this direction of thought can be used to improve the attention mechanism in transformer neural nets, including purely neural and symbolic waveformers.

32.1 Initial Predictive Coding Architecture

In this section we specify a precise predictive-coding-based neural architecture and update rule-set, which we will use for convergence and weakness-minimization analysis. For simplicity, we adopt the canonical hierarchical predictive-coding model of Rao and Ballard [RB99], implemented as a deep feed-forward network with local error units. The conceptual conclusions apply beyond this particular architecture but we need to specify something concrete to get simple, tractable results. A little later we will extend this simple model to include lateral connections.

32.1.1 Network Structure

Let the network have $L + 1$ layers indexed by $l = 0, 1, \dots, L$, where layer 0 is the input and layer L the topmost representation. For each input example n , denote

$$x_0^{(n)} = \text{input data}_n, \quad x_l^{(n)} \in \mathcal{R}^{d_l} \quad (l = 1, \dots, L)$$

the activation vectors at each layer. Each layer l maintains two sets of variables:

- A *representation* vector x_l .
- A *prediction error* vector ϵ_l of the same dimension.

The generative (top-down) mapping from layer l to layer $l - 1$ is given by

$$\hat{x}_{l-1} = f_l(W_l x_l),$$

where $W_l \in \mathcal{R}^{d_{l-1} \times d_l}$ are the feed-forward weights (shared between bottom-up and top-down paths) and f_l is a smooth activation (e.g. tanh or ReLU).

32.1.2 Inference (Error Propagation)

At each time step t of the inference phase, activations and errors are updated to minimize local prediction errors. For each layer $l = 1, \dots, L$, we compute:

$$\epsilon_{l-1} = x_{l-1} - f_l(W_l x_l) \quad \text{and} \quad x_l \leftarrow x_l + \eta_x W_l^T f'_l(W_l x_l) \epsilon_{l-1},$$

where f'_l is the elementwise derivative, and $\eta_x > 0$ is a small inference learning rate. The update for x_l uses the local prediction error ϵ_{l-1} and the transpose weights, implementing a local backprop of error.

32.1.3 Learning (Weight Update)

After the inference phase has approximately equilibrated the x_l and ϵ_l , the weights W_l are updated to reduce prediction error and network weakness:

$$\Delta W_l = -\eta_w \left[f'_l(W_l x_l) \epsilon_{l-1} \right] x_l^T - \eta_w \lambda \partial_{W_l} w(H_W),$$

where

$$H_W = \{(i, j) \mid (W_l)_{ij} \neq 0 \text{ for some } l\}$$

is the neuron-connection relation, $w(H_W)$ its quantale weakness, and $\partial_{W_l} w(H_W)$ the McBride derivative providing the per-weight sensitivity. Here $\eta_w > 0$ is the weight-learning rate and $\lambda \geq 0$ trades off prediction-error fidelity against simplicity.

32.1.4 Summary of Update Cycle

Each predictive-coding iteration consists of:

1. *Inference sweep*: for $l = 1, \dots, L$, update (x_l, ϵ_{l-1}) as above until convergence.
2. *Learning sweep*: for $l = 1, \dots, L$, update W_l via the combined prediction-error and weakness gradient.

Under mild smoothness and boundedness assumptions on f_l and the quantale operations, this alternating scheme provably decreases the composite loss

$$J(W, x) = \sum_{l,n} \|\epsilon_{l-1}^{(n)}\|^2 + \lambda w(H_W),$$

and converges to a blockwise stationary point.

32.2 PC as Descent on a Weakness-Incorporating Objective

We show that the standard predictive coding (PC) algorithm in neural networks, as outlined above, can be viewed as performing block-coordinate descent on a composite objective that combines prediction-error fidelity with a quantale-based weakness penalty. This interpretation bypasses the usual route through minimum description length (MDL) and ties PC directly to our categorical notion of simplicity.

32.2.1 Network Weakness Measure

Let D be a dataset and consider a feed-forward network with L layers. Denote by $\theta = (\theta_1, \dots, \theta_L)$ the layer-wise weight matrices, by $x_l^{(n)}$ the activation at

layer l for example n , and by $f_l(\theta_l, x_{l-1})$ the forward map at layer l . We view the network's connectivity as a relation

$$H_\theta = \{(i, j) \mid \theta_{l,ij} \neq 0 \text{ for some } l\} \subseteq N \times N,$$

where N is the set of all neurons. Assign to each neuron i a valuation $\mu(i)$ (for instance, its average activation), and – as a first simple pass – define the weakness of the network by

$$w(H_\theta) = \bigoplus_{(i,j) \in H_\theta} \mu(i) \otimes \mu(j),$$

with $(V, \otimes, \oplus)$ the chosen quantale.

32.2.2 Composite Loss Function

As a first simple stab in the right direction, we can introduce a combined loss

$$J(\theta) = \sum_{l=1}^L \sum_n \|x_l^{(n)} - f_l(\theta_l, x_{l-1}^{(n)})\|^2 + \lambda w(H_\theta),$$

where the first term is the usual sum of squared prediction errors and the second term penalizes network complexity via weakness.

32.2.3 Leveraging Local Weakness

The above mode for incorporating weakness into PC networks "works" but has one serious problem; it violates the PC principle of locality. In the predictive-coding architecture, penalizing the complexity of the entire network at once can obscure local credit assignment. Instead, it may be better to regularize only the *active* subgraph of connections used in each prediction step. Concretely, let $H_t \subseteq H_\theta$ be the set of synapses and neurons participating in the update at time t . We replace the global penalty

$$\lambda w(H_\theta)$$

with the local penalty

$$\lambda w(H_t) = \lambda \bigoplus_{e \in H_t} \phi(e),$$

where $\phi(e)$ is the weight of edge e in the weakness quantale.

Local Loss Formulation Unrolling across timesteps and layers, the combined loss becomes

$$L(\theta) = \sum_{t,l} \left\| x_{l-1}^{(t)} - f_l(\theta_{t,l}, x_l^{(t)}) \right\|^2 + \lambda \sum_{t,l} w(H_{t,l}),$$

where $H_{t,l}$ is the active subgraph at layer l and time t .

Benefits of Local Weakness

- **Locally Biased Credit Assignment.** Each weight update is regularized only by the complexity of the subgraph that actually generated the error.
- **Scalability.** Computing $w(H_{t,l})$ requires work proportional to the active subgraph size, and yields interpretable, per-layer complexity diagnostics.

Implementation in MeTTa/MORK In practice, $H_{t,l}$ can be defined as all neurons whose activations or errors exceed a small threshold at layer l , or simply the receptive-field subgraph. MORK’s PathMap indices allow efficient extraction of $H_{t,l}$, and MeTTa’s quantale-fold routines compute $w(H_{t,l})$ on the fly. During weight update for layer l , we backpropagate both the prediction-error gradient and

$$\lambda \nabla_{\theta_{t,l}} w(H_{t,l}),$$

affecting only parameters in the active neighborhood.

By focusing the Occam penalty on local neighborhoods, we ensure each update applies the right simplicity bias to the subnetwork that actually contributed to the prediction, improving both learning fidelity and regularization efficiency.

32.2.4 Predictive Coding Updates as Coordinate Descent

A single PC iteration alternates:

1. *Inference step:* update activations $\{x_l\}$ to reduce prediction error,

$$x_l \leftarrow x_l - \eta_x (x_l - f_l(\theta_l, x_{l-1})).$$

2. *Learning step:* update weights θ_l by

$$\Delta\theta_l = -\eta_\theta [\epsilon_l \odot f'_l(\theta_l, x_{l-1})] x_{l-1}^\top, \quad \epsilon_l = x_l - f_l(\theta_l, x_{l-1}).$$

Under mild smoothness assumptions on f_l , each of these substeps decreases $J(\theta)$ if λ is chosen so that the weight update aligns with a descent direction for $w(H_\theta)$. Thus PC implements an inexact block-coordinate descent on the composite objective J .

32.2.5 Choice of Neuron Valuation in Weakness-Based Predictive Coding

According to the above scheme, the quantale requires a valuation map

$$\mu : \{\text{neurons}\} \rightarrow V$$

that assigns each neuron a nonnegative “weight” reflecting its importance. Multiple criteria guide this choice, e.g.:

- **Reflective of Usage.** Neurons that contribute more to data reconstruction should receive higher valuation.
- **Smoothness.** The valuation should vary gradually with small changes in activity to avoid brittle weakness scores.
- **Computational Efficiency.** It must be updateable online with minimal overhead during inference.
- **Semantic Interpretability.** The measure should admit a clear probabilistic or geometric meaning for downstream modules.

Average Activation as a Baseline Average activation over a sliding window is proposed as an initial rough stab to start with, as on first impression it seems to meet all four criteria:

- *Usage.* A neuron with higher mean activation carries more predictive load.
- *Smoothness.* The running mean filters out spikes and noise, yielding stable valuations.
- *Efficiency.* A single add/multiply per neuron per time-step maintains the running mean incrementally.
- *Interpretability.* It matches classical notions of empirical occupancy or duty-cycle in information geometry.

Alternative Measures More complex options would include:

- *Activation Variance or Entropy*, to focus on unpredictability but requiring two-pass or higher storage.
- *Mutual Information*, measuring predictive power but demanding density estimation.
- *Graph-Centrality*, valuing neurons by connectivity but incurring global graph computations.

For initial experiments, average activation seems like the simplest viable default: it is trivial to implement, seems like it will produce nondegenerate weakness signals on simple tasks, and feels like it should interact smoothly with McBride-driven updates, yielding rapid convergence without oscillation. Experimentation will tell a fuller story!

32.2.6 McBride Derivative and Local Sensitivity

The McBride derivative of the weakness functor enumerates one-hole contexts: for each connection $(i, j) \in H_\theta$, it associates the residual context $H_\theta \setminus \{(i, j)\}$ and weight $\phi(i, j) = \mu(i) \otimes \mu(j)$. This provides the local sensitivity of $w(H_\theta)$ to adding or removing each connection. In the learning step, the gradient w.r.t. individual weight $\theta_{l,ij}$ includes a term

$$\frac{\partial}{\partial \theta_{l,ij}} w(H_\theta) = \phi(i, j) \frac{\partial}{\partial \theta_{l,ij}} \mathbf{1}_{\theta_{l,ij} \neq 0},$$

which guides synaptic updates to prune or reinforce connections according to their contribution to network weakness.

32.2.7 Convergence and Implications

Standard results on alternating minimization (to be elaborated below) imply:

- *Monotonicity*: PC substeps yield $J(\theta^{(t+1)}) \leq J(\theta^{(t)})$.
- *Blockwise stationarity*: limit points θ^{ast} satisfy $\partial_{\theta_l} J(\theta^{\text{ast}}) \ni 0$ for each layer l .

Hence PC emerges as a principled weakness-guided learning algorithm. This unifies energy-based neural learning with the categorical notion of weakness, suggesting new hybrid schemes in which McBride-derived sensitivities directly inform synaptic plasticity.

32.3 Convergence Guarantees for Alternating Minimization

Let us now dig into how the "alternating minimization" arguments can actually work here.

Conceptually, predictive coding alternates between two coordinate-descent steps—updating activations to reduce prediction error, and updating weights to reduce prediction error plus a weakness penalty. Under standard regularity assumptions, each substep decreases the overall objective, and any accumulation point of the sequence of iterates is blockwise stationary.

Assumptions

1. Each per-layer prediction-error term

$$E_l(\theta_l, x_l, x_{l-1}) = \|x_l - f_l(\theta_l, x_{l-1})\|^2$$

is continuously differentiable and has Lipschitz-continuous gradients in both θ_l and x_l .

2. The weakness penalty

$$w(H_\theta) = \bigoplus_{(i,j) \in H_\theta} \mu(i) \otimes \mu(j)$$

is nonnegative and lower-bounded.

3. The composite objective

$$J(\theta, x) = \sum_{l=1}^L \sum_n E_l(\theta_l, x_l^{(n)}, x_{l-1}^{(n)}) + \lambda w(H_\theta)$$

is coercive or level-bounded in θ for fixed x .

Monotonic Decrease Let $(\theta^{(t)}, x^{(t)})$ denote the parameters and activations at iteration t . One full PC sweep performs:

$$x^{(t+\frac{1}{2})} = \arg \min_x J(\theta^{(t)}, x), \quad \theta^{(t+1)} = \arg \min_\theta J(\theta, x^{(t+\frac{1}{2})}).$$

By construction,

$$J(\theta^{(t)}, x^{(t+\frac{1}{2})}) \leq J(\theta^{(t)}, x^{(t)}), \quad J(\theta^{(t+1)}, x^{(t+\frac{1}{2})}) \leq J(\theta^{(t)}, x^{(t+\frac{1}{2})}).$$

Hence each full iteration satisfies

$$J(\theta^{(t+1)}, x^{(t+1)}) \leq J(\theta^{(t)}, x^{(t)}),$$

establishing monotonic convergence of the objective values.

Blockwise Stationarity Assume that the sequence $(\theta^{(t)}, x^{(t)})$ has an accumulation point $(\theta^{\text{ast}}, x^{\text{ast}})$. By continuity and the optimality of each substep, one shows

$$0 \in \partial_x J(\theta^{\text{ast}}, x^{\text{ast}}), \quad 0 \in \partial_\theta J(\theta^{\text{ast}}, x^{\text{ast}}),$$

where ∂ denotes the limiting subdifferential (for nonsmooth w) or ordinary gradient. In particular, for each layer l ,

$$0 \in \partial_{\theta_l} J(\theta^{\text{ast}}, x^{\text{ast}}), \quad 0 \in \partial_{x_l} J(\theta^{\text{ast}}, x^{\text{ast}}).$$

Thus no single block update can further decrease the objective, i.e. $(\theta^{\text{ast}}, x^{\text{ast}})$ is blockwise stationary.

Discussion These results justify predictive coding as a weakness-guided alternating minimization: each PC substep strictly reduces the composite loss, and the algorithm converges (in objective value) to a point satisfying first-order optimality conditions in each coordinate block. Extensions to inexact or stochastic updates follow by standard analyses of block coordinate proximal descent.

32.4 Lateral Connections in Predictive Coding

Let us now consider enhancing our basic predictive coding neural set up by adding lateral connections within each layer. Conceptually, adding lateral interactions within each layer of a predictive-coding network allows units to share context, reduce redundancy, and stabilize inference. Lateral links can implement competition, cooperation, or attractor dynamics, all of which help drive down the combined prediction-error plus weakness objective more efficiently.

Benefits of Lateral Connections

- *Error decorrelation:* Inhibitory laterals suppress redundant activations, reducing overlap in residuals and lowering total weakness.
- *Contextual completion:* Excitatory laterals enable partial patterns to recruit neighbors, filling in missing structure and shrinking residual error.
- *Attractor stability:* Auto-associative lateral networks clean noisy activations, yielding more stable fixed points and faster convergence.

32.4.1 Learning Rules for Lateral Weights

We learn lateral weights L_{ij} by including their effect on the composite loss

$$J = \|\epsilon\|^2 + \lambda w(H_{\text{ff}} \cup H_{\text{lat}}),$$

where H_{ff} and H_{lat} are the feed-forward and lateral link relations.

Hebbian-Style Update

$$\Delta L_{ij} \propto x_i x_j - \alpha \frac{\partial}{\partial L_{ij}} w(H_{\text{lat}}),$$

so that co-activated units strengthen their connection unless its contribution to weakness exceeds a threshold.

Hopfield-Style Update Treat the lateral net as an auto-associative energy

$$E_{\text{lat}}(x) = -\frac{1}{2} x^\top L x,$$

and learn L by gradient descent on

$$J_{\text{lat}} = E_x[\|\epsilon(x)\|^2] + \lambda w(H_{\text{lat}}),$$

using the McBride derivative to compute $\partial w / \partial L_{ij}$.

Integration into Predictive Coding Each PC iteration with lateral learning proceeds:

```

1. Inference:
  for t = 1..T do
    x <- x + eta_x*(W^T*epsilon + L*x - x)
    epsilon <- bottom_up_error(x)
2. Learning:
  for each layer l do
    W <- W - eta_w*(dE/dW + lambda*d w(H_ff)/dW)
    L <- L - eta_l*(dE/dL + lambda*d w(H_lat)/dL)

```

where dE/dL is the gradient of the prediction-error after lateral settling, and $dw(H)/dL$ is given by the McBride derivative.

Expected Outcomes

- *Faster error reduction*: Lateral completion captures predictable structure in one step.
- *Sparser codes*: Competitive laterals prune redundant activity, lowering weakness.
- *Robust inference*: Attractor dynamics mitigate noise, improving convergence and stability.

By learning lateral connections under the same weakness-minimizing objective, predictive coding becomes a fully integrated framework in which both feed-forward and lateral architectures are optimized for simplicity and accuracy.

32.4.2 Formal Result on Lateral Connections Improving Convergence

We now give a precise statement and proof showing that adding positive-definite lateral connections strictly accelerates convergence of the predictive-coding inference step.

Demonstration Under Linear Assumptions We begin with a simplified demonstration that holds under linearized inference dynamics and suitable spectral assumptions,

Theorem 27. *Consider a single layer with linear generative model*

$$\hat{x} = W x,$$

prediction error $\epsilon = x_{\text{data}} - \hat{x}$, and inference update

$$x \leftarrow x + \eta_x (W^T \epsilon - L x),$$

where $W^T W$ and L are symmetric positive semidefinite, and $0 < \eta_x < 2/\|W^T W + L\|$. Let

$$M_{\text{no lat}} = I - \eta_x W^T W, \quad M_{\text{lat}} = I - \eta_x (W^T W + L).$$

If L is nonzero positive semidefinite then

$$\rho(M_{\text{lat}}) < \rho(M_{\text{no lat}}),$$

where $\rho(\cdot)$ denotes the spectral radius. Consequently, the inference iteration converges faster with lateral matrix L than without.

Proof. Since $W^T W$ and L commute (they are jointly diagonalizable by assumption), the eigenvalues of $M_{\text{no lat}}$ are

$$\lambda_i^{\text{no}} = 1 - \eta_x \sigma_i, \quad \sigma_i \in \text{spec}(W^T W),$$

and those of M_{lat} are

$$\lambda_i^{\text{lat}} = 1 - \eta_x (\sigma_i + \ell_i), \quad \ell_i \in \text{spec}(L).$$

Because L is positive semidefinite and nonzero, for each i we have $\ell_i \geq 0$ and at least one $\ell_j > 0$. Hence

$$|\lambda_i^{\text{lat}}| = |1 - \eta_x (\sigma_i + \ell_i)| < |1 - \eta_x \sigma_i| = |\lambda_i^{\text{no}}|,$$

for each i (using $0 < \eta_x < 2/\|W^T W + L\|$ to keep all eigenvalues in $(-1, 1)$). It follows that

$$\rho(M_{\text{lat}}) = \max_i |\lambda_i^{\text{lat}}| < \max_i |\lambda_i^{\text{no}}| = \rho(M_{\text{no lat}}),$$

proving the claim. $\square$

Remark This result shows that, at least in the linear regime, adding a symmetric positive-definite lateral matrix L yields a strictly smaller contraction factor for the inference dynamics. More generally, nonlinear activation functions and noncommuting matrices complicate the analysis, but the same principle—lateral connections act as a preconditioner—carries over and motivates their inclusion whenever one seeks faster error reduction under the combined prediction-error plus weakness objective.

Nonlinear Convergence via Contraction under Lipschitz Conditions

We now explore going beyond the linear case – which makes things a little messier but not qualitatively different. Conceptually, even in the presence of nonlinear activation functions, inference with lateral connections can be shown to converge faster by viewing the update as a fixed-point iteration and applying the Banach contraction principle. We only require mild assumptions on smoothness and boundedness of the nonlinearities, avoiding case-by-case analyses of specific nonlinearities and their implications.

Setup and Assumptions Consider a single layer with activation vector $x \in \mathcal{R}^d$, data input x_{data} , weight matrix $W \in \mathcal{R}^{d \times d}$, and lateral weight matrix $L \in \mathcal{R}^{d \times d}$. The generative prediction is

$$\hat{x} = f(Wx),$$

where $f : \mathcal{R}^d \rightarrow \mathcal{R}^d$ acts coordinate-wise with derivative $f'(z)$ satisfying

$$0 \leq m \leq f'(z) \leq M \quad \forall z \in \mathcal{R}^d.$$

We assume:

1. $W^T W$ and L are symmetric positive semidefinite and commute.
2. The step-size $\eta_x > 0$ satisfies

$$\eta_x < \frac{2}{M^2 \|W^T W\| + \|L\|}.$$

Inference Map and Jacobian The inference update with lateral term is

$$F(x) = x + \eta_x (W^T (x_{\text{data}} - f(Wx)) - Lx).$$

Its Jacobian at x is

$$DF(x) = I - \eta_x L - \eta_x W^T \text{diag}(f'(Wx))W,$$

where $\text{diag}(f'(Wx))$ is a diagonal matrix with entries in $[m, M]$.

Contraction Constant Since $W^T \text{diag}(f'(Wx))W \succeq m W^T W$ and $\preceq M W^T W$, we have

$$\|DF(x)\| \leq 1 - \eta_x \lambda_{\min}(L) - \eta_x m \lambda_{\min}(W^T W),$$

where $\lambda_{\min}$ denotes the smallest nonzero eigenvalue. By assumption on η_x , the right-hand side is strictly less than 1. Moreover, because $DF(x)$ is uniformly bounded in norm by a constant $\rho < 1$ for all x , F is a contraction on $\mathcal{R}^d$.

Convergence and Acceleration By the Banach fixed-point theorem, the iteration $x^{(t+1)} = F(x^{(t)})$ converges exponentially fast to the unique fixed point $\bar{x}$. Comparing to the case $L = 0$, the contraction constant

$$\rho(L) = 1 - \eta_x [\lambda_{\min}(L) + m \lambda_{\min}(W^T W)]$$

is strictly smaller for any nonzero L . Hence lateral connections accelerate convergence of the nonlinear inference dynamics under these general Lipschitz and spectral assumptions.

Discussion This argument cleanly extends the linear spectral-radius proof to nonlinear activations, replacing $W^T W$ by its lower bound $m W^T W$ and showing that any positive-definite L further reduces the contraction rate. It holds without specifying particular forms of f , relying only on its uniform derivative bounds, and applies layer by layer in a deep network.

33 The Role of Commutativity in Continual Learning in Predictive Coding Networks

One issue commonly observed in neural nets applied in a continuous or lifelong learning manner is "catastrophic forgetting" – it can be counterintuitively hard to get neural nets to keep on learning and keep on acquiring new knowledge and memories while gracefully rather than messily and suddenly letting its memory of older things fade as it reaches capacity limits. Adding lateral activations to PC networks—implementing competition or inhibition within layers—can help somewhat, improving feature disentanglement and stability in continual learning scenarios. However, naively interleaving inference and lateral updates can lead to order-dependent weight trajectories and catastrophic interference.

In this subsection we explore a principled potential remedy to these issues based on a *commutativity design principle*, proposed by Daniel MacDonald [Mac25] but fleshed out here in a more general and mathematical way. The core idea here is to require that the quantized update actions for PC inference and for lateral inhibition lie in a commutative subquantale. Under this assumption, we prove here that (i) inference and lateral updates commute on the network's parameter module, (ii) the associated enriched-category of layers has all relevant diagrams strictly commuting, and (iii) the parameter space cleanly factorizes into feedforward and lateral submodules, yielding disentangled learning.

We then show how the same quantale-enriched formalism applies to PC-driven reinforcement learning within the MetaMo motivational framework. By identifying sensation and decision updates as commuting quantale actions, we obtain commuting appraisal and decision functors, exactly matching MetaMo's desideratum. We conclude with a discussion of failure modes when commutativity is absent, illustrating risks in both image classification and social-context RL.

33.1 Simple Predictive-Coding Classifier with Lateral Activations

First we outline a simple reference PC network, to use for initial exploration of our ideas on commutativity.

We consider a feedforward network for image classification with L hidden layers. At each layer $\ell = 1, \dots, L$ we maintain:

- A latent activation vector $z_\ell \in \mathbb{R}^{n_\ell}$,

- A feedforward prediction $\hat{z}_\ell = f_\ell(z_{\ell-1}; W_\ell)$, where z_0 is the input image and W_ℓ are the layer weights,
- A lateral prediction $\tilde{z}_\ell = V_\ell z_\ell$, where $V_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$ implements lateral inhibition.

Define the total energy over latents $z_{1:L}$ and label y by

$$E(z_{1:L}, y) = \sum_{\ell=1}^L \frac{1}{2} \|z_\ell - \hat{z}_\ell\|^2 + \frac{1}{2} \|z_\ell - \tilde{z}_\ell\|^2 + \frac{1}{2} \|y - g(z_L; W_{L+1})\|^2,$$

where $g(\cdot; W_{L+1})$ is the final classification readout. Inference is performed by iterating, for each ℓ ,

$$z_\ell \leftarrow z_\ell - \alpha \left[(z_\ell - \hat{z}_\ell) + (z_\ell - \tilde{z}_\ell) \right],$$

and similarly updating y . Learning then updates

$$\Delta W_\ell = \eta (z_\ell - \hat{z}_\ell) z_{\ell-1}^T, \quad \Delta V_\ell = \eta (z_\ell - \tilde{z}_\ell) z_\ell^T.$$

33.2 Commutativity Principle in Continual Learning

The design principle of commutativity demands that inference updates (driven by reconstruction error) and lateral updates (driven by inhibition error) can be applied in either order without affecting the final state. Concretely, if we denote by q_{inf} the update action corresponding to feedforward inference and by q_{lat} the action for lateral inference, commutativity requires

$$q_{\text{inf}} \star (q_{\text{lat}} \star \theta) = q_{\text{lat}} \star (q_{\text{inf}} \star \theta),$$

where $\theta = (W_{1:L}, V_{1:L})$ are the parameters. This ensures that alternating inference and lateral steps in arbitrary sequences does not introduce order-dependent drift, a key property for stable continual learning.

33.2.1 Quantale-Enriched Formalization

Let us formalize this commutativity principle in a more abstract way, so as to more thoroughly understand its implications.

Let $Q = (\mathbb{R}_{\geq 0} \cup \{\infty\}, \min, +)$ be the commutative quantale of energy levels, and let Θ be the Q -module of all network parameters under the action

$$q \star \theta = \theta + \Delta\theta(q),$$

where $\Delta\theta(q)$ is the parameter change induced by an update of energy q . In particular we identify

$$q_{\text{inf}} \in Q, \quad q_{\text{lat}} \in Q,$$

and require $q_{\text{inf}} \otimes q_{\text{lat}} = q_{\text{lat}} \otimes q_{\text{inf}}$ in the commutative subquantale of valid updates.

Organize the L layers as a small Q -enriched category $\mathcal{L}$ with objects $1, \dots, L$ and hom-objects

$$\mathcal{L}(\ell, \ell) = e, \quad \mathcal{L}(\ell, \ell + 1) = q_{\text{inf}}, \quad \mathcal{L}(\ell, \ell) = q_{\text{lat}},$$

so that each square

$$\begin{array}{ccc} \ell & \xrightarrow{q_{\text{inf}}} & \ell + 1 \\ q_{\text{lat}} \downarrow & & \downarrow q_{\text{lat}} \\ \ell & \xrightarrow{q_{\text{inf}}} & \ell + 1 \end{array}$$

commutes strictly in Q . Finally, decompose Θ as a symmetric-monoidal tensor of submodules

$$\Theta \cong W \otimes_Q V,$$

where W are feedforward weights and V are lateral weights. Updates from q_{inf} act only on W , and updates from q_{lat} act only on V .

Given this setup, we can then draw conclusions such as:

Theorem 28. *In the above PC-based classifier with lateral activations, if*

1. *the inference and lateral update energies $q_{\text{inf}}, q_{\text{lat}} \in Q$ commute,*
2. *all hom-objects in the layer category $\mathcal{L}$ associated to inference and lateral edges lie in the commutative subquantale,*
3. *the parameter module Θ factorizes as $W \otimes_Q V$ with disjoint update actions,*

then

- *inference and lateral updates commute on Θ ,*
- *all enriched-category diagrams for layer transitions commute strictly,*
- *the factorization yields disentangled learning of feedforward and lateral parameters.*

Proof. Commutativity of q_{inf} and q_{lat} implies

$$q_{\text{inf}} \star (q_{\text{lat}} \star \theta) = (q_{\text{inf}} \otimes q_{\text{lat}}) \star \theta = (q_{\text{lat}} \otimes q_{\text{inf}}) \star \theta = q_{\text{lat}} \star (q_{\text{inf}} \star \theta).$$

Strict commutation of each square in $\mathcal{L}$ follows because the corresponding hom-objects multiply in Q in either order to the same result. Finally, since q_{inf} acts only on the W factor and q_{lat} acts only on the V factor in the tensor product, their updates never interfere, yielding a clean disentanglement of the two parameter subspaces. $\square$

33.2.2 MetaMo Motivational Framework and Commutativity in PC-RL Networks

We now show that a PC-driven RL network whose inference and learning updates commute in the quantale sense instantiates MetaMo commutativity at the category level.

Theorem 29. *Let $Q = (Q, \otimes, \vee)$ be a commutative quantale of energy costs, and let Θ be a Q -module of network parameters. Identify two distinguished updates:*

$$q_{\text{sens}}, q_{\text{dec}} \in Q$$

corresponding respectively to predictive-coding inference (“sensation”) and RL reward-driven learning (“decision”). Suppose

$$q_{\text{sens}} \otimes q_{\text{dec}} = q_{\text{dec}} \otimes q_{\text{sens}}$$

so that sensation and decision updates commute on Θ . Define two endofunctors on the category $\mathcal{S}$ of network states:

$$\text{App}(\theta) = q_{\text{sens}} \star \theta, \quad \text{Dec}(\theta) = q_{\text{dec}} \star \theta.$$

Then

$$\text{App} \circ \text{Dec} = \text{Dec} \circ \text{App}$$

as endofunctors on $\mathcal{S}$. Conversely, if q_{sens} and q_{dec} do not commute, then $\text{App} \circ \text{Dec} \neq \text{Dec} \circ \text{App}$.

Proof. For any network state $\theta \in \Theta$, the composite actions satisfy

$$(\text{App} \circ \text{Dec})(\theta) = q_{\text{sens}} \star (q_{\text{dec}} \star \theta) = (q_{\text{sens}} \otimes q_{\text{dec}}) \star \theta,$$

and

$$(\text{Dec} \circ \text{App})(\theta) = q_{\text{dec}} \star (q_{\text{sens}} \star \theta) = (q_{\text{dec}} \otimes q_{\text{sens}}) \star \theta.$$

Commutativity $q_{\text{sens}} \otimes q_{\text{dec}} = q_{\text{dec}} \otimes q_{\text{sens}}$ implies the two composites coincide on every object, hence the functors are equal. If the updates do not commute, then the two composite actions differ for some θ , so the functors differ. $\square$

33.2.3 When Commutativity Fails

If predictive-coding inference and RL updates do not commute, then appraisal and decision no longer commute in the MetaMo sense. This can lead to practical failures, such as:

Image Classification A non-commutative PC-classifier may yield different parameter trajectories depending on the scheduling of inference and reward updates. For example, applying a classification-error correction before cleaning up hidden activations can lead to weights that overfit spurious noise. The result is unstable accuracy and greater sensitivity to update order.

Social-Context RL In a complex social environment, decision updates encode learned policies from reward signals, while inference updates integrate new observations of other agents. If these updates do not commute, then learning a policy before fully inferring new social cues can misalign the agent’s beliefs, causing erratic or unsafe actions. Conversely, over-inference before policy adjustment can slow learning or entrench outdated models of other agents. Lack of commutativity thus risks both poor adaptation and unintended social consequences.

33.2.4 Commutativity Regularization in Predictive-Coding Networks

We have argued strenuously for the value of commutativity in PC networks. However, that doesn’t mean commutativity is necessarily easy to achieve. Designing a predictive-coding network so that inference and lateral (or learning) updates commute by construction often requires exact algebraic structures or dual projection steps, which can be computationally expensive and difficult to scale. As MacDonald points out a lightweight *commutativity regularizer* provides a practical alternative: it softly penalizes order-dependent dynamics, encouraging updates to become order-independent without enforcing rigid constraints.

MacDonald’s Basic Commutativity Regularizer MacDonald proposes adding to the loss a term that measures the difference between two update operators applied in opposite orders. If U and V are the inference and lateral update maps on parameters θ , then

$$R_{\text{Mac}}(\theta) = \frac{1}{2} \|U(V(\theta)) - V(U(\theta))\|^2.$$

Minimizing this term alongside the original objective drives $U \circ V \approx V \circ U$, reducing interference in continual learning.

Fisher Information Geometry Regularizer One can go even further in this direction using an information geometry approach.

In a Fisher-information geometry view, U and V induce natural-gradient vector fields X_U and X_V on parameter space with metric $G(\theta)$. Commutativity is equivalent to

$$[X_U, X_V](\theta) = 0,$$

where $[\cdot, \cdot]$ is the Lie bracket. Define the Fisher regularizer

$$R_F(\theta) = \frac{1}{2} \langle [X_U, X_V](\theta), [X_U, X_V](\theta) \rangle_{G(\theta)},$$

with $\langle v, w \rangle_G = v^T G w$. Adding λR_F encourages the two natural-gradient flows to commute.

Wasserstein Information Geometry Regularizer Alternatively, one can view updates as gradient flows in the space of parameter distributions under the 2-Wasserstein metric. Let Φ_U and Φ_V be the JKO-step operators for the two energies. Form the two compositions

$$p^{UV} = \Phi_V(\Phi_U(p)), \quad p^{VU} = \Phi_U(\Phi_V(p)).$$

Define the Wasserstein regularizer

$$R_W(p) = \frac{1}{2} W_2^2(p^{UV}, p^{VU}).$$

Minimizing λR_W alongside the inference and lateral objectives aligns the two flows in the optimal-transport geometry, promoting order-independent updates.

33.3 Causal Weakness-Augmented Transformer Attention

The connection of PC neural learning with weakness is broad in applicability, and can be deployed in the context of many different neural architectures aimed at various different objectives. As a particular and potentially useful example, we explore how to apply it in a transformer context.

Conceptually, we wish to bias transformer self-attention toward past tokens that admit a simple, predictive (causal) relationship to the current token. We define a *causal weakness score* between each context position i and target position t by fitting a tiny "weak" model and measuring its residual weakness. This score is then converted into a causal salience prior, which modulates the transformer's attention logits.

Causal Weakness Score For each pair of positions (i, t) , let x_i be the embedding at position i and x_t the target embedding. Fit a simple predictor (e.g. a linear model or one-layer MLP) $\hat{x}_t^{(i)} = g(x_i; \phi_i)$, and define the residual set

$$R_{i \rightarrow t} = \{(c, x_t) \mid g(x_i; \phi_i) \text{ fails to predict } x_t\}.$$

Assign a valuation μ on events (e.g. token frequency or embedding norm) and measure

$$w_{i \rightarrow t} = \bigoplus_{(c, x_t) \in R_{i \rightarrow t}} \mu(c) \otimes \mu(x_t),$$

the weakness of the residual.

Causal Salience Prior Invert weakness to a salience score

$$s_{i \rightarrow t} = \exp(-\gamma w_{i \rightarrow t}), \quad \gamma > 0.$$

Low residual weakness (strong simple predictor) yields high $s_{i \rightarrow t}$.

Attention Logit Modulation In a standard Transformer layer, the attention logits are

$$A_{i,t} = \frac{q_t^T k_i}{\sqrt{d}}.$$

We augment this with the causal salience prior:

$$A'_{i,t} = A_{i,t} + \alpha \log s_{i \rightarrow t},$$

where $\alpha \geq 0$ controls the strength of the causal bias. The attention weights become

$$\text{Attn}(x_t) = \sum_i \text{softmax}_i(A'_{i,t}) v_i.$$

End-to-End Training All parameters—including the weak-model parameters $\{\phi_i\}$, the transformer weights, and the bias scale α —are trained jointly to minimize the overall language-modeling (or prediction) loss. Optionally one adds a complexity penalty on the weak models, for example an ℓ_1 norm on ϕ_i , to enforce simplicity directly.

Extensions

- *Multi-head causal priors:* Compute distinct $w_{i \rightarrow t}^h$ and $s_{i \rightarrow t}^h$ per attention head h .
- *Cross-attention:* Apply the same causal weakness modulation when attending over encoder tokens in an encoder-decoder model.

By integrating a quantale-based causal weakness score into the attention mechanism, the model “pays attention” preferentially to those past tokens that support a simple causal explanation for the current prediction.

33.4 Integrating Causal Weakness into WaveFormer

A variation on the above ideas would be to integrate weakness-oriented PC learning into the Waveformer wavelet-based transformer architecture. Conceptually, WaveFormer applies a discrete wavelet transform (DWT) to layer activations, processes low-frequency (coarse) and high-frequency (detail) subbands via self-attention, then recombines via an inverse DWT (IDWT). To bias attention toward causally important tokens, we introduce a causal-weakness prior in parallel with the symbolic multiresolution transform and inject it into the WaveFormer’s attention of both subbands.

Causal Weakness Subband Priors At each layer and for each attention head h , compute causal weakness scores $w_{i \rightarrow t}^h$ and salience $s_{i \rightarrow t}^h = \exp(-\gamma w_{i \rightarrow t}^h)$ as we have looked at above, but restricted to low-freq and high-freq token streams:

$$D_{i \rightarrow t}^{\text{low}}, \quad D_{i \rightarrow t}^{\text{high}},$$

where $D_{i \rightarrow t}^{\text{low}}$ contains tokens in the low-frequency subband and $D_{i \rightarrow t}^{\text{high}}$ the detail subband. We compute separate salience priors

$$s_{i \rightarrow t}^{\text{low},h} = \exp(-\gamma^{\text{low}} w^h(D_{i \rightarrow t}^{\text{low}})), \quad s_{i \rightarrow t}^{\text{high},h} = \exp(-\gamma^{\text{high}} w^h(D_{i \rightarrow t}^{\text{high}})).$$

Modified WaveFormer Attention Let q_t^h, k_i^h, v_i^h be the queries, keys, and values for head h , and let

$$A_{i,t}^{\text{low},h} = \frac{q_t^{hT} k_i^h}{\sqrt{d_k}} + \alpha^{\text{low}} \log s_{i \rightarrow t}^{\text{low},h},$$

$$A_{i,t}^{\text{high},h} = \frac{q_t^{hT} k_i^h}{\sqrt{d_k}} + \alpha^{\text{high}} \log s_{i \rightarrow t}^{\text{high},h}.$$

We then compute separate attention weights

$$\text{Attn}^{\text{low},h}(x_t) = \sum_i \text{softmax}_i(A_{i,t}^{\text{low},h}) v_i^h, \quad \text{Attn}^{\text{high},h}(x_t) = \sum_i \text{softmax}_i(A_{i,t}^{\text{high},h}) v_i^h.$$

The layer’s output is obtained by inverse DWT combining these two streams:

$$y_t = \text{IDWT}\left(\{\text{Attn}^{\text{low},h}(x_t)\}_h, \{\text{Attn}^{\text{high},h}(x_t)\}_h\right).$$

End-to-End Training All parameters—including the transforms, the weak-model predictors for w^h , and the scaling factors $\alpha^{\text{low}}, \alpha^{\text{high}}$ —are learned jointly by minimizing the standard task loss (e.g. language modeling or classification). Optionally one adds a complexity penalty on the weak predictors to enforce Occam bias.

Expected Benefits

- *Causal focus in both subbands:* Low-frequency attention prioritizes tokens that provide simple global context; high-frequency attention amplifies detail tokens with strong simple causal links.
- *Interpretable multi-scale causality:* Salience priors at each scale reveal which coarse or fine patterns drive predictions.
- *Efficiency and accuracy:* By gating attention with causal weakness, the WaveFormer can allocate computation to genuinely predictive features and reduce spurious high-frequency noise.

33.5 Integrating Multiresolution Weakness into the Neural-Symbolic Transformer

Conceptually, the Neural-Symbolic Transformer marries a symbolic multiresolution weakness model with a neural WaveFormer backbone. The symbolic branch produces a hierarchy of coarse and fine ”weak” models of the input,

while the WaveFormer processes the same input via discrete wavelet subbands. At each layer and for each head, we fuse the two streams so that the neural attention is guided by symbolic weakness scores, yielding an end-to-end trainable, interpretable multiscale architecture.

Symbolic Multiresolution Branch

- Input tokens $x_1, \dots, x_T$ are fed to a multiresolution weakness transform, yielding subobject probabilities $p_{i,t}^{\text{sym}}$ and residual weakness coefficients $c_{i,t}^{\text{sym}}$ at levels $i = 0, \dots, N$.
- For each level i we compute salience priors

$$s_{i,t}^{\text{sym}} = \exp(-\gamma_i c_{i,t}^{\text{sym}}).$$

- We embed these priors into per-token "symbolic embeddings" $e_{i,t}^{\text{sym}}$ via a small projection network.

Neural WaveFormer Branch

- Each layer applies a DWT to the token features, splitting them into low-frequency $L_{i,t}^h$ and high-frequency $H_{i,t}^h$ subbands per head h .
- Standard self-attention computes

$$A_{i,t}^{\text{low},h} = \frac{q_t^{hT} k_i^h}{\sqrt{d_k}}, \quad A_{i,t}^{\text{high},h} = \frac{q_t^{hT} k_i^h}{\sqrt{d_k}}.$$

Fusion of Symbolic and Neural Streams We augment each attention logit with the corresponding symbolic salience:

$$A_{i,t}^{\text{low},h} = A_{i,t}^{\text{low},h} + \alpha^{\text{low}} \log s_{i,t}^{\text{sym}}, \quad A_{i,t}^{\text{high},h} = A_{i,t}^{\text{high},h} + \alpha^{\text{high}} \log s_{i,t}^{\text{sym}}.$$

The combined attention outputs are

$$\text{Attn}^{\text{low},h}(x_t) = \sum_i \text{softmax}_i(A_{i,t}^{\text{low},h}) v_i^h, \quad \text{Attn}^{\text{high},h}(x_t) = \sum_i \text{softmax}_i(A_{i,t}^{\text{high},h}) v_i^h.$$

These are then merged via the inverse DWT:

$$y_t = \text{IDWT}(\{\text{Attn}^{\text{low},h}(x_t)\}_h, \{\text{Attn}^{\text{high},h}(x_t)\}_h).$$

End-to-End Learning All parameters—including the symbolic transform’s thresholds and projections, the WaveFormer weights, and the fusion scales $\alpha^{\text{low}}, \alpha^{\text{high}}$ —are trained jointly to minimize the task loss (e.g. language modeling cross-entropy). Optional regularizers on the symbolic branch enforce simplicity of the multiresolution models via additional weakness penalties.

This tight integration ensures that the neural WaveFormer attention is directly guided by Occam-style symbolic simplicity measures at every scale, yielding a unified neural-symbolic transformer that is both efficient and interpretable.

33.6 Weakness-AIXI via Predictive Coding and Reinforcement Learning

A different – and less immediately practical – sort of application of weakness-integrated PC neural nets is to provide a twist on our above ideas about Universal AI. We can replace AIXI^{tl}’s Solomonoff-like mixture over computable models with a single predictive-coding network that embodies an Occam bias via weakness regularization, and use that network for both world-model learning and policy search.

Predictive-Coding World Model Let $P_\theta(h_{t+1} \mid h_t, a_t)$ be a predictive-coding network that, given history h_t and action a_t , predicts the next observation and reward. Train it online by minimizing

$$L_{\text{model}}(\theta) = \sum_t \|o_{t+1} - \hat{o}_{t+1}\|^2 + \lambda w(\ker P_\theta),$$

where $w(\ker P_\theta)$ is the quantale-folded weakness of the network’s connection relation (computed via the McBride derivative).

Weakness-Regularized Planning At decision time, generate candidate action sequences $a_{t:t+H}$ and simulate trajectories using P_θ . Score each sequence by

$$V(a_{t:t+H}) = \sum_{k=1}^H r_{t+k} - \beta w(\ker P_\theta \text{ on simulated data}),$$

and select the sequence with maximal V , executing its first action. This is an Occam-weighted expectimax in a single learned model.

Policy Extraction via Predictive Coding RL Augment P_θ with a policy head $\pi_\phi(a \mid h)$ and train (θ, ϕ) jointly on

$$L_{\text{policy}}(\phi) = -E_{a \sim \pi_\phi} \left[\sum_{k=1}^H r_{t+k} - \beta w(\ker P_\theta) \right] + \gamma w(\ker \pi_\phi),$$

so that action proposals minimize complexity as well as maximize reward.

Relation to AIXI

- AIXI averages value across all models weighted by $2^{-K(M)}$; our predictive-coding network is the analog of a single model M whose prior arises from the weakness penalty.
- Planning with a weakness penalty on imagined trajectories implements the equivalent of an Occam-weighted expectimax within P_θ .

Implementation Sketch

1. *Model update*: alternate predictive-coding inference (activations) with weight update on $L_{\text{model}}(\theta)$.
2. *Action selection*: sample action sequences, evaluate $V(a_{t:t+H})$, pick best.
3. *Policy learning (optional)*: update ϕ by gradient descent on $L_{\text{policy}}(\phi)$.

This yields a practical, differentiable approximation to AIXI where the role of the Solomonoff prior is played by a quantale-based weakness regularizer.

33.6.1 Convergence Conditions for Predictive-Coding WAIXI and Hybrid Architectures

What would be general characteristics of environments and reward functions under which this predictive-coding WAIXI agent might be expected to work especially well – e.g. to converge relatively rapidly to an optimal or near-optimal policy?

Favorable Environment Properties It seems rapid convergence is most likely when the environment satisfies:

- *Low Descriptive Complexity*. The true transition and reward dynamics admit a compact predictive-coding model. A small network with a strong weakness penalty can capture the structure with few samples.
- *Fast Mixing / Limited Memory*. Future observations depend only on a short history (low Markov order). Bounded receptive fields suffice for accurate prediction, reducing sample requirements.
- *Dense, Smooth Reward*. Frequent reward feedback that varies smoothly with state/action changes yields informative gradients for both model learning and planning under the weakness-regularized objective.
- *Stationarity and Ergodicity*. The environment’s statistics remain stable over time and all relevant states are eventually visited, ensuring that empirical averages converge to true expectations under the Occam-biased loss.
- *Alignment of Predictability and Reward*. High-reward regions coincide with low-weakness (highly predictable) structure. The Occam bias then guides the agent to focus on those regions first, accelerating policy improvement.

Hybrid WAIXI with Evolutionary Programming In domains where one or more of the above conditions fail—e.g. highly non-stationary, sparse-reward, or algorithmically random environments—a pure predictive-coding WAIXI may learn too slowly or overfit noise. To address this, a *hybrid* architecture can be employed:

- **Predictive-Coding WAIXI Module:** As before, learns a weakness-regularized world model and performs Occam-weighted planning for regions of the environment that are amenable to compact modeling.
- **EDA/Evolutionary Module:** Runs an estimation-of-distribution algorithm (EDA) or genetic search over policy/program spaces, guided by the same weakness prior when evaluating candidate controllers. This module excels at exploring highly irregular, low-predictability regions.
- **Meta-Controller Arbitration:** A higher-level MeTTa routine monitors model confidence and weakness gradients to allocate control:
 - Use WAIXI planning when the predictive model achieves low residual weakness and high confidence.
 - Switch to EDA search when residual weakness remains high or reward is too sparse for reliable gradient signals.

This hybrid scheme combines the sample-efficiency of weakness-AIXI in well-behaved subdomains with the exploration power of evolutionary methods in challenging regions, yielding a more universally robust AGI agent.

33.7 Conditional Weakness Regularization with Background Knowledge

In many settings we wish to regularize a predictive-coding network not only by its internal weakness $w(H_\theta)$, but also by how its structure aligns with an external knowledge base. We consider two paradigms for presenting that knowledge:

Option A: Knowledge as Predictive Desirability Function In this case the knowledge base is given by a fixed function

$$d(\theta) = \text{Desirability}(\theta)$$

which scores any candidate network parameterization θ according to how desirable or plausible the subnetwork is, based on its unregularized performance. We then define a *conditional weakness* penalty

$$w_A(H_\theta) = \bigoplus_{(i,j) \in H_\theta} [\varphi(i,j) \otimes d(\theta)],$$

so that each synapse (i,j) is down-weighted when $d(\theta)$ is low. The PC learning objective becomes

$$\mathcal{L} = \sum_{t,l} \|\text{pred-err}\|^2 + \lambda w_A(H_\theta).$$

Option B: Weak Meta-Model of Capability Here the knowledge base itself is a small ' predictive model B with parameters β . It takes a summary $s(\theta)$ of the main network's structure and predicts its expected performance:

$$B : s(\theta) \mapsto b(\theta).$$

We train B by minimizing its own composite loss

$$\mathcal{L}_B = \sum_i (B(s(\theta_i)) - \text{perf}(\theta_i))^2 + \lambda_B w_B(H_\beta),$$

where $w_B(H_\beta)$ is the quantale-weakness penalty on B 's graph. Once trained, B defines a background score $b(\theta) = B(s(\theta))$, and we set

$$w_B(H_\theta) = \bigoplus_{(i,j) \in H_\theta} [\varphi(i,j) \otimes b(\theta)].$$

The PC-net then minimizes

$$\mathcal{L}_{\text{PC}} = \sum_{t,l} \|\text{pred-err}\|^2 + \lambda_{\text{PC}} w_B(H_\theta).$$

Elegance and Synergy of Option B Option B leverages weakness throughout both learning stages:

- The meta-model B is itself trained with a quantale-weakness penalty $w_B(H_\beta)$, yielding a concise, high-weakness predictor of network capability.
- The main PC-net uses $b(\theta)$ from B as a conditional valuation in its own weakness penalty $w_B(H_\theta)$, focusing regularization on synapses that B deems implausible.
- Both levels share the same commutative, associative quantale structure, so their penalties compose cleanly and admit smooth, closed-form gradients.
- One can even optimize β and θ jointly in a single composite objective, allowing B to adapt as the PC-net learns, and the PC-net to follow B 's updated guidance.
- The CoDD Weakness Schema Theorem extends to this two-tier system, providing provable convergence guarantees under evolutionary or hybrid search.

Thus Option B delivers a fully compositional, algebraically elegant, and provably convergent framework in which weakness drives both the meta-model induction and the PC-net regularization.

34 Toward Financial-Modeling Applications of Quantale Information, Pattern Theory, and McBride Calculus

We now return to the financial quantale introduced in Chapter 2.15 above, and explore how patterns and derivatives in this quantale might relate to the practical matters pursued in the quantitative finance domain.

We work in the quantale

$$V = ([0, \infty), \otimes = \times, \oplus = +),$$

and track a portfolio or scenario payoffs $m : X \rightarrow [0, \infty)$ with risk-normalized valuation

$$\varphi(x) = \frac{p(x)}{r(x)^\gamma}, \quad \sum_x p(x) = 1, \quad \gamma \geq 0,$$

so that total weakness (risk-adjusted expected payoff) is

$$W = w(m) = \sum_{x \in X} \varphi(x) m(x).$$

Quantale Entropy for Diversification Define the probability law over instruments

$$P(x) = \frac{\varphi(x) m(x)}{W},$$

and the corresponding (Shannon-style) quantale entropy

$$H(P) = - \sum_{x \in X} P(x) \ln P(x).$$

Use case: maximize $H(P)$ subject to $W \geq R_{\min}$ to obtain the most diversified, risk-adjusted portfolio achieving a target return $R_{\min}$. The McBride derivative $\partial_{m(x)} W = \varphi(x)$ and $\partial_{m(x)} H = -\varphi(x)(1 + \ln P(x))$ drive gradient-based rebalancing.

Mutual Information for Factor Exposure Let F be a discrete market factor (regime) with joint law

$$P(x, f) \propto \varphi(x) m(x) Q(f \mid x),$$

and marginals P_X, P_F . The quantale mutual information

$$I(X; F) = \sum_{x, f} P(x, f) \ln \frac{P(x, f)}{P_X(x) P_F(f)}$$

measures *risk-factor overlap*. *Use case:* constrain $I(X; F) \leq I_{\max}$ to limit unintended concentration in certain regimes.

Pattern Intensities and Emergent Clusters For an asset cluster $S \subset X$ define the indicator predicate

$$y_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

Then

$$w(y_S) = |S|, \quad I_S(m) = w(y_S) \multimap W = \frac{W}{|S|}.$$

For two clusters S, T ,

$$I_{S,T}(m) = \frac{W}{|S||T|}, \quad \sigma_{S,T} = \frac{I_{S,T}}{I_S I_T} = \frac{1}{W}.$$

Use case: monitor $\sigma_{S,T}$ to detect *emergent correlation* between clusters S, T early—high σ signals unusual joint movement before it becomes widespread.

McBride-Driven Sensitivity and Dynamics The McBride derivative gives

$$\partial_x W = \varphi(x), \quad \partial_x I_S = \varphi(x) / |S|, \quad \partial_x \sigma_{S,T} = -\frac{\varphi(x)}{W^2}.$$

Use case: allocate incremental capital to asset x if $\partial_x H$ is large (improves diversification most), or reduce exposures where $\partial_x \sigma_{S,T}$ is negative and large (mitigates emerging cluster risk).

Dynamic Rebalancing via ODEs Model continuous weight updates $\dot{m}_x = \eta \partial_x H$ to solve $\dot{P}(x) = \eta \varphi(x)(-1 - \ln P(x))$, driving the portfolio toward maximum entropy. Similarly, $\dot{m}_x = -\kappa \partial_x \sigma_{S,T}$ reduces joint cluster synergy.

Together, these tools form a *quantale-based risk-management toolkit*:

- **Entropy maximization** for diversification,
- **Mutual information constraints** for factor-tilt control,
- **Pattern theory** to spot emergent risk clusters,
- **McBride calculus** to compute marginal sensitivities and drive gradient ODEs for real-time rebalancing.

34.1 Quantale-Valued Stochastic Processes for Risk-Normalized Returns

Stochastic processes like Brownian motions and Levy processes are commonly used to model financial time series. However these often fail to model the characteristics of real financial systems, especially in cases such as emerging markets or cryptocurrencies. This brings up the question: Can we use some generalization of stochastic processes over these financial quantales to create

more flexible versions of these sort of stochastic processes for generalizing time series?

We work in the financial quantale

$$V = ([0, \infty), \otimes = \times, \oplus = +), \quad \varphi(x) = \frac{p(x)}{r(x)^\gamma},$$

so that for any payoff function $m : X \rightarrow [0, \infty)$ the weakness

$$W(t) = w(m(t)) = \sum_{x \in X} \varphi(x) m_x(t)$$

gives the risk-normalized aggregate at time t .

V-Valued Random Variables and Noise A V -valued random variable Z is an ordinary nonnegative real random variable, but we interpret its moments and convolutions via the quantale operations. In particular a *quantale-Levy noise* $L(t)$ can be defined by its characteristic exponent in V :

$$E[\exp(-u L(t))] = \exp(-t \Psi(u)), \quad u \geq 0,$$

with $\Psi(u)$ chosen to model heavy tails or jumps.

Quantale-Geometric Brownian Motion (QGBM) Generalize the classical SDE

$$dS = \mu S dt + \sigma S dW$$

to the quantale setting by replacing addition of increments with $\oplus$ and multiplication with $\otimes$. We write in differential form

$$dS(t) = (\mu \otimes S(t)) dt \oplus (\sigma \otimes S(t)) dQ(t),$$

where $dQ(t)$ is an increment of a quantale-Levy process (e.g. a compound-Poisson or stable noise). Equivalently, in integral form

$$S(t) = S(0) \otimes \exp_{\otimes}(\mu t) \oplus \int_0^t (\sigma \otimes S(s)) \oplus dQ(s).$$

Here $\exp_{\otimes}$ is the quantale-exponential (i.e. real $\exp$).

Risk-Normalized State Define the risk-normalized aggregate

$$W(t) = w(m(t)) = \sum_{x \in X} \varphi(x) m_x(t),$$

and suppose each instrument $m_x(t)$ follows its own QGBM:

$$dm_x = (\mu_x \otimes m_x) dt \oplus (\sigma_x \otimes m_x) dQ_x(t),$$

with possibly correlated noises dQ_x . Then by linearity of $\oplus$

$$dW = \sum_x \varphi(x) dm_x = \left(\sum_x \varphi(x) \mu_x m_x \right) dt \oplus \sum_x \varphi(x) \sigma_x m_x dQ_x.$$

This V -valued SDE captures portfolio-level drift and heavy-tailed shock aggregation.

Quantale-OU Process for Mean Reversion A mean-reverting "Ornstein-Uhlenbeck" style process in V is

$$dX = \kappa (\theta \multimap X) dt \oplus \sigma dQ(t),$$

where $\theta \multimap X = (X/\theta)$ is the residuated "ratio" in V . This yields mean-reversion toward level θ with jumps or heavy tails.

Flexible Calibration and Features

- *Heavy tails*: choose $\Psi(u) \sim u^\alpha$ ($0 < \alpha < 2$) to model stable-law increments.
- *Volatility clustering*: let σ_x depend on a quantale-OU process on past $|dQ_x|$.
- *Risk coupling*: encode cross-instrument correlation by letting $dQ_x = \rho_{xy} dL_0 + \sqrt{1 - \rho_{xy}^2} dL_x$.

Benefits This quantale-valued framework allows

- direct aggregation of heavy-tailed shocks via $\oplus$,
- compounding via $\otimes$ that respects risk-normalized weights,
- flexible calibrations by choosing quantale-Levy exponents,
- consistent portfolio-level SDEs preserving the quantale algebra.

Such models can better capture jump clustering, volatility regimes, and extreme events in emerging or crypto markets than classical Brownian or Levy models.

34.2 Quantale-Valued Stochastic Models for Cryptocurrency Time Series

To make the above more concrete, we flesh out some speculative possible details of how to apply these modeling methods to a practical use-case, cryptocurrency time series.

34.2.1 Data Ingestion

1. *Universe of assets*:

$$X = \{\text{BTC}, \text{ETH}, \dots\}.$$

2. *Price history*: retrieve $P_x(t_k)$ at discrete times t_k .

3. *Log-returns or payoffs*: define

$$m_x(t_k) = \ln \frac{P_x(t_k)}{P_x(t_{k-1})} \quad \text{or} \quad m_x(t_k) = \frac{P_x(t_k)}{P_x(t_{k-1})} - 1.$$

34.2.2 Financial Quantale

We work in the commutative quantale

$$V = ([0, \infty), \otimes = \times, \oplus = +), \quad e = 1, 0 = 0.$$

Fix a risk-normalization exponent $\gamma \geq 0$ and per-asset risk r_x (e.g. historical volatility). Define weights

$$\varphi_x = \frac{1}{r_x^\gamma}, \quad \sum_x \varphi_x = 1.$$

Then the *total weakness* (risk-normalized portfolio return) is

$$W(t) = \sum_{x \in X} \varphi_x m_x(t).$$

34.2.3 Quantale-Lévy Increments $dQ_x(t)$

For each asset x we model jumps and heavy tails via a Lévy process $Q_x(t)$ whose Laplace transform in V is

$$\mathbb{E}[e^{-u Q_x(t)}] = \exp(-t \Psi_x(u)), \quad u \geq 0,$$

where the *characteristic exponent* $\Psi_x(u)$ is calibrated from data:

- Fit the empirical tail of $|m_x(t_k)|$ to a power law $\Pr[|m_x| > y] \approx c_x y^{-\alpha_x}$, yielding tail index $\alpha_x \in (0, 2)$.
- Then choose a stable-law exponent $\Psi_x(u) = C_x u^{\alpha_x}$ or a compound-Poisson form $\Psi_x(u) = \lambda_x(1 - \phi_x(u))$ where ϕ_x is the jump-size Laplace transform.

An infinitesimal increment $dQ_x(t)$ is the quantale-Lévy noise driving jumps for m_x .

34.2.4 Quantale-GBM for Individual Assets

We posit for each x a *quantale-geometric Brownian motion* with jumps:

$$dm_x(t) = (\mu_x \otimes m_x(t)) dt \oplus (\sigma_x \otimes m_x(t)) dQ_x(t).$$

Equivalently, in integral form,

$$m_x(t) = m_x(0) \exp(\mu_x t) + \int_0^t \sigma_x m_x(s) \oplus dQ_x(s).$$

34.2.5 Portfolio-Level Aggregation

Define the portfolio return process

$$W(t) = \sum_{x \in X} \varphi_x m_x(t).$$

By linearity of $\oplus$,

$$dW(t) = \sum_x \varphi_x dm_x(t) = \left(\sum_x \varphi_x \mu_x m_x(t) \right) dt \oplus \sum_x \varphi_x \sigma_x m_x(t) dQ_x(t).$$

34.2.6 Simulation and Calibration

1. *Estimate parameters:*

$$\hat{\mu}_x = \frac{1}{T} \sum_k m_x(t_k), \quad \hat{\sigma}_x^2 = \frac{1}{T} \sum_k (m_x(t_k) - \hat{\mu}_x)^2,$$

and fit α_x, C_x for $\Psi_x(u)$.

2. *Euler-Maruyama scheme in V :*

$$m_x(t + \Delta) = m_x(t) \oplus (\hat{\mu}_x \Delta) \oplus (\hat{\sigma}_x m_x(t)) \Delta Q_x,$$

drawing ΔQ_x from the calibrated Lévy increment.

3. *Monte Carlo:* generate many paths of $W(t)$ to compute risk metrics (VaR, CVaR).

34.2.7 Real-Time Monitoring and Intervention

- Compute on sliding windows: $W(t)$, portfolio entropy, mutual information with market regimes.
- Detect emergent co-movement patterns via $\sigma_{S,T}(t) = 1/W(t)$.
- Trigger alerts when quantale diagnostics exceed thresholds.

34.3 Financial Quantale Based Attention Allocation

The Economic Attention Allocation (ECAN) subsystem of the Hyeron design applies economic principles to the allocation of attention in an AGI system's memory metagraph. Here we explore how this system might be upgraded to leverage the more sophisticated approach to finance embodied in a financial quantale.

34.4 A Quantale-ECAN for Risk-Normalized Financial Attention

We begin by simplistically adapting the Economic Attention Network (ECAN) of [PISG09] to our financial quantale

$$V = ([0, \infty), \otimes = \times, \oplus = +), \quad e = 1, 0 = 0,$$

with risk-normalized weights $\varphi_x = p_x / r_x^\gamma$ on assets $x \in X$.

States: STI and LTI as Quantale Values For each asset x at time t we maintain

$$\text{STI}_x(t), \text{LTI}_x(t) \in V,$$

interpreted as short-term and long-term "importance-currencies."

Rent Generation Each asset x generates a "rent" from its STI:

$$\text{rent}_x(t) = \rho_{\text{rent}} \otimes \text{STI}_x(t), \quad \rho_{\text{rent}} \in V,$$

analogous to interest on importance funds.

Importance Diffusion A fraction $\rho_{\text{spread}} \in V$ of each asset's STI is re-distributed to others via a normalized adjacency $w_{x \rightarrow y}$ ($\sum_y w_{x \rightarrow y} = e$):

$$\text{STI}_y(t + \Delta) = (\text{STI}_y(t) \otimes (e \ominus \rho_{\text{spread}})) \oplus \bigoplus_{x \in X} [\rho_{\text{spread}} \otimes w_{x \rightarrow y} \otimes \text{STI}_x(t)].$$

Here $e \ominus \rho$ is the (residuation) "complement" of ρ in V .

Long-Term Importance Update A fraction $\rho_{\text{LTI}} \in V$ of each rent is added to LTI:

$$\text{LTI}_x(t + \Delta) = \text{LTI}_x(t) \oplus [\rho_{\text{LTI}} \otimes \text{rent}_x(t)].$$

Attentional Focus At each step, rank assets by $\text{STI}_x(t)$ in the quantale order and select the top K as the *Attentional Focus*.

Risk-Weighted Stimulus When new market data arrives (e.g. a price shock s_x), we *stimulate* STI by

$$\text{STI}_x = \text{STI}_x \oplus [\varphi_x \otimes s_x],$$

so that high-return, low-risk assets receive proportionally larger attention boosts.

Interpretation and Use

- Assets in the Attentional Focus have the greatest STI and thus attract portfolio rebalancing or risk-management actions.
- The rent/diffusion dynamics spreads attention from highly "active" assets to their neighbors in the asset-graph (e.g. sector or correlation network).
- LTI accumulates persistent importance, guiding long-horizon investment decisions.
- Risk-normalized stimulation $\varphi_x s_x$ ensures that attention aligns with both return potential and risk exposure.

This *Quantale-ECAN* framework thus provides a principled, algebraic mechanism—fully expressed in the $(\times, +)$ financial quantale—to allocate "attention-capital" across assets, blending short-term market shocks with long-term sectoral influence.

34.5 Quantale Information, Pattern Theory, and Emergence on STI/LTI

We next ask: What might quantale information, quantale pattern and emergence look like on the quantales associated with STI and LTI? And what might this "attentional information" tell us about a cognitive system?

Let X be a set of assets and let

$$V = ([0, \infty), \otimes = \times, \oplus = +)$$

be our financial quantale. At time t each asset $x \in X$ has

$$\text{STI}_x(t), \text{LTI}_x(t) \in V.$$

Define the *short-term and long-term totals*

$$W_{\text{STI}}(t) = \sum_{x \in X} \text{STI}_x(t), \quad W_{\text{LTI}}(t) = \sum_{x \in X} \text{LTI}_x(t).$$

Quantale Entropy (Attention Diversity) Form the induced distributions

$$P_{\text{STI}}(x) = \frac{\text{STI}_x(t)}{W_{\text{STI}}(t)}, \quad P_{\text{LTI}}(x) = \frac{\text{LTI}_x(t)}{W_{\text{LTI}}(t)}.$$

Define Shannon-style entropies

$$H_{\text{STI}}(t) = - \sum_x P_{\text{STI}}(x) \ln P_{\text{STI}}(x), \quad H_{\text{LTI}}(t) = - \sum_x P_{\text{LTI}}(x) \ln P_{\text{LTI}}(x).$$

These measure how *broadly* attention (STI) or importance (LTI) is spread.

Mutual Information (STI/LTI Coupling) Define the joint ' measure

$$M(x) = \text{STI}_x(t) \times \text{LTI}_x(t), \quad W_C(t) = \sum_x M(x),$$

and the joint distribution

$$P_C(x) = \frac{M(x)}{W_C(t)}.$$

Then the mutual information

$$I(\text{STI}; \text{LTI}) = H_{\text{STI}}(t) + H_{\text{LTI}}(t) - H_C(t),$$

with $H_C(t) = -\sum_x P_C(x) \ln P_C(x)$, quantifies the degree to which short-term and long-term attention *agree* on which assets are important.

Pattern Intensities for Asset Clusters For any subset $S \subseteq X$, define the indicator predicate

$$y_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

Its weakness in STI is

$$w_{\text{STI}}(y_S) = \sum_{x \in X} y_S(x) = |S|,$$

and the *STI-pattern intensity* is

$$I_S^{\text{STI}}(t) = w_{\text{STI}}(y_S) \multimap W_{\text{STI}}(t) = \frac{W_{\text{STI}}(t)}{|S|}.$$

Similarly for LTI,

$$I_S^{\text{LTI}}(t) = \frac{W_{\text{LTI}}(t)}{|S|}.$$

For the combined predicate $m(x) = M(x)$ on X , its pattern intensity is

$$I_S^C(t) = |S| \multimap W_C(t) = \frac{W_C(t)}{|S|}.$$

Emergent Coupling of Patterns Define the *synergy* of cluster S as

$$\sigma_S(t) = (I_S^{\text{STI}}(t) \otimes I_S^{\text{LTI}}(t)) \multimap I_S^C(t).$$

When $\sigma_S(t) \geq 1$, the joint intensity I_S^C exceeds what would be predicted from individual STI and LTI intensities, indicating an *emergent* focus on cluster S that is not explained by short- or long-term attention alone.

Cognitive Interpretation

- H_{STI} and H_{LTI} measure *breadth* of a cognitive system's short-term focus versus long-term priorities.
- Mutual information $I(\text{STI}; \text{LTI})$ captures *alignment* between rapid attentional shifts and stable importance beliefs.
- Pattern intensity I_S^{STI} (resp. I_S^{LTI}) measures average attention (importance) per asset in cluster S .
- Synergy σ_S reveals when a cluster attracts disproportionate *combined* attention and importance—i.e. an emergent thematic focus or cognitive "hot-spot."

These quantale-information and pattern-theory diagnostics provide a rich, algebraic description of how a cognitive system allocates attention over assets (or concepts) both in the moment and over time, and how new, unexpected clusters of focus can emerge.

34.6 Attention as an Emergent Strange Attractor via McBride Pattern-Dynamics

The next natural step in this line of investigation is to ask: Can one use the notion of McBride-differential equations on patterns on the STI/LTI quantale to articulate the notion of the attentional-focus as an emergent strange attractor of the ECAN dynamics? That is: Can we present ECAN's ongoing updating of attention values as a nonlinear dynamic in which the strange/chaotic attractor of attentional focus emerges?

We show here how the ECAN update laws for short-term importance (STI) and long-term importance (LTI) can be given a continuous-time, McBride-derivative form whose nonlinear coupling admits a strange (chaotic) attractor, interpreted as the emergent attentional focus.

34.6.1 Continuous-Time ECAN via McBride Derivatives

Recall the discrete ECAN update from above:

$$\text{STI}_x(t + \Delta) = (\text{STI}_x(t) \otimes (e \ominus \rho_s)) \oplus \bigoplus_y \rho_s \otimes w_{y \rightarrow x} \otimes \text{STI}_y(t) \oplus \varphi_x s_x(t),$$

$$\text{LTI}_x(t + \Delta) = \text{LTI}_x(t) \oplus \rho_L \otimes (\rho_r \otimes \text{STI}_x(t)).$$

Define McBride derivatives by

$$\dot{\text{STI}}_x(t) = \lim_{\Delta \rightarrow 0} \frac{\text{STI}_x(t + \Delta) - \text{STI}_x(t)}{\Delta}, \quad \dot{\text{LTI}}_x(t) = \lim_{\Delta \rightarrow 0} \frac{\text{LTI}_x(t + \Delta) - \text{LTI}_x(t)}{\Delta}.$$

Using residuation and linearization of $\oplus, \otimes$ near the identity, one obtains the ODEs

$$\dot{\text{STI}}_x = -\lambda \text{STI}_x + \rho_s \sum_y w_{y \rightarrow x} \text{STI}_y + \varphi_x s_x(t),$$

$$\dot{\text{LTI}}_x = \rho_L \rho_r \text{STI}_x,$$

where λ arises from $(e \ominus \rho_s) \approx (1 - \rho_s)$.

34.6.2 Pattern-Intensity Coordinates

Choose a small collection of candidate clusters $S_i \subset X$, $i = 1, 2, 3$. Define their pattern intensities

$$I_i^{\text{STI}}(t) = |S_i| \multimap \sum_{x \in S_i} \text{STI}_x(t), \quad I_i^{\text{LTI}}(t) = |S_i| \multimap \sum_{x \in S_i} \text{LTI}_x(t).$$

Let $P_i(t) = I_i^{\text{STI}}(t) \otimes I_i^{\text{LTI}}(t)$. By McBride differentiation,

$$\dot{I}_i^{\text{STI}} = |S_i| \multimap \sum_{x \in S_i} \dot{\text{STI}}_x, \quad \dot{I}_i^{\text{LTI}} = |S_i| \multimap \sum_{x \in S_i} \dot{\text{LTI}}_x.$$

Substitute the ECAN ODEs to get

$$\begin{aligned} \dot{I}_i^{\text{STI}} &= -\lambda I_i^{\text{STI}} + \rho_s \sum_y \left(\sum_{x \in S_i} w_{y \rightarrow x} \right) \text{STI}_y + \sum_{x \in S_i} \varphi_x s_x(t), \\ \dot{I}_i^{\text{LTI}} &= \rho_L \rho_r I_i^{\text{STI}}. \end{aligned}$$

34.6.3 Reduced Nonlinear System and Chaos

Focus on the three-dimensional system in the pattern-intensities:

$$\dot{I}_i = -a I_i + b P_i + u_i(t), \quad \dot{P}_i = c I_i - d P_i, \quad i = 1, 2, 3,$$

where $P_i = I_i^{\text{STI}} \otimes I_i^{\text{LTI}}$ and $u_i(t)$ is external stimulus. For suitable parameters (a, b, c, d) this is a well-known Lorenz-like system admitting a *strange attractor* in (I_1, I_2, I_3) -space.

34.6.4 Emergent Strange Attractor as Attentional Focus

The chaotic attractor in the (I_1, I_2, I_3) dynamics corresponds to a *persistent, structured locus* in pattern-intensity space—i.e. the cognitive system's *attentional focus* emerges not as a fixed point but as a strange attractor. Trajectories on this attractor represent continual exploration of a repertoire of patterns, with sensitivity to stimuli $s_x(t)$ yet bounded by the attractor's geometry.

34.6.5 Interpretation

- The *McBride-derived ODEs* translate discrete ECAN updates into a smooth, residuated dynamical system.
- Pattern-intensity coordinates reduce high-dimensional ECAN to a low-dimensional nonlinear system.
- Under generic coupling and feedback parameters, the resulting ODE exhibits a *strange attractor*, modeling sustained, non-repeating, but structured attentional focus.
- This accounts for cognitive phenomena like *persistent exploration*, *sensitivity to novelty*, and *bounded rationality* as manifestations of chaotic attractor dynamics in pattern space.

Part VI

Weakness Is All You Need (Except For All That Other Stuff): Leveraging Weakness for AGI

35 Weakness as a Unifying Bias in PRIMUS

The ideas regarding weakness and AI unfolded throughout this paper are in most cases general in scope rather than tightly tied to any particular AGI approach or software architecture. However, the Hyperon architecture has nonetheless been a thread running throughout, and in this section we will do a quick re-cap of the various ideas presented with a view toward their utility for shaping, improving and realizing the PRIMUS⁵ cognitive architecture for the Hyperon platform.

PRIMUS [GBD⁺23] interleaves two cognitive meta-dynamics over a shared collection of Atomspaces (largely stored in MORK instances in-RAM, and managed across machines by the Distributed Atomspace (DAS)), both driven by MeTTa scripts:

- **Goal-driven cognition:** MetaMo maintains goals and invokes planning, inference, learning and decision routines.
- **Background cognition:** Unsupervised pattern mining, concept formation and evolution continuously enrich the Atomspace.

⁵Formerly known as CogPrime

The various AI processes underlying these meta-dynamics can all be viewed in one way or another as guided by quantale-based Occam simplicity measures. The following bullets highlight, for some of the subsystems of PRIMUS, some of the ways it benefits from the weakness framework:

- **PLN with Weakness-Guided or SVM-Guided Inference**
 - Goal-driven: MetaMo requests proofs or classifiers ranked by probabilistic or linear weakness, ensuring only the simplest high-confidence derivations are chosen to execute serving current goals.
 - Background: Continuous proof-mining and classifier retraining prune low-prominence inference paths, keeping the belief graph lean and reusable.
- **Logical Entropy and Graphropy**
 - Goal-driven: Concept refinement for a subgoal uses partition-weakness or graphropy to split or merge categories with minimal complexity.
 - Background: Unsupervised concept discovery applies entropy-based weakness to identify compact clusters and distinction subgraphs.
- **Evolutionary Learning (MOSES/CoDD)**
 - Goal-driven: To synthesize a controller or program, MetaMo triggers CoDD evolution under a Kraft-type weakness prior to drive evolution in rapid convergence to minimal-complexity solutions.
 - Background: Offline CoDD runs generate libraries of simple, high-utility codelets guided by weakness, ready for later use.
- **Neural Learners and Predictive Coding**
 - Goal-driven: Decision-transformers and neural planners include McBride-based weakness regularizers to produce generalizable, simple policies. Similar combinations of methods can be used for perceptual pattern recognition across different modalities.
 - Background: Continuous background training of Transformers and WaveFormers uses weakness-driven gradient updates to compress representations.
- **CoDD Schema Theorem**
 - Goal-driven: MetaMo relies on exponential convergence guarantees for weakness-guided genetic operators in time-critical synthesis.
 - Background: Unbounded background evolution exploits the same theorem to build extensive libraries of compact programs.
- **Clear, Disentangled Representations**

- Goal-driven: MetaMo requests "clear" subgraphs (simplicity + causal-strength fusions) for robust planning models.
- Background: Disambiguation processes extract disentangled concept graphs to improve downstream tasks.
- **Weakness-Based Decision Theories**
 - Goal-driven: MetaMo invokes WCDT, WUDT or WCUDT policies to optimize weakness-regularized expected utility in paradoxical decisions.
 - Background: Offline simulation refines weakness priors in these policies for faster real-time deployment.
- **MetaMo Motivational Bias**
 - Goal-driven: Short-term importance and confidence incorporate local weakness gradients to focus on under-explained subgoals.
 - Background: Anomaly detection flags high-weakness regions as candidates for future goals.
- **Aesthetic Generality in Creativity**
 - Goal-driven: For creative subgoals (e.g. "compose a melody"), weakness-based generality metrics select broad-appeal motifs.
 - Background: Continuous aesthetic evaluation populates a library of simple yet novel patterns.
- **WAIXI and Universal Planning**
 - Goal-driven: MetaMo's AIXI^{tl} analogue uses a CoDD-weakness prior to bias model search toward simple predictors for real-time planning.
 - Background: Offline hypothesis enumeration under weakness priors builds a cached mixture of environment models.
- **Controller Synthesis and Complexity**
 - Goal-driven: MetaMo invokes CoDD-synthesis for control subgoals, obtaining minimal-complexity controllers.
 - Background: Background synthesis populates an Atomspace index of minimal-weakness controllers for common subtasks.
- **Quantum Weakness Extensions**
 - Goal-driven: When Hyperon is run on combined classical/hardware, MetaMo invokes quantum-WAIXI with multiperspectival weakness priors for quantum subgoals.
 - Background: Quantum model mining uses quantum graphentropy to compress operator algebras for later retrieval.

- **Galois Connection Acceleration**
 - Goal-driven: MetaMo’s dynamic programming and RL leverage Galois-fusion of abstractions for near-real-time multi-scale planning.
 - Background: Background RL on coarse abstractions refines complexity where weakness gradients signal the need for finer detail.
- **Integrated Information via Weakness**
 - Goal-driven: MetaMo can use weakness-based integrated information (Φ) as a utility subgoal.
 - Background: Continuous monitoring computes Φ via whole-vs-part weakness differences to detect emergent states.
- **Cognitive Synergy Metrics**
 - Goal-driven: MetaMo consults formal synergy scores (lax-monoidal weakness) to decide which modules to engage jointly in which situations.
 - Background: Analysis of module interactions populates an Atom-space synergy map for future workflows.
- **Categorical Concept Blending**
 - Goal-driven: MetaMo invokes MeTTa blending routines parameterized by chosen weakness semirings to invent new concepts.
 - Background: Background blends existing ontologies under multiple weakness measures, expanding the concept base.
- **Multiresolution Weakness Transform**
 - Goal-driven: MetaMo requests a multiresolution weakness decomposition of the current world model to guide hierarchical planning.
 - Background: Multiscale pattern mining continuously populates MORK with per-scale detail Atoms for improved awareness.
- **McBride Derivative and Unified Learning**
 - Goal-driven: MetaMo’s adaptation calls trigger McBride-driven gradient and natural-gradient updates to achieve minimal-complexity explanations of new data.
 - Background: Continuous predictive coding applies the same McBride-guided updates to compress representations and prune links.

36 Toward a Quantale Weakness Library for MeTTa

Given the applicability of quantale weakness and associated ideas to the various cognitive algorithms in the PRIMUS cognitive architecture, it seems desirable to implement various commutative quantales with enriching measures on them in the MeTTa language, which lives at the core of the Hyperon software architecture being used to implement PRIMUS. One would then use the assorted quantale operations to implement various AI algorithms in MeTTa, making use of tools such as weakness, quantale-specific information and pattern and emergence, McBride derivatives of weakness and information and pattern, etc.. It would be desirable to implement all this efficiently at the MeTTa-IL and MORK level [F1R25], but also have it easily accessible to users of MeTTa at the full MeTTa-syntax level.

With all this in mind, in this section we very roughly outline what a quantale-weakness MeTTa software library might look like, embodying a variety of desired features.

36.0.1 A Basic Quantale-Weakness Library for MeTTa

We might organize such a library in four layers:

Core Primitives (MeTTa-IL / MORK) Define the bare quantale operations as built-in primitives:

```
(module Quantales.Core
  ;; q-join : Q Q ? Q
  (prim q-join (Q a b)      ;; associative, commutative, idempotent
    ...)

  ;; q-times : Q Q ? Q
  (prim q-times (Q a b)     ;; associative, commutative, unit = 1
    ...)

  ;; q-unit : Q
  (prim q-unit (Q)          ;; the multiplicative unit
    ...)

  ;; q-resid : Q Q ? Q
  (prim q-resid (Q a b)     ;; residual for implication
    ...))
```

All heavy loops over join/times compile to optimized MORK code.

Weakness Module (MeTTa Syntax) Provide user-facing macros and functions:

```

(module Quantales.Weakness
  ;; define a new quantale by naming its operations
  (macro define-quantale (name join times unit resid?)
    `(do
      (def ,name
        { :join ,join
          :times ,times
          :unit ,unit
          :resid ,resid? })
        (register-quantale ,name)))

  ;; weakness : given Q, weighting phi : X->Q, model m : X->Q
  ;; w(m) = reduce join [ times(phi(x), m(x)) | x in keys(phi) ]
  (defn weakness (Q phi m)
    (let ((j (Q :join)) (t (Q :times)))
      (reduce j
        (for x in (keys phi)
          (t (phi x) (m x))))))

  ;; alias for convenience
  (alias q-weakness weakness))

```

Information & Pattern Analytics Build entropy, mutual-info, pattern-intensity on top:

```

(module Quantales.Information
  ;; quantale-entropy:  $H(m) = \text{reduce join } [ \text{times}(m(x), \ln(m(x))) ]$ 
  (defn entropy (Q m)
    (let ((j (Q :join)) (t (Q :times)))
      (reduce j
        (for x in (keys m)
          (t (m x) (ln (m x))))))

  ;; quantale-mutual-info:  $I(X;Y)$ 
  (defn mutual-info (Q mxy px py)
    (let ((j (Q :join))
          (t (Q :times))
          (r (Q :resid)))
      (reduce j
        (for (x y) in (keys mxy)
          (let ((p (mxy x y))
                (c (t (px x) (py y))))
            (t p (r p c))))))

  ;; pattern-intensity:  $I_S(m) = \text{resid}(\text{join}_{\{x \in S\}} m(x), \text{join}_{\{x\}} m(x))$ 

```

```

(defn pattern-intensity (Q m S)
  (let ((j (Q :join)) (r (Q :resid)))
    (r
     (reduce j (for x in S (m x)))
     (reduce j (vals m))))))

```

McBride Derivatives Expose partial-derivatives for sensitivity:

```

(module Quantales.Differentiation
  ;; partial_x weakness = phi(x)
  (defn mcbride-weakness-deriv (Q phi m x)
    (phi x))

  ;; partial_{x,y} mutual-info = resid-partial...
  (defn mcbride-mi-deriv (Q mxy px py x y)
    ...)
)

```

Example Usage (Rough Sketch) Users import and instantiate quantales, then call analytics:

```

(import Quantales.Core
        Quantales.Weakness
        Quantales.Information
        Quantales.Differentiation)

;; create the [0,1] probabilistic quantale
(define-quantale prob-quantale
  Quantales.Core/q-join    ;; max
  Quantales.Core/q-times  ;; multiply
  1                        ;; unit
  nil)                    ;; no resid

;; define weight phi and model m on keys "a","b"
(def phi {"a" 0.2, "b" 0.8})
(def m    {"a" 0.5, "b" 0.3})

;; compute weakness, entropy, mutual-info
(def w (weakness prob-quantale phi m))
(def H (entropy prob-quantale m))
;; for mutual-info, supply joint mxy and marginals px,py
;; (def I (mutual-info prob-quantale mxy px py))

;; McBride derivative of weakness w.r.t. "a"
(def dw (mcbride-weakness-deriv prob-quantale phi m "a"))

```

36.0.2 A Quantale Factor-Graph Sub-Library for MeTTa

As an example of how to possibly develop such libraries one step deeper, we sketch how one might build on the above code infrastructure to make a quantale factor-graph library in three layers.

Core Primitives (MeTTa-IL / MORK) Define built-in primitives to manipulate graphs of variables and factors:

```
(module Quantales.FactorGraphs.Core
  ;; fg-create : ? Graph
  (prim fg-create ()                ;; allocate new empty graph
   ...)
  ;; fg-add-var : Graph Symbol Set?Q -> Graph
  (prim fg-add-var (G name domain)  ;; add variable node
   ...)
  ;; fg-add-fctr: Graph (X?Q)^n Symbol^n -> Graph
  (prim fg-add-fctr (G factor-fn var-list)  ;; add factor tying variables
   ...)
  ;; fg-infer : Graph Symbol^m -> (Symbol -> Q)
  (prim fg-infer (G query-vars)           ;; run belief/update
   ...))
```

User-Facing Syntax (MeTTa level) Provide macros and functions for ease of use:

```
(module Quantales.FactorGraphs
  ;; define a new empty graph
  (macro define-graph (name)
    `(do (def ,name (fg-create)) ,name))

  ;; add a variable with domain in Q
  (defn add-variable (G name domain)
    (fg-add-var G name domain))

  ;; add a factor (potential) over given variables
  (defn add-factor (G factor-fn vars)
    (fg-add-fctr G factor-fn vars))

  ;; perform marginal or MAP inference
  (defn infer (G queries)
    (fg-infer G queries)))
```

Inference Module Wrap common routines over commutative quantale Q :

```
(module Quantales.FactorGraphs.Inference
  (defn compute-marginals (Q G)
```

```

;; e.g. run sum-product or max-product over Q
...)
(defn belief-update (Q G evidence)
  ;; condition G on observed evidence
  ...))

```

CoDD Models as a Special Case of Quantale Factor-Graphs In this sort of framework, Combinatory Decision DAGs (CoDD) can be encoded directly as factor graphs over a suitable quantale (e.g. logical or weakness quantale) by treating each decision node as a variable with domain {branches} and each combinator constraint as a factor. In MeTTa syntax:

```

;; 1. Instantiate a graph
(define-graph codd-graph)

;; 2. For each decision node d in the CoDD, add a variable
(add-variable codd-graph d '(choice1 choice2 ...))

;; 3. For each combinator (e.g. K, S) impose a factor constraining
;;   child variables to match the combinator semantics
(add-factor codd-graph k-combinator-factor '(d1 d2))

;; 4. Optionally add weakness weights or logical potentials
;;   (domain Q chosen to reflect CoDD logical entropy)

;; 5. Perform inference to extract most likely program/path
(def result (infer codd-graph '(root-node)))

```

Because all CoDD operations – pattern-memoization, SK combinators, logical entropy constraints – can be expressed as quantale factors and variables, CoDD evolution, mutation and synthesis become instances of generic factor-graph inference and learning over Q .

36.1 Quantale-Based Crossover and Mutation for CoDD Models

As an example of how the nitty-gritty of AI operations might be expressed in terms of a quantale software library, let's consider the case of GA evolution of CoDD models.

Suppose each CoDD program is interpreted as a model

$$m : X \rightarrow Q$$

in a commutative quantale $(Q, \otimes, \oplus, e)$. We can then define offspring models directly via the quantale operations:

Pointwise Join-Crossover Given two parent models $m_1, m_2 : X \rightarrow Q$, define their *join-crossover*

$$m_{\text{child}}(x) = m_1(x) \oplus m_2(x), \quad x \in X.$$

This yields a model that, for each datum x , adopts the *weaker* (larger) of the two parents' predictions.

Pointwise Times-Crossover Alternatively, define

$$m_{\text{child}}(x) = m_1(x) \otimes m_2(x).$$

This produces a model whose confidence is the *intersection* (strongest common prediction) of the parents.

Masked Crossover via a Predicate Let $p : X \rightarrow \{0, 1\} \subset Q$ be a predicate (e.g. a random mask), with complement $\bar{p} = e \multimap p$. Then

$$m_{\text{child}} = (p \otimes m_1) \oplus (\bar{p} \otimes m_2),$$

i.e.

$$m_{\text{child}}(x) = p(x) \otimes m_1(x) \oplus (e \multimap p(x)) \otimes m_2(x).$$

This generalizes the usual GA mask-based crossover to arbitrary quantales.

Mutation as Quantale Perturbation Given a small "perturbation" model $\delta : X \rightarrow Q$ (e.g. drawn at random from a low-weakness distribution), define two natural mutations:

$$m_{\text{mut}}^+(x) = m(x) \oplus \delta(x), \quad m_{\text{mut}}^\times(x) = m(x) \otimes \delta(x).$$

The "additive" mutation $m \oplus \delta$ relaxes the model on those inputs where δ is large; the "multiplicative" mutation $m \otimes \delta$ sharpens it where δ has high support.

Residual-Based Correction For a desired adjustment δ , one can also use the residuum to "remove" or "correct" parts of m :

$$m_{\text{corr}}(x) = m(x) \otimes (m(x) \multimap \delta(x)).$$

This pulls the model's prediction at x toward $\delta(x)$ in a residuated manner.

MeTTa-Level Definition In MeTTa pseudocode, one might write:

```
(defn crossover-join (Q m1 m2)
  (fn (x) ((Q :join) (m1 x) (m2 x))))

(defn crossover-mask (Q p m1 m2)
  (let ((join (Q :join)) (times (Q :times)) (resid (Q :resid)))
```

```

(fn (x)
  (join
    (times (p x) (m1 x))
    (times (resid (Q :unit) (p x)) (m2 x))))))

(defn mutate-plus (Q m delta)
  (let ((join (Q :join)))
    (fn (x) (join (m x) (delta x)))))

(defn mutate-times (Q m delta)
  (let ((times (Q :times)))
    (fn (x) (times (m x) (delta x)))))

```

These definitions eliminate ad-hoc tree-rewriting for crossover/mutation and instead leverage the quantale’s built-in $\oplus, \otimes, \multimap$. This yields a uniform, algebraic GA in which all variation operators are instances of the same underlying quantale calculus.

36.2 MeTTa-like Pseudocode for GA Evolution of Weak CoDD Models

We can then give quite elegant pseudocode for a GA evolving CoDD programs to identify weak models in data.

```

;;-----
;; 1. Import core quantale & weakness library
;;-----
(import Quantales.Core
  Quantales.Weakness
  Quantales.Crossover
  Quantales.Mutation)

;;-----
;; 2. Define the CoDD quantale
;;   - q-join implements quantale join (e.g. max/union)
;;   - q-times implements quantale product (e.g. composition/intersection)
;;   - q-unit is the unit for q-times
;;   - q-resid is the residuation (implication) operator
;;-----
(define-quantale codd-quantale
  Quantales.Core/q-join ;; q-join
  Quantales.Core/q-times ;; q-times
  Quantales.Core/q-unit
  Quantales.Core/q-resid)

;;-----

```



```

;; 3. Random model generator: produce m : X -> Q
;;-----
(defn random-model (domain Q)
  (let ((j (Q :join)))      ;; to combine random component values
    (fn (x)
      ;; pick a random Q-element, e.g. q-unit or a small random multiple
      (Quantales.Core/q-times Q
        (Q :unit)
        (rand))))          ;; rand in Q sampled

;;-----
;; 4. Fitness: weakness w(m)
;;   w(m) = q-join over x of [ phi(x) q-times m(x) ]
;;-----
(defn fitness (m phi)
  ;; 'weakness' uses q-times for phi(x)*m(x) and q-join to aggregate
  (weakness codd-quantale phi m))

;;-----
;; 5. Crossover & Mutation using quantale ops
;;   - join-crossover: m1 join m2 pointwise
;;   - mutate-plus:    m join delta pointwise
;;-----
(defn crossover (m1 m2)
  ;; uses q-join
  (crossover-join codd-quantale m1 m2))

(defn mutate (m)
  (let ((delta (random-model (domain m) codd-quantale)))
    ;; uses q-join for mutation
    (mutate-plus codd-quantale m delta)))

;;-----
;; 6. Initialize population of M models
;;-----
(defn initialize-population (domain M)
  (for i in (range M)
    (random-model domain codd-quantale)))

;;-----
;; 7. Select top K by weakness
;;-----
(defn select-top (pop phi K)
  (let ((scored (map (fn (m) [m (fitness m phi)]) pop)))
    (map first
      (take K

```

```

        (sort-by second scored :descending))))))

;;-----
;; 8. Reproduce next generation of size M
;;   Crossover + mutation use quantale ops
;;-----
(defn reproduce (parents M)
  (let ((offspring '()))
    (while (< (length offspring) M)
      (let ((m1 (choice parents))
            (m2 (choice parents))
            (child (crossover m1 m2)))
        (push (mutate child) offspring)))
    offspring))

;;-----
;; 9. GA loop: evolve for G generations
;;   Selection & fitness use quantale weakness
;;-----
(defn evolve (domain phi M K G)
  (let ((pop (initialize-population domain M)))
    (loop ((gen 1) (population pop))
      (if (> gen G)
        population
        (let ((parents (select-top population phi K))
              (children (reproduce parents M)))
          (recur (inc gen) children))))))

;;-----
;; 10. Run GA on dataset
;;-----
(def domain (keys dataset))
(def phi (fn (x) ((dataset x) :phi)))
(def final-population (evolve domain phi 100 20 50))

```

37 Conclusion

We have developed a unified categorical theory of *weakness* as an Occam-style simplicity measure, showing how quantale-enriched categories recover and generalize classical notions of simplicity in various fields – from probabilistic weakness and SVM margins to logical entropy, graphentropy, and quantum multiperspectival priors.

If you have read this treatise through from the start, it's been a long and somewhat winding journey through a lot of rather interesting and varied territory. We have linked weakness to description-length bounds in Combinatorial

Distinction Dags, formulated weakness-guided inference control for Probabilistic Logic Networks, and interpreted predictive-coding and information-geometric updates as direct weakness minimization via the McBride derivative. We proved a schema theorem for CoDD genetic algorithms, constructed a CoDD-weakness analogue of AIX^{tl} ("WAIXI") with universality and regret-vanishing guarantees, and demonstrated efficient implementation of a multiresolution weakness transform in graph databases (e.g. MORK). Among many other things....

Given the conceptual breadth of Occam's Razor as a concept, and the mathematical generality of quantale weakness as a formal instantiation of Occam's Razor, the applications are both wide and deep and this book could have been either much shorter or much longer...

Looking ahead, several directions stand out:

- **Deepening the mathematics.** Further development of the theory of weakness—particularly in noncommutative, multiperspectival, and categorical-analytic settings—can yield new insights into Occam's Razor and its relation to measure-theoretic and homotopical notions of complexity.
- **Hyperon AGI applications.** Embedding weakness as a first-class quantale in the Atomspace/MORK store and MeTTa interpreter promises tighter cognitive synergy in PRIMUS, unifying symbolic, probabilistic, evolutionary, neural, and quantum modules under a single simplicity objective.
- **Quantum computing and hybrid systems.** The extension of weakness to operator quantales and quantum AIX^{tl} opens paths to principled priors in quantum machine learning and decision making, as well as to new hybrid classical-quantum AGI architectures.
- **Cognitive science and consciousness.** Weakness-based integrated-information measures and cognitive-synergy functors suggest a formal bridge between category-theoretic simplicity and phenomenological richness. Developing these notions may advance mathematical models of attention, awareness, and consciousness.
- **Practical algorithms and benchmarks.** Implementing and evaluating weakness-driven predictive coding, wavelet-like decompositions, and natural-gradient updates across real-world tasks—ranging from language modeling to robotics—will test and refine the theory in grounded settings.

By continuing to develop the mathematical, computational, and empirical aspects of quantale weakness theory, we aim to forge new connections between simplicity theory, cognitive science and AGI algorithmics, ultimately guiding the construction of more efficient, interpretable, and general AI systems as well as perhaps gaining new understanding into biological minds.

References

- [Ahm14] Arif Ahmed. Evidential decision theory and the win-win solution. *Mind*, 123(490):451–474, 2014.
- [Ben25] Michael T Bennett. How to build conscious machines, May 2025.
- [BLM96] Paul Busch, Pekka Lahti, and Peter Mittelstaedt. *The Quantum Theory of Measurement*. Springer, 1996.
- [BS12] John C. Baez and Mike Stay. Algorithmic thermodynamics. *Mathematical Structures in Computer Science*, 22(5):771–787, 2012.
- [CL20] Yifan Chen and Wuchen Li. Optimal transport natural gradient for statistical manifolds with continuous sample space. *CoRR*, abs/1805.08380, 2020.
- [CV95] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995.
- [Dai09] Wei Dai. Updateless decision theory. LessWrong discussion post, 2009. URL: <https://www.lesswrong.com>.
- [DH16] David Darais and David Van Horn. Programming with galois connections. In *Proceedings of the 43rd ACM SIGPLAN–SIGACT Symposium on Principles of Programming Languages*, pages 616–629, 2016.
- [Ell13] David Ellerman. An introduction to logical entropy and its relation to shannon entropy. *International Journal of Semantic Computing*, 7(1):33–79, 2013.
- [F1R25] F1R3FLY.io. Mettacycle architecture proposal. Yellow paper, F1R3FLY-io, 2025.
- [GBD⁺23] Ben Goertzel, Vitaly Bogdanov, Michael Duncan, Deborah Duong, Zarathustra Goertzel, Jan Horlings, Matthew Ikle’, Lucius Greg Meredith, Alexey Potapov, Andre’ Luiz de Senna, Hedra Seid Andres Suarez, Adam Vandervorst, and Robert Werko. Opencog hyperon: A framework for agi at the human level and beyond, 2023.
- [GDD⁺20] Ben Goertzel, Mike Duncan, Debbie Duong, Nil Geisweiller, Hedra Seid, Abdulrahman Semrie, Man Hin Leung, and Matthew Iklé. Embedding vector differences can be aligned with uncertain intensional logic differences. *CoRR*, abs/2005.12535, 2020.
- [GL25a] Ben Goertzel and Ruiting Lian. From metamo to openpsi: Embodying abstract motivational principles in concrete agi systems. In *Proceedings of the Twenty-Fifth International Conference on Artificial General Intelligence (AGI-25)*. Springer, 2025.

- [GL25b] Ben Goertzel and Ruiting Lian. Metamo: A robust motivational framework for open-ended agi. In *Proceedings of the Twenty-Fifth International Conference on Artificial General Intelligence (AGI-25)*. Springer, 2025.
- [Goe08] Ben Goertzel. *Probabilistic Logic Networks: A Comprehensive Framework for Uncertain Inference*. Springer, 2008.
- [Goe10] Ben Goertzel. Infinite-order probabilities and their application to modeling self-referential semantics. In *Proceedings of the 2010 International Conference on Advanced Intelligence (ICAI-10)*, Beijing, China, 2010.
- [Goe13] Ben Goertzel. A mind-world correspondence principle. In *2013 IEEE Symposium on Computational Intelligence for Human-like Intelligence (CIHLI)*, pages 68–73, 2013.
- [Goe14] Ben Goertzel. Characterizing human-like consciousness: An integrative approach. In *Proceedings of the Fifth Annual International Conference on Biologically Inspired Cognitive Architectures (BICA-14)*, volume 41 of *Procedia Computer Science*, pages 152–157. Elsevier, 2014.
- [Goe19] Ben Goertzel. Distinction graphs and graphtropY: A formalized phenomenological layer underlying classical and quantum entropy, observational semantics and cognitive computation. *CoRR*, abs/1902.00741, 2019.
- [Goe20a] Ben Goertzel. Combinatorial decision dags: A natural computational model for general intelligence. *CoRR*, abs/2004.05268, 2020.
- [Goe20b] Ben Goertzel. Grounding occam’s razor in a formal theory of simplicity. *CoRR*, abs/2004.05269, 2020.
- [Goe20c] Ben Goertzel. Simplificational causal decision theory. Blog post, <https://multiverseaccordingtoben.blogspot.com/2020/06/simplificational-causal-decision-theory.html>, 2020. Accessed: 2025-07-09.
- [Goe21a] Ben Goertzel. The general theory of general intelligence: A pragmatic patternist perspective. *CoRR*, abs/2103.15100, 2021.
- [Goe21b] Ben Goertzel. Paraconsistent foundations for probabilistic reasoning, programming and concept formation. *CoRR*, abs/2012.14474, 2021.
- [Goe21c] Ben Goertzel. Paraconsistent foundations for quantum probability. *CoRR*, abs/2101.07498, 2021.

- [Goe21d] Ben Goertzel. Patterns of cognition: Cognitive algorithms as galois connections fulfilled by chronomorphisms on probabilistically typed metagraphs. *CoRR*, abs/2102.10581, 2021.
- [Goe25a] Ben Goertzel. Efficient quantum-safe homomorphic encryption for quantum computer programs. *CoRR*, abs/2504.21235, 2025.
- [Goe25b] Ben Goertzel. Homomorphic encryption of intuitionistic logic proofs and functional programs: A categorical approach inspired by composite-order bilinear groups. *CoRR*, abs/2503.05779, 2025.
- [Goe25c] Ben Goertzel. An information-theoretic approach to quantifying and creating emergent and coherent concept blends, 2025. Preprint.
- [GT25] Ben Goertzel and Hyperon Pattern Miner Team. Introduction. Online tutorial at <https://pm-tutorial-3.onrender.com/tutorials/introduction>, 2025. Accessed: 2025-07-09.
- [Hof85] Douglas R. Hofstadter. Superrationality: An answer to the prisoner’s dilemma. In *Metamagical Themas: Questing for the Essence of Mind and Pattern*, pages 153–162. Basic Books, New York, NY, 1985.
- [Hol75] John H. Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, 1975.
- [Hol90] Craig Stewart Holman. *Elements of an Expert System for Determining the Satisfiability of General Boolean Expressions*. Ph.d. thesis, Northwestern University, 1990. UMI Order No. ADG90-31921.
- [Hut05] Marcus Hutter. *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, 2005.
- [KCLS25] Akarsh Kumar, Jeff Clune, Joel Lehman, and Kenneth O. Stanley. Questioning representational optimism in deep learning: The fractured entangled representation hypothesis. *arXiv preprint arXiv:2505.11581*, 2025.
- [Ke25] Shujing Ke. Build type-based hierarchical named functional embeddings, 2025. Unpublished note.
- [KT98] Anil K. Kaufmann and R.-C. Tsaur. Fuzzy cross-entropy and divergence measures. *Fuzzy Sets and Systems*, 96(1):105–123, 1998.
- [Loo06] Moshe Looks. *Competent Program Evolution*. Ph.d. thesis, Washington University in St. Louis, 2006.
- [Loo10] Moshe Looks. A critique of schmidhuber’s theory of mathematical aesthetics. In *Proceedings of the 3rd Conference on Artificial General Intelligence*, pages 211–222, 2010.

- [LT72] Andrea De Luca and Settimo Termini. A definition of non-probabilistic entropy in the setting of fuzzy sets theory. *Information and Control*, 20(4):301–312, 1972.
- [Mac25] MacDonald. Commutativity as a design principle for continual learning. Unpublished research report, 2025.
- [McB01] Conor McBride. The derivative of a regular type is its type of one-hole contexts. Technical report, Unpublished manuscript, 2001. Available online at <https://strictlypositive.org/diff.pdf>.
- [Mer25] Greg Meredith. Metta-Calculus. <https://github.com/leithaus/rho4u/blob/main/metta-calculus/metta-calculus.pdf>, 2025. Accessed: 2025-07-08.
- [MP94] S. Mitra and S. K. Pal. Mutual information in fuzzy joint relations. *International Journal of Fuzzy Sets and Systems*, 27(2):117–128, 1994.
- [nL02] Pedro Larra naga and José A. Lozano. *Estimation of Distribution Algorithms: A New Tool for Evolutionary Computation*. Springer, 2002.
- [Pat98] Anna L. Patterson. *Implicit Programming and the Logic of Constructible Duality*. PhD thesis, University of Illinois at Urbana–Champaign, January 1998.
- [Pea00] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2000.
- [Pei92] Charles Sanders Peirce. *The Essential Peirce, Volume 1: Selected Philosophical Writings (1867?1893)*. Indiana University Press, Bloomington, IN, 1992.
- [PISG09] Joel Pitt, Matthew Ikle, George Sellmann, and Ben Goertzel. Economic attention networks: Associative memory and resource allocation for general intelligence. In *Proceedings of the 2nd Conference on Artificial General Intelligence*, Advances in Intelligent Systems Research, pages 88–93. Atlantis Press, June 2009.
- [PMZM25] Benjamin Prada, Shion Matsumoto, Abdul Malik Zekri, and Ankur Mali. Bridging predictive coding and mdl: A two-part code framework for deep learning. *CoRR*, abs/2505.14635, 2025.
- [RB99] Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra?classical receptive?field effects. *Nature Neuroscience*, 2(1):79–87, 1999.
- [Ris78] Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

- [Ros03] Marshall B. Rosenberg. *Nonviolent Communication: A Language of Life*. PuddleDancer Press, Encinitas, CA, second edition, 2003.
- [Ros10] Paul Rosenbloom. Rethinking cognitive architecture via graphical models. Technical report, USC Institute for Creative Technologies, 2010. Draft submitted to Cognitive Systems Research, January 6, 2010.
- [Sch97] Jürgen Schmidhuber. Low-complexity art. *Leonardo*, 30(2):97–103, 1997.
- [Sch07] Jürgen Schmidhuber. Gödel machines: Self-referential universal problem solvers that self-improve. In *Proceedings of the 2nd Conference on Artificial General Intelligence*, pages 199–226, 2007.
- [SF14a] Javier Snaider and Stan Franklin. Modular composite representation. *Cognitive Computation*, 6(3):510–527, 2014.
- [SF14b] Javier Snaider and Stan Franklin. Modular composite representation. *Cognitive Computation*, 6(3):510–527, 2014.
- [She81] Rupert Sheldrake. *A New Science of Life: The Hypothesis of Formative Causation*. William Morrow & Company, New York, NY, 1981.
- [Smo13] Lee Smolin. *Time Reborn: From the Crisis in Physics to the Future of the Universe*. Houghton Mifflin Harcourt, Boston, MA, 2013.
- [Ton04] Giulio Tononi. An information integration theory of consciousness. *BMC Neuroscience*, 5(42):1–22, 2004.
- [War22] Jonathan Warrell. Formalizing higher-order probabilistic type systems in metta. Invited talk at the OpenCog Hyperon Workshop, 2022.
- [WHSX16] Andrew G. Wilson, Zhiting Hu, Ruslan Salakhutdinov, and Eric P. Xing. Deep kernel learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, pages 370–378, 2016.
- [Yag79] Ronald R. Yager. On the measure of fuzziness and negation, part i: Membership in the unit interval. *International Journal of General Systems*, 5(3):221–229, 1979.
- [Yud10] Eliezer Yudkowsky. Timeless decision theory. Technical report, Machine Intelligence Research Institute, 2010. Technical Report.

A Appendix: Review of CMMC and CoDD Computational Models

We review in this Appendix two computational models that play roles in the body of the paper. Both of these have been described in previous papers but in slightly different formal languages, and are here re-presented in a notation and framework more immediately amenable to incorporation in the weakness perspective.

A.1 The CMMC model of Computation

Next we review a "biological-ish" model of computation proposed in [Goe20b], which has been used among other purposes to formalize a species of Occam's Razor, and which we will show later on to connect very elegantly with weakness.

This can be thought of as a "dynamic Lego" mode of computation, wherein one pieces together a computational process from components by sticking them together in the ways they allow – but unlike ordinary lego blocks, once one block is appropriately connected to others, it can produce some output – including the option to produce additional blocks which then get stuck onto the existing ones in the growing structure.

These Legos are not necessarily embedded in any sort of ambient 3D space (though one *could* do it that way), so the Lego-blocks metaphor should not be overstretched. Rather, each of the simple building blocks involved can be thought of as a tiny "macro" that lives in a non-restrictive abstract space and takes some inputs and produces some output (which may be a value like a number or may be another combinator), without explicitly naming or manipulating variables. Instead of writing code that says "let x be this value and let y be that value, then compute $f(x,y)$ ", you simply have a collection of these little Legos/macros (often called *combinators*) and you glue them together in various ways. The result is a network of interconnected pieces. When you feed an initial input into that network, the combinators pass messages, combine them, split them, or ignore them, and eventually an output emerges.

Intelligent computation can be built up from these combinator operations in a highly flexible and self-adaptive way. Rather than thinking in terms of variables that store numbers or symbols, you think in terms of combinators that "already know" how to combine their inputs. This leads to a style of programming (and of reasoning) that is very compact yet powerful: you only needs a few primitive combinators to get started, and then every more complex function arises by combining them. (In theory you can build everything from a single well-chosen combinator, though in practice it's simpler to start with a slightly larger set than this.). One can extend this idea to richer data structures – graphs, metagraphs, or networked representations – and still remain within the same basic framework; basically, one builds the primitives of these data structures (e.g. nodes and edges) into the assumed primitive combinators.

To summarize in plain language:

- We replace the usual notion of *variables and assignments* with a collection of *self-sufficient modules* (combinators), each of which knows how to process its inputs.
- Computation proceeds by repeatedly *applying one module to another*—for instance, telling combinator *S* to combine two other combinators to form a new one, or telling *K* to pick one of its two inputs, and so on.
- Because each combinator is just a tiny piece of logic, when you assemble many of them together, you can build arbitrarily complex functions. There is no need for explaining how data “flows through variables”—you only need to specify how these small pieces interconnect.
- The extension of this combinatory approach to handle *graph-based data structures* and *self-modifying rewrite rules* yields what can be thought of as a sort of metagraph-based “algorithmic chemistry”, which is appealing for modeling highly dynamic AI systems.

With that intuitive picture in mind, we can now turn to a more formal discussion of the combinational model.

A.1.1 Basic Definitions

We begin by recalling some standard notions from *combinatory logic*. Afterwards we will indicate how to extend these ideas to more general graph-based structures.

Definition 11 (Combinator Terms). *Let $\mathcal{C} = \{S, K, I, \dots\}$ be a set of primitive combinators. We define the set of terms, denoted $\mathcal{T}$, as follows:*

- : $C \in \mathcal{C} \implies C \in \mathcal{T}$,
- : *If $M \in \mathcal{T}$ and $N \in \mathcal{T}$, then $(M N) \in \mathcal{T}$.*

In other words, every primitive combinator is a term, and if M and N are terms then the application of M to N , written $(M N)$, is also a term. We often omit parentheses by adopting the convention that application is left-associative. Thus XYZ stands for $((X Y) Z)$.

Definition 12 (Reduction Rules). *Each primitive combinator $C \in \mathcal{C}$ has an associated rule that specifies how to reduce certain application forms. The most common examples are:*

$$\begin{aligned} K x y &\longrightarrow x, \\ S x y z &\longrightarrow x z (y z), \\ I x &\longrightarrow x. \end{aligned}$$

(note that this uses curried notation in which a term first binds the argument immediately to its right, and then the combinator produced by this binding goes on to bind other things). In more detail:

- The K-combinator extracts its first argument: $(K x) y = x$.
- The S-combinator distributes its third argument to both x and y : $((S x) y) z = (x z) (y z)$.
- The I-combinator is the identity: $(I x) = x$.

A reduction is any finite sequence of applying these rewrite rules until no further rule can apply, at which point the term is in normal form.

Remark. It is a classical result that combinatory logic with just the S and K combinators is Turing-complete. That is, any computable function on (suitably encoded) data can be represented by some combinator term built from S and K alone.

A.1.2 Extending Graph-Based Computation to Metagraphs

For many AI applications (especially those involving knowledge graphs, concept networks, or self-referential rule sets), the simple tree-structured terms of classical combinatory logic can be awkwardly restrictive. It is desirable to generalize the notion of a “term” from a rooted tree to a *labeled directed graph* (sometimes called a “metagraph”). The key ideas here are:

Definition 13 (Metagraph). A metagraph $G = (V, E, \ell_V, \ell_E)$ consists of:

- A finite set V of vertices (or nodes).
- A set $E \subseteq V \times V$ of directed edges.
- A vertex-labeling function $\ell_V : V \rightarrow \Sigma_V$ that assigns each vertex a label drawn from some finite alphabet Σ_V .
- An edge-labeling function $\ell_E : E \rightarrow \Sigma_E$ that assigns each edge a label from a finite alphabet Σ_E .

Each vertex label might encode a primitive combinator (like S , K , or I), or it might encode some data symbol (an integer, string, etc.). Edge labels typically indicate the “role” of that edge in a larger computational pattern (for instance, whether it represents a function-argument link, a type annotation, or a pointer to a subgraph).

Definition 14 (Graph Rewrite Rule). A rewrite rule is a pair $(L \rightarrow R)$, where L (the left side) and R (the right side) are both metagraphs. Intuitively:

- L specifies a pattern to be matched somewhere in the “host” metagraph G .
- R specifies the subgraph that replaces the matched portion of G .

More formally, a rewrite step applies whenever there is a graph isomorphism ϕ from L into a subgraph of G . One removes the image $\phi(L)$ from G and then inserts a (possibly renamed) copy of R , connecting R to the rest of G in the same way that L was connected. The result is a new metagraph G' .

Definition 15 (Combinational Metagraph Model of Computation). *One then arrives at a "simple combinational computational model of computation" (CMMC) based on the following components:*

1. **Atoms and Molecules:** *An Atom is a single labeled vertex v (with no edges). A Molecule is a connected metagraph.*
2. **Agent State:** *At any moment, the agent's state is given by a (usually large) metagraph G . Nodes in G can represent data, primitive combinators, inference rules, goals, etc.*
3. **Reaction Rules:** *A finite (or countable) set $\mathcal{R}$ of graph rewrite rules, each of which matches a small subgraph of G and replaces it with a different subgraph. Some of these rules may correspond to inference steps ("if you see pattern P , replace it by pattern Q "), others may correspond to learning rules ("upgrade a rule's parameters based on error signals"), or goal-driven modifications ("when you reach a certain subgraph, trigger a new process").*
4. **Computation Dynamics:** *Computation proceeds by repeatedly applying any applicable rewrite rule from $\mathcal{R}$ to the current metagraph G . One may choose the rule to apply:*
 - Non-deterministically, leading to many possible rewrite paths.
 - Stochastically, with probabilities assigned to each rule.
 - Goal-driven, selecting rules that move the metagraph toward satisfying a particular objective.

Over time, G evolves through a search space of metagraphs, ideally converging (or oscillating) to configurations where the agent's computational or motivational goals are met.

Remark. *Note that in classical combinatory logic each term is a tree of nested applications (e.g. $(S K)(K I)$). In this CMMC model, those trees are replaced by arbitrary directed graphs. Thus, one can encode not only simple function application but also shared substructures, cyclic references, and self-modifying rule sets. This added flexibility is crucial for modeling rich cognitive processes such as meta-learning, self-reflection, and open-ended creativity.*

A.1.3 Formal Properties

1. Expressiveness. Since classical S, K -combinatory logic is Turing-complete, and since any tree can be seen as a special case of a graph, it follows that CMMC is also Turing-complete. Moreover, the presence of graph rewrites allows direct modeling of operations that are awkward in tree-based systems, such as merging identical subterms (graph unification) or explicitly encoding cyclic dependencies.

2. Self-Modification. We may include *meta-rules* in the rule set $\mathcal{R}$ that match and rewrite portions of $\mathcal{R}$ itself (since $\mathcal{R}$ can be encoded as a subgraph of G). Formally, let $\mathcal{R}(G)$ denote the subgraph of G that encodes the current set of rewrite rules. A meta-rule might have the form

$$(L_{Rmrule} \longrightarrow R_{Rmrule}),$$

where L_{Rmrule} matches some pattern within $\mathcal{R}(G)$, thereby altering or extending the agent's own inference/learning strategies at runtime. This ability to rewrite one's own rules is the cornerstone of self-reflection and meta-learning.

3. Confluence and Termination. In general, graph rewrite systems need *not* be confluent (different orders of applying rules lead to different final graphs) and need *not* terminate (one can keep rewriting forever). For many AI applications, nontermination is acceptable (the agent runs indefinitely). Confluence may or may not be desired: if one wants *deterministic* outcomes, one must restrict $\mathcal{R}$ to a confluent subset. If one wants *divergent creative exploration*, permitting non-confluence can be advantageous.

A.1.4 Example: Simple Graph-Based Combinator Application

To illustrate concretely how this sort of combinator computation works, consider the following minimal metagraph G :

- **Nodes:**

$$V = \{S, K, X, Y, U, V\},$$

where S, K are *combinator* nodes and X, Y, U, V are *data* nodes.

- **Edges:** Four directed edges indicating the combinator-data matches:

$$S \rightarrow X, \quad S \rightarrow Y, \quad K \rightarrow U, \quad K \rightarrow V.$$

Here each arrow $C \rightarrow D$ means -combinator C will match data node D in its two-argument pattern.

In this setup:

1. The rewrite rule for S , say $R_S : (P_S \rightarrow T_S)$, will match the subgraph consisting of node S together with its two outgoing edges to X and Y . After matching, it replaces that pattern by its template T_S .
2. Likewise, K 's rule R_K matches $K \rightarrow U$ and $K \rightarrow V$ and rewrites accordingly.
3. No graphics are needed: the entire combinational application is captured by listing nodes and edges, and by explaining that $C \rightarrow D$ - encodes a combinator-pattern match.

This example is trivial and lacks sophisticated AI-related metagraph rewrite operations, which would make the example substantially larger, but may be valuable in illustrating how the basic rewrite dynamics works.

A.1.5 Summary of Formal Treatment

To reiterate the essence of the CMMC model, then:

- We start with *primitive combinator nodes* (for example S , K , I) and *data nodes* (integers, strings, etc.) all embedded in a labeled directed graph.
- We have a (possibly dynamic) set of *graph rewrite rules* that specify how small matched subgraphs are replaced by other subgraphs. These rules encode:
 - *Combinator reduction* (e.g. S , K reductions).
 - *Learning or inference steps* (e.g. modifying weights or rule parameters based on local errors).
 - *Goal-driven transformations* (e.g. -if you reach a certain configuration, insert a new subgoal-).
 - *Meta-rules* that rewrite other rules, enabling self-reflection.
- Computation is simply *repeatedly apply any applicable rule* to the current metagraph, yielding a (possibly nonterminating, possibly nondeterministic) sequence of graph states.
- This framework generalizes ordinary combinatory logic (which uses tree-structured terms) to arbitrary graphs, allowing direct representation of shared substructures, cyclic references, and self-modifying code.
- Standard properties like Turing-completeness, confluence, and termination must be revisited in the context of graph rewrites, but the fundamental expressive power follows from the fact that tree terms embed into graphs, and S , K combinators are already Turing-complete.

Putting these pieces together, we arrive at a unified, graph-based generalization of classical combinatory logic, enriched with the ability to rewrite its own rewrite rules, thereby capturing powerful forms of self-organizing and self-modifying computation.

A.2 The CoDD Model: A Specific Example of CMMC with Interesting Properties

The CMMC envisions a system as a graph where both *data* and *combinators* (small rewrite modules) are nodes, and computation proceeds by matching certain subgraph patterns and rewriting them into new subgraphs. In this abstract model:

- Each *combinator* is defined by a pair $(P \rightarrow T)$, where P is a pattern subgraph and T is a template for replacement.
- When the pattern P appears in the current graph, it is replaced by T , producing new data or further computational structure.

- Because combinator definitions themselves are stored as parts of the graph, the system can include *meta-rules* that modify combinator definitions at run time-enabling self-modification and learning.

In [Goe20a] the *CoDD* (*Combinatory and Data-Driven Dynamics*) model of computation is presented as a specific interesting *instance* of this general combinational model. In CoDD:

- Every node is labeled either by a *data symbol*, a *combinator name*, or the special label **rule** (indicating a rule encoding).
- Each *combinator* C has its rewrite rule ($P_C \text{Rightarrow} T_C$) stored under a node labeled **rule**, so that matching and rewriting follow the general CMMC scheme.
- Meta-rules are simply other **rule**-labeled nodes whose patterns match these combinator rule subgraphs, allowing CoDD to modify its own combinator set.

CoDD *implements* CMMC in a concrete way, specifying exactly how to encode patterns, templates, and meta-rules in a metagraph.

A.2.1 Specifying the CoDD Model

We now describe CoDD formally:

Definition 16 (CoDD Metagraph). *A CoDD metagraph is again a quadruple*

$$G = (V, E, \ell_V, \ell_E),$$

with the following conventions on labels:

- $\Sigma_V = \Sigma_D \cup \Sigma_C \cup \{\mathbf{rule}\}$, where:
 - Σ_D is the set of data labels (e.g. numeric constants, symbolic tokens).
 - Σ_C is the set of combinator names (e.g. *plus*, *mult*, or higher-order combinators).
 - The label **rule** identifies a node that encodes a rewrite rule.
- Σ_E is the set of edge labels, including roles such as *arg1*, *arg2*, *result*, *pattern*, and *template*.

Definition 17 (Encoding a Combinator Rule in CoDD). *For each combinator $C \in \Sigma_C$, CoDD stores its rewrite rule $R_C = (P_C \text{Rightarrow} T_C)$ as follows:*

- Create a node $r_C \in V$ labeled **rule**.
- Add an edge ($r_C \xrightarrow{\text{comb}} v_C$) where v_C is a node labeled C . This attaches the rule to its combinator name.

- Under r_C , attach two distinguished subgraphs: one labeled **pattern** whose structure is isomorphic to P_C , and one labeled **template** whose structure is isomorphic to T_C . Variables in P_C are nodes labeled with question-marks (e.g. $-x$, $-y$).

In this way, R_C is itself represented as a subgraph rooted at r_C .

Definition 18 (CoDD Pattern Matching). *To apply a combinator C in CoDD:*

1. Locate the node r_C labeled **rule** and find the **pattern** subgraph P_C under it.
2. Find an injective embedding $\sigma : P_C \rightarrow S \subseteq G$ that matches node labels (instantiating variables) and edge labels.
3. Record variable bindings $\sigma(-x) \mapsto v_x$, etc.

If such S exists, R_C applies to S .

Definition 19 (CoDD Rewrite Step). *Given a match $\sigma : P_C \rightarrow S$:*

1. Remove all nodes and edges in S from G .
2. Insert a copy of the **template** subgraph T_C under r_C , instantiating each variable node $-x$ by $\sigma(-x)$.
3. Reconnect any edges in G that formerly pointed to nodes in S to the corresponding nodes in the instantiated T_C , according to edge labels in T_C .

Denote this rewrite by

$$G \xrightarrow{r_C} G'.$$

Definition 20 (CoDD Meta-Rule). *A CoDD meta-rule M is similarly encoded as a node $m \in V$ labeled **rule**. Its **pattern** subgraph P_M is designed to match another rule encoding (i.e. a subgraph rooted at some $r_{C'}$). When P_M matches that rule encoding, apply the rewrite to replace the matched subgraph under $r_{C'}$ by the **template** T_M , thereby modifying the combinator C' 's rule definition.*

Definition 21 (CoDD Computation). *A CoDD computation is any sequence of metagraphs*

$$G_0 \xrightarrow{R_1} G_1 \xrightarrow{R_2} G_2 \xrightarrow{R_3} \dots,$$

where each R_i is either a combinator rule node r_C or a meta-rule node m . If at some step no rule applies, the computation halts in normal form.

A.2.2 CoDD as a Special Case of CMMC

We now list explicitly how each component of the general CMMC model is instantiated by CoDD:

1. **Node Labels:**

- In the general model, Σ_V may include arbitrary combinator names and data labels.
- CoDD specifies $\Sigma_V = \Sigma_D \cup \Sigma_C \cup \{\mathbf{rule}\}$. Every combinator rule is rooted at a **rule**-labeled node.

2. Encoding of P_C and T_C :

- In the abstract model, each combinator C has an associated pattern P_C and template T_C .
- CoDD stores P_C and T_C as subgraphs under the node r_C . Variables in P_C are explicit nodes labeled with question-marks, and the structure of T_C indicates how matched data variables propagate.

3. Pattern Matching:

- The general CMMC model requires an injective graph isomorphism $\sigma : P_C \rightarrow S \subseteq G$.
- CoDD implements exactly this: find a subgraph $S \subseteq G$ matching the **pattern** subgraph under r_C .

4. Rewrite Step:

- Abstractly, remove subgraph S and insert T_C .
- CoDD removes $S \subseteq G$ and instantiates the **template** subgraph T_C , reconnecting external edges according to the labels in T_C .

5. Meta-Rules:

- In the general model, meta-rules match and replace rule definitions.
- CoDD's meta-rules are nodes labeled **rule** whose **pattern** subgraphs match the encoding of another $r_{C'}$. Applying a meta-rule rewrites that encoding-modifying combinator definitions at run time.

6. Computation Sequence:

- Both models describe a computation as a sequence of metagraphs connected by rewrite steps.
- CoDD's computation $G_0 \rightarrow G_1 \rightarrow \dots$ follows the same pattern, with each step labeled by the node r_C or m whose rule is applied.

7. Universality:

- CMMC can simulate Turing-complete systems if combinators are chosen appropriately (e.g. to simulate S and K combinators).
- CoDD inherits that universality, since one can encode the same combinators and their rules under **rule**-nodes.

None of this tells us why CoDD might be especially interesting compared to other instantiations of CMMC – but our burden in this section has been just to explain what CMMC is. The relation between CoDD and weakness is covered in Section 2.3.

A.2.3 Universality of CoDD

As a footnote, if one wanted to construct a more detailed demonstration that CoDD has universal computational power— means it can simulate any Turing machine — one would proceed roughly as follows (leveraging the classic tape-based TM construct):

- First, one encodes the Turing-machine tape, head position, and finite-state control as nodes and edges in a CoDD metagraph.
- Each Turing transition becomes a single CoDD rewrite rule that updates the state, writes a new symbol, and moves the head.
- By repeatedly applying these rewrite rules, CoDD mimics each step of the Turing machine.

Since a Turing machine can compute anything a conventional computer program can, CoDD can also compute it. In other words, any computable function can be implemented in CoDD by writing the appropriate metagraph and rewrite rules.

A.2.4 Elegant Normal Form for CoDD Programs

Finally, we ask: What would an ENF for CoDD look like?

An *Elegant Normal Form* (ENF) for a Combinational Decision DAG (CoDD) program is a unique, minimal DAG representation satisfying the following conditions [Hol90]:

1. Primitive Nodes.

- Every leaf is labeled by an `Atom(a)` for some atomic test or action `a`.
- Every internal node is labeled `Combine_i(v_1, ..., v_n_i)`, where `ast_i` is the i th combinator of arity n_i and each v_j is a pointer to a child node.

2. DAG Structure.

- The graph is acyclic, with a single designated *root* node.
- Each node has a unique integer ID, and for every edge $u - v$ we require $ID(u) < ID(v)$.

3. Maximal Sharing.

- If two subgraphs rooted at nodes u and v are isomorphic in labels and structure, they are represented by the same node (i.e. $u = v$).

4. Canonical Ordering.

- For each combinator `ast_i` that is commutative, its child pointers $(v_1, \dots, v_{n_i})$ are sorted by ascending node ID.
- For non-commutative combinators, the ordering of children is as in the original program.

Conversion to ENF. Given any CoDD program, its ENF is obtained by:

1. *Unfold* the CoDD into a tree of combinator applications plus atoms.
2. *Assign* provisional node IDs by a postorder traversal.
3. *Merge* identical subtrees: traverse bottom-up, replacing isomorphic subtrees by a single shared node.
4. *ReID*: renumber nodes so that each node's ID is one more than the maximum of its children's IDs.
5. *Sort children* of each commutative combinator by child ID.

Uniqueness. By maximal sharing, ID-ordering, and canonical argument sorting, the ENF is *unique* for each equivalence class of CoDD programs under renaming and associative restructuring of commutative combinators.

Benefits.

- Guarantees a one-to-one correspondence between CoDD semantics and their DAG representation.
- Enables efficient equality tests by simple pointer comparison.
- Minimizes storage by maximal subgraph reuse.
- Provides a clear measure of program complexity via the number of nodes.